The Holographic Alaya-Field Framework

Worlds selected by physical access

Complete Collected Volume

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Introduction

Which relations constitute a world that an apparatus can read and sustain? The HAFF papers approach this question through channels, couplings, records and interventions. They distinguish a reconstructed description from a change in the matter that carries it. The gravitational proposal is therefore tested twice: through consistency of geometric data and through sources produced by the interactions that change access.

Read A and B together to establish the operational objects, then E for their measurement realization. C and F develop agency and temporal direction. D addresses the source and geometry interfaces; G returns those interfaces to revisable organization. The postscript discusses what persistence and participation mean for a finite system.

Reading map

HAFF-A *Physical Access and Correlation Geometry: The Holographic Alaya–Field Framework.* Physical channels and their record laws distinguish operational access from an abstract algebra. One initial preparation and fixed reconstruction rule can give non-isometric correlation geometries after different physical reductions.

HAFF-B *Coupling, Records, and Stable Relations: Formation, Protected Content and the Lifetime of a Geometry.* Couplings select what survives. Protected content, slowly decaying relations and physically acquired records have distinct roles in maintaining a description.

HAFF-C *Causation, Agency, and Existence: Historical Control and Maintained Intervention.* Agency is tested by interventions allowed by a retained history, including the maintenance needed for those interventions to remain available.

HAFF-D *Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return.* Selection reaches gravitational response through an actual source. The paper joins finite physical returns to modular constraints and develops a common-action conjecture for the missing general mechanism.

HAFF-E *Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout.* An implementable measurement, a stable pointer and a public record need not coincide. Explicit permissions and instruments determine their connections.

HAFF-F *Records, Dissipation and Intervention: Conditions for a Temporal Direction.* A temporal direction requires an appropriate record and dissipation account. Causal order, entropy production and an agent's stored history are not interchangeable.

HAFF-G *Internal Selection and Revisable Organization: How History Changes Permissions and Physical Sources.* History can revise the operations a system selects. A target operation remains distinct from its larger executor; physical programming and source changes have costs.

HAFF-PS *Postscript: Persistence, Participation and an Unfinished Self.* Persistence and participation are interpreted through the finite constructions, without equating a useful self-model with a complete self-description.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.

PHYSICAL ACCESS AND CORRELATION GEOMETRY: THE HOLOGRAPHIC ALAYA–FIELD FRAMEWORK

A PREPRINT

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ABSTRACT

Physical access has a resolution that an observable algebra does not specify. We describe a reduction by its channel and distinguish its implementable effects from their linear span, their generated algebra and the observables stable under the dynamics. For qubit reductions we characterize equality of output entropy on every pure input. The nonzero branch is separated by every mixed input; the zero branch includes operationally incomparable channels with equal entropy on every input. A nine-probe finite-copy test bounds violations. We then connect physical implementation to geometric certification in one protocol: local unitary couplings reduce a fixed three-qubit GHZ state, further local collisions produce environmental records, and measurements on those records estimate the pair correlations. Calibrated readout errors and sampling errors propagate to confidence intervals for a fixed shortest-path metric. Two noise allocations with the same expected flip count produce non-isometric metrics, and a finite experiment separates their diameters without direct access to the source qubits. The reconstruction rule, source preparation and coupling architecture remain explicit inputs. The results show how finite measurement permissions, rather than an algebra label or exact entropy access, determine which correlation geometry can be established.

1 Introduction

Decoherence explains why interference between some degrees of freedom stops being visible [1]. Entanglement-based approaches to quantum gravity read geometric data from correlations between regions, most sharply through the Ryu–Takayanagi formula [2, 4, 3]. Both start after the system has been divided: into system, apparatus and environment in the first case, into boundary regions or tensor factors in the second. A generic Hilbert space carries no canonical division [5, 6, 7], and the observables that define a division can be chosen so that a given pure state has any Schmidt decomposition one likes [8]. Any geometry read from correlations therefore depends on how the state is reduced.

A reduction can preserve the algebra generated by accessible observables while changing the accuracy of every readout. This is the issue studied here. We first distinguish channel, effect set and stable structure (Section 2), then determine when qubit reductions are indistinguishable by entropy and how finite data detect a violation (Section 3). Section 4 connects a local interaction to an actual environmental record, a measured correlation and a certified finite geometry. Appendix A gives the factorization background and the precise typicality statement.

The geometry in the worked protocol is a metric on three sites under a declared rule. Its non-isometry is proved from the source dynamics and certified through the record dynamics. A Lorentzian metric requires further physical and reconstruction input. Keeping that distinction allows the finite experiment to answer a real part of the world-structure problem: which geometric differences survive the operations a reader can actually perform?

We call the source structure the Alaya-Field, a term taken from Yogācāra philosophy, where it names the storehouse consciousness. Here it is used only as a technical name for the triple $(\mathcal{H}_U, \mathcal{A}_U, |\Omega\rangle)$, with $\mathcal{A}_U$ a von Neumann algebra on $\mathcal{H}_U$ and $|\Omega\rangle$ cyclic and separating, so that the pair is in standard form. The point of the name is the absence of a canonical factorization, not a particular choice of state. In an algebraic quantum field theory $\mathcal{A}_U$ may be a Type III_1

factor [9]. Such a factor has no normal trace, so its states have no intrinsic density operators and no von Neumann trace entropy. The factorization, Haar-measure, partial-trace and entropy results below therefore use their own finite-dimensional representations. There a pure state of the full matrix algebra is cyclic but not separating, so these finite models do not realize every assumption placed on the infinite source structure, and none of their entropy statements is asserted in the Type III setting. The passage from entropy or connectivity data to a spacetime metric is a separate, model-dependent input throughout.

2 Coarse-graining structures, algebras and channels

2.1 Definition

An observable algebra does not by itself reinstate a tensor factorization: operators can act irreducibly on $\mathcal{H}_{\mathrm{total}}$, and in algebraic quantum field theory local algebras are assigned to regions without any global tensor product [9]. A coarse-graining here is meant to be realized by a physical interaction, not chosen by an agent.

Definition 2.1 (Coarse-graining structure)**.** *A coarse-graining structure is a pair* $\mathbf{c} = (\mathcal{A}_{\mathrm{sel},\mathbf{c}}, \Phi_{\mathbf{c}})$*, where*

1. $\mathcal{A}_{\mathrm{sel},\mathbf{c}} \subset \mathcal{B}(\mathcal{H}_U)$ *is a selected unital* $*$*-subalgebra used to define the effective representation. It is not identified with Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set [26]; a physical model must state any map between them.*
2. *In the finite-dimensional results,* $\Phi_{\mathbf{c}}$ *is a completely positive trace-preserving (CPTP) map on states that implements an operational reduction; its Heisenberg dual* $\Phi_{\mathbf{c}}^*$ *is unital and completely positive. Partial trace over a specified factor and channels induced by a conditional expectation are examples. In a general von Neumann setting the corresponding object is a normal channel with a normal unital completely positive dual, and no entropy result below is claimed for that extension.*

No theorem guarantees a unique maximal observable algebra for a given physical system without additional structure. Several admissible algebras can coexist, and for a fixed algebra several channels can implement different losses of information, such as tracing out inaccessible degrees of freedom or effective decoherence. The two entries of $\mathbf{c}$ must therefore be related explicitly.

2.2 The algebra generated by a channel

For a channel $\Phi : \mathcal{B}(\mathcal{H})_* \to \mathcal{B}(\mathcal{K})_*$ put

$$\mathcal{S}_\Phi = \operatorname{ran} \Phi^*, \qquad \mathcal{A}_\Phi = C^*(\mathcal{S}_\Phi).$$

$\mathcal{S}_\Phi$ is a unital self-adjoint subspace, not necessarily closed under products. The effects that the channel actually reads out are $\Phi^*([0, I])$, which is in general smaller than the set of positive contractions in $\mathcal{S}_\Phi$. We call $\mathbf{c}$ weakly compatible if $\mathcal{S}_{\Phi_{\mathbf{c}}} \subseteq \mathcal{A}_{\mathrm{sel},\mathbf{c}}$ and minimally compatible if $\mathcal{A}_{\mathrm{sel},\mathbf{c}} = \mathcal{A}_{\Phi_{\mathbf{c}}}$.

Proposition 2.2 (Algebras and post-processing)**.**

1. *If two states agree on* $\mathcal{S}_\Phi$*, in particular if they agree on* $\mathcal{A}_\Phi$*, then their outputs under* Φ *coincide.*
2. *If* $\Phi_2 = \Lambda \circ \Phi_1$ *for a channel* Λ*, then* $\mathcal{S}_{\Phi_2} \subseteq \mathcal{S}_{\Phi_1}$ *and* $\mathcal{A}_{\Phi_2} \subseteq \mathcal{A}_{\Phi_1}$*.*
3. *The converse of (2) fails. For the depolarizing channels* $D_p(\rho) = p\rho + (1-p)I/2$ *on a qubit with* $0 < p < q \leq 1$*, both* $\mathcal{A}_{D_p}$ *and* $\mathcal{A}_{D_q}$ *equal the full matrix algebra,* $D_p = D_{p/q} \circ D_q$*, and* D_q *is not a post-processing of* D_p*.*

Proof. (1) For any output operator Y, $\mathrm{Tr}[\Phi(\rho)Y] - \mathrm{Tr}[\Phi(\sigma)Y] = \mathrm{Tr}[(\rho - \sigma)\Phi^*(Y)]$, and $\Phi^*(Y) \in \mathcal{S}_\Phi$. (2) $\Phi_2^* = \Phi_1^*\Lambda^*$, so the range of Φ_2^* lies in the range of Φ_1^*, and generated algebras inherit the inclusion. (3) Both maps are invertible on the operator space, so their dual ranges are everything. The identity $D_{p/q}(D_q(\rho)) = p\rho + (1-p)I/2$ is direct. Two orthogonal pure inputs have outputs at trace distance p under D_p and q under D_q; a channel cannot increase trace distance, so no Λ satisfies $\Lambda \circ D_p = D_q$. □

Two reductions with the same algebra can thus have different operational resolution. An algebra label is not a substitute for the channel and its allowed post-processing. If a conditional expectation is chosen as the implementation, that is an additional structure, not a synonym for every CPTP reduction.

2.3 Matching the dynamical and measurement interfaces

Paper B's commuting-coupling dynamics gives $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$ in the joint eigenbasis [25]. At finite t all multipliers are nonzero: $\mathcal{A}_{\Phi_t} = \mathcal{B}(\mathcal{H})$. Its block algebra $\mathcal{B}_\pi$ selected by a stability window is therefore an auxiliary description of persistent capability. Unless it is the full algebra, $(\mathcal{B}_\pi, \Phi_t)$ is not even weakly compatible in Definition 2.1. The minimally compatible finite-time pair is $(\mathcal{B}(\mathcal{H}), \Phi_t)$, supplemented by the window, tolerance and $\mathcal{B}_\pi$. In the long-time limit the channel is instead the pinching $\mathcal{E}_{\pi_\infty}$ and $(\mathcal{B}_{\pi_\infty}, \mathcal{E}_{\pi_\infty})$ is minimally compatible. The finite description and its limiting algebra answer different questions.

A measurement channel has a different output. In Paper E's pointer-controlled collision, the source remains a system while an environmental fragment receives a conditional state $|e_z\rangle$, $z = 0, 1$ [26]. Its record channel is

$$\Gamma(\rho) = \sum_{z=0}^{1} \mathrm{Tr}(P_z\rho)|e_z\rangle\langle e_z|, \qquad \Gamma^*(F) = \sum_z \langle e_z|F|e_z\rangle P_z.$$

For overlap $0 < r = |\langle e_0|e_1\rangle| < 1$, the dual range spans the pointer algebra, but the implementable effects omit its sharp projections. The source dephasing channel and Γ arise from the same unitary, although their ranges, outputs and uses differ. The geometric protocol below uses the reduction to set the correlations and Γ to read them. It never replaces $\Gamma^*([0, I])$ by all positive contractions of its span. Persistence is relative to the specified dynamics: a record effect fixed by the recording interaction need not be fixed by the preceding noise interaction.

2.4 Refinement

Definition 2.3 (Refinement)**.** *For coarse-graining structures $\mathbf{c}_1, \mathbf{c}_2$ write $\mathbf{c}_1 \succeq \mathbf{c}_2$ if there is a CPTP map Λ from states on the output space of $\Phi_{\mathbf{c}_1}$ to states on the output space of $\Phi_{\mathbf{c}_2}$ with $\Phi_{\mathbf{c}_2} = \Lambda \circ \Phi_{\mathbf{c}_1}$. Then $\mathbf{c}_2$ is a further coarse-graining of $\mathbf{c}_1$.*

Proposition 2.4 (Order on classes)**.** *The relation $\succeq$ is reflexive and transitive but not antisymmetric. Writing $\Phi \sim \Psi$ when each is a post-processing of the other, $\succeq$ induces a partial order on equivalence classes.*

Proof. Reflexivity uses $\Lambda = \mathrm{id}$, transitivity uses composition of channels. Two different replacer channels, which output fixed states $\omega_1 \neq \omega_2$, simulate each other by a further replacement, so antisymmetry fails. On classes the relation is well defined because composing with a simulating channel does not change membership, and antisymmetry holds by construction. □

If the allowed post-processings are restricted, for example to local, energy-bounded or specified control operations, the restricted class must be closed under composition before the same quotient can be formed. The comparison of channels by approximate post-processing has an operational characterization through success probabilities of discrimination tasks [14]; we use only exact post-processing.

2.5 Mutual information under refinement

Proposition 2.5 (Local and global post-processing)**.** *For channels Λ_A, Λ_B acting on A and B separately,*

$$I(A' : B')_{(\Lambda_A \otimes \Lambda_B)(\rho)} \leq I(A : B)_\rho.$$

A global CPTP post-processing can increase mutual information: the channel that replaces every two-qubit input by a Bell state takes $|00\rangle$, with $I = 0$, to a state with $I = 2\ln 2$.

Proof. $I(A : B)_\rho = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, and $\Lambda_A \otimes \Lambda_B$ maps $\rho_A \otimes \rho_B$ to the product of the new marginals, so data processing for relative entropy gives the inequality. The Bell example is a direct computation. □

A requirement that mutual information decrease under refinement is therefore a theorem for local refinements and false for global ones. A geometric reading of mutual information must say which refinements it allows.

3 When do two reductions give the same entropies?

3.1 Generic separation

Theorem 3.1 (Generic separation for a fixed pair of channels)**.** *Let $\mathcal{H}$ and $\mathcal{K} = \mathcal{K}_A \otimes \mathcal{K}_{A^c}$ be finite dimensional, and let $\Phi_1 \neq \Phi_2$ be CPTP maps from states on $\mathcal{H}$ to states on $\mathcal{K}$. The pure inputs on which Φ_1 and Φ_2 agree form a proper*

real-algebraic subset of Haar measure zero. If, in addition, the real-analytic function

$$f_A(\Psi) := S(\mathrm{Tr}_{A^c}\,\Phi_1(|\Psi\rangle\langle\Psi|)) - S(\mathrm{Tr}_{A^c}\,\Phi_2(|\Psi\rangle\langle\Psi|))$$

is not identically zero on each connected component under consideration of the open set where both reduced outputs have full rank, then $f_A(\Psi) \neq 0$ for generic inputs in those components.

Proof. Pure-state projectors span the Hermitian operators, so $\Phi_1 - \Phi_2$ cannot vanish on all of them. The equality condition is polynomial in the real and imaginary parts of $|\Psi\rangle$, so its solution set is a proper real-algebraic subset and has Haar measure zero. On the full-rank domain the von Neumann entropy, and hence f_A, is real analytic, and a nonzero real-analytic function has a zero set of measure zero. □

The nondegeneracy hypothesis cannot be dropped: channels that differ by a unitary acting on A alone give different outputs with identical entropies on every input. Channel inequivalence alone therefore does not yield an entropic or geometric difference. The rest of this section decides, for qubit reductions, exactly when the hypothesis fails.

3.2 Qubit reductions: the criterion

A qubit channel acts on Bloch vectors as an affine map

$$\Phi_i :\ r \mapsto M_i r + t_i, \qquad i = 1, 2,$$

with M_i a real 3×3 matrix and $t_i \in \mathbb{R}^3$, subject to the complete-positivity conditions [15]. Put

$$A = M_1^T M_1 - M_2^T M_2, \qquad b = M_1^T t_1 - M_2^T t_2, \qquad c = \|t_1\|^2 - \|t_2\|^2.$$

The difference of the squared output Bloch lengths is

$$q(r) = \|M_1 r + t_1\|^2 - \|M_2 r + t_2\|^2 = r^T A r + 2b^T r + c. \tag{1}$$

The entropy of a qubit state is a strictly decreasing function of its Bloch length: $S = h\big((1+\|s\|)/2\big)$ with h the binary entropy in nats.

Theorem 3.2 (Equal entropy on all pure inputs)**.** *Φ_1 and Φ_2 give equal output entropy on every pure input if and only if there is a real number α with*

$$A = \alpha I, \qquad b = 0, \qquad c = -\alpha. \tag{2}$$

Proof. Equal entropy on pure inputs means $q(r) = 0$ on the unit sphere. Comparing $q(r)$ and $q(-r)$ gives $b^T r = 0$ for all unit r, so $b = 0$. A real symmetric quadratic form that is constant on the unit sphere is a multiple of the identity, so $A = \alpha I$ and then $c = -\alpha$. Conversely, (2) gives $q(r) = \alpha(\|r\|^2 - 1)$, which vanishes on the sphere. No rank condition on the outputs is needed, and pure outputs on the boundary are included. □

Together with positivity of the Choi matrices, (2) decides membership for any given pair. It is a decision rule, not yet a classification of all solutions into physically distinct normal forms.

3.3 The two branches

Proposition 3.3 (Zero branch)**.** *Let $X_i = [M_i \mid t_i]$ be the 3×4 matrix with t_i appended. If $\alpha = 0$, then $X_2 = OX_1$ for an orthogonal $O \in O(3)$. If $\mathrm{rank}\, X_1 \leq 2$, O can be chosen with $\det O = +1$, so Φ_2 is Φ_1 followed by a unitary conjugation of the output. If $\mathrm{rank}\, X_1 = 3$, O is unique; $\det O = +1$ again gives a unitary conjugation, and $\det O = -1$ gives a unitary conjugation composed with transposition.*

Proof. For $\alpha = 0$, (2) says exactly $X_1^T X_1 = X_2^T X_2$. Equal Gram matrices define an isometry between the column spaces of X_1 and X_2, which extends to an orthogonal map of $\mathbb{R}^3$. If the column space has dimension at most two, the extension can be composed with a reflection in an unused direction to make the determinant positive. Rotations of the Bloch ball are unitary conjugations, and the reflection $\mathrm{diag}(1, -1, 1)$ is transposition in the standard basis. □

Transposition is not completely positive, but both endpoints of a transposed pair can still be channels.

Example 3.4 (Equal entropies everywhere, no simulation either way)**.** *Take $M_1 = pI$, $M_2 = p\,\mathrm{diag}(1,-1,1)$ and $t_1 = t_2 = 0$. The first map is a depolarizing channel for $0 \le p \le 1$; by the conditions for unital qubit channels [16], the second is completely positive exactly when $0 \le p \le 1/3$. They give equal output entropy on every input, pure or mixed. For $0 < p \le 1/3$ neither is a post-processing of the other. Indeed both are invertible on the Bloch ball, so a post-processing Λ with $\Lambda \circ \Phi_1 = \Phi_2$ would have to act as $r \mapsto \mathrm{diag}(1,-1,1)r$ on an open set and hence everywhere; that map is transposition, whose normalized Choi matrix has smallest eigenvalue $-1/2$. The same argument excludes the reverse direction.*

Identical entropy functions on the whole input space therefore do not imply operational equivalence, and the two channels are incomparable in the order of Proposition 2.4.

Solutions with $\alpha \neq 0$ are not rare. Every pair of complementary qubit channels satisfies (2): if $V : \mathcal{H} \to \mathcal{K} \otimes \mathcal{E}$ is an isometry with $\mathcal{K}$ and $\mathcal{E}$ two-dimensional, $\Phi(\rho) = \mathrm{Tr}_{\mathcal{E}}\, V\rho V^\dagger$ and $\Phi^c(\rho) = \mathrm{Tr}_{\mathcal{K}}\, V\rho V^\dagger$, then for a pure input the joint output is pure and its two marginals have the same entropy. For amplitude damping with decay probability γ and its complementary channel, $\alpha = 1 - 2\gamma$, which is nonzero unless $\gamma = 1/2$. A family with full Choi rank is

$$\Phi_{a,b} : r \mapsto ar + b\hat{z}, \qquad \Phi_{b,a} : r \mapsto br + a\hat{z}, \qquad a, b > 0,\ a + b < 1.$$

$\Phi_{a,b}$ is a convex combination of the identity and the replacer with Bloch vector $b\hat{z}/(1-a)$, so it is CPTP, and likewise $\Phi_{b,a}$. Both give squared pure-output length $a^2 + b^2 + 2abr_z$, and $\alpha = a^2 - b^2$. For $a = .3$, $b = .2$ both normalized Choi matrices have smallest eigenvalue .125.

3.4 Mixed inputs expose the nonzero branch

Proposition 3.5 (Entropy gap on mixed inputs)**.** *For any two qubit channels and any input,*

$$|S_1(r) - S_2(r)| \ge \tfrac{1}{2}|q(r)|. \tag{3}$$

If (2) *holds, then $q(r) = \alpha(\|r\|^2 - 1)$ on the whole Bloch ball, so*

$$|S_1(r) - S_2(r)| \ge \frac{|\alpha|}{2}\big(1 - \|r\|^2\big), \tag{4}$$

and $\mathrm{sgn}(S_1 - S_2) = \mathrm{sgn}\,\alpha$ for every strictly mixed input.

Proof. Write the entropy as a function of the squared Bloch length, $f(x) = h\big((1+\sqrt{x})/2\big)$. Then $-f'(x) = \mathrm{artanh}(\sqrt{x})/(2\sqrt{x}) \ge \frac{1}{2}$, with the continuous extension at $x = 0$ and the integral limit at $x = 1$. The mean value theorem gives (3), and substituting (2) into (1) gives (4). A larger Bloch length means smaller entropy, which fixes the sign. □

When all pure inputs agree, a single strictly mixed input with equal entropies therefore forces $\alpha = 0$ and the orthogonal relation of Proposition 3.3. The argument uses only the one-to-one relation between qubit entropy and Bloch length, not a polynomial form of the entropy.

3.5 Probes and random inputs

Corollary 3.6 (Nine pure probes and one mixed probe)**.** *Let the probe set consist of the nine Bloch vectors*

$$\pm e_1,\ \pm e_2,\ \pm e_3,\ (e_1+e_2)/\sqrt{2},\ (e_1+e_3)/\sqrt{2},\ (e_2+e_3)/\sqrt{2}.$$

If the two channels give equal output entropies on these nine inputs, they give equal output entropies on every pure input. If they also agree at $r = 0$, they agree on every input.

Proof. The six axial probes give $b_i = 0$ and $A_{ii} + c = 0$; the three diagonal probes then give $A_{ij} = 0$ for $i \neq j$. This is (2). At $r = 0$, $q = c = -\alpha$, so agreement there gives $\alpha = 0$. Then q vanishes on the entire Bloch ball, hence the output lengths and entropies agree everywhere. □

The corollary concerns exact entropy values on nine inputs, not nine measurements. On the unit sphere $q(r) = r^T A_0 r + 2b^T r + d$ with $\alpha = \frac{1}{3}\,\mathrm{tr}\,A$, $A_0 = A - \alpha I$ and $d = c + \alpha$, which is linear in nine real parameters.

Proposition 3.7 (Random pure inputs)**.** *For Haar-random pure inputs,*

$$Q := \mathbb{E}\, q(r)^2 = \frac{2}{15}\|A_0\|_F^2 + \frac{4}{3}\|b\|^2 + d^2. \tag{5}$$

$Q = 0$ *exactly when* (2) *holds. If* $Q > 0$*, then* $\mathbb{E}(S_1 - S_2)^2 \geq Q/4$*, and for* $0 < \tau < 1$

$$\Pr\Big\{|S_1 - S_2| > \tfrac{1}{2}\sqrt{\tau Q}\Big\} \geq \Pr\{q^2 > \tau Q\} \geq \frac{(1-\tau)Q}{1-\tau Q}. \tag{6}$$

Proof. Use $\mathbb{E} r_i r_j = \delta_{ij}/3$ and $\mathbb{E} r_i r_j r_k r_l = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$; odd moments and the cross terms with the traceless A_0 vanish. The second claim follows from (3). Both outputs are states, so $0 \leq q^2 \leq 1$; splitting $\mathbb{E}q^2 = Q$ at $q^2 = \tau Q$ gives $Q \leq \tau Q(1-P) + P$ with $P = \Pr\{q^2 > \tau Q\}$, which is (6). □

With n independent Haar inputs, the probability of missing the difference at the level $\frac{1}{2}\sqrt{\tau Q}$ is at most $(1 - p_\tau)^n$, where p_τ is the right side of (6). This assumes exact entropies; the next proposition removes that assumption. The average change of purity over pure inputs is also the basis of the unitarity measure of noise [17]; Eq. (5) concerns the difference between two channels rather than the coherence of one.

3.6 Finitely many copies

For a qubit, purity and entropy are both monotone functions of the Bloch length, so equal entropy is equivalent to equal purity, and purity can be measured directly. A SWAP test on two copies of a state σ returns the outcome 0 with probability $\frac{1}{2}\big(1 + \operatorname{Tr}\sigma^2\big)$ [18]. For a qubit with Bloch vector s this is $\frac{3}{4} + \frac{1}{4}\|s\|^2$. We assume that every channel use acts on a freshly prepared, known probe state, that the channels do not drift between uses, and that two output copies can be measured jointly.

Finite data cannot establish that the two entropy functions are exactly equal. They can separate equality from a violation of prescribed size, and they can bound how large a violation can be. Put

$$\kappa_9 = \frac{75 - 3\sqrt{465}}{8} = 1.28855\ldots.$$

Proposition 3.8 (Finite-copy test and confidence bound)**.** *For each of the nine probes* r_k *of Corollary 3.6, send* $2m$ *copies of the input through* Φ_1*, apply a SWAP test to each of the* m *output pairs, and do the same for* Φ_2*. Let* D_k *be the difference of the two observed frequencies of the outcome* 0*.*

1. *Given* $\gamma > 0$*, declare the channels different on pure inputs if* $\max_k |D_k| > \gamma/8$*. If* (2) *holds, a difference is declared with probability at most* $18\, e^{-m\gamma^2/64}$*. If* $\max_k |q(r_k)| \geq \gamma$*, the difference is missed with probability at most* $2\, e^{-m\gamma^2/64}$*.*

2. $\sum_k q(r_k)^2 \geq \kappa_9\, Q$*, and* κ_9 *is the largest constant with this property. Hence* $Q \geq Q_0 > 0$ *forces* $\max_k |q(r_k)| \geq (\kappa_9 Q_0/9)^{1/2}$*, and*
$$m \geq \frac{576\, \ln(18/\delta)}{\kappa_9\, Q_0}$$
output pairs per probe and channel keep both error probabilities below δ *for the hypotheses* $Q = 0$ *and* $Q \geq Q_0$*.*

3. *For every* $t > 0$*, with probability at least* $1 - 18\, e^{-mt^2}$*,*
$$Q \leq \frac{16}{\kappa_9} \sum_{k=1}^{9} \big(|D_k| + t\big)^2.$$

Proof. The expected value of D_k is $q(r_k)/4$. D_k is a sum of $2m$ independent Bernoulli variables with coefficients $\pm 1/m$, so Hoeffding's inequality [20] gives $\Pr\{|D_k - \mathbb{E}D_k| \geq t\} \leq 2e^{-mt^2}$. Under (2) every $\mathbb{E}D_k$ vanishes, and $t = \gamma/8$ with a union bound over the nine probes gives the first error bound. If $|q(r_k)| \geq \gamma$ for some k, then $|\mathbb{E}D_k| \geq \gamma/4$, and missing it requires $|D_k - \mathbb{E}D_k| \geq \gamma/8$, which gives the second.

The smallest generalized eigenvalue of the two quadratic forms $\sum_k q(r_k)^2$ and Q is κ_9, as computed in Appendix C. The largest of nine terms is at least their mean; setting $\gamma^2 = \kappa_9 Q_0/9$ and requiring $18e^{-m\gamma^2/64} \leq \delta$ gives item 2. For item 3, on the event that all nine deviations are at most t, which fails with probability at most $18e^{-mt^2}$, we have $|q(r_k)| = 4|\mathbb{E}D_k| \leq 4(|D_k| + t)$, and $Q \leq \kappa_9^{-1} \sum_k q(r_k)^2$. □

If the pure-input criterion is known to hold, the same argument at the mixed probe $r = 0$ tests the size of α: the expected difference there is $-\alpha/4$, and $m \geq 64\ln(2/\delta)/\alpha_0^2$ pairs per channel separate $\alpha = 0$ from $|\alpha| \geq \alpha_0$ with error at most δ in each direction. The pure test uses $18m$ channel uses per channel, $36m$ in all. The constants are conservative. Beyond qubits, purity and entropy are different functionals and this reduction no longer applies.

4 Connectivity and geometry

4.1 Mutual information as connectivity

To compute correlations, declare a subsystem or region representation of the channel output and its relation to the selected source algebra. The channel and state alone do not determine that representation. For output factors $A, B, \ldots$ in this representation, set

$$I_{\mathbf{c}}(A : B) = S_{\mathbf{c}}(A) + S_{\mathbf{c}}(B) - S_{\mathbf{c}}(AB),$$

with entropies computed after coarse-graining.

Definition 4.1 (Connectivity)**.** *For disjoint output factors A, B in the declared representation, the connectivity is $\mu_{\mathbf{c}}(A, B) := I_{\mathbf{c}}(A : B)$. It defines a weighted graph but is not a distance: large mutual information denotes strong correlation rather than large separation, and the metric axioms generally fail.*

A state-parameter metric, such as the SLD Fisher metric in Appendix B, is a different construction. Neither it nor the graph weights alone supply a Lorentzian metric.

4.2 A finite reconstruction rule

For a declared family of finite-dimensional subsystems, a positive connected correlation graph supports a shortest-path metric once an edge-length rule has been chosen. We use $\ell_e = 1/I_e$ on edges with $I_e > 0$. An edge with $I_e = 0$ is absent; distinct connected components then have infinite separation under this rule. Mutual information itself is not a distance. The rule assigns longer distances to weaker correlations, but its physical interpretation depends on the model.

A positive lower bound on the used mutual informations makes this rule stable. No assumption of exponential decay is needed for this finite statement. Exponential decay relative to a pre-existing locality structure can help construct a spatial interpretation [24, 23]; it would supply an additional input rather than derive that locality from the graph.

Proposition 4.2 (Stability of a graph metric)**.** *On the complete graph with v vertices, give the edge e the length $1/I_e$ and use the shortest-path metric d. If every $I_e \geq m > 0$ and the perturbed weights satisfy $|I'_e - I_e| \leq \varepsilon < m$, then*

$$\max_{i,j} |d'_{ij} - d_{ij}| \leq (v-1)\frac{\varepsilon}{m(m-\varepsilon)}.$$

Proof. For each edge, $|1/I'_e - 1/I_e| = |I_e - I'_e|/(I_e I'_e) \leq \varepsilon/(m(m-\varepsilon))$. A shortest path with positive weights uses at most $v-1$ edges. Comparing the new shortest path with the old lengths, and the old shortest path with the new lengths, bounds the difference in both directions. □

4.3 From entropy data to a spacetime metric

In holography, the entropy of a boundary region equals the area of an extremal surface in Planck units [4], and the connectivity of spacetime has been argued to track the pattern of entanglement [3]. In tensor-network models, graph distance in an entanglement graph can approximate continuum distance [22]. These are reconstruction dictionaries for particular classes of states.

For a comparison of spacetime geometries, a dictionary $\mathfrak{G}$ must also state which entropy profiles it identifies. A sufficient condition is

$$\mathfrak{G}[S_1] \simeq_{\text{diff}} \mathfrak{G}[S_2] \implies S_1 \simeq_{\mathfrak{G}} S_2.$$

Its contrapositive says that separated profiles produce separated geometries. This is a requirement on the dictionary, not a derivation of geometry from a channel. The next result replaces that requirement by a fixed finite rule for which the distances and their measurement uncertainties can be computed.

4.4 One state, one operation family, one rule

Proposition 4.3 (Non-isometric finite geometries)**.** *Fix the three-qubit GHZ state. Allow only local bit-flip channels*

$$\Phi_{\mathbf{p}} = \bigotimes_{i=1}^{3} \big[(1-p_i)\,\mathrm{id} + p_i\,\mathrm{Ad}_{X_i}\big], \qquad 0 < p_i < \tfrac{1}{2},$$

which are generated by local bit-flip semigroups with $p_i = (1-e^{-2\gamma_i t})/2$. Fix the reconstruction rule: on the complete graph of the three qubits, the edge $\{i,j\}$ has length $1/I_{ij}$, and distances are shortest-path lengths. Then

$$I_{ij} = \ln 2 - h\Big(\frac{1+c_{ij}}{2}\Big) > 0, \qquad c_{ij} = (1-2p_i)(1-2p_j),$$

and different flip probabilities can give metric spaces that are not isometric under any relabelling of the vertices:

flip probabilities	*mutual informations of the three edges*	*diameter*
$(.1, .1, .1)$	$(.221753694,\ .221753694,\ .221753694)$	4.509507747
$(.05, .1, .2)$	$(.288183695,\ .153870839,\ .120090263)$	8.327069751
$(.05, .1, .15)$	$(.288183695,\ .214256042,\ .166239219)$	6.015427679

The first and third rows have the same expected number of applied flips, $\sum_i p_i = .3$, when each channel is implemented by applying X_i with probability p_i.

Proof. Every two-qubit marginal of the GHZ state is an equal mixture of $|00\rangle$ and $|11\rangle$. Local flips leave both bits uniform and make them agree with probability $(1+c_{ij})/2$, so $I_{ij} = 2\ln 2 - \big(\ln 2 + h((1+c_{ij})/2)\big)$. All edge lengths are finite and positive, so the rule defines a metric. The diameters differ, so no relabelling makes the spaces isometric. The numbers were also computed from the full eight-dimensional density matrix with the channels applied and partial traces taken; the analytic formula agrees to better than 3×10^{-16}. □

Every entry is fixed in advance: the state, the class of allowed operations, the interpretation of the vertices and the reconstruction rule. Nothing about the target geometry is written into the channels. The local flip channels are invertible on the operator space, so all of them generate the full matrix algebra; the algebra label is the same in every row while the geometry is not, which is Proposition 2.2(3) in geometric form. The equal flip budget in rows one and three shows that the difference comes from how a fixed resource is distributed among the qubits, not from its total. This is a one-step budget in a stated implementation; it does not claim equal cost on all hardware or under continuous-time noise. By Proposition 4.2, perturbing the three mutual informations by ε changes each distance by at most $2\varepsilon/(m(m-\varepsilon))$ with m the smallest mutual information. The model is a finite existence result for a fixed rule, not a derivation of the rule from mutual information and not a spacetime.

4.5 The reduction is a local interaction

For each source qubit prepare a noise ancilla N_i in $|0\rangle$ and apply

$$U_i(\theta_i) = e^{-i\theta_i X_i \otimes Y_{N_i}}, \qquad U_i(\theta_i)|\psi\rangle|0\rangle = \cos\theta_i|\psi\rangle|0\rangle + \sin\theta_i X_i|\psi\rangle|1\rangle.$$

Discarding N_i realizes exactly the local flip channel with $p_i = \sin^2\theta_i$. One possible interaction Hamiltonian is $\hbar g_i X_i \otimes Y_{N_i}$ applied for θ_i/g_i. The angles, tensor factors and timing are supplied by the apparatus. They do not encode a target metric.

Next attach, at each site, a fragment of k_i fresh memory qubits. Use Paper E's collisions $P_{0,i} \otimes I + P_{1,i} \otimes R_{i,a}$, with real conditional overlaps $c_{i,a}$, so the fragment overlap is $r_i = \prod_{a=1}^{k_i} c_{i,a}$. These collisions dephase the source in the Z basis. Every two-site marginal of the noisy GHZ state is already diagonal, so its probabilities and mutual information are unchanged by recording. The post-recording source therefore has the same graph metric as before, although its three-site coherence may have changed. This is preservation of the data used by this rule, not preservation of the entire source state.

The noise gate ends before recording begins. This sequencing and its switching resources matter: the flip couples through X, while the recorder uses Z. Their common fixed algebra is only the scalars. Indeed the composed single-site source channel sends a Bloch vector to $(r_i x, r_i(1-2p_i)y, (1-2p_i)z)$; for $0 < p_i < 1/2$ and $0 < r_i < 1$ it has no fixed nonidentity direction, although its linear range is still the full matrix algebra. The record channel nevertheless

reads the final pointer probabilities with nonzero resolution. This supplies an actual finite-access example beyond an exact stable-algebra label.

Tracing out the source after the collisions gives the exact joint fragment state

$$\rho_{\mathrm{rec}} = \sum_{\mathbf{z}} P_{\mathbf{p}}(\mathbf{z}) \bigotimes_{i=1}^{3} |e_{z_i}^{(i)}\rangle\langle e_{z_i}^{(i)}|, \tag{7}$$

where $z_i = \pm 1$ labels the local pointer and $P_{\mathbf{p}}$ is the actual diagonal distribution of the noisy GHZ state. Thus conditional product structure follows from the unitary couplings and fresh product memories; it is not an extra assumption about independent answers. Each site has equal pointer priors.

For $r_i < 1$, the equal-prior Helstrom measurement on fragment i has a symmetric binary error probability $\epsilon_i = (1 - \sqrt{1 - r_i^2})/2$ [19, 26]. To see the symmetry directly, choose real vectors $e_0 = (1, 0)$ and $e_1 = (r, \sqrt{1 - r^2})$. The sign of $|e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|$ has expectations $+\sqrt{1 - r^2}$ and $-\sqrt{1 - r^2}$ in the two states. Writing $Y_i = \pm 1$ for the actual outcome and $v_i = 1 - 2\epsilon_i = \sqrt{1 - r_i^2}$ therefore gives

$$\mathbb{E}(Y_i \mid \mathbf{z}) = v_i z_i, \qquad \mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}. \tag{8}$$

Local measurement on (7) gives these identities even though the records are correlated before conditioning.

4.6 Finite records certify the geometric difference

Let $f(c) = \ln 2 - h((1 + c)/2)$, $-1 \le c \le 1$. Consider K noise configurations, with independently prepared source states and fresh ancillas in each run. For every configuration collect n triples (Y_1, Y_2, Y_3) from the local measurements above. Calibration supplies estimates $\widehat{v}_i \in [0, 1]$ with $\widehat{v}_i \ge \nu > 0$ and $|\widehat{v}_i - v_i| \le \xi$. The estimate

$$\widehat{c}_{ij} = \mathrm{clip}_{[-1,1]} \frac{n^{-1} \sum_{s=1}^{n} Y_i^{(s)} Y_j^{(s)}}{\widehat{v}_i \widehat{v}_j}$$

uses the measurement outcomes, not hidden pointer values.

Theorem 4.4 (Record-to-metric confidence certificate)**.** *Suppose the calibration event has failure probability at most δ_{cal}, and the record law is* (7) *with the stipulated local measurements. For every $\tau > 0$, with probability at least $1 - \delta_{\mathrm{cal}} - 6Ke^{-n\tau^2/2}$, all $3K$ correlations satisfy*

$$|\widehat{c}_{ij} - c_{ij}| \le e_c := \frac{\tau + 2\xi}{\nu^2}. \tag{9}$$

Intersect each interval $[\widehat{c}_{ij} - e_c, \widehat{c}_{ij} + e_c]$ with $[-1, 1]$ and minimize and maximize f on it, obtaining I_{ij}^-, I_{ij}^+. If every $I_{ij}^- > 0$, form shortest-path metrics d^- and d^+ from edge lengths $1/I_{ij}^+$ and $1/I_{ij}^-$. The true metric obeys $d^- \le d \le d^+$ entrywise, and its diameter lies in $[\operatorname{diam} d^-, \operatorname{diam} d^+]$. Disjoint diameter intervals for two configurations certify non-isometry on the same confidence event.

For a sample-size guarantee, if true and estimated correlations obey $|c_{ij}|, |\widehat{c}_{ij}| \le b < 1$, set $e_I = \operatorname{artanh}(b) e_c$. When $I_{ij} \ge \mu > e_I$,

$$\|\widehat{d} - d\|_\infty \le R := \frac{2e_I}{\mu(\mu - e_I)}. \tag{10}$$

For two configurations with true diameter gap $\Delta > 4R$, the intervals $[\operatorname{diam} \widehat{d} - R, \operatorname{diam} \widehat{d} + R]$ are disjoint on that event.

Proof. Each $Y_i Y_j \in [-1, 1]$ has expectation $v_i v_j c_{ij}$. Hoeffding's inequality gives a two-sided tail $2e^{-n\tau^2/2}$; a union bound covers the $3K$ edge estimates. The three edges within a run need not be independent. On the calibration event, $|v_i v_j - \widehat{v}_i \widehat{v}_j| \le 2\xi$, by telescoping and the bounds $v_i, \widehat{v}_i \le 1$. Dividing by $\widehat{v}_i \widehat{v}_j \ge \nu^2$, using $|c_{ij}| \le 1$, and clipping cannot increase the error, proving (9). The interval construction contains the true mutual informations. Shortest-path lengths and their diameter are monotone in positive edge lengths, proving the metric enclosure. Finally $f'(c) = \operatorname{artanh} c$, so $|\widehat{I} - I| \le e_I$. Proposition 4.2, with three vertices, gives (10). Each estimated diameter differs from its true value by at most R, so the estimated gap is at least $\Delta - 2R$; two radius-R intervals are disjoint when $\Delta > 4R$. □

For the equal-flip-count configurations $(.1, .1, .1)$ and $(.05, .1, .15)$, take one recording qubit per site with overlap $r_i = .2$. Then $v_i = \sqrt{.96}$ and the correlations lie between .56 and .72. With exactly specified v_i, use $\nu = \sqrt{.96}$, $\xi = 0$, $\tau = .003$, $b = .8$, $\mu = .16$ and $\delta = .05$. The choice

$$n = \left\lceil \frac{2\ln(12/\delta)}{\tau^2} \right\rceil = 1\,217\,920$$

runs per configuration gives $e_c = .003125$, $e_I = .0034332$ and $R < .275$. The true diameter gap is $1.505919931 > 4R$. Thus a finite record-only experiment separates the two geometries with confidence at least .95, under the stated source and apparatus model. The sample bound is conservative and does not claim optimal tomography.

A known unitary specification is not free empirical calibration. If the v_i are unknown, prepare reference sources in $Z = +1$ and apply the same record couplings and readers. Each reference outcome has mean v_i. For N independent reference runs per site, clipping the empirical mean to $[0, 1]$ yields

$$\Pr\{\max_i |\widehat{v}_i - v_i| > \xi\} \le 6e^{-N\xi^2/2}$$

for three shared readers; separately calibrated readers for K configurations replace 6 by $6K$. Thus $N \ge \lceil 2\ln(6/\delta_{\mathrm{cal}})/\xi^2 \rceil$ per site suffices. Reject calibrations below ν. For the same two geometries, take $\xi = .001$, $\delta_{\mathrm{cal}} = .025$, $\nu = .978$ and $\tau = .002$. Then $N = 10\,961\,278$ reference runs per site and $n = 3\,086\,894$ main runs per configuration give total failure at most .05, $e_c < .004182$, $R < .36955$ and $4R < 1.4782 < \Delta$. On the calibration event the true $v_i = \sqrt{.96}$ ensures $\widehat{v}_i > \nu$; the correlation bound also stays inside $b = .8$. The reference budget is therefore $3N$ additional preparations and readouts, not part of the $2n$ main runs. Equation (9) includes their residual error. Prepared reference pointers, their basis labels and the stability of the reader between calibration and use are additional physical resources.

The comparison uses $2n$ preparations of the same GHZ source, $6n$ fresh noise ancillas, $6n$ fresh recording qubits, $6n$ local binary readouts and communication of the outcomes to a comparator, plus any calibration runs. The expected noise-flip count is .3 per source in both configurations; their recording and readout schedule is identical. Preparation, switching work, storage, communication and renewal do not vanish because the counts agree. No heat or total-energy equality follows from the flip count, and this finite protocol supplies no reservoir reset.

4.7 Limits of record inversion

At $r_i = 1$ the record states coincide and $v_i = 0$: algebraic inversion of their span supplies no source information. Even for $v_i > 0$ the error grows as the inverse product visibility. A narrow effect set can span the pointer algebra while requiring too many trials for the available budget.

Calibration of individual error rates is also insufficient when the readers share an unmodelled error. Let a common random sign B multiply all otherwise ideal outcomes, with $\mathbb{E}B = v$. Each local reference test then reports visibility v, while the pair products are unchanged, since $B^2 = 1$. Dividing by v^2 overestimates the true correlations. A common inversion has $v = -1$ and is invisible to agreement. Joint reference tests, conditional error modelling or a verified local apparatus are needed to exclude such contamination. The confidence certificate is conditional on the joint record law, rather than an inference of that law from single-site calibration.

A spacetime reconstruction must solve a further problem: finite measurements determine a profile within an error region, not an exact profile. The relevant dictionary must be stable on that region and must quotient its coordinate freedom. Entropy equality from Section 3 supplies a genuine obstruction, including operationally different channels with identical entropy profiles. More accurate readout cannot repair a dictionary's failure to distinguish such profiles.

Conjecture 4.5 (Finite permissions approaching geometric access)**.** *In a source family with an independently justified local reconstruction dictionary, histories that enlarge the available record fragments and calibrate their readout can make the inferred geometry converge while the generated observable algebra remains constant. The required growth of visibility, preparation count and dictionary stability bounds must jointly dominate sampling and model error.*

The finite certificate supplies the mechanism and the costs of this candidate. For E edges with visibility at least ν, error in the correlations scales as $\nu^{-2}\sqrt{\ln(2E/\delta)/n}$ before calibration error. If reconstructed edge mutual informations have lower bound μ, distance error carries another factor of order μ^{-2} times the number of edges in a shortest path. Consequently, a growing graph or an approaching zero-information edge can defeat convergence even while all fixed-algebra labels are unchanged. The conjecture would fail in a proposed model if its dictionary is unstable, if common error survives calibration, or if its available budget cannot overcome these factors. This is a route from finite permissions to geometric access, not an assumption that a continuum or its dynamics is already generated.

The spatial/eigenmode example in Appendix B illustrates dependence of connectivity on factorization. Unlike Proposition 4.3, its empty eigenmode graph has no finite reconstructed paths.

4.8 Not a change of coordinates

A diffeomorphism acts within a fixed algebra of observables and leaves unchanged what can be measured. A change of coarse-graining changes the accessible observables, which correlations are visible and how subsystems are defined. Algebraic difference alone does not prove that two reconstructed metrics are not diffeomorphic, because a dictionary may discard it; the conclusion needs a separating dictionary or an explicit computation such as Proposition 4.3.

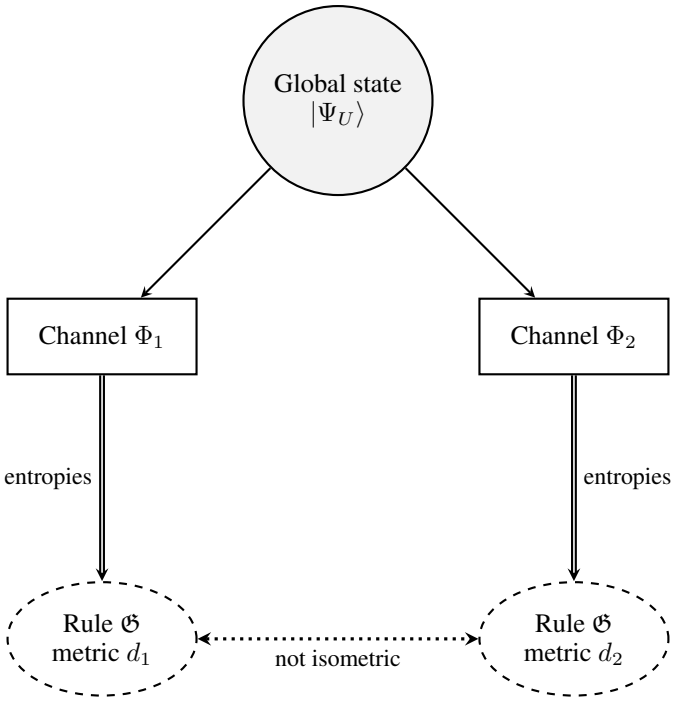


Figure 1: One global state is reduced by two channels from the same allowed family. The same reconstruction rule turns their entropy data into metrics. Section 3 decides when the entropy data differ; Proposition 4.3 gives a case where the metrics are not isometric.

5 Discussion

5.1 Relation to other frameworks

QBism treats the quantum state as an agent's degree of belief. Here a single objective global state is assumed throughout, and coarse-grainings are realized by physical interactions. The only point of contact is that no subsystem decomposition is privileged. Decoherence and many-worlds accounts explain classical behaviour and branching relative to a fixed split into system, apparatus and environment. The present framework leaves that analysis unchanged and asks where the split comes from. The closest relative is algebraic quantum field theory, where algebras are primary and states are functionals on them [9], and the closest technical precursor is the result that tensor-product structures are induced by the accessible observables [6]. What the present paper adds to that line is the channel level: a reduction has a resolution that its algebra does not record (Proposition 2.2), entropy-based readings have exact blind spots (Theorem 3.2, Example 3.4), those blind spots can be tested with finite resources (Proposition 3.8), and a fixed rule can turn a different distribution of the same noise budget into a different finite geometry (Proposition 4.3). The affine form of qubit channels, the complete-positivity conditions and the comparison of channels by post-processing are known tools [15, 16, 14]; the criterion, the branch analysis and the tests are derived here, and a full comparison with the literature on channels with equal output spectra is still to be done.

5.2 Series interfaces and the world-structure question

The protocol connects three distinct responsibilities. Paper B derives a reduction and its persistent structures from specified couplings. Its window theorem selects a block partition when within-block and between-block rates permit one; this does not make the finite channel's full range equal to that block algebra. Paper E derives the fragment's implementable effects and its public readout properties from a collision. Here their composition reaches a certified geometric difference. The source-channel algebra can stay fixed while recording permission and the attainable precision change. Imprint's cross-scale work develops the related question of which reading is worth its cost [27].

The coupling operators, factorization into sites, source preparation and reconstruction rule are still inputs. The new connection derives what happens between those inputs and a finite reader's geometric conclusion; it does not derive the architecture from an unstructured Hilbert space. Histories that alter actual coupling or record access, as studied in the subject and maintenance lines of the trilogy, can change this capability. A claim that they also generate locality or a spacetime dictionary requires its own dynamics and bridge, including the error conditions of Conjecture 4.5.

The finite models use trace entropy, whereas the infinite source description introduced above does not. Neither the SLD Fisher metric nor a shortest-path metric is identified with Lorentzian spacetime. The task is to determine which operational structures can constrain a physical geometric reading, and which geometric claims fail because a finite context cannot distinguish the data.

6 Conclusion

A physical reduction carries more information than its generated algebra. Two reductions may share the algebra and differ in resolution; they may agree on every entropy and remain operationally incomparable. For qubits these possibilities are decided by a quadratic criterion, and finite-copy tests bound its violation.

The worked protocol carries the argument through actual operations. A local unitary produces the source noise, local collisions create environmental records without changing the pair data used by the rule, and the measured records certify non-isometric shortest-path metrics. Noise allocation changes geometry at equal expected flip count. The finite sample guarantee includes imperfect readout, calibration and a common-error failure mechanism, so the conclusion is attainable by the declared readers.

The resulting dependence is physical and quantitative within the model: changing an implemented coupling changes a state, its recorded correlations and the reconstructed metric. Explaining the coupling architecture, deriving a spacetime dictionary and its continuum dynamics remain the larger world-structure problem. Conjecture 4.5 gives a mechanism and failure conditions for one part of that extension. No empirical departure from quantum theory is claimed; the protocol itself gives testable statements.

A No canonical tensor factorization

For a Hilbert space $\mathcal{H}_{\mathrm{total}}$ of dimension $d_A d_B$ with $d_A, d_B > 1$, there is no unique way to write $\mathcal{H}_{\mathrm{total}} = \mathcal{H}_A \otimes \mathcal{H}_B$ without further physical input. Given any pure state $|\Psi_U\rangle$, choose a unitary U with $U(|0\rangle_A \otimes |0\rangle_B) = |\Psi_U\rangle$ and pull the reference factors through U; in the resulting factorization $|\Psi_U\rangle$ is a product. More generally, observables can be chosen so that a given pure state has any prescribed Schmidt decomposition [8].

What survives is typicality for a family of factorizations fixed in advance.

Theorem A.1 (Entanglement typicality for a specified family)**.** *Let $\mathcal{H}_{\mathrm{total}}$ be finite dimensional and let $\mathfrak{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be a finite family of nontrivial tensor factorizations selected by a physical rule that does not depend on the sampled state. Write $\mathcal{H}_{\mathrm{total}} = \mathcal{H}_{A_j} \otimes \mathcal{H}_{B_j}$ with $d_{A_j} \le d_{B_j}$. For $\delta > 0$ let*

$$p_j(\delta) := \Pr_{\Psi \sim \mathrm{Haar}} \left[S(\rho_{A_j}) < \log d_{A_j} - \delta \right].$$

Then

$$\Pr\left[S(\rho_{A_j}) \ge \log d_{A_j} - \delta \text{ for every } j \right] \ge 1 - \sum_{j=1}^{m} p_j(\delta),$$

and standard concentration estimates make each $p_j(\delta)$ small at large dimension under the usual conditions on the dimension ratio.

Proof. For each fixed $\mathbf{f}_j$, Page's formula gives [10]

$$\mathbb{E}[S(\rho_{A_j})] = \sum_{k=d_{B_j}+1}^{d_{A_j}d_{B_j}} \frac{1}{k} - \frac{d_{A_j}-1}{2d_{B_j}}, \tag{11}$$

and concentration of measure bounds the lower tail [11]; this tail is $p_j(\delta)$. The union bound over the finite, preselected family proves the inequality. □

The size of the statement matters. For a balanced split $d_A = d_B$, Eq. (11) gives a mean entropy of $\log d_A - \frac{1}{2} + o(1)$, so typical states are close to maximally entangled but keep a gap of about half a nat. Drawing the factorization at random with the state fixed is equivalent to drawing the state at random with the factorization fixed. Hence for a fixed state, product factorizations form a set of measure zero, while factorizations with small but nonzero entanglement have small positive measure.

Theorem A.1 says nothing about all factorizations and does not pick out any member of $\mathfrak{F}$. Which family is admissible, and why, is a question of quantum mereology [12, 13]. Earlier work showed that subsystem structure is induced by the accessible observables rather than given by the Hilbert space [5, 6]. The channel-level results in the main text retain the accessible-observable view while distinguishing its algebra from its resolution.

B Other connectivity constructions

When a metric on a parameter space of coarse-grained states is wanted, the finite-dimensional construction is the quantum Fisher metric. For a smooth faithful family $\lambda \mapsto \rho_{\mathbf{c}}(\lambda)$, define the symmetric logarithmic derivatives by $\partial_a \rho_{\mathbf{c}} = \frac{1}{2}(L_a \rho_{\mathbf{c}} + \rho_{\mathbf{c}} L_a)$ and

$$g^{\mathrm{QF}}_{ab}(\lambda) := \frac{1}{2}\,\mathrm{Tr}[\rho_{\mathbf{c}}(\lambda)\{L_a, L_b\}]\,. \tag{12}$$

This is a positive-semidefinite Riemannian metric on identifiable parameter directions [21]. It is not a Lorentzian spacetime metric, and the pairwise numbers $\mu_{\mathbf{c}}(A, B)$ do not determine it without an additional map from graph or state parameters to geometry.

B.1 Spatial and eigenmode descriptions of a free-fermion chain

Example B.1 (A dimerized free-fermion chain)**.** *Take a periodic chain of $2L$ sites with alternating hopping amplitudes $t_1 = 1$ and $t_2 = 1/2$,*

$$H = -\sum_{c=0}^{L-1} \big(t_1\, c^\dagger_{2c} c_{2c+1} + t_2\, c^\dagger_{2c+1} c_{2c+2} + \mathrm{h.c.}\big),$$

at half filling. The single-particle spectrum has two bands separated by the gap $2|t_1 - t_2| = 1$, and the ground state fills the lower band, $|\Psi_0\rangle = \prod_k a^\dagger_{k,-}|0\rangle$, where $a_{k,-}$ are the lower-band Bloch modes. In the description by occupation numbers of the band-resolved Bloch modes the ground state is a product of filled and empty modes: every set of modes has a pure reduced state, disjoint sets have zero mutual information, and the connectivity graph is empty. In the spatial description blocks of sites are correlated. For $L = 60$ and blocks of four sites, the block entropy saturates at $.3554$ as the block grows, as the area law for gapped one-dimensional systems requires [24, 23]. The mutual information between two blocks falls from $.333$ for adjacent blocks to 9.0×10^{-5} at a separation of ten sites and 5.2×10^{-8} at twenty, close to $e^{-r/1.32}$. The connectivity graph is dominated by edges between neighbouring blocks.

The second description has to be chosen with care. Plain Fourier modes are not enough when the Hamiltonian is not translation invariant: for two disconnected dimers on four sites, $H = -(c^\dagger_0 c_1 + c^\dagger_2 c_3 + \mathrm{h.c.})$ at half filling, the gap at fixed particle number is 2, yet the Fourier modes $k = \pi/2$ and $3\pi/2$ share one particle in superposition and have mutual information $2\ln 2$. A translation-invariant single-band chain with particle conservation is gapless unless its band is full or empty, so a gapped example whose ground state is a product of mode occupations needs band-resolved modes, as above, or in general the single-particle eigenmodes of the Hamiltonian.

The example shows that connectivity depends on the factorization: the spatial description has short-range edges and the eigenmode description has none. It does not exhibit two reconstructed geometries, since its zero-connectivity graph supplies no finite paths under the chosen rule; Proposition 4.3 is the example in which both descriptions produce geometries.

C The nine-probe constant

On the unit sphere, $\sum_k q(r_k)^2$ and Q are quadratic forms in the nine parameters (A_0, b, d), and both are positive definite; the first because the nine probe values determine the parameters, as in the proof of Corollary 3.6. Their generalized eigenvalues do not depend on the basis chosen for the parameters, and a direct computation gives the characteristic polynomial, up to a constant factor,

$$(\lambda - 6)\,(4\lambda^2 - 75\lambda + 90)\,(8\lambda^3 - 225\lambda^2 + 1620\lambda - 2700)^2.$$

Its roots are 1.2886, 2.3962, 6, 7.9001, 17.4614 and 17.8287, the three roots of the cubic each counted twice. The smallest is the smaller root of the quadratic factor, κ_9, and it is attained.

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COUPLING, RECORDS, AND STABLE RELATIONS: FORMATION, PROTECTED CONTENT AND THE LIFETIME OF A GEOMETRY

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ABSTRACT

A physical coupling determines which distinctions decay, which can be recorded and which remain available for action. For finite systems with commuting Hermitian noise operators, we derive the fixed block algebra and its long-time pinching, then prove a finite-window selection theorem: a separated set of decoherence rates selects at most one block partition, and its algebra contains every stable block algebra in the supplied joint eigenbasis. Finite-time channels still have full operator range. Block-constant recording couplings report only sector labels while protecting the quantum content within each sector; leakage through independent fragments costs coherence multiplicatively, giving a sharp $R^{-1/2}$ symmetry threshold for R records. Geometry read from mutual information receives both sector and protected-content contributions, which we separate by an exact classical–quantum chain identity. Neither contribution is confined to the noise hypergraph: disconnected decoherence-free sectors can preserve an initially shared logical Bell pair. We therefore distinguish preservation from formation. A collective recording interaction creates a flagged protected Bell branch from a product input, with every branch and the nonlocal recording cost retained; a physical Hamiltonian link creates a deterministic Bell pair between two noise-disconnected protected sectors. For the GHZ state, collective and local couplings yield a triangle, one edge or no edge under one reconstruction rule, with an explicit lifetime under symmetry breaking. Beyond commuting noise, complete finite-time record laws and risks vary continuously even when the exact fixed algebra jumps. These results identify coupling-selected storage, record-conditioned formation and finite operational capability without equating them. The ambient carrier, available interactions and reconstruction dictionary remain physical inputs.

1 Introduction

A stable distinction is a physical achievement. It must survive the interactions that would erase it, be readable through an available instrument, and last long enough to matter. An algebra of invariant observables describes one part of that achievement; it does not specify a reader, a preparation history or a set of available gates.

Paper A [1] shows why the distinction matters for geometry. Channels generating the same observable algebra can resolve different states and produce non-isometric finite correlation geometries. Its local bit-flip family has a Stinespring realization and a finite fragment-readout certificate. Here we ask a complementary question: what structure a coupling protects, how actual records and interactions form usable relations within that structure, and when those relations lose their support.

Commuting Hermitian noise gives an exact starting point. The fixed observables are the commutant of the coupling operators, familiar from einselection and noiseless codes [9, 10, 12, 13]. The infinite-time channel pinches onto this algebra. At finite times its operator range is full, so a protected block algebra is additional stability data rather than the

range of the actual channel. A window and a tolerance give a unique block partition when intra-block and inter-block rates are separated (Theorem 3.2). The maximality statement concerns block algebras in the specified joint eigenbasis, not every rotated algebra or every useful finite task.

Recording adds a different restriction. A coupling constant on each block can record its label without learning its quantum content. Independent fragments that resolve two states within a block destroy their coherence according to a multiplicative visibility budget (Theorem 4.2). The centre and the blocks thus perform different jobs within one stable structure. They are not disjoint sources of geometry: classical sector weights can sustain mutual information even when every block is one-dimensional. Section 5 gives the exact decomposition needed to distinguish the contributions.

Selection alone also says too little about formation. Two disconnected collective-noise components can preserve a logical Bell pair shared at the start. Conversely, local independent processes cannot create cross-component correlations from independent inputs. Section 6 supplies the missing process distinction and two explicit formation mechanisms. One retains a collective measurement flag and all its branches; the other uses an actual cross-component Hamiltonian link. Both generate relations from product inputs, and both expose the interaction resources responsible for them. Foundation III [6] supplies a further mechanism in which acquired records select an encoder and decoder for the same pre-existing quantum content. Storage, formation and learned use then have separate, testable meanings.

The GHZ calculation in Section 7 shows how a coupling changes a finite correlation geometry and how weak symmetry breaking ends its plateau. For noncommuting perturbations, Section 3.1 bounds complete adaptive record laws and task risks without requiring a stable algebra. A transverse qubit exhibits a diverging record lifetime despite a scalar exact fixed algebra.

The supplied Hilbert space, coupling permissions and site factorization are retained throughout. Consciousness I [4] selects a Virasoro fixed net inside a given current-net representation; Foundation's hulls concern closure under declared operations. Neither result derives the ambient carrier. Our formation results address how relations and usable protected sectors arise on that carrier. The stronger question of effective subsystem formation returns in Section 9, with a process-sensitive conjecture rather than a noise-only hypergraph rule.

2 Reductions selected by a coupling

2.1 Setting

Let $\mathcal{H}$ be a finite-dimensional system Hilbert space. The environment couples to the system through Hermitian operators $A_1, \dots, A_K$ with $[A_k, A_l] = 0$, at rates $\gamma_k > 0$. In the Markovian limit the reduced dynamics is generated by

$$\mathcal{L}(\rho) = \sum_{k=1}^{K} \gamma_k \Big(A_k \rho A_k - \tfrac{1}{2}\{A_k^2, \rho\} \Big), \qquad \Phi_t = e^{t\mathcal{L}}. \tag{1}$$

A system Hamiltonian that commutes with every A_k only rotates states inside the blocks defined below; we work in the corresponding interaction picture and return to Hamiltonians that do not commute in Remark 2.2. Section 4 treats the same couplings in an exact unitary model with environmental fragments.

Choose an orthonormal basis $\{|x\rangle : x \in \mathcal{X}\}$ of joint eigenvectors, $A_k|x\rangle = a_k(x)|x\rangle$, and put

$$D(x,y) = \tfrac{1}{2} \sum_k \gamma_k \big(a_k(x) - a_k(y)\big)^2 = \|u(x) - u(y)\|^2, \qquad u(x) = \big(\sqrt{\gamma_k/2}\, a_k(x)\big)_{k=1}^{K} \in \mathbb{R}^K. \tag{2}$$

We call $u(\mathcal{X}) \subset \mathbb{R}^K$ the rate image of the basis; decoherence rates are squared Euclidean distances in it. Write $x \approx y$ when $a_k(x) = a_k(y)$ for every k, let π_∞ be the partition of $\mathcal{X}$ into $\approx$-classes a, and let P_a be the projection onto $\mathcal{H}_a = \operatorname{span}\{|x\rangle : x \in a\}$. For any partition π of $\mathcal{X}$ into classes C, put

$$\mathfrak{B}_\pi = \bigoplus_{C \in \pi} \mathcal{B}(\mathcal{H}_C), \qquad \mathbb{E}_\pi(\rho) = \sum_{C \in \pi} P_C \rho P_C.$$

$\mathfrak{B}_\pi$ is a unital $*$-algebra, $\mathbb{E}_\pi$ is a channel (the pinching), and $\mathbb{E}_\pi$ is its own dual, the conditional expectation onto $\mathfrak{B}_\pi$.

Proposition 2.1 (The reduction selected by a commuting coupling)**.** *1. Φ_t acts as a Schur multiplier in the eigenbasis: $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$, and $\Phi_t^* = \Phi_t$.*

2. *The observables fixed by every Φ_t^* form the algebra $\mathfrak{B}_{\pi_\infty} = \{A_1, \dots, A_K\}'$.*

3. *$\Phi_t \to \mathbb{E}_{\pi_\infty}$ as $t \to \infty$, and the algebra generated by the dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$. In the terminology of Paper A, $(\mathfrak{B}_{\pi_\infty}, \mathbb{E}_{\pi_\infty})$ is a minimally compatible coarse-graining structure.*

4. *For every finite t, Φ_t is invertible on the operator space, so the algebra generated by its dual range is $\mathcal{B}(\mathcal{H})$.*

Proof. (1) In the eigenbasis, $(A\rho A - \frac{1}{2}\{A^2,\rho\})_{xy} = (a(x)a(y) - \frac{1}{2}a(x)^2 - \frac{1}{2}a(y)^2)\rho_{xy} = -\frac{1}{2}(a(x)-a(y))^2\rho_{xy}$, so $\mathcal{L}$ multiplies the (x,y) entry by $-D(x,y)$. The multiplier is real and symmetric, hence $\Phi_t^* = \Phi_t$. (2) X is fixed for all t exactly when $X_{xy} = 0$ whenever $D(x,y) > 0$, that is, whenever $x \not\approx y$. An operator commutes with every diagonal A_k exactly when it has no entries between basis vectors with different eigenvalue tuples. (3) Entries with $D(x,y) > 0$ tend to zero and the others are constant. The dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$, which is already an algebra. (4) All multipliers $e^{-tD(x,y)}$ are nonzero. □

Part (2) is the elementary commuting case of the known structure of fixed points of quantum dynamical semigroups [14, 15]. For this series the consequence matters more than the statement: the given coupling determines both its channel and its invariant algebra. Part (4) is Paper A's result that an algebra label does not record resolution, now in dynamical form. Before the long-time limit the channel generates everything, and the structure that a coupling makes accessible has to be identified by how well observables survive over time.

The objects of Paper E [3] can be located in this class. Its environmental fixed set $\mathrm{Fix}(\mathcal{E})$ is $\mathfrak{B}_{\pi_\infty}$. In Paper E's collision model, with a qubit pointer, the pointer algebra equals the fixed algebra of the induced dephasing and contains every effect a fragment can report. The centre of $\mathfrak{B}_{\pi_\infty}$ is generated by the block projections P_a; it is commutative, and Section 4 shows that it is what the environment records, so it serves as Paper E's pointer or record algebra. The inclusion of the record algebra in the fixed set is thus a conclusion of the model, as Paper E requires of any such overlap. The fixed set is strictly larger than its centre as soon as some block has dimension greater than one; a nontrivial two-sector qubit pointer has only one-dimensional blocks. Such blocks are decoherence-free subspaces [12, 13]. They retain quantum content; sector weights can retain classical correlations as well.

Remark 2.2 (Hamiltonians that do not commute with the couplings)**.** *If the system Hamiltonian H has a part H_{off} connecting different blocks, $\mathfrak{B}_{\pi_\infty}$ need not remain exactly invariant. Strong dephasing can then confine the dynamics to the blocks, with leakage on a time scale of order $\Gamma_{\mathrm{out}}/\|H_{\mathrm{off}}\|^2$, where Γ_{out} is the smallest between-block rate; this is the mechanism of quantum Zeno subspaces [16]. Section 3.1 gives an exact two-level lifetime and a finite-time stability bound without a commutativity assumption. Paper D's chain with a transverse field requires its own many-body estimate.*

3 Selection over a finite window

Definition 3.1 (Stable algebra on a window)**.** *A unital $*$-subalgebra $\mathfrak{B} \subseteq \mathcal{B}(\mathcal{H})$ is ε-stable on $[0,T]$ if*

$$\sup_{0\le t\le T}\ \sup_{X\in\mathfrak{B},\ \|X\|\le 1} \|\Phi_t^*(X) - X\| \le \varepsilon.$$

The diagnostic tests the whole unit ball of an algebra, rather than selected states or a list of slow observables. For a partition π define the width of a class C in the direction k, $w_k(C) = \max_{x\in C} a_k(x) - \min_{x\in C} a_k(x)$, and the two rates

$$\Gamma_{\mathrm{in}}(\pi) = \max_{C\in\pi}\ \tfrac{1}{2}\sum_k \gamma_k w_k(C)^2, \qquad \Gamma_{\mathrm{out}}(\pi) = \min\{D(x,y):\ x,y \text{ in different classes of } \pi\}. \tag{3}$$

The minimum over an empty set is $+\infty$, so a single-block partition has $\Gamma_{\mathrm{out}} = +\infty$. If every coupling is scalar, the full algebra is fixed and selected for every window. In the rate image, $\Gamma_{\mathrm{in}}(\pi)$ is the largest squared diagonal of the axis-parallel boxes enclosing the classes, and $\Gamma_{\mathrm{out}}(\pi)$ is the smallest squared distance between points of different classes.

Theorem 3.2 (Selection by a window)**.** *Let $0 < \varepsilon < 1$ and $T > 0$.*

1. *For every $X \in \mathfrak{B}_\pi$ and $t \ge 0$, $\|\Phi_t^*(X) - X\| \le t\,\Gamma_{\mathrm{in}}(\pi)\|X\|$. For every state $\rho \in \mathfrak{B}_\pi$, $\|\Phi_t(\rho) - \rho\|_1 \le t\,\Gamma_{\mathrm{in}}(\pi)$. In particular $\mathfrak{B}_\pi$ is ε-stable on $[0,T]$ when $T\Gamma_{\mathrm{in}}(\pi) \le \varepsilon$.*
2. *If $1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$, no unital $*$-algebra that contains $\mathfrak{B}_\pi$ properly is ε-stable on $[0,T]$.*
3. *Call π selected by (T,ε) if*

$$T\,\Gamma_{\mathrm{in}}(\pi) \le \varepsilon < 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)}. \tag{4}$$

At most one partition is selected by (T,ε). If π is selected by (T,ε) and π' by (T',ε) with $T \le T'$, then π' refines π, so $\mathfrak{B}_{\pi'} \subseteq \mathfrak{B}_\pi$.

4. *For $T > -\ln(1-\varepsilon)/\Gamma_{\mathrm{out}}(\pi_\infty)$ the selected partition exists and is π_∞, so the selected algebra is the fixed algebra of Proposition 2.1.*

Proof. (1) $\mathfrak{B}_\pi$ is invariant under $\mathcal{L}$, and on it $\mathcal{L}(X) = -\frac{1}{2}\sum_k \gamma_k [A_k,[A_k,X]]$. On a class C, replacing A_k by $A_k - m_k$ with m_k the midpoint of its eigenvalues on C does not change the commutator and gives an operator of norm $w_k(C)/2$; hence $\|[A_k,X_C]\| \le w_k(C)\|X_C\|$ and $\|[A_k,[A_k,X_C]]\| \le w_k(C)^2\|X_C\|$. Since $\|X\| = \max_C \|X_C\|$, $\|\mathcal{L}(X)\| \le \Gamma_{\mathrm{in}}(\pi)\|X\|$. Then $\Phi_t^*(X) - X = \int_0^t \Phi_s^*(\mathcal{L}(X))\,ds$ and Φ_s^* is a contraction in operator norm. For a state $\rho \in \mathfrak{B}_\pi$ and $\|Y\| \le 1$, $\mathrm{Tr}[(\Phi_t(\rho)-\rho)Y] = \mathrm{Tr}[\rho(\Phi_t^* - \mathrm{id})(\mathbb{E}_\pi Y)]$, because $\rho = \mathbb{E}_\pi \rho$, $\mathbb{E}_\pi$ is self-dual and commutes with the Schur multiplier Φ_t^*; since $\mathbb{E}_\pi Y \in \mathfrak{B}_\pi$ and $\|\mathbb{E}_\pi Y\| \le 1$, the first statement bounds this by $t\Gamma_{\mathrm{in}}(\pi)$.

(2) Let $\mathfrak{B} \supsetneq \mathfrak{B}_\pi$ and $X \in \mathfrak{B} \setminus \mathfrak{B}_\pi$. Some entry $X_{x'y'}$ with $x' \in C$, $y' \in C'$, $C \ne C'$, is nonzero. For any $x \in C$, $y \in C'$ the matrix units $E_{xx'}$ and $E_{y'y}$ lie in $\mathfrak{B}_\pi \subseteq \mathfrak{B}$, so $E_{xx'} X E_{y'y} = X_{x'y'} E_{xy} \in \mathfrak{B}$. Then $\|\Phi_T^*(E_{xy}) - E_{xy}\| = 1 - e^{-TD(x,y)} \ge 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$.

(3) Let π and π' be selected by (T,ε) and (T',ε), $T \le T'$. If x, y lie in one class of π', then $D(x,y) \le \Gamma_{\mathrm{in}}(\pi') \le \varepsilon/T' \le \varepsilon/T$. If x, y lay in different classes of π, then $D(x,y) \ge \Gamma_{\mathrm{out}}(\pi) > -\ln(1-\varepsilon)/T \ge \varepsilon/T$. Hence every class of π' lies in a class of π. For $T = T'$ the same argument with the roles exchanged gives $\pi = \pi'$.

(4) $\Gamma_{\mathrm{in}}(\pi_\infty) = 0$ and $\Gamma_{\mathrm{out}}(\pi_\infty) > 0$, so π_∞ satisfies (4) for such T; uniqueness is part (3). □

A window and a tolerance therefore pick out at most one algebra of the form $\mathfrak{B}_\pi$, and they pick one out when the rate image of the basis splits into clusters that are small compared with ε/T and far apart compared with $-\ln(1-\varepsilon)/T$. This is a separation of time scales, the mechanism behind metastability in open quantum systems [17]; for commuting couplings it can be stated exactly. The selected algebra is moreover the largest ε-stable algebra of block form: if $\mathfrak{B}_\sigma$ is ε-stable on $[0,T]$, every matrix unit E_{xy} with x, y in one class of σ satisfies $1 - e^{-TD(x,y)} \le \varepsilon$, so x and y lie in one class of the selected partition π, and $\mathfrak{B}_\sigma \subseteq \mathfrak{B}_\pi$. As the window lengthens, the selected algebra shrinks in discrete steps located at the gaps of the rate spectrum and ends at the fixed algebra. In the order of Paper A this is a chain of further coarse-grainings, since $\mathbb{E}_{\pi'} = \mathbb{E}_{\pi'} \circ \mathbb{E}_\pi$ when π' refines π.

Without a gap there is no selected algebra at that window. Selection uses the sufficient bound of part (1), so between that bound and the necessary bound of part (2) a unique maximal stable block algebra can still exist; but the window may also fail to determine one, and two incomparable maximal stable block algebras can occur. The set of ε-stable matrix units can then be an operator system whose generated algebra contains unstable elements: if $D(x,y)$ and $D(y,z)$ are small, $D(x,z)$ can still be as large as $(\sqrt{D(x,y)} + \sqrt{D(y,z)})^2$, and a chain of small steps can connect distant points of the rate image. Theorem 3.2 is a sufficient selection certificate. Failure of its gap condition does not by itself decide existence or uniqueness of a maximal stable algebra. Nor does stability of a block algebra supply the instruments needed to use every element as a gate.

3.1 Continuous capabilities when the fixed algebra jumps

A weak coupling can change the exact commutant discontinuously. It cannot change every finite experiment discontinuously. The distinction matters when a selected algebra is used to describe a finite observer.

Proposition 3.3 (Finite-time operational stability)**.** *Let $\mathcal{L}_0, \mathcal{L}_1$ be finite-dimensional Lindblad generators, and put $v = \|\mathcal{L}_1 - \mathcal{L}_0\|_\diamond$. Then*

$$\|e^{t\mathcal{L}_1} - e^{t\mathcal{L}_0}\|_\diamond \le \min\{2, tv\}. \tag{5}$$

Consider any experiment with the same initial states, retained ancillas and permitted instruments in both models. Between its instruments the system evolves for prescribed intervals $t_1, \dots, t_m$, with total $T = \sum_i t_i$. Instruments may depend on earlier outcomes. The total variation distance between its complete record laws is at most $\min\{1, Tv/2\}$. Consequently, for a common decision problem with loss in $[0, L]$, the two optimal risks differ by at most $L\min\{1, Tv/2\}$. In particular, a Hamiltonian perturbation $-i\epsilon[K,\cdot]$ gives $v \le 2|\epsilon|\|K\|$.

Proof. Duhamel's formula writes the channel difference as $\int_0^t e^{(t-s)\mathcal{L}_1}(\mathcal{L}_1 - \mathcal{L}_0)e^{s\mathcal{L}_0}\,ds$. CPTP maps have diamond norm one, giving tv; two channels differ by at most two. A hybrid argument replaces the evolution intervals one at a time, keeping outcome-controlled instruments as CPTP maps on the system and classical record. Contractivity and the triangle inequality give a trace-norm difference at most Tv. The bounded-loss estimate holds uniformly over each common policy, so it also holds for the infima. Finally, $\|[K,A]\|_1 \le 2\|K\|\|A\|_1$ holds with every retained ancilla. □

The estimate needs neither a gap nor a selected algebra. It bounds distinguishing, storing and acting through the permitted experiment, not just the state of an isolated system. The available instruments and the total time remain

the same on both sides. A discontinuous change of permissions, or an unbounded observation time, is a different comparison.

For an exact noncommuting example, take

$$\mathcal{L}_\epsilon(\rho) = -i\epsilon[X,\rho] + \kappa(Z\rho Z - \rho), \qquad \kappa > 0. \tag{6}$$

The fixed algebra is $\mathrm{span}\{I, Z\}$ at $\epsilon = 0$ and only $\mathbb{C}I$ at every $\epsilon \neq 0$. To see the latter, the Hilbert–Schmidt dissipative form of a fixed observable forces $[Z, O] = 0$; the remaining Hamiltonian equation then forces $[X, O] = 0$. Nevertheless, the two initially perfect records $(I \pm Z)/2$ remain readable for a long time. If $0 < 2|\epsilon| < \kappa$, put $\omega = \sqrt{\kappa^2 - 4\epsilon^2}$. Their Bloch vectors at time t are $\pm(0, y(t), z(t))$, where

$$z(t) = e^{-\kappa t}\left(\cosh\omega t + \frac{\kappa}{\omega}\sinh\omega t\right), \qquad y(t) = -\frac{2\epsilon}{\omega}e^{-\kappa t}\sinh\omega t. \tag{7}$$

These expressions solve $\dot{y} = -2\kappa y - 2\epsilon z$, $\dot{z} = 2\epsilon y$, with $(y, z) = (0, 1)$ initially. Reading only Z has error $(1 - z(t))/2$; allowing the optimal system measurement gives error $(1 - \sqrt{y(t)^2 + z(t)^2})/2$. The slow decay rate is exactly

$$r = \kappa - \sqrt{\kappa^2 - 4\epsilon^2} = \frac{2\epsilon^2}{\kappa} + O(\epsilon^4/\kappa^3). \tag{8}$$

Thus arbitrarily weak transverse coupling erases the infinite-time record, while its finite-time lifetime diverges as $\kappa/(2\epsilon^2)$. Exact hulls and fixed algebras describe the ideal endpoint; finite-time risks describe what remains available on the way to it. This is also the distinction needed by Foundation's hulls and the maintenance tasks of T-DOME and Brain Anchor. Neither the general stability estimate nor this example asserts that every slow operator space is an algebra or a physical projection.

4 Records live on the centre

4.1 A recording model

To see what the environment learns we replace the Markovian generator by an exact unitary model. Environmental fragments $E_1, \ldots, E_K$ start in states ω_k, and the interaction is

$$U = \sum_{x\in\mathcal{X}} |x\rangle\langle x| \otimes V_1(x) \otimes \cdots \otimes V_K(x), \tag{9}$$

with unitaries $V_k(x)$ that depend on the basis state through the coupling, for example $V_k(x) = \exp[-i\theta_k a_k(x) G_k]$ for a fragment generator G_k. The joint state after the interaction is

$$\sigma = \sum_{x,y} \rho_{xy}|x\rangle\langle y| \otimes \bigotimes_k V_k(x)\,\omega_k\,V_k(y)^\dagger,$$

so the system undergoes the Schur multiplier

$$\rho_{xy} \mapsto c(x,y)\,\rho_{xy}, \qquad c(x,y) = \prod_k c_k(x,y), \qquad c_k(x,y) = \mathrm{Tr}\big[\omega_k V_k(y)^\dagger V_k(x)\big]. \tag{10}$$

The generator (1) arises from such models in weak-coupling and repeated-collision limits. Here the collision statements are exact for the specified unitary and an initial product $\rho \otimes \bigotimes_k \omega_k$. Fresh independent fragments are an input: recollisions or shared initial fragment correlations do not obey the product formula automatically.

Proposition 4.1 (Records carry only the block label)**.** *Let π be a partition of $\mathcal{X}$ such that every $V_k(x)$ depends on x only through its class $a(x) \in \pi$. Then:*

1. *$c(x, y) = 1$ whenever x and y lie in one class, so $\mathbb{E}_\pi(\rho)$, and with it the state on every block, is unchanged.*
2. *For every $O \in \mathfrak{B}_\pi$ and every operator F on any collection $\mathcal{F}$ of fragments,*

$$\mathrm{Tr}[\sigma(O \otimes F)] = \sum_{a\in\pi} \mathrm{Tr}[P_a \rho P_a\, O]\ \ \mathrm{Tr}[\omega_{\mathcal{F}}(a) F], \qquad \omega_{\mathcal{F}}(a) = \bigotimes_{k\in\mathcal{F}} V_k(a)\,\omega_k\,V_k(a)^\dagger.$$

The state of the fragments is $\sum_a p(a)\,\omega_{\mathcal{F}}(a)$ with $p(a) = \mathrm{Tr}\,P_a\rho$; it depends on ρ only through the distribution of the label, and given the label, block observables and fragments are uncorrelated.

Proof. (1) $V_k(x) = V_k(y)$ when x, y lie in one class, so $c_k(x, y) = 1$. (2) $O_{yx} \neq 0$ only when x and y lie in one class a, and then the fragment factor is $\omega_{\mathcal{F}}(a)$ for both. Summing $\rho_{xy}O_{yx}$ over $x, y \in a$ gives $\mathrm{Tr}[P_a \rho P_a O]$. □

Whatever the environment learns under such couplings is the block label, which generates the centre of $\mathfrak{B}_\pi$. Many fragments can hold readable copies of that label while leaving the blocks alone. Block dependence alone does not guarantee distinguishability: the conditional states $\omega_k(a)$ must also be different, and a permitted measurement must resolve them. Records and the correlations inside the blocks are thus carried by one selected algebra, the first by its centre and the second by its blocks. That the redundantly available information is classical, and is information about one measurement, holds far more generally [18]; a two-qubit system with a decoherence-free subspace that proliferates part of its information while the remaining subspace keeps non-classical features was studied in [19]. What the commuting class adds is the exact location of both kinds of information in one algebra and, next, the price of imperfect symmetry.

4.2 Leakage costs coherence, and the costs multiply

Theorem 4.2 (Leakage and coherence)**.** *For basis states x, y and a fragment k, let $D_k(x, y) = \frac{1}{2}\big\|V_k(x)\omega_k V_k(x)^\dagger - V_k(y)\omega_k V_k(y)^\dagger\big\|_1$ be the fragment's distinguishability of x and y, so that an optimal measurement on the fragment identifies one of the two equally likely states with probability $\frac{1}{2}(1 + D_k)$ [20]. Then*

$$|c_k(x, y)|^2 + D_k(x, y)^2 \leq 1, \tag{11}$$

with equality when ω_k is pure. Consequently

$$|c(x, y)| \leq \prod_k \big(1 - D_k(x, y)^2\big)^{1/2} \leq \exp\Big(-\tfrac{1}{2}\sum_k D_k(x, y)^2\Big), \tag{12}$$

and if the coherence between x and y must stay above $1 - \varepsilon$, then

$$\sum_k D_k(x, y)^2 \leq 2 \ln \frac{1}{1 - \varepsilon}. \tag{13}$$

Proof. Purify $\omega_k = \mathrm{Tr}_R\, |\Omega\rangle\langle\Omega|$. Then $c_k(x, y) = \langle\Omega|(V_k(y)^\dagger V_k(x) \otimes I)|\Omega\rangle$ is the overlap of the pure states $(V_k(x) \otimes I)|\Omega\rangle$ and $(V_k(y) \otimes I)|\Omega\rangle$, whose trace distance is $(1 - |c_k|^2)^{1/2}$. Partial trace over R does not increase trace distance, which gives (11), with equality when there is nothing to trace out. Inequality (12) follows from (10) and $\ln(1 - u) \leq -u$, and (13) by taking logarithms. □

Inequality (11) is Englert's relation between visibility and which-way distinguishability [21], applied to one fragment. It bounds what a fragment learns by the coherence it destroys, not the reverse: a fragment in a mixed state can destroy coherence while learning nothing, as in Paper E's maximally mixed memory example. Purity is one useful preparation resource; mixed fragments can also record when their conditional states differ. For many fragments it becomes the budget (13): the sum of their squared pairwise distinguishabilities is at most $2\ln(1/(1 - \varepsilon))$ for a protected pair of basis states. This is not a bound on their total Shannon information about an arbitrary block.

Corollary 4.3 (Redundancy against symmetry)**.** *Suppose that R fragments each distinguish two states x, y of one block with $D_k(x, y) \geq \beta$. Then $|c(x, y)| \leq (1 - \beta^2)^{R/2}$, and keeping the coherence above $1 - \varepsilon$ requires*

$$\beta \leq \Big(\frac{2\ln(1/(1 - \varepsilon))}{R}\Big)^{1/2}. \tag{14}$$

In a sequence of models with $R \to \infty$ and $D_k(x, y) = \beta_R = \lambda R^{-\alpha}$, $\alpha > 0$, for the R fragments that couple to the block, the coherence tends to zero if $\alpha < \frac{1}{2}$. If the fragment states are pure and no other fragment distinguishes x from y, it tends to 1 for $\alpha > \frac{1}{2}$ and to $e^{-\lambda^2/2}$ for $\alpha = \frac{1}{2}$.

Proof. The bound and (14) are (12) and (13). For pure fragments $|c(x, y)| = (1 - \beta_R^2)^{R/2}$ by the equality case, and $\frac{R}{2}\ln(1 - \lambda^2 R^{-2\alpha}) \to 0,\ -\lambda^2/2,\ -\infty$ for $\alpha > \frac{1}{2}, = \frac{1}{2}, < \frac{1}{2}$. □

A coupling of the form (9) is block-constant only by symmetry: the fragments must respond identically to all states of a block. Real couplings break such a symmetry slightly, and each recording fragment then leaks a little. Corollary 4.3 says that objectivity by redundancy and protection of the blocks can coexist, but only if the symmetry error per fragment falls as $R^{-1/2}$. In the limit of many records this is a sharp threshold in the exponent α. It is the closest analogue of a transition in this class. It is a statement about how a symmetry error scales with the number of records, and it is not a phase transition in the thermodynamic sense.

5 Sector records and protected correlations

A mutual-information geometry does not require entanglement. Even local dephasing preserves $\rho = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$, for which $I(1:2) = \ln 2$. With local Z couplings every selected block is one-dimensional and every sector state is a product. Its entire edge comes from sector weights.

For a selected state $\bar{\rho} = \mathbb{E}_\pi(\rho) = \sum_k p_k \rho_k$, with $\rho_k = P_k \rho P_k / p_k$, retain the label in a classical register:

$$\Omega_{SK} = \sum_k p_k \rho_k \otimes |k\rangle\langle k|_K. \tag{15}$$

This extension is physically realized by a perfectly resolving block-constant recorder followed by measurement or dephasing of its label register. It is not the coherent source–memory state before that readout. If the actual fragment states overlap, K is still a useful mathematical label, but it is not a freely readable record. Zero-weight sectors are omitted.

Proposition 5.1 (The two contributions to correlation)**.** *For any supplied disjoint subsystems A, B of S,*

$$I(A:B)_{\bar{\rho}} = \sum_k p_k I(A:B)_{\rho_k} + I(A:K)_\Omega + I(B:K)_\Omega - I(AB:K)_\Omega. \tag{16}$$

The last three terms need not have a nonnegative sum.

Proof. For $X = A, B, AB$, the classical–quantum entropy formula gives $I(X:K) = S(\bar{\rho}_X) - \sum_k p_k S((\rho_k)_X)$. Substitution in $S(A) + S(B) - S(AB)$ proves the identity. □

The first term locates correlations protected within sectors; it can include classical as well as quantum correlations. The correction describes what mixing sector-conditioned states does to those correlations. In the locally dephased example above it equals $\ln 2$. In the next recording example it is negative. Thus neither a noncommutative block nor a classical centre alone determines the geometry. The sector states, their weights, the site marginals and the dictionary all enter.

6 How protected relations are formed

A noise commutant specifies what can be stored. It does not determine whether that content was inherited or created. The distinction already invalidates a noise-only mereology rule.

6.1 Disconnected noise can preserve a shared Bell pair

Take four qubits with only $A_{12} = Z_1 + Z_2$ and $A_{34} = Z_3 + Z_4$. Each pair has a zero-magnetization sector with logical basis $|0_L\rangle = |01\rangle$, $|1_L\rangle = |10\rangle$. The state

$$|\Psi_L\rangle = \frac{|0_L\rangle_{12}|0_L\rangle_{34} + |1_L\rangle_{12}|1_L\rangle_{34}}{\sqrt{2}} \tag{17}$$

is annihilated by both noise operators and is fixed exactly by (1). Yet $I(12:34) = 2\ln 2$. The two noise-hypergraph components are disconnected. Applying a Hadamard to each qubit gives the same counterexample for the X couplings of Section 7. Classical and quantum correlations can both cross disconnected noise components when they are present initially.

6.2 A collective recorder forms a flagged protected relation

Start two qubits in the physical product $|+_X\rangle|+_X\rangle$ and a fresh three-level memory in $|0\rangle$. Let $M = Z_1 + Z_2$ and let P_+, P_0, P_- project onto its eigenvalues $2, 0, -2$. For the shift $S_3|k\rangle = |k+1 \bmod 3\rangle$, use

$$U_{\mathrm{rec}} = P_+ \otimes I + P_0 \otimes S_3 + P_- \otimes S_3^2. \tag{18}$$

This is a block-constant recording interaction, not a product of independent local readers. It has a finite Hamiltonian implementation: choose a Hermitian logarithm G with $e^{-iG} = S_3$ and $\|G\| = 2\pi/3$, put $a(M) = (2I - M)/2$, and apply $H_{\mathrm{rec}} = (\hbar/t_g)a(M) \otimes G$ for a time $t_g > 0$. The same memory couples to both sites. Its preparation, shared physical port, interaction time, control and subsequent readout are resources.

The complete output is

$$|\Omega\rangle = \tfrac{1}{2}|00\rangle|0\rangle + \tfrac{1}{\sqrt{2}}|\Psi^+\rangle|1\rangle + \tfrac{1}{2}|11\rangle|2\rangle, \qquad |\Psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}. \tag{19}$$

A computational-basis measurement of the memory produces the classical–quantum state with probabilities $(1/4, 1/2, 1/4)$ and conditional source states $(|00\rangle\langle 00|, |\Psi^+\rangle\langle\Psi^+|, |11\rangle\langle 11|)$. All branches remain in the protocol. Flag 1, of probability $1/2$, certifies a Bell pair in the $M = 0$ protected sector; flags $0, 2$ certify product states.

If the flag is discarded, the source is instead

$$\rho_* = \tfrac{1}{4}|00\rangle\langle 00| + \tfrac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \tfrac{1}{4}|11\rangle\langle 11|, \qquad I(1:2)_{\rho_*} = \tfrac{1}{2}\ln 2. \tag{20}$$

It is separable. Indeed it is the uniform collective Z-rotation twirl of $|+_X, +_X\rangle\langle +_X, +_X|$, a mixture of product rotations. Its partial-transpose eigenvalues are $0, 1/4, 1/4, 1/2$. The unconditional source therefore gains a mutual-information edge without entanglement; separability does not imply that every correlation measure is classical. The retained flagged ensemble permits a conditional protected Bell use, with herald probability and every unsuccessful branch charged. The pure source–memory state (19), the measured flagged state and the unconditional source are three different outputs.

For this flagged ensemble, $\sum_k p_k I(1:2)_{\rho_k} = \ln 2$, $I(1:K) = I(2:K) = \frac{1}{2}\ln 2$ and $I(12:K) = H(K) = \frac{3}{2}\ln 2$. Equation (16) recovers the unconditional $\frac{1}{2}\ln 2$ with correction $-\frac{1}{2}\ln 2$. Collective dephasing of the product input alone reaches the same ρ_* but supplies no accessible flag. A flag requires the specified recording interaction and actual readout.

6.3 A real link forms a deterministic protected Bell pair

The distinction is sharper in the four-qubit example. Define

$$L^x_{ab} = \tfrac{1}{2}(X_aX_b + Y_aY_b), \qquad H_{\rm link} = \hbar J L^x_{12}L^x_{34}, \quad J > 0.$$

L^x_{ab} exchanges $|01\rangle$ and $|10\rangle$ and vanishes on $|00\rangle, |11\rangle$. Consequently $[H_{\rm link}, A_{12}] = [H_{\rm link}, A_{34}] = 0$. Starting from the physical product $|01\rangle_{12}|01\rangle_{34}$, the full noisy evolution remains in the joint zero sector and is exactly

$$|\psi(t)\rangle = \cos(Jt)|0_L0_L\rangle - i\sin(Jt)|1_L1_L\rangle. \tag{21}$$

At $t = \pi/(4J)$ it has $I(12:34) = 2\ln 2$ and one ebit across the pair partition. The noise term vanishes throughout the trajectory.

This is new protected correlation from an initially uncorrelated state. It uses an actual cross-pair Hamiltonian link, here a four-body control term. The noise graph remains disconnected but the complete interaction graph does not. The factorization, prepared input, gate permissions, coupling strength and duration are supplied; there is no inferred free implementation or heat price for the link.

6.4 The process constraint and learned use

Proposition 6.1 (Correlation provenance across independent components)**.** *Let a supplied partition of a carrier have initially independent states and auxiliary resources. Suppose its full channel, including record registers, factors across these components. Then it creates no cross-component mutual information. If the input is correlated, a product channel cannot increase the mutual information across any bipartition of the components, but it can preserve initially shared classical or quantum correlations.*

Proof. A product channel sends a product input to a product output, including its local records. For a correlated input, $I(A:B) = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, so the second statement is data processing under the product channel. □

An implementation satisfies the factor premise when its Hamiltonian and noise generators act separately on the components, its fresh ports are independent, and every control and instrument is component-local. Shared random controls, communication or common memories must be added to the process description: they can correlate unconditional outputs. Conditioning on a joint event can also change correlations and is not a product unconditional channel. The proposition is a diagnostic, not a claim that every process with a disconnected noise graph factors.

The constructions above do more than add initial-state qualifications to a conjecture. They exhibit how the missing relations can form: a collective port records an indistinguishability sector, while an interaction inside protected sectors builds usable quantum content. The other indispensable step is learning which encoding an unknown physical plant

protects. Foundation III [6] implements that step for $A_s = n_1 + sn_2$ with unknown $s = \pm 1$. A diagnostic sharing the plant's actual sign writes a record; that record, not s, selects odd or even encoding and the matching decoder. The same logical content and its old reference survive from before diagnosis. The record-controlled frame changes the later contact, rather than merely changing a description of the fixed commutant.

That result assumes a physical relation between diagnostic and storage couplings. A diagnostic carrying an unrelated sign would not learn the protected frame. Its retained-history scalar/JT implementation also contains real body forces, a source and a second acquisition; it is not identified with the full quantum material model. The complementary contribution here is the separation of formation, protection and use: each can fail while the others hold. A noiseless block can be unreadable, a readable label can outlive its quantum content, and a learned encoder can protect content without supplying every logical gate.

7 A geometry selected by the coupling

We now apply these results to the state and the reconstruction rule of Paper A. The rule acts on the complete graph of the sites: the edge $\{i, j\}$ has length $1/I_{ij}$, where I_{ij} is the mutual information of sites i and j, and distances are shortest-path lengths. We extend it to $I_{ij} = 0$ by giving that edge infinite length, so that a site whose edges all have zero mutual information is disconnected. The state is the three-qubit GHZ state $(|000\rangle + |111\rangle)/\sqrt{2}$. In the eigenbasis $|\pm\rangle$ of the Pauli operator X it reads

$$|\mathrm{GHZ}\rangle = \tfrac{1}{2}\big(|+{+}{+}\rangle + |+{-}{-}\rangle + |-{+}{-}\rangle + |-{-}{+}\rangle\big).$$

All couplings below are sums of X_i, hence diagonal in this basis, and Sections 2–4 apply. A local coupling $A = X_i$ with rate γ_i is the bit-flip noise of Paper A, with flip probability $p_i = (1 - e^{-2\gamma_i t})/2$. A collective coupling $A = X_1 + X_2 + X_3$ dephases between different values of the X-magnetization $M = \sum_i x_i$ and leaves states of equal M coherent.

Proposition 7.1 (Three couplings, three geometries)**.** *Let* $\Psi^+ = (|+-\rangle + |-+\rangle)/\sqrt{2}$ *and* $\tilde{W} = (|+{-}{-}\rangle + |-{+}{-}\rangle + |-{-}{+}\rangle)/\sqrt{3}$*. The long-time selected states* $\mathbb{E}_{\pi_\infty}(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|)$ *and their geometries are:*

1. *Collective coupling* $X_1 + X_2 + X_3$*:* $\frac{1}{4}|{+}{+}{+}\rangle\langle{+}{+}{+}| + \frac{3}{4}|\tilde{W}\rangle\langle\tilde{W}|$*. Each pair has* $I_{ij} = \frac{1}{2}\ln 2$*; the geometry is an equilateral triangle with side* $2/\ln 2 = 2.8854$*.*
2. *Collective coupling* $X_1 + X_2$ *and local coupling* X_3*:* $\frac{1}{4}|{+}{+}{+}\rangle\langle{+}{+}{+}| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+|_{12} \otimes |-\rangle\langle-|_3 + \frac{1}{4}|{-}{-}{+}\rangle\langle{-}{-}{+}|$*. Here* $I_{12} = \frac{1}{2}\ln 2$ *and* $I_{13} = I_{23} = 0$*; the geometry is a single edge, and site* 3 *is disconnected.*
3. *Local couplings* X_1, X_2, X_3*: the uniform mixture of the four even strings* $|{+}{+}{+}\rangle, |{+}{-}{-}\rangle, |{-}{+}{-}\rangle, |{-}{-}{+}\rangle$*. Every* $I_{ij} = 0$*; no edge survives.*

Proof. The pinching keeps the amplitudes inside each class of equal eigenvalue tuples. (1) The classes met by the GHZ state are $M = 3$ and $M = -1$. The marginal of sites $1, 2$ is $\frac{1}{4}|{+}{+}\rangle\langle{+}{+}| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \frac{1}{4}|{-}{-}\rangle\langle{-}{-}|$, with eigenvalues $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ and entropy $\frac{3}{2}\ln 2$; each single-site marginal is $I/2$; so $I_{12} = 2\ln 2 - \frac{3}{2}\ln 2$, and the other pairs follow by symmetry. (2) The classes are labelled by $(x_1 + x_2, x_3)$; the strings $|+{-}{-}\rangle$ and $|-{+}{-}\rangle$ share the label $(0, -)$. The marginal of sites $1, 2$ is as in (1). The marginal of sites $1, 3$ is uniform on the four strings $|{+}{+}\rangle, |{+}{-}\rangle, |{-}{-}\rangle, |{-}{+}\rangle$, and so is that of sites $2, 3$. (3) Every pair marginal of the uniform mixture of even strings is uniform. Direct evaluation of the full eight-dimensional density matrices gives the same values. □

The state and the rule are the same in all three cases; only the coupling differs. At zero time each pair has $I_{ij} = \ln 2$, Paper A's GHZ geometry at zero noise. Under local noise Paper A's triangles relax toward the empty graph; a collective coupling stops the relaxation at an equilateral triangle, and a coupling that is collective on one pair keeps exactly that edge. The surviving correlations come from blocks of dimension greater than one: the three-dimensional block $M = -1$, which contains the entangled state $\tilde{W}$, in case (1), and the two-dimensional block spanned by $|+{-}{-}\rangle$ and $|-{+}{-}\rangle$ in case (2). In case (3) the selected algebra is commutative with product minimal projections, every selected state is classically correlated, and for this state the classical pair correlations vanish as well. By Proposition 2.1(4) the channels at any finite time generate the full matrix algebra in all three cases, so an algebra label cannot tell them apart.

7.1 The lifetime of the selected geometry

Real couplings are not exactly collective. Add weak local couplings X_i with rates γ_i to the collective coupling with rate γ_c. The exact fixed algebra is then commutative and the long-time geometry is empty, but on a window the

magnetization blocks are still selected. With π the partition by M, the rates (3) are $\Gamma_{\rm in}(\pi) = 2(\gamma_1+\gamma_2+\gamma_3)$ and $\Gamma_{\rm out}(\pi) = 2\gamma_c + 2\min_i \gamma_i$, while the coherences of the GHZ state between its two blocks decay at the rates $8\gamma_c + 2(\gamma_i+\gamma_j)$.

Lemma 7.2 (Plateau bound)**.** *For a state ρ and a partition π, let $\Gamma_{\rm out}(\rho,\pi)$ be the smallest $D(x,y)$ over pairs in different classes with $\rho_{xy}\neq 0$, and let r be the number of basis vectors with $\rho_{xx}\neq 0$. Then for all $t\geq 0$*

$$\|\Phi_t(\rho) - \mathbb{E}_\pi(\rho)\|_1 \leq \sqrt{r}\, e^{-t\Gamma_{\rm out}(\rho,\pi)}\, \|\rho - \mathbb{E}_\pi(\rho)\|_2 + t\,\Gamma_{\rm in}(\pi). \tag{22}$$

Proof. Split $\Phi_t(\rho) - \mathbb{E}_\pi\rho = \Phi_t(\rho - \mathbb{E}_\pi\rho) + (\Phi_t(\mathbb{E}_\pi\rho) - \mathbb{E}_\pi\rho)$. The second term is bounded by Theorem 3.2(1). The first is the Schur product of $\rho - \mathbb{E}_\pi\rho$ with multipliers of modulus at most $e^{-t\Gamma_{\rm out}(\rho,\pi)}$; its rank is at most r, because a row of a positive matrix with zero diagonal entry vanishes, and $\|Z\|_1 \leq \sqrt{\operatorname{rank} Z}\,\|Z\|_2$. □

The bound turns into a certified geometry through continuity of entropy. If two states of a qubit pair are at trace distance $\delta = \frac{1}{2}\|\rho-\sigma\|_1 \leq \frac{1}{2}$, then by the Fannes–Audenaert inequality [22] and Winter's bound for conditional entropy [23] their mutual informations differ by at most

$$f(\delta) = h(\delta) + 2\delta\ln 2 + (1+\delta)\,h\Big(\frac{\delta}{1+\delta}\Big), \tag{23}$$

with h the binary entropy in nats. Partial traces do not increase trace distance, and Paper A's stability bound for graph metrics converts a uniform change $f < m$ of mutual informations bounded below by m into a change of at most $2f/(m(m-f))$ of every distance among three sites.

t	$\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1$	bound (22)	distances (d_{12}, d_{13}, d_{23})
0	0.8660	1.2247	(1.4427, 1.4427, 1.4427)
1	0.0010	0.0018	(2.8968, 2.9034, 2.9066)
10	0.0094	0.0140	(2.9728, 3.0221, 3.0458)
100	0.0897	0.1400	(3.5620, 3.9855, 4.2034)
300	0.2442	0.4200	(4.8943, 6.5032, 7.4508)
1000	0.5969	1.4000	(12.559, 28.867, 41.426)

Table 1: GHZ state under a collective coupling with $\gamma_c = 1$ and local couplings with $(\gamma_1,\gamma_2,\gamma_3) = 10^{-4}(1,2,4)$. The plateau value of every distance is $2/\ln 2 = 2.8854$.

Proposition 7.3 (A certified plateau)**.** *For $\gamma_c = 1$ and $(\gamma_1,\gamma_2,\gamma_3) = 10^{-4}(1,2,4)$, the bounds* (22) *and* (23) *certify that every distance of the GHZ geometry lies within 25% of the plateau value $2/\ln 2$ for $0.716 \leq t \leq 3.58$. Numerical root finding of the exact evolution gives the approximate 25% window $0.053 \lesssim t \lesssim 53.6$ and 5% window $0.163 \lesssim t \lesssim 8.86$. If the local rates are scaled to $s(1,2,4)$ with $s \leq 10^{-3}$, the late edges scale to leading order as $1/s$: approximately $5.36\times 10^{-3}/s$ for the 25% window, $8.86\times 10^{-4}/s$ for the 5% window and, when $s \leq 10^{-4}$ so that it exists, $3.58\times 10^{-4}/s$ for the certified 25% window. The windows begin at times of order $1/\gamma_c$.*

Proof. For this state $r = 4$, $\|\rho - \mathbb{E}_\pi\rho\|_2 = \sqrt{6}/4$, $\Gamma_{\rm out}(\rho,\pi) = 8.0006$ and $\Gamma_{\rm in}(\pi) = 1.4\times 10^{-3}$. The right side of (22) is the convex function $b(t) = (\sqrt{6}/2)e^{-8.0006t} + 0.0014t$. On $[0.716, 3.58]$ its maximum is at an endpoint and is less than 0.0050121. The trace distance used in (23) is $\delta(t) = \frac{1}{2}\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1 \leq b(t)/2$, not $b(t)$. The increasing entropy and graph bounds therefore give a distance error less than $0.72130 < \frac{1}{4}\cdot 2/\ln 2$ throughout that interval. The exact-evolution window edges were located by root finding of the Schur multipliers. Once the coherences between blocks have decayed, the in-block multipliers depend on t only through st, which gives the late-edge scaling. The root values are numerical diagnostics, distinct from the conservative analytic interval. □

Table 1 shows the three stages. The GHZ geometry, with side $1/\ln 2$, relaxes on the time scale $1/\gamma_c$ to the selected triangle with side $2/\ln 2$; the triangle persists while $t\Gamma_{\rm in}(\pi)$ is small; then the unequal local rates deform it into scalene triangles, which are not isometric to the plateau, before every edge diverges. This is Paper A's non-isometry produced by one time-ordered dynamics rather than by a comparison of separately chosen channels. Proposition 7.3 gives a state-specific stability certificate for Paper A's declared graph dictionary on an explicit window. The plateau exists when its end lies after its beginning, that is, when the ratio of in-block to between-block rates is small, as Theorem 3.2 requires.

The constants in Proposition 7.3 are small. At the end of the exact 25% window the three in-block coherences have fallen by only 3.2, 5.2 and 6.2 per cent. The selected pair marginal has a zero eigenvalue, and in-block decoherence populates it, so pair entropies respond to a coherence loss δ like $\delta \ln(1/\delta)$. Geometries read from entropies are therefore far more fragile than the coherences that carry them, and the gap between the certified and the exact window comes mostly from this logarithmic response together with worst-case continuity bounds.

7.2 Imperfect redundant records detach a vertex

Finally we let fragments record the magnetization with a small asymmetry. Each of K fragment qubits starts in $|0\rangle$ and interacts through $V_k(x) = \exp[-i\theta(M(x)+\eta x_1)Y]$ with $\theta = \pi/8$, where Y is the Pauli operator of the fragment. For $\eta = 0$ the coupling is block-constant. The two magnetizations present in the GHZ state, $M = 3$ and $M = -1$, then produce orthogonal fragment states, so every fragment holds a perfect copy of the label, and by Proposition 4.1 the equilateral triangle survives for any number of fragments. For $\eta > 0$ each fragment distinguishes strings of the block $M = -1$ that differ at site 1, with $D_k = |\sin 2\theta\eta|$.

η	K	coherence of $\|+--\rangle, \|-+-\rangle$	$I_{12} = I_{13}$	I_{23}	label error
0.02	1	0.99988	0.34631	0.34657	6.2×10^{-5}
0.02	100	0.98774	0.32790	0.34657	6.2×10^{-5}
0.02	1000	0.88393	0.23581	0.34657	6.2×10^{-5}
0.05	1	0.99923	0.34525	0.34657	3.9×10^{-4}
0.05	100	0.92577	0.26724	0.34657	3.9×10^{-4}
0.05	1000	0.46243	0.05555	0.34657	3.9×10^{-4}

Table 2: Recording the magnetization with a site-1 asymmetry η. The last column is the minimal error probability with which a measurement on one fragment identifies the label M, whose prior is $(\frac{1}{4}, \frac{3}{4})$. Coherences are relative to their initial values; the coherence between $|-+-\rangle$ and $|--+\rangle$ stays equal to 1.

Table 2 shows the effect. Every fragment is a nearly perfect and readable record of the label for both asymmetries, so the loss of geometry is not a price paid for better records. It is the small leakage about site 1, multiplied over fragments as in Theorem 4.2. The coherences between strings that differ at site 1 decay as $(1 - D_k^2)^{K/2}$, the edges to site 1 weaken, and as $K \to \infty$ the state approaches $\frac{1}{4}|+++\rangle\langle+++| + \frac{1}{4}|+--\rangle\langle+--| + \frac{1}{2}|-\rangle\langle-|_1 \otimes |\Psi^+\rangle\langle\Psi^+|_{23}$. Then $I_{12} = I_{13} = 0$ and $I_{23} = \frac{1}{2}\ln 2$: site 1 is detached and the geometry becomes the single edge of Proposition 7.1(2), now on sites $2, 3$. With $\eta_K = 0.5K^{-\alpha}$ the coherence after $K = 10^6$ fragments is 0 for $\alpha = \frac{1}{4}$, $0.9258 = e^{-(\pi/8)^2/2}$ for $\alpha = \frac{1}{2}$, and 0.9999 for $\alpha = \frac{3}{4}$, as Corollary 4.3 predicts with $\lambda = 2\theta \cdot 0.5$ to leading order.

8 What selection settles

The selected partition is fixed by the supplied dynamics, window and tolerance, and is robust to perturbations preserving the rate gap. This differs in emphasis from QBism's state assignments and relational quantum mechanics' reference-system dependence [26, 25], without deciding their wider interpretations. Public agreement further needs actual fragments and calibrated readers; Paper A provides one finite realization. A stable algebra alone does not grant those permissions.

The GHZ comparison changes a state under one graph dictionary, not the coordinates of an unchanged metric. Sector mixing can sustain edges after quantum content is lost; the leaking GHZ block instead supports the specific edges destroyed in Table 2. Formation requires the collective port or Hamiltonian link of Section 6, not noise selection alone. Removing either resource supplies its corresponding opposing control.

AQFT and Consciousness I start with their local or current-net carriers [24, 4]. Observable-induced factorization and quantum mereology address when retained relations support an effective subsystem description [28, 29, 30]. Decoherence and Everettian accounts share the underlying dynamics [10, 27]; the issue here is whether its resulting structure is readable, operable and persistent for a finite task.

9 Beyond the commuting class

Conjecture 9.1 (Process-selected effective subsystems)**.** *On an initially specified carrier, robust effective subsystems form through a history of interactions that both builds relations and makes a noncommuting set of internal operations*

reproducible across a finite task window. The relevant connectivity is that of the complete physical process, including recording ports and Hamiltonian controls, not the collective-noise hypergraph alone. Within a separated-rate regime, a learned encoding can turn the process's indistinguishable sectors into stable operational components; their boundaries change when the reproducible operations or their lifetime change.

The conjecture concerns effective factorization, not the existence of elementary sites. It predicts more than correlation alone: successful formation should yield an encoder, decoder and repeatable operations on the formed component, with calibration records and finite error bounds. The exact formation examples give protected relational content; Foundation III adds record-controlled encoding of unknown content. Together they support a route from coupling through history to changed capability. They do not yet supply a general autonomous acquisition of a complete internal gate set. A proposed component that has only a state-dependent correlation but no reproducible internal operations would fail this criterion. The disconnected Bell counterexample rules out noise-only connectivity as a necessary condition, while Proposition 6.1 identifies the real no-creation obstruction for product processes.

For general Lindblad dynamics, a separated spectrum defines slow states and an effective metastable evolution [17]. A finite-window extension should select the operational structure of this manifold rather than assume that slow observables close under multiplication. Even complete positivity and idempotence do not ensure closure in the original multiplication. For example,

$$P(X) = \operatorname{diag}\big(x_{11}, x_{22}, (x_{11} + x_{22})/2\big) \tag{24}$$

is a unital completely positive idempotent on M_3. It is completely positive because it is a sum of two positive evaluation maps into positive diagonal matrices. Its range contains $E = \operatorname{diag}(1, 0, 1/2)$ but not $E^2 = \operatorname{diag}(1, 0, 1/4)$. A completely positive projection gives its range a Choi–Effros product $x \circ y = P(xy)$ [7, 8]. This abstract product does not make xy a physically permitted observable or control in the original representation. An ordinary conditional expectation onto a represented subalgebra needs the additional algebraic closure. A metastable spectral projection may not even be completely positive. Thus a general selection theory must provide actual instruments or recovery maps, their errors and their operational closure, rather than infer them from a slow spectral space.

The physical price of that capability is the next link to Q-RAIF and the gravitational line. A passive GHZ geometry has lifetime of order $1/s$ under symmetry breaking, while independent redundant fragments require an $R^{-1/2}$ protecting symmetry. Q-RAIF B realizes correction as field coupling with work and stress [5]; HAFF D distinguishes entropy removal, usable side information and supplied purity, and gives specified source–response constructions [2]. These results suggest a maintained-structure mechanism: actual records learn a protected frame, operations install it, maintenance changes the physical source, and the response changes subsequent records. Foundation III realizes a finite retained-history scalar/JT return, whereas the full quantum material source requires its own support and preparation account. Connecting those implementations and renewing their resources is a substantive requirement, not an irreducible heat flux inferred from algebra inclusion.

Many-site and continuum selection require further work when rate gaps close. Consciousness I's ideal fixed-net closure and finite-time noise estimates are not already a continuum uniqueness theorem for window-selected blocks. Likewise, the graph dictionary is supplied here. A general emergence of spacetime needs a physical family of readers and a reconstruction law that remains controlled as the operational structure grows.

10 Conclusion

A commuting coupling determines its fixed block algebra and its long-time pinching. On a separated finite window it selects a unique block partition in the supplied joint eigenbasis; the selected algebra contains every stable block algebra of that form. The actual finite-time channel has full operator range. Block-constant record couplings report sector weights without resolving protected content. Independent leakage multiplies the coherence loss and imposes a sharp redundancy–symmetry threshold.

The relation to geometry has two parts. Sector mixing and within-sector content contribute through the exact identity (16); neither alone determines the mutual information graph. For the GHZ source, collective, pair-collective and local couplings yield a triangle, one edge and no edge under one rule. Weak asymmetry gives a finite plateau and then destroys its supporting edges. Noncommuting perturbations can erase the exact fixed algebra without abruptly erasing a finite record or task capability.

Formation requires a history. An initially shared logical Bell pair can survive between noise-disconnected protected components, so a noise graph cannot delimit all surviving quantum relations. A collective recorder forms a flagged protected Bell branch from a product input, with all branches retained; an actual Hamiltonian link forms a deterministic Bell pair protected throughout its generation. These constructions reduce the missing formation step to identifiable physical operations without hiding their costs. Record-controlled encoding supplies the further route from learned

relations to changed capability. The remaining general problem is to derive reproducible subsystem operations and their sustained maintenance from such histories, rather than merely rename correlated states as subsystems.

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Causation, Agency, and Existence: Historical Control and Maintained Intervention

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Abstract

A cause changes what an allowed intervention can bring about; a record alone need not identify that change. We define interventional influence relative to a process, an intervention set and a readout. It contracts under post-processing and along channel chains. A direct cause and a common cause can have identical declared record laws and influences 1 and 0.

Agency requires more than a persistent register. In a binary tracking task, the memoryless failure probability is bounded by $\frac{1}{2}[1-(1-2q)\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}]$. Three retained readings beat every present-reading controller in the specified sensor menu, despite a smaller influence of the present world on the action. We then give the same agent a damageable body: an actual mixed-port diagnosis controls repair, and the repaired body determines the next sensing channel. The finite reversible protocol maintains tracking error at most 0.15820 from a healthy start, against a long-run lower bound 0.22172 for damage-independent maintenance calendars at the same repair rate. All spent ports and old records remain; fresh blank supply is not a thermal reset. This is a mechanism for maintaining intervention capability, not a derivation of the initial body or its couplings.

These results support relational existence: structures can be objective within their coupling, history and observation window without being intrinsic to a state alone. The essay develops implications for will, self-maintenance and structural realism, then compares traces and dependence with Yogācāra and Mādhyamaka. These comparisons have their own arguments and do not establish phenomenal consciousness.

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1 Introduction

A register can remain unchanged while doing nothing. An agent can change almost every internal value while continuing to sense, act and preserve the conditions for doing so again. The distinction matters when a quantum state does not by itself select a division into things, causes and agents.

The earlier papers provide different parts of the physical account. Paper A [1] connects actual channel and record permissions to different correlation geometries for a fixed source and dictionary. Paper B [2] derives stable block content in a commuting coupling class and distinguishes pre-existing relations from relations formed by actual recording or Hamiltonian links. Block-constant couplings record central labels; arbitrary environmental records need not have that form. Paper E [3] derives measurement permissions, partly mixed records and calibrated public readout. Paper F [4] accounts for record formation, retention and updating; Paper G [5] asks how much of the selection can be performed from inside the process.

The present question is how these structures acquire causal and agentive roles. Section 2 separates observation from intervention. Section 3 bounds viable control with and without history, then closes a missing step: records act on the same body whose condition controls later sensing. Section 4 develops relational objectivity rather than identifying reality with a reader's answer. Section 5 compares the argument with Buddhist accounts of traces and dependence. The initial partition and gate permissions are stated inputs to the finite mechanism; their primitive formation remains a broader question.

2 Part I: Causation, records and interventions

2.1 Observational asymmetry and its limits

Standard quantum mechanics assigns no joint density operator to non-commuting observables at two times without a process formalism, so a cross-time mutual information is defined only relative to declared instruments.

Definition 2.1 (Instrument-relative directional asymmetry)**.** *Fix a two-time process $\mathcal{N}_{t_1\to t_2}$, an initial preparation and instruments $\mathcal{I}_A, \mathcal{I}_B$ with classical outcomes $X_A^{t_i}, X_B^{t_i}$, measured in separate forward and reverse arrangements. Put*

$$\Delta^{\mathcal{I}}_{A\to B} = I(X_A^{t_1} : X_B^{t_2}) - I(X_B^{t_1} : X_A^{t_2}),$$

with Shannon mutual informations of the outcome distributions.

$\Delta^{\mathcal{I}}$ is a well-defined observational diagnostic. Its sign depends on the instruments, the preparation and the process, and it can arise from a common cause or from asymmetric noise. Other conventions for quantum two-time objects exist [9, 10]; none turns a correlation into a cause without an account of intervention, which we now give.

2.2 Interventional influence

Definition 2.2 (Interventional influence)**.** *Fix a process in which a declared set $\mathcal{J}_A$ of interventions can be applied at A, each replacing the mechanism that ordinarily determines A while everything else is held fixed, and fix a readout at B. Let ρ_B^J be the state, or the outcome distribution, at B under the intervention J. The interventional influence of A on B is*

$$\kappa_{\mathcal{J}}(A \to B) = \sup_{J,J' \in \mathcal{J}_A} D\big(\rho_B^J, \rho_B^{J'}\big),$$

where D is the trace distance, or the total-variation distance for classical outcomes.

For several steps, with memory in the environment and controls at several times, the natural setting is a quantum comb or process tensor [11, 12]; the definition applies to any slot of such an object.

Proposition 2.3 (Properties of interventional influence)**.**

1. *$0 \le \kappa \le 1$, and $\kappa = 0$ exactly when no allowed intervention changes the state at B.*
2. *Post-processing the readout at B by any channel, or restricting $\mathcal{J}_A$, cannot increase κ.*
3. *If the interventions at A affect B only through M, in the sense that $\rho_B^J = \Lambda(\rho_M^J)$ for a channel Λ that does not depend on J, then*

$$\kappa(A \to B) \le \eta(\Lambda)\, \kappa(A \to M), \qquad \eta(\Lambda) = \sup_{\rho \neq \sigma} \frac{D(\Lambda\rho, \Lambda\sigma)}{D(\rho, \sigma)}.$$

For a classical channel with kernel $\Lambda(\cdot \mid m)$, $\eta(\Lambda) = \max_{m,m'} D(\Lambda(\cdot \mid m), \Lambda(\cdot \mid m'))$, Dobrushin's coefficient [13], which is the influence of M on B under all point interventions on M.

Proof. (1) is immediate. (2) Trace and total-variation distances do not increase under channels, and a supremum over a smaller set is smaller. (3) $D(\Lambda\rho_M^J, \Lambda\rho_M^{J'}) \le \eta(\Lambda) D(\rho_M^J, \rho_M^{J'})$ by the definition of η. For classical kernels the supremum is attained on pairs of point masses, which is Dobrushin's formula. □

Part (3) is the causal counterpart of the data-processing inequality: influence passed along a chain is at most the product of the influences of its links. Section 3 uses it to show that an agent needs both sensing and control.

2.3 Records do not reveal causes

Proposition 2.4 (Same observations, different causes)**.** *Consider two processes on bits A, B observed in the computational basis at times $t_1 < t_2$.*

1. *Direct cause: A is a uniform bit at t_1, $B = 0$ at t_1, and a controlled NOT from A to B acts between t_1 and t_2.*
2. *Common cause: a hidden uniform bit C is copied into A before t_1 and into B between t_1 and t_2; $B = 0$ at t_1.*

The joint distributions of $(X_A^{t_1}, X_B^{t_1}, X_A^{t_2}, X_B^{t_2})$ *coincide, both equal to* $\frac{1}{2}$ *on* $(0,0,0,0)$ *and on* $(1,0,1,1)$*, and both give* $\Delta^{\mathcal{I}}_{A\to B} = \ln 2$*. For the interventions "set* A *to* 0*" and "set* A *to* 1*" at* t_1*, the influence on* B *at* t_2 *is* 1 *in the first process and* 0 *in the second.*

Proof. In both processes A keeps its value, B starts at 0 and ends equal to the uniform bit carried by A. Under the intervention, B at t_2 equals the set value in the first process and the hidden bit in the second. □

Every statistic of the declared observed variables is the same in the two processes, so accumulating more of those records cannot settle which structure produced them. For quantum variables the situation is partly different: some causal structures can be inferred from observations alone when entanglement or coherence is available [14]. That is an advantage of particular quantum protocols, not a general replacement for interventions.

2.4 Records and interventions in one process

A record carries a recoverable difference from an earlier process. An intervention changes a declared mechanism or operation. They can occur within the same interaction without being the same role. Paper E shows that a memory need not be pure to record: the relevant budget is its purity deficit $\ln d - S(\eta)$, and partly mixed pointer ports have nonzero readable information. Records of a commuting pointer label are a specific acquisition model, not a classification of every record.

For an already detached port, a later trace-preserving operation on the source leaves its marginal unchanged, by no-signalling. A renewed interaction with the port, or a direct operation on it, can change its state. Reversible relabelling, repair, transfer into another memory and reset have different resource balances. Paper F treats them through the joint state; they cannot all be priced as erasure of a marginal entropy. The Imprint Chain [6] makes a complementary distinction: a passive record and a feedback can carry the same information about an interface while only the feedback changes the world state.

In Proposition 2.4, the recorded correlation between A and B does not make A a cause of B. Conversely, an influence can change a state without a record being retained. Memory consists here of physical records that remain available to later decisions. Their availability may itself depend on repair, storage and permission.

The process already has a time order. Under fresh-port and specified interaction conditions, it can acquire an operational direction of record accumulation and intervention. Repeated contact can return correlations and restore source coherence; local entropy need not increase at every step. These mechanisms explain a direction within the process, not the origin of its time parameter.

3 Part II: Historical control and maintained capability

3.1 What a boundary criterion must allow

A tempting boundary criterion asks every internal observable to stay nearly unchanged: $\sup_t \sup_{\|O\|\le 1} \|\mathcal{E}_t^*(O) - O\| \le \epsilon$. It admits an isolated frozen register and excludes a changing tracker. In the example below, the internal $Z_r = (-1)^r$ has deviation $1+(1-2\nu)(1-2q) = 1.768$, although the tracker succeeds. What must persist is a capability and the physical organization that supports it, not every stored value.

A stable algebra of the kind derived in Paper B can hold part of that organization. It does not by itself provide the sensor, actuator or maintenance procedure. We first specify the task and then construct the maintenance mechanism.

Definition 3.1 (Viable agent on a window)**.** *Fix a declared partition into a world variable W, an agent register R and an actuated variable b, declared interventions on W and on R, a viability condition $v(W, b)$ and a window. The register is a viable agent on the window, with margin $\eta < \frac{1}{2}$, if*

1. *the viability condition fails with probability at most η at every step of the window;*
2. *the sensing influence $\kappa_{\mathrm{s}} = \kappa(W \to R)$ is positive;*
3. *the control influence $\kappa_{\mathrm{c}} = \kappa(R \to b)$ is positive.*

The agent is memoryless if its action at a step depends on the world only through its reading at that step.

3.2 Sensing and control multiply

Consider a world bit W_t that flips with probability $q < \frac{1}{2}$ at each step. The agent reads it into R, acts on b, and is viable when $b_{t+1} = W_{t+1}$.

Proposition 3.2 (Viability needs sensing and control)**.** *Let a uniform world bit be read through a fixed sensor channel with independent fresh randomness. The actuator receives the world only through that reading, uses independent randomness, and the next world flip is independent of both. Thus the sensor's ordinary conditional output law equals its output under interventions setting W_t, with no initial world–controller common cause or retained world-correlated side input. For this memoryless menu, with κ_{s} computed for interventions setting W_t to 0 or 1 and κ_{c} for all point interventions on R,*

$$\Pr(b_{t+1} \neq W_{t+1}) \;\geq\; \tfrac{1}{2}\big(1 - (1 - 2q)\,\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}\big). \tag{1}$$

For a sensor that errs with probability ν and an actuator that errs with probability μ, both symmetric, equality holds and the misalignment is $\frac{1}{2}(1 - (1 - 2\nu)(1 - 2\mu)(1 - 2q))$.

Proof. Let K be the channel from W_t to b_{t+1}. With $W_{t+1} = W_t \oplus f$, $\Pr(f = 1) = q$, the success probability is $q + \frac{1}{2}(1 - 2q)(K(0|0) + K(1|1))$, and $K(0|0) + K(1|1) \leq 1 + D(K(\cdot|0), K(\cdot|1)) = 1 + \kappa(W_t \to b_{t+1})$. By Proposition 2.3(3), $\kappa(W_t \to b_{t+1}) \leq \kappa_{\mathrm{s}}\kappa_{\mathrm{c}}$. For symmetric channels $\kappa_{\mathrm{s}} = 1 - 2\nu$, $\kappa_{\mathrm{c}} = 1 - 2\mu$, and the misalignment is the probability that an odd number of the three independent events sensor error, actuator error and flip occurs. □

Corollary 3.3 (Records without control, control without records)**.** *In this fresh-sensor task, a passive recorder with $\kappa_{\mathrm{c}} = 0$, an open-loop actuator with $\kappa_{\mathrm{s}} = 0$, and an initially uncorrelated frozen register fail with probability $\frac{1}{2}$.*

The channel condition is necessary. If a hidden fair bit is copied into W_t and a frozen R before the slot, then $b_{t+1} = R$ has ordinary error q, although intervening on current W_t does not change R: $\kappa_{\mathrm{s}} = 0$. The common-cause record is useful without a new sensing influence. It is not a fresh sensor channel from the intervened world. Treating observed $\Pr(R \mid W_t)$ as that channel would erase the distinction proved in Proposition 2.4. Retained world-correlated inputs must instead be included as memory, as in the next subsection.

The magnitudes $\kappa_{\mathrm{s}}, \kappa_{\mathrm{c}}$ alone do not certify the labels needed for the directional task: a reversed sensor has the same influence as a correctly oriented one. T-DOME III shows that acting cannot reveal parameters moved by a symmetry of the complete record law. Certifying this directional task therefore needs a symmetry-breaking labelled operation or retained standard [31]. It may be physically implemented inside the agent; the requirement concerns the information it supplies, not its location outside the loop.

With $\nu = 0.1$, $\mu = 0.05$ and $q = 0.02$ the memoryless agent fails with probability 0.1544, while its register changes value in one step with probability 0.193. It is viable in Definition 3.1 and fails the pointwise-invariance criterion; an initially uncorrelated frozen register passes that criterion and has no sensing route.

3.3 Memory changes what an agent can do

Let the agent instead act on the majority of its last three readings. Exact enumeration gives a misalignment of 0.1082. The influence of the present world state on the action falls at the same time: an intervention on W_t changes only the newest reading, which decides the majority only when the two older readings disagree, so the influence is $(1-2\nu)(1-2\mu)\Pr(\text{older readings disagree}) = 0.139$, against 0.72 for the memoryless agent.

Corollary 3.4 (Memory is necessary for this level of viability)**.** *In the specified drifting-bit menu with one fresh sensor reading per slot, every memoryless agent fails with probability at least* $\frac{1}{2}(1-(1-2q)(1-2\nu))$*, which is* 0.116 *for* $\nu = 0.1$*,* $q = 0.02$*. The majority-of-three agent fails with probability* 0.1082 *even with an imperfect actuator,* $\mu = 0.05$*. Within this menu, a failure probability below* 0.116 *therefore requires dependence on past readings.*

Proof. A memoryless agent's register depends on W_t only through the reading, so $\kappa_{\mathrm{s}} \le 1-2\nu$ by Proposition 2.3(2); Proposition 3.2 with $\kappa_{\mathrm{c}} \le 1$ gives the bound; the majority value follows from enumerating the 2^8 values of the initial world bit, three flips, three reading errors and actuator error. The exact value is 0.10822784; Appendix A gives the verification method. □

The memory agent tracks the world better while the present world influences it less. Its success rests on records of past world states, which are correlated with the present one because the world persists; an intervention on the present world cannot reach those records. Agency therefore involves two different relations to the world: causal influence through current sensing and control, and correlation carried by records. Proposition 3.2 bounds what the first can achieve alone. This is the task-level form of the memory resource studied in T-DOME I [8], and it is the step at which the self-reading of Imprint W [7] and the internal selection rules of Paper G become relevant: an agent that also keeps a model of its own sensor and actuator, and revises it from its records, can choose which readings to take.

3.4 Records maintain the body that reads next

The tracker so far has a supplied sensor. Give it a body bit D_t: 0 is functional and 1 damaged. The same physical body remains throughout the run; its condition modulates the world-sensing coupling, and it can be repaired before sensing. Damage is independent of the world, arrives after each task slot with probability $0 < d < 1$, and is absorbing until repair. This is a specified failure mechanism, not a biological model.

Each slot begins with diagnosis. A fresh port in $\eta_{\mathrm{d}} = [I + (1-2e)Z]/2$, $0 < e < 1/2$, interacts by controlled X with D_t. Its Z record is $Y_t = D_t \oplus N_t$, $\Pr(N_t = 1) = e$, from the mixed-port instrument of Paper E. Exactly when $Y_t = 1$, the controller swaps a fresh $|0\rangle$ into D_t; the old body state goes into that port and remains in the spent bank. The post-repair body is $D'_t = D_t(1-Y_t)$. The decision uses Y_t, not hidden D_t.

The world sensor then gives

$$R_t = W_t \oplus N_t^{\mathrm{s}} \oplus D'_t Q_t, \qquad \Pr(N_t^{\mathrm{s}} = 1) = \nu, \quad \Pr(Q_t = 1) = \tfrac{1}{2}. \tag{2}$$

For a healthy body it is the original sensor; for a damaged body its reading is fair. Two gates realize this law: controlled X from W_t to a port in $[I+(1-2\nu)Z]/2$, followed by a flip controlled by D'_t and a fresh fair bit Q_t. The spent Q_t is retained but outside the controller's readout menu. Joint access to it would change the sensing permission.

The reading drives the same actuated variable b. A fresh blank is swapped into b, retaining its old value; copying R_t and adding independent actuator error gives $b_{t+1} = R_t \oplus N_t^{\mathrm{a}}$, $\Pr(N_t^{\mathrm{a}} = 1) = \mu$. The world then flips with probability q, and finally the body suffers damage: $D_{t+1} = D'_t \vee J_t$, $\Pr(J_t = 1) = d$. Fresh ports are independent of the retained body, world and history. Diagnostic orientation and error are known or separately calibrated; a common unknown sign is not eliminated by a self-certifying reference.

Proposition 3.5 (Maintenance of a sensing route)**.** *Let* $x_t = \Pr(D_t = 1)$*,* $z_t = \Pr(D'_t = 1)$*, and* $r_t = \Pr(Y_t = 1)$*. Then*

$$z_t = ex_t, \qquad r_t = e + (1-2e)x_t, \qquad x_{t+1} = d + (1-d)ex_t, \tag{3}$$

$$x_* = \frac{d}{1-(1-d)e}, \qquad z_* = ex_*, \qquad r_* = e + (1-2e)x_*. \tag{4}$$

With $\rho = (1-d)e$*,* $x_t = x_* + \rho^t(x_0 - x_*)$*. For a uniform world independent of the initial body, the raw sensing influence and tracking failure are*

$$\kappa_{\mathrm{s},t} = (1-z_t)(1-2\nu), \tag{5}$$

$$\mathcal{R}_t = \mathcal{R}_0 + (\tfrac{1}{2} - \mathcal{R}_0)z_t, \qquad \mathcal{R}_0 = \tfrac{1}{2}[1-(1-2q)(1-2\nu)(1-2\mu)]. \tag{6}$$

A healthy start gives $z_t \le z_*$ *and* $\mathcal{R}_t \le \mathcal{R}_0 + (\frac{1}{2} - \mathcal{R}_0)z_*$ *at every slot. Without repair, sensing influence tends to zero and tracking failure tends to* $1/2$*.*

Proof. Only a damaged body missed by diagnosis remains damaged, with probability ex_t. The diagnostic is positive on a healthy body with probability e and on a damaged body with probability $1-e$. Independent damage gives $x_{t+1} = z_t + d(1-z_t)$. Solving the recurrence gives the fixed point and convergence. Repair and damage do not depend on the world, so the body remains independent of W_t. The sensor mixes the original channel, with weight $1-z_t$, and a fair channel, with weight z_t, giving its influence and risk. Without repair, $x_t = 1-(1-x_0)(1-d)^t$. □

This is intervention rather than relabelling. Replacing diagnosis by *never repair* or *always repair*, with the pre-slot distribution fixed, changes post-repair damage from x_t to zero and risk by $(\frac{1}{2} - \mathcal{R}_0)x_t$. Setting the body to healthy or damaged after repair changes the available sensing channel. The record maintains a physical condition of later intervention, not merely confidence in a previous answer.

Proposition 3.6 (A matched calendar comparison)**.** *Consider repair calendars independent of damage and of all information about it, including diagnostic or world-sensor histories. They may use private randomness and have positive long-run repair frequency* r*, with empirical mean gap* $1/r$ *and vanishing end effects in the occupancy average. In the same present-reading control menu, their long-run mean post-repair damage is at least*

$$z_{\mathrm{cal}}(r) = r[(1-\omega)G(L) + \omega G(L+1)], \quad L = \lfloor r^{-1} \rfloor, \quad \omega = r^{-1} - L, \tag{7}$$

where

$$G(\ell) = \sum_{j=0}^{\ell-1}[1-(1-d)^j] = \ell - \frac{1-(1-d)^\ell}{d}.$$

Gaps $L, L+1$ *in these proportions attain the bound. Frequency at most* r *gives the same lower bound.*

Proof. A gap of length ℓ, starting with repair, has expected damaged occupancy $1-(1-d)^j$ at age j, with sum $G(\ell)$. Conditioning on an independent calendar preserves the damage law. The differences $G(\ell+1) - G(\ell) = 1-(1-d)^\ell$ increase. The convex envelope on the integers minimizes average gap occupancy at fixed mean gap $1/r$ by balancing adjacent lengths. Multiplication by gaps per slot gives (7). Adding repairs cannot increase occupancy, so the allowance is optimally used in full. □

For $d = e = 0.1$ and $q = 0.02, \nu = 0.1, \mu = 0.05$,

$$x_* = \frac{10}{91}, \quad z_* = \frac{1}{91}, \quad r_* = 0.18791209, \quad \mathcal{R}_* = 0.15819780.$$

The optimal independent calendar at this repair frequency has $z_{\mathrm{cal}} = 0.19479225$ and mean error 0.22172020. Both receive the same diagnostic and world-reading ports and the same repair and actuator menu. One unit is charged per actual repair, with other per-slot inputs equal. A controller inferring damage from sensor history is another record-guided strategy, outside the calendar class. Equal long-run rate does not mean equal repair counts on every finite trajectory.

3.5 Finite stock and a maintained boundary

For n slots, the realization has three continuing bits D, W, b and nine fresh ports per slot: diagnostic, repair blank, world sensor, fair sensor-noise, actuator-noise, world-flip, damage coin, damage-dilation blank and actuator blank. Maximum repair count is n; a healthy start gives expected count at most nr_*. Supplying the bank, storing spent ports and routing records belong to the finite-window budget. Mean activation count is not stock capacity.

Each of the three blank ports per slot supplies initial purity deficit $\ln 2$. Diagnostic and sensor ports supply $\ln 2 - h(e)$ and $\ln 2 - h(\nu)$; actuator, world-flip and damage coins supply $\ln 2 - h(\mu), \ln 2 - h(q), \ln 2 - h(d)$; fair noise supplies none. Here h is binary entropy in nats. Unused repair blanks may remain pure; supplied purity is not necessarily exhausted on every slot. Old states and correlations remain; the reversible circuit in Appendix A conserves joint entropy. No bath is reset and no work or heat price follows from counting swaps. For the numerical parameters, the nine-port input purity is $8 \ln 2 - 3h(0.1) - h(0.05) - h(0.02) = 4.27337417$ nats per slot. For degenerate port Hamiltonians, $k_B T$ times this quantity is the free-energy difference from a maximally mixed preparation at temperature T, not heat dissipated or net work consumed by the run. The initialized healthy body and blank actuator in the explicit check add two initial pure bits; the world starts fair. Gate strengths, preparation, duration, clock, storage and later bank renewal remain physical inputs.

The body's condition regulates whether a reading is informative, and a record of that condition regulates the operation preserving the route. Organization survives while its values change. Continued intervention capability is produced by record-conditioned action, rather than merely imposed as a viability condition. Repair uses the current diagnosis only; earlier diagnoses remain stored but are not read by this policy. The majority tracker establishes the value of longer history, whereas this circuit establishes the maintenance of later permissions. Neither result alone constructs a long-term self-model.

Remark 3.7 (Maintained organization)**.** *A broader agency hypothesis is that an effective agent boundary is the organization of sensing and action routes whose continued availability is causally regulated by retained history. Disrupting use of a bodily diagnostic should impair later intervention capability even when current discrimination is spared. At a matched budget, an independent calendar should fail where a condition-dependent policy succeeds.*

The binary body establishes one instance. Formation must also explain how routes and the repair menu arise from earlier interactions and how histories coordinate several demands. The hypothesis explains no additional boundary if a fixed external controller accounts for every apparent self-maintenance effect, or if manipulating history never changes later permissions. BRAIN supplies learned protection and routing for a different body model [32]; Imprint W and Imprint SH price reliability-based action [7, 33]. Their dynamics and budgets must match before they form a larger construction; they are not assumed to be this circuit.

3.6 Will, degrees of agency and free will

On this account, "will" names the control route from an agent's records to the world: the influence κ_{c}, together with the dependence of the action on the agent's records. A thermostat has a sensing route, a control route and a viability condition, and it is a viable agent in the weak sense of Definition 3.1. Organisms differ from it in degree along several axes: the length of the

window over which viability is maintained, the margin, the strength of sensing and control, the depth of memory, and whether the agent can revise its own selection rules. Agency is graded along these axes and is not an all-or-nothing property.

The traditional free-will question presupposes well-defined agents and causal histories. The present account supplies both relative to a declared partition and intervention set, and it turns part of the question into questions that can be answered: how strongly the agent's records, rather than the present state of the world, determine what it does, and whether its selection rules are themselves revised from its records. It does not settle the metaphysical question and is not offered as doing so.

The account says nothing about what it is like to act. The explanatory gap between functional organization and experience [15, 16] remains. One empirical route connects these notions to neuroscience, where habitual and flexible control have been associated with distinct cortico-basal ganglia loops [17, 18] and the sense of agency with motor prediction [19]; the relevant times there are classical control times, not quantum coherence times [20]. Whether the quantities defined here are useful for that research is open.

4 Part III: Existence without substance

4.1 Relational existence

Parts I and II treat causes and agents as real within declared contexts and dependent on them. This raises the question of their ontological status. Two extremes are available and both are mistaken: treating context-dependent structures as fundamental, and denying them objective reality because they depend on context. We propose structural realism about relational patterns [21, 22].

Definition 4.1 (Relational existence)**.** *An entity X exists relative to a context $\mathcal{S}$ if X is well defined and stable within $\mathcal{S}$, participates in objective relational patterns there, and may be absent or differently constituted in another context $\mathcal{S}'$.*

Definition 4.2 (Emptiness, technical sense)**.** *An entity is empty in the technical sense if it lacks being independent of every context while exhibiting stable patterns within specified contexts. Emptiness in this sense entails neither non-existence nor illusion.*

Paper B supplies physical instances of a context. In its commuting-coupling class, the stated stability and observability criteria select a unique largest admissible block algebra on a window. Smaller admissible algebras also exist. Selected content depends on coupling and window, yet the same inputs define the same criterion for all readers. Readable redundant labels additionally require actual recording couplings, fragment preparation and calibrated readout, as in Paper E; stability alone does not make them public.

The maintained body adds history and budget to the context. The sensor's operational distinction survives under record-guided repair and disappears under unmaintained damage. That difference is real in the task outcomes even for a reader unaware of the controller's policy. Relational objectivity concerns these reproducible consequences, not agreement on a chosen label.

4.2 Examples

Temperature depends on which degrees of freedom are treated as macroscopic, yet given that choice it is measurable and predictive. Quasiparticles such as phonons and magnons are collective excitations with no counterpart among the microscopic constituents; operationally they are stable peaks in spectral functions, dispersion relations, lifetimes and scattering amplitudes, all of which depend on temperature, doping and resolution. Subsystems are the case closest to

this framework. A state alone does not privilege one of its many factorizations, and Paper A shows that entropies, mutual informations and reconstructed geometries depend on the channel through which the state is read; Paper B shows how couplings select one structure over a window.

4.3 What is invariant

Objectivity does not require independence from every context. It requires that patterns be stable and predictive within a context and that the transformations between contexts be stated. What stays invariant depends on that class of transformations. Under a change of factorization, implemented by a unitary relabelling of the Hilbert space, the spectrum of the global state is invariant, while subsystem entropies, mutual informations and the geometries built from them are not. For readers applying the same selection criterion with couplings and window fixed, the selected algebra is unchanged. Access to its labels requires the declared record protocol; readers with different measurement permissions need not obtain the same information. A mere unitary change of coordinates is also different from physically changing a coupling.

Kind	Characterization	Example
Intrinsic existence	independent of every context	classical particles, if fundamental
Relational existence	dependent on context, objective within it	temperature, selected subsystems, records
Conventional existence	dependent on context and on agreement among agents	money, legal rights
Non-existence	absent from every relevant context	phlogiston, luminiferous ether

Table 1: Kinds of existence claim. The causal influences, agents and geometric features of this series belong to the second kind.

5 Part IV: Interpretive bridges to Buddhist philosophy

5.1 Method

The comparison below is between formal patterns. It does not claim historical influence in either direction, endorsement of Buddhist doctrine, or that physics confirms any religious view; "Buddhism" is not treated as one doctrine, and the discussion is limited to some Yogācāra and Mādhyamaka texts. One limitation must be stated plainly. The framework was named after the ālaya-vijñāna, so a resemblance between the two was partly built in by the naming and cannot count as independent evidence for either. The mathematical results of Parts I–III do not depend on the comparison; the comparison depends on them.

5.2 Yogācāra: traces that condition later appearance

Yogācāra texts describe the ālaya-vijñāna, the store consciousness, as the stream in which the impressions (*vāsanā*) of past actions are kept as seeds (*bīja*) that condition later experience [23, 24, 26, 27]. Vasubandhu's *Triṃśikā* says that it proceeds like a torrent; it is a changing stream, not a fixed container. The following correspondences are formal analogies.

The analogy concerns the role of traces. In both pictures what can appear later depends on what earlier processes have left behind, and the traces are not inert stores but conditions on later access. The differences are as important. The Yogācāra substrate is described as consciousness; the HAFF state is physical and the framework is neutral on mind and matter. The analogy does not establish panpsychism, idealism or the claim that quantum information is conscious.

Yogācāra notion	Counterpart in this series (analogy only)
Ālaya-vijñāna, a stream that keeps traces	the global state together with the records its history has formed
Bīja and vāsanā, seeds and impressions	records that change what later readouts can distinguish (Papers E, F; the Imprint papers)
Pravṛtti-vijñāna, the active consciousnesses	selected readouts and the effective subsystems they define (Papers A, B)
Āśraya-parāvṛtti, transformation of the basis	a change of couplings that changes which structure is selected, as opposed to a change of description (Paper B)

Table 2: Formal analogies. Yogācāra concerns consciousness and liberation; HAFF concerns physical structure, and its global state carries no phenomenal properties.

5.3 Mādhyamaka: dependence without a foundation

For Nāgārjuna, whatever arises dependently is empty of intrinsic nature (*svabhāva*), and that emptiness is itself a dependent designation, the middle way (*Mūlamadhyamakakārikā* 24:18) [25, 28, 29]. The technical emptiness of Definition 4.2 has the same form: denial of intrinsic being together with affirmation of functional reality.

Dependence also applies to the apparatus of selection. The structure that selects is itself conditioned: the selected algebra depends on couplings and on a window (Paper B), and a system that selects its own couplings does so through dynamics that its own registers do not select (Paper G). In a finite model the chain of selectors ends at that dynamics. The endpoint is not a foundation in Nāgārjuna's sense, an independently existing ground, because it is only the part of the dynamics that the given model leaves outside, and a larger model can take it in. Dependence therefore holds at every level without an infinite regress and without an independent base. Whether this resembles the Mādhyamaka middle way is a matter of interpretation. The resemblance is in form, and Mādhyamaka uses the insight to loosen attachment to views, whereas the physical account here describes how structure is selected and maintained.

5.4 Karma: actions, traces and later access

Buddhist accounts of karma hold that intentional actions leave traces that condition future experience [30, 34]. In the stated fresh-port process, an intervention on the source changes later records while detached earlier ports retain their marginals (Paper F). Records then condition later reading and action. In Section 3.4, they condition whether later sensing remains informative. The analogy is formal and stops at normativity. Karma classifies actions as wholesome or unwholesome; interventional influence and records carry no ethical valuation.

5.5 Where the traditions and the framework differ

The differences are substantive. Buddhist philosophy starts from first-person experience and aims at liberation from suffering; the physical results used here begin with third-person structure. Rebirth, karmic continuity across lives, cosmological realms and meditative attainment have no counterpart here, and the comparison neither supports nor undermines them. Yogācāra readings that give consciousness priority differ from the framework's neutrality. And the methods differ: contemplative inquiry, textual interpretation and dialectic on one side, mathematics and experiment on the other. The comparison is worth making because both traditions and the framework deny intrinsic subsystem structure while affirming stable relational patterns, and because precise statements on one side can sharpen questions on the other.

6 Conclusion

Causal influence is a property of a declared intervention process, not of correlation alone. It contracts through links and distinguishes processes whose observed records coincide. In the tracking task, sensing and control jointly limit memoryless viability; history can surpass that limit without raising the influence of the current world on the action.

The same finite task now includes a damageable body. An actual diagnostic controls repair of that body, whose condition determines the next sensor channel. The recurrence and matched calendar comparison show how a record maintains later intervention capability. Fresh blanks and spent history are explicit resources; this is a finite-stock mechanism, not a perpetual thermal cycle or the primitive origin of the initial body.

Agency so understood is a maintained relational organization. Partition, loss and permissions specify the task; success and failure are physical consequences, not matters of agreement. A fuller formation account must generate and coordinate the routes rather than supply all of them. The Buddhist comparison clarifies a related philosophical possibility: traces and dependent structures can have functional reality without an independently existing substrate. Its experiential and ethical questions remain independent of the physical mechanism.

A Reversible realization and checks

The body protocol is an explicit basis-permutation unitary on twelve bits. Besides continuing D, W, b, use fresh $(C, B, R, Q, A, F, J, G, V)$ with initial one-probabilities

$$(e, 0, \nu, \tfrac{1}{2}, \mu, q, d, 0, 0).$$

Here C, B are the diagnostic and repair ports; R, Q the sensor and fair-noise ports; A, F, J actuator, world and damage randomness; G, V blank damage and actuator ports. On diagonal states perform, in order:

1. $C \leftarrow C \oplus D$; swap D, B conditional on $C = 1$.
2. $R \leftarrow R \oplus W$; $R \leftarrow R \oplus DQ$.
3. Swap b, V; $b \leftarrow b \oplus R \oplus A$.
4. $W \leftarrow W \oplus F$.
5. $G \leftarrow G \oplus D$; $D \leftarrow D \oplus [(1 - G)J]$.

Each gate is reversible on all basis inputs, not just the initialized subspace. Since G starts at zero, the last step stores the post-repair body in G and gives $D_{\text{next}} = D' \vee J$. Repair uses $C = Y$ alone. The physical control of noise by the body does not grant a hidden diagnostic read. All ports remain in the spent bank, and the next slot uses fresh ones. Z reading of C, R is compatible with the diagonal protocol and does not change its joint law. Realizing these gates and their known orientations is a charged hardware permission.

The check enumerates all 4096 input strings and verifies that the composite is a permutation. For diagonal product inputs with $x = \Pr(D = 1)$ equal to $0, 1/5, 10/91, 1$, exact rational probabilities reproduce every term of (3) and (6). Old b does not affect the output: it remains in V. Joint entropy is unchanged. This is a circuit distribution check for the stated classical task, not unknown quantum-source or bath certification.

The calendar check verifies discrete convexity of G and the optimal $5, 6$ gap mixture at the stated rate. Independently, enumeration of the original three-reading task gives 0.10822784. The probability that the two older readings disagree is $\frac{1}{2}[1 - (1 - 2q)(1 - 2\nu)^2] = 0.1928$, giving present-world influence 0.138816. Analytic contraction and calendar proofs, rather than numerical sampling, establish the general statements.

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Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return

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Abstract

Physical selection changes a gravitational source when it changes an actual coupling, preparation or maintenance process. We develop this route alongside a modular route that constrains which geometric laws can agree with selected algebraic data. The two routes have different inputs: modular stability does not select a unique algebra, and a supplied gravitational action does not explain the origin of that law.

Calibrated correction, projection selection and learned protection produce stress through the interactions that change access. We compute its Jackiw–Teitelboim (JT) response and construct a finite semiclassical mean-action return in which acquisition changes a material source and later sensing changes the next force. Quantum history, old references, support, synchronization, clock work and recoil are retained. Joint perturbation estimates allow bounded controller coupling to an unbounded common Hamiltonian on domain-preserving strong solutions. A separate full-mode two-field model matches its stress to a work-corrected source at rate $O(I^{-1})$ as clock inertia I grows; the energy correction leaves a nonzero pressure response. For a real-quartet carrier, an exact nonlinear normal-order conversion couples its actual full-mode quantum state, physical work clock and material director to constrained JT evolution. The resulting bulk solution has a prepared positive causal domain.

Maintenance is priced by the entropy removed from the full retained state. For a thermal chiral exhaust of capacity c_{ex}, its long-window heat rate obeys $\dot{Q} \geq k_B T \dot{s} + 3\hbar \dot{s}^2/(\pi c_{\mathrm{ex}})$. The complete output's relative entropy and entropy slack measure its excess above this floor. Direct export of the current purifier retains cross-species redundancy, so accurate renewal alone cannot make that protocol sharp. Matched horizon transport gives $\overline{\dot{S}_h} \geq \dot{s} + 6\dot{s}^2/c_{\mathrm{ex}}$ in the thermal rapidity clock. The actual thermal-source construction includes the running clock's nonextensive reference stress; physical realizations use finite operating windows. Pointer-aligned dephasing gives different complete renewal debits for the two selected axes. This identifies a candidate mechanism for a gravitational maintenance reading to favor the stable pointer algebra. Sustainable renewal, sharpness of the complete adaptive output and persistent causal program contrast form a joint conjecture, with drive routing, moving-clock conversion and support work charged. A common source action and the origin of geometric selection remain further targets.

1 The problem: what makes selection gravitational?

An apparatus changes which distinctions it can preserve and which operations it can execute. That change may reorganize a reconstructed geometry; it may also change the state of matter. A gravitational claim must say which of these changes it describes. Forgetting a record after an operation changes what can be inferred, while the emitted stress persists. Changing the interaction can change both access and the physical source.

The question of this paper is whether these two descriptions can meet: can the structures selected by physical operations supply geometric data whose consistency requires a gravitational law, and can the same operations produce sources whose response changes subsequent selection? HAFF-A, B and E provide the distinction between channels, stable content and actual records [19, 20]; Foundation I provides a hull relative to a readable seed [54]. Here the object to explain is the relation between these physical selections and gravitational dynamics.

One route starts with modular data. A supplied standard half-sided modular family generates translations. With a holographic dictionary and the required ball first laws, its variations match the linearized Einstein equation. Consciousness II supplies a different, same-spacetime conditional consistency law for a boosted ball family [24]. Neither route derives its geometric dictionary or consistency condition from modular stability alone. The SLD metric of distinguishability is positive semidefinite; it is neither the BKM Hessian of relative entropy nor a Lorentzian spacetime metric.

The other route starts with actual interactions. We derive sources from calibrated correction, selected projections and learned protection, then compute their response in a specified JT theory. The JT metric has fixed curvature in the minimally coupled sector; its dilaton and invariant respond. Direct dilaton-coupled support changes that sector explicitly. The resulting finite return retains a quantum reader and its old reference while its acquired record drives mechanics and new radiation. Its complete mean-action work and recoil are included. This realizes a causal source mechanism without assuming the proposed origin of the gravity law.

Sections 2–4 develop the derived sources, finite physical return and maintenance price. The modular route follows in Sections 5–8; the final conjectures ask how these routes could meet. The appendices retain the instruments, profiles, clocks and apparatus proofs. In particular, Table 1 states the objects and the interfaces between them. It separates established results from the growing-window conjecture; it is not a decomposition of one already completed apparatus.

2 From a Selected Interface to a Physical Source

A calibrated interface acts through a field coupling whose outgoing stress sources the supplied JT theory. This gives a physical selection–source map; the origin of the gravitational law in Conjecture 7.2 is a separate question.

2.1 What records can and cannot do to a source

Let a classical record $R \sim \mathrm{Bernoulli}(p)$ control whether a physical operation prepares the matter state ψ or leaves the reference ω. Without reading R, the matter state is $\rho_p = (1-p)\omega + p\psi$ and its stress is the mixture of the two stresses. Deleting or ignoring R after the preparation does not act on the matter, so ρ_p and its source are unchanged.

Object and supplied inputs	Result	Remaining interface
Modular family, dictionary and ball laws	Conditional translations and linearized geometric consistency (Sections 5–8)	Formation of the candidate family and its geometric dictionary.
Calibrated field operations and a JT action	Derived outgoing source, retarded response and finite echo/contact readout (Section 2; Appendix C)	Total apparatus stress and physical references; the action is an input.
Quantum core with classical mechanics and mean-action source	Finite retained-history return and strong-domain perturbation bounds (Section 3; Foundation III)	Fixed-window mean-action result; protected content, old references and initial work remain charged. No quantum stress tensor for the scalar fields or support is supplied.
Spatial quadratic fields and real-quartet bulk	Work-corrected full-mode stress matching, a finite spatial return, and a separately constrained quartet interior	Joining their receivers, moving endpoints, preparation and thermal ports in one material model.
Complete retained entropy and paid finite carrier	Long-window heat floor and neutral-sector sorting of a resettable source (Section 4; Proposition 9.3)	Actual adaptive source concentration, protected-content compatibility and local delayed readout.
One reused body over growing runs	Conjecture 9.2 jointly requires complete-output sharpness, normalized window contrast and causal transition gain	Sustained inlet, support, sensing and resource account in that same dynamics; none is supplied by sorting alone.

Table 1: Objects, results and interfaces. Finite constructions retain their stated inputs; the joint growing-window target is a conjecture.

Changing the preparation probability p changes the source. The information side has a matching statement. For the joint state $\rho_{RM} = (1-p)|0\rangle\langle 0| \otimes \omega + p|1\rangle\langle 1| \otimes \psi$ and the reference $\pi_R \otimes \omega$, with π_R the marginal of R, additivity of relative entropy on direct sums and data processing give

$$D(\rho_{RM}\|\pi_R \otimes \omega) = p\, D(\psi\|\omega), \qquad D(\rho_p\|\omega) \le p\, D(\psi\|\omega) \tag{1}$$

[51]. Losing the record lowers the distinguishability available to a reader; it does not remove the matter source. A gravitational consequence of selection must therefore pass through a preparation or interaction, and the conditional geometry of a branch, the average of branch geometries and the geometry of the averaged source are three different objects.

2.2 The source of a calibrated feedback actuator

The companion paper [21] realizes a corrective flip of a degenerate bit as a field coupling. A right-moving free massless scalar with null coordinate u, current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$ is coupled through a real smooth switch g. We require a finite nonzero one-particle norm $\lambda^2 = \int_0^\infty \omega |\hat{g}(\omega)|^2 d\omega/(4\pi)$ and $\int (1+u^2)|g'(u)|^2 du < \infty$, where $\hat{g}(\omega) = \int g(u) e^{-i\omega u} du$ in the L^2 sense when needed. For example, $g \in H^1(\mathbb{R})$ together with the weighted derivative condition is sufficient. Smooth compact switches and the full-line packet below both belong to this class. The smeared current defines a Weyl unitary,

$$U = \exp\Big(-iX \int g(u) j(u)\, du\Big) = \exp\big[-i\lambda X (b + b^\dagger)\big], \tag{2}$$

where b is a normalized field mode with mean frequency $\bar{\omega}$ fixed by g, and $\lambda^2 \bar{\omega} = \frac{1}{4}\int g'^2 du$. With the drive mode in the coherent state $|\sqrt{N}\rangle$ and $2\lambda\sqrt{N} = \pi/2$, the coupling flips the bit with success probability $s_N = (1 + e^{-\pi^2/(8N)})/2$. From the current commutator $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$ one has $U^\dagger j(u) U = j(u) + \frac{1}{2} X g'(u)$ and

$$U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u) X j(u) + \tfrac{1}{4} g'(u)^2. \tag{3}$$

The cross term has zero expectation if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis; a Z-diagonal bit marginal alone does not suffice [21]. Writing $j = \psi(u) b + \psi^*(u) b^\dagger + j_\perp$, the incoming coherent beam has stress $T_{\mathrm{in}} = 4N[\mathrm{Re}\,\psi(u)]^2$ and each activation adds

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2. \tag{4}$$

This added stress is a property of the coupling. Its integral is the switching work $\lambda^2 \bar{\omega}$ supplied by the external switching agent.

The feedback task is that of the Imprint Chain [22] with the flip realized by this coupling. A hidden instrument label is estimated from calibration records with mismatch probability q; each task step reads the bit with reliability ν, accepts feedback with probability h, switches g on when accepted and the reading is negative, and lets a bath with contraction c act. Each opportunity receives its own scheduled incoming beam, independent of the bit and earlier records. In the companion paper the stationary loss $L_\infty(q)$ and the stationary coupling rate satisfy [21]

$$p(q) = h\nu L_\infty(q) + \tfrac{h(1-\nu)}{2}. \tag{5}$$

Proposition 2.1 (Source–loss relation)**.** *At fixed drive, switch and task parameters, the stationary mean source per opportunity outside the apparatus is* $\langle T_{uu}\rangle_q = T_{\rm in} + p(q)F_g$*. For two calibrations,*

$$\langle T_{uu}(u)\rangle_q - \langle T_{uu}(u)\rangle_{q'} = h\nu\,[L_\infty(q) - L_\infty(q')]\,F_g(u). \tag{6}$$

Proof. The incoming beam is present at every opportunity. A coupling occurs with stationary probability $p(q)$ and, under the joint initial condition, adds F_g in expectation by Eq. (3). Equation (5) gives the difference. □

Better calibration lowers the task loss and, through it, the source. The coefficient comes from the same interaction that performs the correction; no separate rule maps a calibration error to an emitted pulse. The relation does not assign stress to knowledge. Deleting the calibration record after the commands have been executed leaves the outgoing source unchanged, in line with Section 2.1. The relation is a stationary average per opportunity; a single trajectory, with its fixed hidden match label, does not follow the population mean step by step.

2.3 A source set by learned quantum protection

The activation-rate mechanism is not the only way that retained records can change a source. Brain Anchor [57] constructs two degenerate qubits with a programmable coupling

$$A_a = Z_1 + \alpha a Z_2, \qquad U_a = e^{-iA_aQ_f}, \quad Q_f = \int f(u)j(u)du = \sqrt{k}(b + b^\dagger). \tag{7}$$

The real smooth compact switch f and positive gain interval are supplied. The gain α is unknown, and each encounter receives a fresh independent vacuum field. On $\mathrm{span}\{|01\rangle, |10\rangle\}$ the coupling is $(1 - \alpha a)Z_L$. Calibration records from the commands $\pm h$ have means $r_\pm = e^{-2k(1\mp\alpha h)^2}$, so $\log(r_+/r_-) = 8k\alpha h$. The finite-record certificate in that paper bounds $|1 - \alpha a|$, including command rounding and known drift. It does not promise an exact fixed algebra after finite training.

Proposition 2.2 (One residual controls protection and source)**.** *Fix the acquired command* a*, write* $m = 1 - \alpha a$ *and* $F_f = f'^2/4$*, and keep the initial field independent of the logical body and retained ancillas. For every logical input, one encounter has added mean stress* m^2F_f *and switching work* m^2E_f*, with* $E_f = \int F_f du$*. After* M *independent identical encounters, with the output ports discarded and no interleaved gates or active recovery, its passive protection error is*

$$\epsilon_M = \tfrac{1}{2}(1 - e^{-2Mkm^2}). \tag{8}$$

For a single pulse with stress moments $E_j = \int u^j F_f du$ *and the initial data of Proposition 2.4, the late invariant is*

$$\mathcal{C}(m) = H_0^2 + 2\kappa H_0 m^2 E_1 + \kappa^2 m^4 (E_1^2 - E_0E_2). \tag{9}$$

Thus retained history can change the physical protection and the gravitational response at fixed pulse shape and activation schedule.

Proof. The current shift is $j \mapsto j + A_a f'/2$. The centered independent field input makes the stress cross term vanish for every logical state, and $A_a^2 = m^2 I_L$ on the code. Its characteristic function multiplies the logical off-diagonal by e^{-2km^2} per encounter; a dephasing channel's half-diamond error is its phase-flip probability. Time ordering may give a factor $e^{i\vartheta_f A_a^2}$, which is scalar on the code but not on the whole two-qubit register. Finally substitute $m^2 F_f$ in the fixed-initial-data source solution below. □

This link does not require a rule that emits a pulse only after a wrong answer. The same interaction produces both the residual error and its source, even when the logical body is entangled with retained memory. At $m = 0$ the body decouples in the code; a noiseless stored qubit need not continuously export entropy. The finite-rate entropy-export bound of Section 4 applies to actual renewal, not to every protected state. Calibration itself adds field energy $2N(1 + \alpha^2 h^2)E_f$ for N independent tests at each sign. Preparations, measurement, the controller and switching mechanism add further charges. The source suppression during operation alone is not a total-energy saving.

Brain Anchor also treats an unknown centered Gaussian variance K_ω. Logical records at $0, \pm h$ give $K_\omega = -\log r(0)/2$ and $\alpha = [\log r(h) - \log r(-h)]/(8hK_\omega)$; two settings suffice when the signs of $1 \mp \alpha h$ are known. Its finite-record mismatch certificate needs no supplied temperature label. The coherence is now $e^{-2K_\omega m^2}$, but the coupling-added source remains $m^2 F_f$ on centered independent input.

Incoming stress must still be added. A u-clock chiral KMS input has background $\pi/(12\beta^2)$ in natural units. Within a supplied bounded one-temperature family, strictly decreasing K_β permits a finite-error background estimate; a general Gaussian variance does not determine incoming energy. At common preparations and initial data, the linear dilaton equation compares coupling-source differences. Equation (9) cannot be used for the full source after dropping its thermal background, nor is u thermality the rapidity-clock reference of Section 4.

The same command can also operate on one reused vacuum field. For nonnegative compact switches f_j on disjoint intervals, with no noncommuting logical gates, the smeared currents commute but have negative correlations. Brain Anchor gives exact coherence $e^{-2m^2 k_M}$ with $k_M = \langle (\sum_j Q_{f_j})^2 \rangle \le Mk$, and added source $m^2 \sum_j f_j'^2/4$ of energy $Mm^2 E_f$. Its JT response uses the actual pulse-train moments and initial coefficients in a valid operating domain. No independent-port approximation, thermal input or infinite-lifetime worldline is inferred.

There is a finite causal pulse in this construction. For example, set $f(u) = A \exp[-1/(1 - (u-3)^2)]$ on $2 < u < 4$, zero outside, and normalize its one-particle norm to $k = .2$. Numerical quadrature gives $(E_0, E_1, E_2) = (.4568522, 1.3705565, 4.3215400)$. For $H_0 = 1$, $\kappa = .05$ and $\alpha = 1.1$, the supplied command $a = 48/55$ has residual $m = .04$ and gives $\mathcal{C} \simeq 1.00021929$; the comparison $a = 1$ has $m = -.1$ and $\mathcal{C} \simeq 1.00137053$. These are deterministic evaluations of the displayed pulse, with code and outputs in `supplemental/pulse_debit`, not statistics of a calibration pilot. For a random learned command, the invariant of the mean source uses $(\mathbb{E}m^2)^2$, whereas the mean branch invariant uses $\mathbb{E}m^4$ in the quadratic term. The compact source ends before a later boundary echo is scheduled, but the calibrated wall and timing instrument remain the inputs of Section C.1.

Stress readback also has an exact blind direction. At fixed $a_0 > 0$, the gains $\alpha_\pm = (1 \pm d)/a_0$ give residuals $\mp d$ and therefore the same protection channel and source. Field parity conjugates their full code interactions while leaving the vacuum and stress invariant. No

parity-covariant stress instrument with invariant outcome labels can distinguish them, nor can their mean JT responses at common initial data. Their correcting commands $1/\alpha_\pm$ differ. Probes at $a_0 \pm h$ break this ambiguity through the opposite signs of $8k\alpha h(1-\alpha a_0)$ in their log contrast. A gravitational response is therefore not a complete calibration reference for the interface that produced it. Off-frame coupling probes, or a signed field quadrature, are additional access, with their own sources and costs. Full output-field parity and a conditional logical pulse can instead restore the stored state without identifying the gain, as shown in Ref. [57]. This recovery leaves the emitted source unchanged. Active state correction and learning a match that suppresses the actual source are therefore distinct operations.

The gain is a physical actuator parameter, not a freely changed description. The selected finite code is not the geometric algebra of Section 8, and the supplied JT action still supplies the gravitational response. A link to continuous modular selection requires that further interface; it is not obtained by identifying a protected M_2 sector with a half-sided modular inclusion.

2.4 A physical seed changes the source

The rate-controlled model changes how often a fixed interaction occurs. The coupled observable can itself change, giving a direct relation between an exactly represented algebra and a source. Keep the incoming field, switch and activation schedule fixed. On a qubit put $P_\theta = (I + \cos\theta\, Z + \sin\theta\, X)/2$ and use $U_\theta = \exp\bigl[-i\lambda P_\theta(b + b^\dagger)\bigr]$. The projection selects a pointer algebra and its source population under this fixed schedule.

Proposition 2.3 (Selected projection and outgoing stress)**.** *For $\lambda \neq 0$, an initially independent qubit state ρ and normal field state σ, the exactly represented algebra of the induced qubit channel is $\{P_\theta\}'$. Let $r_\theta = \mathrm{tr}(\rho P_\theta)$. The stress formula below requires its displayed current moments to exist. The outgoing field is*

$$(1 - r_\theta)\sigma + r_\theta e^{-i\lambda Q}\sigma e^{i\lambda Q}, \qquad Q = b + b^\dagger,$$

and its stress is

$$T_\theta(u) = T_{\mathrm{in}}(u) + r_\theta \left[g'(u)\langle j(u)\rangle_\sigma + \tfrac{1}{4}g'(u)^2\right]. \tag{10}$$

Thus two selected projections can change the source at fixed global input state and fixed activation schedule.

Proof. The quadrature has non-atomic spectral distribution in a normal state, so the multiplicative-domain theorem of Q-RAIF C gives $\{P_\theta\}'$ [14]. Expanding U_θ in its two projection blocks and tracing the qubit gives the field mixture; off-diagonal blocks have zero trace even when ρ does not commute with P_θ. In the current identity replace the Pauli X by P_θ and use $P_\theta^2 = P_\theta$. Initial independence gives (10). □

For $\rho = (I + 0.8Z)/2$, the seeds P_0 and $P_{\pi/2}$ select different pointer algebras, with $r_0 = 0.9$ and $r_{\pi/2} = 0.5$. In the worked coherent packet, the cross profile is $8\lambda\sqrt{N}\,\mathrm{Re}\,\psi\,\mathrm{Im}\,\psi$. Its first three moments vanish, so the invariant increment is (62) with p replaced by r_θ. With the table's $N = 16$ and background it is 2.0807581×10^{-5} versus 1.1559770×10^{-5}. The full source also contains the cross profile, which enters the local dilaton even though its first three moments do not enter the final invariant.

Here the source weight is derived from the state's population in the selected projection; it is not assigned to an external mismatch label. Changing the projection means

changing a physical interaction, not renaming an unchanged algebra. The two algebras are abstractly isomorphic but are differently embedded relative to the fixed state and coupling. This supplies a seed–interface–source link. It does not identify that finite-dimensional pointer algebra with a Type III geometric candidate or explain why the coupling selects this seed rather than another. Those are the remaining selection and geometry bridges.

2.5 Response of a given Jackiw–Teitelboim model

Take the JT action [52, 53]

$$I = \frac{1}{16\pi G}\int \sqrt{-g}\,\Phi(R+2/L^2) + I_{\rm matter}, \qquad ds^2 = -\frac{4L^2\,du\,dv}{(1+uv)^2}, \tag{11}$$

with the usual boundary terms, $\kappa = 8\pi G$, $D = 1+uv$ and the vacuum dilaton $\bar\Phi = H_0(1-uv)/D$, $H_0 > 0$. The field equations are $\nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Box\Phi + g_{\mu\nu}\Phi/L^2 = -\kappa T_{\mu\nu}$, and in these coordinates

$$\Phi_{uu} + \tfrac{2v}{D}\Phi_u = -\kappa T_{uu}, \qquad \Phi_{vv} + \tfrac{2u}{D}\Phi_v = -\kappa T_{vv}, \qquad \Phi_{uv} + \tfrac{2}{D^2}\Phi = \kappa T_{uv}. \tag{12}$$

The matter is a conformal theory with matched left and right central charges, and the state-independent anomaly is removed by subtracting the same background. The left-moving input is unchanged, and the state-dependent source is right-moving, $T_{uu} = F(u)$ with $T_{uv} = T_{vv} = 0$.

Proposition 2.4 (Response with fixed initial data)**.** *Let F be integrable with $\int |u|^2|F|du < \infty$. On a connected coordinate domain with $D \neq 0$, write the perturbation as $\Phi - \bar\Phi = (A(u) + vB(u))/D$ and impose the initial coefficient data $A(u), B(u) \to 0$ as $u \to -\infty$. For a source bounded below in support this is equivalent to zero perturbation before the source. The unique solution with these data is*

$$\Phi = \bar\Phi + \frac{A(u)+vB(u)}{D}, \qquad A = -\kappa\int_{-\infty}^{u}(u-z)F(z)dz, \qquad B = -\kappa\int_{-\infty}^{u} z(u-z)F(z)dz. \tag{13}$$

The perturbation is linear in F. With $E_j = \int u^j F\,du$, after the source (or in the asymptotic limit for sources with tails) the invariant $\mathcal{C} = \Phi^2 - L^2(\nabla\Phi)^2$ takes the vacuum value

$$\mathcal{C} = H_0^2 + 2\kappa H_0 E_1 + \kappa^2(E_1^2 - E_0E_2), \tag{14}$$

and inside the support $\partial_u\mathcal{C} = \kappa F\,(2H_0u + uA - B)$.

Proof. Write $\psi = D\,(\Phi - \bar\Phi)$. The vv equation of (12) reads $\psi_{vv} = 0$, so $\psi = A(u) + vB(u)$, and the other two equations reduce to $A'' = -\kappa F$, $B'' = uA''$ and $B' = uA' - A$, which Eq. (13) satisfies. The absolute first and second moments make both retarded coefficients tend to zero at $-\infty$. Two solutions differ by $\alpha + \beta u + \gamma v - \alpha uv$, whose coefficients are $A_h = \alpha + \beta u$ and $B_h = \gamma - \alpha u$. Their two zero limits force $\alpha = \beta = \gamma = 0$. Every vacuum solution has numerator $\alpha + \beta u + \gamma v - \alpha uv$ and invariant $\alpha^2 + \beta\gamma$; after the source $\alpha = H_0 + \kappa E_1$, $\beta = -\kappa E_0$, $\gamma = \kappa E_2$. The balance law follows by differentiation. □

The coefficient condition cannot be replaced by $\Phi - \bar\Phi \to 0$ at each fixed v. The homogeneous solution $\eta v/(1+uv)$ has that limit for every fixed v along nonsingular

points. It even tends to zero uniformly on the subdomain $u < 0$, $v \le 0$, since its absolute value is at most $|\eta|/|u|$; no uniform limit near $D = 0$ is implied. Adding it leaves the source unchanged but changes the late invariant by $-\kappa E_0 \eta$. It is excluded by $B(-\infty) = 0$. Source comparisons below hold H_0, the coordinate matching and both coefficient boundary data fixed.

The JT curvature stays $-2/L^2$; the dilaton and the invariant respond. A dilaton difference is not by itself a higher-dimensional curvature or a mass, and comparing boundary proper times needs a boundary normalization.

The matching condition matters. Set $A_F, B_F \to 0$ at $+\infty$, or exactly to zero after a compact source, instead of the initial data of Proposition 2.4. The final-state-matched solution differs from the retarded solution by

$$A_F = A_R + \kappa(uE_0 - E_1), \qquad B_F = B_R + \kappa(uE_1 - E_2).$$

For the chiral diffeomorphism states of Consciousness I [25], with a Cut at a to the left of the support, the relative entropy is $D_a = 2\pi \int (u-a)F(u)\,du$. The final-state-matched solution gives $\delta\Phi(a,0)/(4G) = -D_a$, while the initial-state-matched solution gives $\delta\Phi(a,0) = 0$ at a Cut before the pulse. The first identity compares an earlier section with a matched final state; it does not say that a pulse removes area before it arrives. The dilaton–modular-energy relation itself is known [58]; what is used here is its type-correct pairing with a specified source and matching condition.

The same equations show that a null-only record of the source is incomplete. A genuine vacuum-density shift $T_{\mu\nu} = -\delta\rho\, g_{\mu\nu}$ has zero null components but gives the particular solution $\delta\Phi = \kappa L^2 \delta\rho$. A probe that reads only null energy cannot certify that a source is absent, which is the source-completeness condition relevant to the vacuum curvature test of Ref. [26].

3 Response that changes the next operation

A source-to-geometry calculation becomes a return only when the geometry changes an actual record and that record changes a later interaction. The readout permission matters. In the minimally coupled JT sector, instruments on the same matter metric with the same inputs have identical adaptive record laws even when the dilaton differs (Proposition C.1). A boundary tied to a fixed dilaton value evades that obstruction. Its echoes reconstruct the three vacuum coefficients, and a source-compensated inverse includes finite probe backreaction (Propositions C.2 and C.3). An exact current contact instead obtains an invariant-dependent bit, including the contact's own source (Propositions C.5 and C.6). The bit selects the next physical projection. Its same-field conditional covariance remains part of the history; detector replacement does not reset the field.

These are finite physical records, not permission to read an exact metric. The worked current contact has total variation only 2.3650×10^{-4} between its two initial geometries. Its fixed-cutoff energy budget also forbids indefinitely many contacts of unchanged positive source charge. Appendix C contains the timing, stress and accumulated-source bounds.

3.1 A finite prefix is a sufficient response object

The source need not have finite second moment over all time to determine a finite episode. This matters for actual material radiation, whose tails are not inferred from a finite

smooth-source certificate.

Proposition 3.1 (Finite-prefix response and its source tolerance)**.** *Fix $u_* < U$ and matched data (a_*, b_*, γ_*) at u_* in the traceless single-chiral JT sector on a common patch with $1 + uv \neq 0$. For $F \in L^1([u_*, U])$, define*

$$a(u) = a_* + \kappa \int_{u_*}^{u} zF(z)dz, \quad b(u) = b_* - \kappa \int_{u_*}^{u} F(z)dz, \quad \gamma(u) = \gamma_* + \kappa \int_{u_*}^{u} z^2 F(z)dz.$$

Then $\Phi = (a + bu + \gamma v - auv)/(1 + uv)$ solves the JT equations on this window with those data, and $\mathcal{C}(u) = a(u)^2 + b(u)\gamma(u)$. No future-source assumption enters either reading. If two sources have $\int_{u_*}^{U}(1 + z^2)|\delta F|dz \leq D$, then

$$\begin{aligned} |\delta a|, |\delta b|, |\delta\gamma| &\leq \kappa D, \\ |\delta\Phi| &\leq \frac{\kappa D(1 + |uv| + |u| + |v|)}{|1 + uv|}, \\ |\delta\mathcal{C}| &\leq \kappa D(2|a| + |b| + |\gamma|) + 2\kappa^2 D^2. \end{aligned} \tag{15}$$

Proof. The coefficient derivatives are $a' = \kappa uF$, $b' = -\kappa F$ and $\gamma' = \kappa u^2 F$. They cancel from the first derivatives of the numerator, giving the field equations in their weak sense. Direct substitution in $\Phi^2 - L^2(\nabla\Phi)^2$ gives $a^2 + b\gamma$. The three integrands are bounded in absolute value by $1 + z^2$. Expanding the numerator and invariant gives the estimates. □

For the complete quantum material, Q-RAIF C bounds smooth interacting stress tests on one positive Hamiltonian with all old references retained [14]. Its cross and creation contributions are essential. The bound is a test-functional certificate, whereas Eq. (15) uses a weighted L^1 certificate. They are not interchangeable. With a smooth preparation cutoff, smearing a finite-resolution dilaton reading gives smooth retarded kernels. Each such kernel can be used as a source test with its own Q-RAIF C form constant. Recovering a sharp prefix additionally requires a bound on unresolved source structure. This gives a direct finite-window interface without claiming an unproved all-time tail estimate or discarding controller and support sources.

3.2 The same quantum reader, support and source

A reflecting boundary alone does not supply its support. The material branch uses a positive dilaton-dependent mass and varies the worldline. It contains the finite reader, history clock, blank and old purification in its mass. Classical scalar fields couple to an angle θ; storage y has positive kinetic and spring energies. Their emitted powers are $P_\pm = g_\pm^2 \dot{\theta}^2$, derived by characteristics, and their reaction is included in the mechanical equation. The two exterior invariants evolve with the proper worldline and obey the mass junction. Appendix F gives this complete mean-action construction and its admissible positive domain.

The quantum core is

$$H_c = H_{\text{hist}} + E_{\text{off}} I + \lambda(\Phi - \phi_0) Z_P + \frac{\chi_R}{2} P_R (y - d)^2.$$

Its first active clock edge writes an actual Y_P outcome; the next swaps the probe with a paid blank. The record spring reacts while acquisition is in progress. Variation gives

$F_y = \chi_R(y-d)\langle P_R\rangle$ and $F_\Phi = \lambda\langle Z_P\rangle$, with $\dot{E}_c = F_y\dot{y} + F_\Phi\dot{\Phi}$. Consequently $M = m_{\rm rest} + U + e + E_c$ obeys $\dot{M} = (U' + F_\Phi)\dot{\Phi} - P_+ - P_-$. These are quantum expectations in a retained pure-state calculation, not a sampled record fed to an independent source generator.

Proposition F.7 proves a nonzero source return through its shortest-path expansion. Either $\lambda = 0$ or $\chi_R = 0$ removes the proper-power difference between its two initial geometries. Both interactions present produce a nonzero coefficient at order t^{2r+7}, where r is the acquisition edge. The finite 33-state example retains its old probe reference and all work balances. This is a mean-worldline realization; quantum support fluctuations are not removed by taking the quantum expectation of its force.

The reader must also remain accurate when the response is allowed to move, rather than being supplied beforehand. The following estimate controls this coupled problem.

Proposition 3.2 (Joint tolerance of an acquired-record source return)**.** *Write the mean-action equations as $\dot{z} = \mathcal{F}(z,\Psi)$ and $i\dot{\Psi} = H_c(z)\Psi$, retaining the old reference in Ψ. Allow $H_c(z) = H_* + V_c(z)$, where H_* is a common semibounded self-adjoint internal Hamiltonian and V_c and its first two coordinate derivatives are bounded on the stated domain. For an unbounded core, require strong initial data in $D(H_*)$ and propagators preserving this common domain along the stated coordinate paths. This includes the finite core above and regular Gaussian internal data with bounded controller coupling; it does not include an unbounded coordinate-force operator merely because its free field is Gaussian. Include the complete mass M among the classical variables, with its actual initial value and work equation*

$$\dot{M} = (U' + \langle\partial_\Phi V_c\rangle)\dot{\Phi} - P_+ - P_-.$$

Let exact and approximate solutions, with the same core and reference space, lie in a compact subset of the positive domain of Proposition F.4, with $|y| \le Y$ and $|\dot{\theta}| \le V$. Use the maximum norm for classical variables nondimensionalized in fixed material and clock units. Suppose the approximate equations have classical residual at most $r_z(t)$, including their mass-work residual. Let $r_H(t)$ bound the Hermitian Hamiltonian error applied to the approximate quantum vector; the propagators and this residual must exist on their specified domains. Choose constants on this domain such that

$$|\mathcal{F}(z,\Psi) - \mathcal{F}(\widehat{z},\widehat{\Psi})| \le L_z|z-\widehat{z}| + L_\Psi\|\Psi - \widehat{\Psi}\|.$$

Here the same reference is compared, not just its reduced probe state. For the displayed sensor and spring the Hamiltonian Lipschitz constant is $L_H = |\lambda| + \chi_R(Y + |d|)$. For any $\sigma > 0$ put $L = \max\{L_z + \sigma L_H, L_\Psi/\sigma\}$ and $E = |z-\widehat{z}| + \sigma\|\Psi - \widehat{\Psi}\|$. Then

$$E(t) \le e^{Lt}E(0) + \int_0^t e^{L(t-s)}[r_z(s) + \sigma r_H(s)]ds. \tag{16}$$

Any terminal record probability differs by at most $2E/\sigma$, and the total proper emitted power differs by at most $2\gamma VE$, where $\gamma = g_+^2 + g_-^2$. These estimates cover the actual quantum-force reaction in the classical response, including its contribution to mass and clock conversion.

Proof. For normalized states, $|\langle A\rangle_\Psi - \langle A\rangle_{\widehat{\Psi}}| \le 2\|A\|\|\Psi - \widehat{\Psi}\|$. In particular the P_R and Z_P expectations have that bound. The force differences are bounded by

$$|\delta F_y| \le \chi_R|\delta y| + 2\chi_R(Y + |d|)\|\delta\Psi\|, \quad |\delta F_\Phi| \le 2|\lambda|\|\delta\Psi\|,$$

The mechanical work $-F_y\dot{y}$ cancels its contribution $F_y\dot{y}$ to the internal energy, leaving the stated mass equation. Thus mass is compared through its initial value and integrated work, not by bounding $\langle H_* \rangle$ with the state-vector error. The mass, square roots $\beta_\pm$ and null-clock denominators are smooth with bounded derivatives on the stated compact domain; these formulas give finite L_z, L_Ψ for the actual coupled equations. All clock and material energy remains in the initial mass. The quadratic spring and the probe term give the displayed L_H. Duhamel comparison using the exact unitary propagator gives

$$\|\delta\Psi(t)\| \le \|\delta\Psi(0)\| + \int_0^t [L_H|\delta z(s)| + r_H(s)]ds.$$

It contains no spurious growth from the norm of a common clock Hamiltonian. Add the classical integral bound with weight σ and apply Gronwall. Finally $P_+ + P_- = \gamma\dot{\theta}^2$ gives the power estimate, and a record effect has norm at most one. □

For the strong-domain solutions just specified, the Schrödinger equation gives

$$\frac{d}{dt}\langle H_* + V_c(z)\rangle = \sum_j \langle \partial_j V_c(z)\rangle \dot{z}_j.$$

The common H_* contributes no explicit coordinate work. In its interaction picture only bounded perturbation differences enter Duhamel's estimate, so the same proof applies without bounding $\langle H_* \rangle$ by a vector-norm error. Extending to all energy-form initial data would additionally require an energy-controlled approximation preserving this work identity. Spatial field observables still require the separate material form bounds of Q-RAIF C. This extension treats the core as internal material of the specified thin worldline; it does not equate spatial field stress with a point source. Smooth tests of the line stress $Mu^\mu u^\nu \delta_\Gamma$ are Lipschitz on the same compact tube, so their errors are bounded by a constant times $\int E(s)ds$. The constants and admissible tube may change when the initial clock mass grows; no uniform-in-clock-size geometric limit follows.

The work equation is necessary even when the retained quantum states are close. Q-RAIF C gives a positive clock-driven oscillator whose actual total energy is mJ, whereas its scheduled product-clock target has energy $mJ + b^2/(2\omega)$ for every m, despite vanishing joint state error [14]. A nearby rotated clock can match that scalar energy, but does not thereby match local stress. For a scheduled comparison with internal Hamiltonian jumps $h_{b-1} \to h_b$, subtract the switching work $W_b = \langle h_b - h_{b-1}\rangle_{t_b}$ from its clock ledger. Together with the continuous coordinate work this gives the mass equation used above. Initial mass mismatch and any missing work belong in $E(0)$ and r_z; they cannot be dropped by declaring the clock state error small. This ledger is not the energy expectation of the unmodified product-clock target.

A source contrast remains nonzero whenever the two power-error bounds sum to less than its ideal contrast at a chosen time. Thus finite implementation errors can be admitted while preserving the causal sensor-removal and feedback-removal tests. The domain condition is verified by a strict ideal margin and an error tube smaller than that margin; it cannot be certified from a reduced-state fidelity alone. These are the fixed-window mean-action bounds in Table 1.

Foundation III uses this action for a frame learned while the original quantum content is held [56]. In a finite positive parameter family, the acquired record changes the apparatus's own source; its geometric response enters an X-basis reading of the old probe into

the already spent cell. The first-order geometry signal is generated after record acquisition, with no new probe or blank port. The matched intervention keeps the complete initial state, mass and geometric constraints fixed and removes only the first record's mechanical use. A sensor-edge diagnostic separates the geometric path from direct probe and cursor effects. This is one finite semiclassical mean-action scalar/JT return. The completed-window geometry-edge contrast, averaged over the stated plant-sign prior, is about -4.98×10^{-9} in the example; its small size is reported separately from the task advantage. It is distinct from the spatial realization below and from the sustainable sharp-exhaust conjecture.

3.3 A complete spatial clock-work source

The work correction can be a spatial tensor rather than a scalar debit. Consider the local quadratic action

$$S_{\chi u} = -\frac{1}{2}\int \sqrt{-g}\left[I(\nabla\chi)^2 + (\nabla u)^2 + I\mu^2\chi^2 + (\omega u + \lambda\chi/\omega)^2\right] d^2x, \tag{17}$$

where $I, \mu, \omega > 0$. In canonical variables $(\sqrt{I}\chi, u)$ its mass matrix is

$$\begin{pmatrix} \mu^2 + \lambda^2/(I\omega^2) & \lambda/\sqrt{I} \\ \lambda/\sqrt{I} & \omega^2 \end{pmatrix}.$$

Its determinant is $\mu^2\omega^2 > 0$. Constant orthogonal diagonalization gives two positive-mass Klein–Gordon fields with their complete spatial mode towers. Choose the interacting static vacuum and a finite-energy Weyl displacement with initial data $\chi = u = \dot{u} = 0$, $\dot{\chi} = \Omega f$, $f \in C_c^\infty$. Its added clock energy in flat space is $I\Omega^2\|f\|_2^2/2$. One fixed local renormalization prescription is used for the interacting vacuum and all displaced states.

Put $L_a = \partial_t^2 - \partial_x^2 + a^2$. In flat space let

$$L_\mu c = 0, \qquad L_\omega u_0 = -\lambda c, \qquad L_\mu \eta_0 = -\lambda u_0 - \lambda^2 c/\omega^2, \tag{18}$$

where c has the prepared clock data and the other two fields have zero data. Define

$$T^{(a)}_{\alpha\beta}[q] = \partial_\alpha q \partial_\beta q - \tfrac{1}{2} g_{\alpha\beta}[(\nabla q)^2 + a^2 q^2], \tag{19}$$

$$B_{\alpha\beta}[c, \eta] = \partial_\alpha c \partial_\beta \eta + \partial_\alpha \eta \partial_\beta c - g_{\alpha\beta}(\nabla c \cdot \nabla \eta + \mu^2 c\eta). \tag{20}$$

The work-corrected comparison source is

$$T_{\mathrm{cmp}} = IT^{(\mu)}[c] + B[c, \eta_0] + T^{(\omega)}[u_0] - g(\lambda c u_0 + \lambda^2 c^2/(2\omega^2)) + T_{\mathrm{vac},I}. \tag{21}$$

The large positive clock source and the same interacting-vacuum source are present on both sides of the comparison.

Proposition 3.3 (Full-mode stress matching)**.** *For each fixed finite window and compact smooth tensor test W, the actual coherent quantum state of* (17) *satisfies*

$$\left| \int \sqrt{-g}\, W^{\alpha\beta}(\langle T_{\alpha\beta}\rangle - T_{\mathrm{cmp},\alpha\beta}) \right| \le C_W/I. \tag{22}$$

The constants are independent of I above a finite threshold. In flat space the comparison obeys the exact local Ward identity. The source difference also tends to zero in every fixed C^k norm on a compact window. The same statements hold on a fixed static metric using its finite-window Klein–Gordon energy constants.

Proof. The actual means obey $L_\mu \chi_I = -\lambda u_I/I - \lambda^2 \chi_I/(I\omega^2)$ and $L_\omega u_I = -\lambda \chi_I$. For $\eta_I = I(\chi_I - c)$ these imply the exact identities

$$L_\mu \eta_I = -\lambda u_I - \lambda^2 \chi_I/\omega^2, \qquad u_I - u_0 = -\lambda G_\omega \eta_I / I,$$

where G_ω is the zero-data retarded operator. Use $\|q\|_{s,a}^2 = \|\dot{q}\|_{H^s}^2 + \|q_x\|_{H^s}^2 + a^2\|q\|_{H^s}^2$. The energy inequality is $\|G_a F(t)\|_{s,a} \le \int_0^t \|F(r)\|_{H^s} dr$. Set $C = |\Omega| \|f\|_{H^s}$, $a_T = |\lambda| T/\mu$, $b_T = |\lambda| T/\omega$, $d_T = \lambda^2 T/(\omega^2 \mu)$ and $h_T = a_T b_T + d_T$. For $I \ge 2h_T$ it gives

$$\sup \|\eta_I\|_{s,\mu} \le 2h_T C, \qquad \sup \|u_I - u_0\|_{s,\omega} \le 2a_T h_T C/I, \tag{23}$$

$$\sup \|\eta_I - \eta_0\|_{s,\mu} \le 2h_T^2 C/I. \tag{24}$$

These are full spatial energy estimates, without a mode cutoff. The exact quantum covariance equals the interacting-vacuum covariance; only the mean is displaced. Metric variation therefore gives

$$\langle T \rangle = I T^{(\mu)}[c] + B[c, \eta_I] + I^{-1} T^{(\mu)}[\eta_I] + T^{(\omega)}[u_I] - g(\lambda \chi_I u_I + \lambda^2 \chi_I^2/(2\omega^2)) + T_{\mathrm{vac},I}.$$

Polarization of each quadratic tensor and the displayed energy estimates prove (22). Smooth initial data, higher Sobolev estimates and the wave equations give the C^k result. Finally $\nabla^\alpha B_{\alpha\beta} = (\lambda u_0 + \lambda^2 c/\omega^2)\partial_\beta c$; the material and contact terms have the opposite divergence. □

The correction has a measurable tensor component. Relative to the common clock and vacuum terms in (21), the flat-space preparation gives

$$\delta T_{tt} = O(t^4), \qquad \delta T_{xx} = -\lambda^2 \Omega^2 f^2 t^2/\omega^2 + O(t^4). \tag{25}$$

The clock pays the order-t^2 energy transfer; pressure does not cancel. This subtraction defines a comparison, not a second experiment with the coupling removed and a different vacuum. For metric-only matter on a fixed AdS patch, the matched sources drive the linear JT dilaton equation with the same initial constraints and boundary data. Its finite-window response inherits the source estimate. A positive dilaton domain may require initial homogeneous data growing with I; that resource and a detector's gain cannot be removed from the account. This is the quadratic clock/material entry of Table 1; its receiver and thermal interfaces are separate.

At finite spatial bandwidth, these source and readback steps can be placed in one mean-action evolution. Proposition F.8 includes an elastic bar, a physical synchronization field, two retained quantum spins and the full oscillator Hilbert space of the chosen modes. The two preparations have the same initial gravitational constraints. Their later pressure contrast produces a dilaton contrast, which changes the retained sensor and the next material force. The synchronization field is varied, not supplied as a prescribed clock. This establishes the finite spatial return listed in Table 1.

The retained quartet admits a full-mode bulk step toward that construction. Proposition F.9 evolves its physical clock, material director and exact renormalized quantum source together with constrained JT data. The normal-order conversion and causal-state identification are nonlinear. The result supplies an interior patch for specified global quantum data; moving endpoints, the other gate windows and the finite-horizon renewal error must still be joined to it.

4 Maintenance and its gravitational price

4.1 Entropy removal and the source it can constrain

A fixed unitary correction can return its control port coherently. Removing entropy injected by unobserved noise is a different task. To price it, the state whose entropy is removed must include the controller's usable side information. The heat account must also distinguish a thermal reservoir from a supply of pure memories.

HAFF G supplies a record-driven program and executor in which selecting a physical coupling changes its source under one fixed executor [55]. The dephased quartet below adds a specified full renewal debit. It does not identify the two material realizations or inherit a thermal controller from that program loop.

Proposition 4.1 (Maintenance entropy account)**.** *Let $V = SC$ include the maintained system and every retained controller and side-information register. A correction acts unitarily on finite-dimensional VAB, initially in $\rho_V \otimes \alpha_A \otimes \tau_B$, where $\tau_B = e^{-\beta H_B}/Z$, $\beta = (k_B T)^{-1} > 0$. Put*

$$s = S(V) - S(V'), \qquad p = S(A') - S(A), \qquad Q = \operatorname{tr} H_B(\rho'_B - \tau_B).$$

Then, with $I_{V:A:B} = S(V) + S(A) + S(B) - S(VAB)$,

$$\beta Q + p = s + I_{V':A':B'} + D(\rho'_B \| \tau_B) \ \geq s. \tag{26}$$

If the marginal entropy of C is returned, s equals the decrease of $S(S|C)$. In a sequence of such corrections, a bounded total purity consumption and a positive asymptotic entropy-removal rate $\dot{s}$ imply $\liminf Q_{[0,t]}/t \geq k_B T \dot{s}$.

Proof. Unitary invariance of total entropy and the initial product give $\Delta S(V) + \Delta S(A) + \Delta S(B) = I_{V':A':B'}$. Also $\beta Q = \Delta S(B) + D(\rho'_B \| \tau_B)$. Combine the identities and sum over corrections. The rate statement follows when the accumulated p divided by elapsed time tends to zero. □

This is the Landauer entropy balance, with the memory-purity term displayed explicitly; the equality and the role of accessible side information are standard [13, Theorem 3 and Section 5.1]. For initially correlated or nonthermal inputs the same calculation instead gives

$$\beta Q + p = s + (I_{\text{out}} - I_{\text{in}}) + D(\rho'_B \| \tau_B) - D(\rho_B \| \tau_B).$$

These consumed correlations and bath free energy can pay for correction. A classical copy of the error permits a reversible conditional flip with $s = 0$, although the system's marginal entropy decreases. Negative quantum conditional entropy gives another reason to include the side information. A pure-memory swap has $p = s$ and can have $Q = 0$. A retained oscillator's closed unitary loop also has no entropy removal by itself. Thus a count of corrections or a marginal noise entropy is not automatically $\dot{s}$. In a cyclic device $\dot{s}$ must be established from its actual noise and controller dynamics. For nondegenerate systems, heat and supplied work are further separated by the change in internal energy.

The memory can also be reused without being decorrelated after each correction. Let correction i act on VA and a fresh thermal port B_i, initially independent of the whole state of VA and all earlier ports, and put $s_i = S(V_i) - S(V'_i)$ and $x_i = S(B'_i) - S(B_i)$. Suppose each intervening noise step is a CPTP map on V, tensored with the identity on A

and the previously exported ports; updates involving A belong to the charged correction. Entropy conservation gives

$$x_i = s_i - [S(A_i') - S(A_i)] + I(V_i' : A_i') - I(V_i : A_i) + I(V_i' A_i' : B_i').$$

Data processing between corrections and telescoping therefore imply

$$\sum_{i=1}^{n} x_i \geq \sum_{i=1}^{n} s_i - [S(A_n') - S(A_1)] - I(V_1 : A_1). \tag{27}$$

For a fixed finite A, the last two terms are bounded. Reused memory purity and initial correlations cannot absorb a positive entropy-removal rate indefinitely. Here x_i refers to each exported port's marginal entropy. Summing those marginals can overcount correlated records. There is a stronger joint-port statement under the same fresh-input and local-noise assumptions:

$$S(B_{1:n}') - \sum_{i=1}^{n} S(\tau_{B_i}) \geq \sum_{i=1}^{n} s_i - [S(A_n') - S(A_1)] - I(V_1 : A_1). \tag{28}$$

To see this, retain all previously exported ports in the mathematical state. A noise step on V, with the identity on those ports, changes its total entropy by $\Delta S(V) - \Delta I(V : AB_{<i}) \geq \Delta S(V)$, by data processing. The corrections preserve total entropy, and each new port adds $S(\tau_{B_i})$. At the end, subadditivity between VA and all exported ports gives Eq. (28) after the V entropies telescope. Thus finite reused memory cannot hide the maintained rate in correlations between outputs.

The same account applies when the noise also acts on the retained controller. Put $W = VA$ and let each correction be a unitary on WB_i with a fresh, independent input τ_{B_i}, uncorrelated with W and the earlier ports. Noise between corrections acts on W alone. With $s_i^W = S(W_i) - S(W_i')$, the exact identity is

$$S(B_{1:n}') - \sum_{i} S(\tau_{B_i}) = \sum_{i} s_i^W + I(W_n' : B_{1:n}') + \sum_{i=1}^{n-1} \ell_i \;\geq\; \sum_{i} s_i^W, \tag{29}$$

where $\ell_i = I(W_i' : B_{1:i}') - I(W_{i+1} : B_{1:i}') \geq 0$ is the information lost to the intervening noise. Indeed, that noise changes the entropy of the joint state by $\Delta S(W) + \ell_i$; the correction preserves it. Summing these changes and expanding the final entropy as $S(W_n') + S(B_{1:n}') - I(W_n' : B_{1:n}')$ gives the identity. Returning an old port to an operation would change these hypotheses. The joint VA rate, rather than a sum of system marginal entropies, is what an apparatus with noisy internal memory must export.

Proposition 4.2 (A finite-rate chiral exhaust floor)**.** *Consider a unitary, zero-chemical-potential chiral conformal exhaust with central charge $c_{\mathrm{ex}} > 0$, thermal input at $T \geq 0$, and an extensive entropy-current account. Energy and temperature are defined by the same translation generator and clock. Assume that the outgoing modes export net dimensionless entropy current $x \geq \dot{s} > 0$, where $\dot{s}$ is the actual maintenance entropy-removal rate after all retained side information and sublinear resource consumption are accounted for. The entropy-current maximum at fixed energy is the thermal one. Then the excess heat current over the incoming thermal stream obeys*

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2. \tag{30}$$

Proof. Write $a = \pi c_{\mathrm{ex}} k_B^2/(12\hbar)$. The thermal energy and dimensionless entropy currents are $J_E(T) = aT^2$ and $J_S(T) = 2aT/k_B$. The first is the standard chiral CFT current [11, Eq. (1)]; the second follows from its extensive Gibbs entropy. Entropy maximization at fixed energy is the Gibbs variational principle, underlying the single-channel information-flow bound [10]. Choose $T_* = T + k_B x/(2a)$ so that the thermal output has the required entropy. Any output with at least that entropy has energy current at least aT_*^2. Subtracting aT^2 gives

$$\dot{Q} \geq k_B T x + \frac{k_B^2 x^2}{4a} = k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}.$$

The expression increases for $x \geq 0$. □

The current account is a physical condition on the exported modes, not the entropy of a Type III local interval. Equation (28) establishes the needed joint entropy rate in a fresh-port account; matching those ports to a conformal current must preserve its capacity and entropy budget. At fixed capacity and positive rate the quadratic term survives even at $T = 0$. Thermal output saturates the channel inequality, but this alone does not construct a correcting apparatus. The currents are long-window, extensive Gibbs limits. For an observation window τ with net exported entropy X_τ, bounded rates and sublinear boundary corrections give

$$Q_\tau \geq k_B T X_\tau + \frac{3\hbar X_\tau^2}{\pi c_{\mathrm{ex}} \tau} + o(\tau).$$

The current formula is not an exact inequality for an arbitrarily short finite packet. Applying the Gibbs bound to the entire output also includes correlations between packets; it does not require each output to be a separate thermal state.

Proposition 4.3 (Output deficit and direct renewal)**.** *Use the complete outgoing carrier B_τ, its physical translation generator H_τ, and the thermal input of Proposition 4.2. Let the actual charged entropy removal satisfy $s_\tau/\tau \to x > 0$, set $T_* = T + 6\hbar x/(\pi c_{\mathrm{ex}} k_B)$, and write $\tau_* = e^{-H_\tau/(k_B T_*)}/Z_\tau$. Assume the extensive Gibbs limits used above and that input, charge, interaction, resource and boundary remainders are $o(\tau)$. If the full entropy account gives $L_\tau = S(\rho_{B_\tau}) - S_T - s_\tau \geq -o(\tau)$, then*

$$Q_\tau - \tau\left(k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}\right) = k_B T_* \left[D(\rho_{B_\tau} \| \tau_*) + L_\tau\right] + o(\tau). \tag{31}$$

Thus asymptotic equality requires, and under these hypotheses is equivalent to, both deficit and entropy-slack rates tending to zero.

For Q-RAIF C's original quartet preparation followed by direct per-species ring renewal [14], let d_{ring} bound its reset error against the actual old outside marginal, η its residual zero-mode amplitude, and e_H the full-joint clock comparison error over the specified horizon. Suppose each round has disjoint emission modes, these bounds hold uniformly, and no later converter changes the exported modes. With $t = d_{\mathrm{ring}} + 2\arcsin\eta + 2e_H < 1/2$, the output of M rounds obeys

$$D(\rho_{B_\tau} \| \tau_*) \geq M\left[h(\epsilon) - 2h(t)\right], \qquad h(p) = -p\ln p - (1-p)\ln(1-p), \tag{32}$$

when the zero-chemical-potential free-rail reference factorizes across the original species. If $M/\tau \to \nu$ and $h(\epsilon) > 2h(t)$, the asymptotic heat excess above the floor is at least $k_B T_* \nu[h(\epsilon) - 2h(t)]$ per unit time. Temporal independence of thermal packets is not required.

Proof. The finite-carrier Gibbs identity is $D(\rho\|\tau_*) = (E_\rho - E_*)/(k_B T_*) - S(\rho) + S_*$. Here $S_* - S_T = x\tau + o(\tau)$ and $E_* - E_T = \tau[k_B T x + 3\hbar x^2/(\pi c_{\mathrm{ex}})] + o(\tau)$, which gives (31).

The current purifier's occupation pair is 00 or 11, with probabilities $1-\epsilon, \epsilon$. Operations on the receiver leave its joint marginal with previous outputs unchanged. The reset certificate compares that pair with the same actual past marginal; it need not be independent of the current readout record. The exact linear scattering column emitting an old zero mode d_j has squared norm $1-\eta^2$. CAR therefore give the extracted mode $b_j = \sqrt{1-\eta^2}\, d_j + \eta e_j$, where e_j is a canonical orthogonal mode in the actual body and incoming field. A two-mode rotation of angle $\arcsin\eta$ compares b_j with d_j, retaining every old reference. The two purifier species cost $2\arcsin\eta$; replacement of the comparison joint state and its outside marginal costs at most $2e_H$.

Measure their even occupations in the disjoint emitted modes, giving $X_{1:M}, Y_{1:M}$. For each round the joint law of $(X_j, X_{<j})$ is within variation t of $\mathrm{Bern}(\epsilon) \otimes P_{X_{<j}}$, and $\Pr(X_j \neq Y_j) \le t$. Conditional binary entropy continuity and the binary Fano bound give $H(X_j|X_{<j}) \ge h(\epsilon) - h(t)$ and $H(X_j|Y_j) \le h(t)$. Hence $I(X_{1:M} : Y_{1:M}) \ge M[h(\epsilon) - 2h(t)]$. The reference measurement law is $q_X q_Y$, even though each sequence may be thermally correlated. Data processing and $D(P_{XY}\|q_X q_Y) = I(X:Y) + D(P_X\|q_X) + D(P_Y\|q_Y)$ prove (32). □

All entropies in this proposition use the finite-carrier account and its extensive limit, not a trace entropy of a Type III interval. A fixed-charge canonical reference needs its sector-weight correction; it has the same rate conclusion only when that correction is sublinear. A conditional account must use the same charged removal. For a complete usable system $W = ZM$ with a conserved classical record Z, write $s^W = S(W) - S(W')$. A unitary interaction with a port B gives

$$S(B'|Z) - S(B|Z) = s^W + I(M' : B'|Z) - I(M : B|Z).$$

When the port is initially independent of W, its thermal entropy replaces $S(B|Z)$ and the last term vanishes. Old purifications remain in the physical state; they are usable side information only if the apparatus has that access. Fresh-port sequences with noise acting only on M also accumulate nonnegative information losses to earlier outputs. Reading an old output or updating Z needs a joint account of that operation.

For a classical record X retained at input and output, put $L_c = S(B_{\mathrm{out}}|X) - S(B_{\mathrm{in}}|X) - s_\tau$. When the input marginal entropy is S_T, the same Gibbs identity is

$$\begin{aligned} Q_\tau - \tau f(x) &= k_B T_* \left[D(\rho_{B_{\mathrm{out}}}\|\tau_*) + L_c + I(X : B_{\mathrm{out}}) - I(X : B_{\mathrm{in}})\right] + o(\tau), \\ f(x) &= k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}. \end{aligned} \tag{33}$$

Indeed $L_\tau = L_c + I_{\mathrm{out}} - I_{\mathrm{in}}$, while $D(\rho_{XB}\|\rho_X \otimes \tau_*) = D(\rho_B\|\tau_*) + I(X:B)$. These are two forms of one account; thermal input marginals alone do not remove old correlations. The direct protocol's deficit lower bound tends to $\nu h(\epsilon) > 0$ as its certified error vanishes. A physical converter that redistributes those correlated records, with its own work and retained memory charged, is therefore necessary for that route to approach the sharp floor.

The incoming state and conserved charges matter. For a nonthermal stream with energy current e_{in} and dimensionless entropy current σ_{in}, put $E_{\mathrm{th}}(\sigma) = 3\hbar\sigma^2/(\pi c_{\mathrm{ex}})$. The same variational argument gives

$$\dot{Q} + e_{\mathrm{in}} - E_{\mathrm{th}}(\sigma_{\mathrm{in}}) \ge k_B T_\sigma x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}, \qquad T_\sigma = \frac{6\hbar\sigma_{\mathrm{in}}}{\pi c_{\mathrm{ex}} k_B}. \tag{34}$$

The nonequilibrium energy above the entropy-matched thermal stream can pay for export. A pure coherent input therefore cannot be charged as though it were a zero-energy thermal reservoir. At nonzero chemical potential the corresponding Gibbs comparison must hold all retained charges fixed. For example, with a level-one $U(1)$ current normalized as in Ref. [12, Eq. (84)], the unitary CFT Gibbs currents in units $\hbar = k_B = 1$ are $e = \pi c_{\rm ex} T^2/12 + \pi\mu^2/2$, $j = \pi\mu$ and $\sigma = \pi c_{\rm ex} T/6$ [12, Eq. (93)]. Equal incoming and outgoing charge currents cancel the chemical energy in their difference; an apparatus that absorbs charge can instead consume it. Equation (30) does not cover that consumption without a charge ledger.

For independent exhausts at a common temperature, with central charges c_j and nonnegative entropy currents x_j, the same argument and Cauchy–Schwarz give

$$\dot{Q}_{\rm tot} \geq k_B T \dot{s} + \frac{3\hbar \dot{s}^2}{\pi \sum_j c_j}, \qquad \sum_j x_j \geq \dot{s}. \tag{35}$$

The capacity term is sharp at the channel level: equality requires thermal outputs and $x_j = c_j \dot{s}/\sum_k c_k$. Additional channels are an additional physical resource. Intermittent export through the same capacity cannot lower the long-window Gibbs floor: apply it to the complete output with its total entropy. If a local-current description is valid, $\overline{x^2} \geq \bar{x}^2$ gives the same conclusion.

Heat alone does not bound every geometric observable. Even a positive E_0 can give zero or negative $\delta\mathcal{C}$ in Eq. (62); its response depends on other moments and on the background. A positive weighted integral $\int K(u)F(u)du \geq K_{\rm min} \int F du$ requires both a lower bound on K over the exhaust and $F \geq 0$. Quantum excess stress need not have that sign. There is, however, a simple JT observable which reads E_0 without either assumption.

Proposition 4.4 (A heat account read through the dilaton)**.** *Compare two JT solutions with the same H_0, boundary coefficient data and coordinate and clock normalization. Suppose their source difference F is the exhaust of the correction in Proposition 4.1, with finite absolute second moment and $\int F du = Q$. This last equality requires that the specified outgoing channel carry all the heat in the same energy normalization. Then its retarded coefficient difference obeys*

$$-\frac{A'(+\infty)}{\kappa} = Q \;\geq\; k_B T (s - p). \tag{36}$$

For a compact exhaust and any $u_2 > u_1$ after it, the left side is exactly $-[A(u_2) - A(u_1)]/[\kappa(u_2 - u_1)]$. No pointwise sign of F is required.

Proof. Differentiate Eq. (13): $A'(u) = -\kappa \int_{-\infty}^{u} F(z)dz$. Take the late limit and apply Eq. (26). □

The reference for Q is the incoming thermal reservoir transported through the same exhaust, before its heat uptake; H_B is its u-translation generator. A thermal background has a nonintegrable stationary flux, so the JT solution here is the difference from that same transported thermal background. The identification $E_0 = Q$ is an energy-transport hypothesis: other exhaust channels, energy stored in the device, work beams crossing the observation surface and redshift must be accounted for separately. With a fixed transport factor $E_0 = \zeta Q$, Eq. (36) acquires the same factor ζ. The coefficient slope is read from

the dilaton in the declared matched coordinate frame, not an invariant under arbitrary changes of boundary clock. The relation links an actual entropy-removal process to a specified dilaton reading. The JT dynamics and heat-routing map are its geometric inputs.

4.2 A horizon reading of the maintenance price

For a matched running single-chiral JT source, let $\mathcal{C} = R^2 > 0$ and $\xi = (bu^2 + 2au - \gamma)/(2R) > 0$ on the regular future outer patch. The marginal horizon has $\Phi_h = R$ and $S_h = S_0 + 2\pi R/\kappa$. In its rapidity clock $d\eta = du/\xi$,

$$\frac{dS_h}{d\eta} = 2\pi F\xi^2.$$

When the charged heat is exactly $F\xi^2$ and the physical exhaust input is thermal at $T_H = 1/(2\pi)$ in this same clock, the long-window channel bound gives

$$\overline{\dot{S}_h} \geq \dot{s} + \frac{6\dot{s}^2}{c_{\mathrm{ex}}}.$$

Appendix D proves the horizon balance and its domain conditions. The entropy here is classical Wald entropy; a generalized-entropy statement needs its matter contribution.

Steady positive flux cannot be extrapolated in one fixed u chart: its boundary dilaton numerator eventually vanishes. Nor does a new clock prepare a reservoir. For an actual source field thermal in η, the vacuum-subtracted physical flux contains $q_{\mathrm{ref}} = -c(\delta' + \delta^2/2)/(24\pi)$, where $\delta = R'/R$. Its self-consistent equations are

$$R' = R\delta, \qquad \delta' = \frac{\kappa q - R\delta}{\kappa c/(24\pi)} - \frac{\delta^2}{2}.$$

The positive growing family of Proposition D.5 has $R = R_0 + \kappa q\eta + O(1)$ and finite accumulated reference correction. Hence its asymptotic horizon rate is $2\pi q = s + 6s^2/c$. A partially transmitting local thermal contact produces a nonthermal output with a strictly larger energy price at the same entropy current; changes of angle add their own pump work. The detailed state, source and clock calculations are in Appendix D. Their global source-clock solution does not grant infinite life to a finite material energy reserve.

5 The Factorization Problem

5.1 No canonical tensor factorization

A Hilbert space of dimension $d = d_1 d_2$ admits a tensor factorization, but its Hilbert structure alone does not select one. Commuting observable algebras can induce the subsystem structure [17, 18]. Which of those algebras an apparatus can use is a physical question about its couplings and resources.

5.2 Coarse-graining and effective descriptions

Paper A [19] gives the following conditional separation.

Proposition 5.1 (Conditional coarse-graining separation)**.** *Let $\Phi_1 \neq \Phi_2$ be finite-dimensional coarse-graining channels with a common output space. Their outputs differ for generic pure inputs. Entropy profiles differ only under the additional nondegeneracy hypothesis stated in Paper A, and metric inequivalence follows only when a supplied reconstruction dictionary separates those profiles modulo diffeomorphisms.*

Channel inequivalence therefore establishes a difference of effective descriptions, not by itself a difference of spacetime metrics. Paper A also decides, for qubit reductions, exactly when two channels give the same output entropy on every pure input, and it constructs one global state, one family of local noise channels and one fixed reconstruction rule that yield non-isometric finite metrics. Metric inequivalence of spacetime geometries remains a conditional model claim throughout this paper.

5.3 Geometry from entanglement

In a holographic model the Ryu–Takayanagi formula [27] relates an extremal area to boundary entropy,

$$S_A = \frac{\mathrm{Area}(\gamma_A)}{4G_N\hbar}, \tag{37}$$

and motivates an entanglement account of connectivity [28]. Such a reconstruction requires its dictionary. Outside a specified model, the accessible data determine operator relations rather than a spacetime metric.

6 A Structural Distinction: Gauge Forces and Gravity

Gauge interactions evolve matter and connection fields within a supplied localization structure. The algebraic proposal asks whether gravity instead describes how physical correlations, accessible operations and their stability select that structure. With a geometric dictionary, changing those relations can change effective distances and causal structure. The preceding sources test a necessary interface: changed access reaches geometry through a specified matter process.

If the proposal holds, quantum gravity must describe selection of localization as well as fields within it. This is not an explanation of why quantization fails. Background independence [29, 30] asks for covariance under changes of selected geometric structure, not equality of the metrics assigned to operationally different algebras. Selection is constrained by interactions and stability, rather than subjective choice [20]; changed knowledge affects stress only through an operation.

A common reconstruction dictionary gives every matter representation in one selected geometric sector the same metric. This conditional universality is not a derivation of free fall or equality of inertial and gravitational mass. Degrees outside the sector cannot be declared geometrically irrelevant while changing its metric: a dark-sector application would need an enlarged source algebra and explicit backreaction map. No dark-matter or dark-energy conclusion follows here.

7 Technical Formulation

7.1 Setup

We separate the operator-algebraic and finite-dimensional regimes result by result. Modular statements use a Type III_1 ambient factor with a faithful normal state and explicit cyclic and separating hypotheses. Density matrices, traces, the quantum Fisher information metric and the qubit chain are finite-dimensional statements. The Rindler example is an algebraic quantum field theory check. The linearized Einstein equation is imported only in the holographic regime of Theorem 8.6, and the JT model of Section 2 is a supplied two-dimensional gravity theory.

Definition 7.1 (Selected geometric algebra)**.** *A* ***selected geometric algebra*** $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ *is a von Neumann subalgebra chosen as input to a specified algebra-to-geometry reconstruction model. It is not Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set. A coarse-graining model must declare the channel or inclusion relating any of those objects to* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$*.*

No canonical tensor factorization is assumed. State restrictions and relative-entropy data on $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ can be associated, as in holography and tensor networks, with an effective distance structure. That structure depends on the choice of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$, on the stability of correlations under coarse-grained dynamics and on the redundancy of the encoding; it is an effective description valid in the context the algebra defines.

7.2 Central conjecture

Conjecture 7.2 (Gravity as adiabatic algebra evolution)**.** *Gravitational dynamics corresponds to an* ***adiabatic flow*** *of selected geometric algebras* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t)$*:*

1. *the global state evolves unitarily,* $|\Psi_U(t)\rangle = U(t)|\Psi_U(0)\rangle$*;*
2. *a model-dependent rule selects a geometric candidate* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \subset \mathcal{M}_U$ *at each time;*
3. *as the state evolves, the selected candidate may shift,* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \to \mathcal{A}_{\mathrm{geo},\mathbf{c}}(t+dt)$*;*
4. *this shift manifests phenomenologically as the dynamical curvature of spacetime.*

Unitary evolution of observables within a fixed $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ *manifests as gauge interaction. Gauge dynamics is evolution within the algebra (fixed stage, moving actors); gravitational dynamics is evolution of the algebra itself (moving stage).*

The conjecture answers one objection: if algebras were kinematical background, gravity could not be dynamical. Here the selection of which algebra is stable is state dependent, and state evolution induces algebra flow. The selection rule that produces the path is not derived in this paper; Section 4 returns to what a physical selection would have to supply. The adiabatic assumption takes algebra transitions to be slow compared with dynamics inside the algebra. Rapid transitions would correspond to strong-field regimes where the effective geometric description breaks down, and the dynamics of non-adiabatic transitions is left open.

7.3 Comparison with quantum information geometry

For comparison consider the quantum Fisher information metric (QFIM) on a smooth finite-dimensional family of faithful effective states $\rho(\lambda)$ [36, 37],

$$g_{\mu\nu}(\lambda) = \frac{1}{2}\mathrm{Tr}\left(\rho(\lambda)\{L_\mu, L_\nu\}\right), \qquad \partial_\mu\rho = \frac{1}{2}(\rho L_\mu + L_\mu\rho). \tag{38}$$

It is coordinate independent, reduces to the classical Fisher metric in commuting limits, and measures how hard nearby states are to distinguish. It is also positive semidefinite, whereas a Lorentzian spacetime metric is indefinite and carries causal structure. Nothing in Eq. (38) supplies the negative direction, so the QFIM is a metric on model space and an analogy, not the spacetime metric. A change $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \to \mathcal{A}_{\mathrm{geo},\mathbf{c}} + \delta\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ may change this state-space metric; calling the change a spacetime perturbation requires an additional Lorentzian reconstruction map. In AdS/CFT, relative entropy and the entanglement first law constrain bulk perturbations [38, 39]; those results use the holographic dictionary and do not make an arbitrary boundary QFIM the bulk metric.

8 Algebraic Foundation: Modular-Stable Candidates

This section places one candidate class in Tomita–Takesaki modular theory. It does not translate measurement accessibility or environmental redundancy into modular conditions, and it does not select a unique algebra.

Standard modular definitions and their two checks are in Appendix A.

8.1 Modular definition of candidate algebras

Let $\mathcal{M}_U$ be a Type III_1 factor in standard form on $\mathcal{H}_U$, let $|\Psi_U\rangle$ be cyclic and separating, and write $\omega(\cdot) = \langle\Psi_U| \cdot |\Psi_U\rangle$. Then ω is faithful and normal and its ambient modular group $\sigma_t^{\omega,U}$ exists. Let $\{\mathcal{E}_t\}_{t\geq 0}$ be a normal unital completely positive (UCP) semigroup acting in the Heisenberg picture on $\mathcal{M}_U$.

Definition 8.1 (Modular-stable geometric candidate)**.** *A von Neumann subalgebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \subset \mathcal{M}_U$ is a **modular-stable geometric candidate** if:*

(U1) ***Faithful restriction.*** *$\omega|_{\mathcal{A}_{\mathrm{geo},\mathbf{c}}}$ is faithful and normal. In the faithful ambient setting this holds for every unital subalgebra, so it is a regularity condition, not a selector.*

(U2) ***Ambient modular stability.*** *$\sigma_t^{\omega,U}(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) = \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all real t.*

(U3) ***Semigroup stability.*** *$\mathcal{E}_t(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) \subseteq \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all $t \geq 0$.*

Remark 8.2 (Typed relation to the measurement objects)**.** *Conditions (U1)–(U3) do not derive Paper E's implementable effects* Eff*, their operator system* $\mathrm{Eff}_{\mathrm{span}}$ *or the pointer algebra $\mathcal{P}_{\mathrm{env}}$, and setwise semigroup stability is not the exact fixed set* $\mathrm{Fix}(\mathcal{E})$*. A downstream model may specify a channel or inclusion from those objects into $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$; without it the relation is motivational. Environmental redundancy does not imply faithfulness of a state restriction. A causal restriction map and a Lindblad evolution are different types, and each model must supply its own $\mathcal{E}_t$ and verify invariance.*

8.2 Selection is not settled by these conditions

Remark 8.3 (No uniqueness from (U1)–(U3))**.** *Conditions (U1)–(U3), even with only scalars fixed by both dynamics, do not determine a unique candidate. There are four independent obstructions.*

1. *Uniqueness of the modular flow for a fixed pair $(\mathcal{A}, \omega)$ does not select the algebra $\mathcal{A}$.*

2. *A normal completely positive map may preserve two subalgebras separately without preserving their join, because it is not multiplicative. Join preservation holds for modular automorphisms but not for a general $\mathcal{E}_t$; it does hold for subalgebras of the fixed algebra of a state-preserving dynamics, on which $\mathcal{E}_t$ acts as the identity [54].*

3. *Pointwise ergodicity constrains fixed elements, while (U2)–(U3) are setwise conditions; no implication to a unique maximal element of the lattice was proved.*

4. *With a faithful ambient state, $\mathcal{M}_U$ itself satisfies (U1)–(U3), so a bare maximality rule selects the ambient algebra unless an independent seed, properness, read-out or minimal-hull condition is supplied.*

Replacing $\mathcal{B}(\mathcal{H}_U)$ by a faithful Type III factor repairs the existence of the ambient modular flow but none of these selection failures. The class of candidates may contain several members that are not unitarily equivalent, and classifying it requires additional structure. A nontrivial selection principle could start from a foreground seed and construct a minimal invariant hull, or impose an explicit read-out constraint. Foundation I carries out the first route for a supplied seed: under state-preserving dynamics the hull of a seed in the fixed algebra lies between that seed and the fixed algebra. For a minimal compact symmetry, it equals the seed-stabilizer fixed algebra when that algebra is preserved by the hull generators and the hull contains the group-fixed algebra [54]. These hypotheses establish uniqueness relative to the seed; the seed's physical origin is a further question.

8.3 Connection to linearized gravity

Theorem 8.4 (Entanglement first law in a density-matrix setting)**.** *Let ρ_0 be a faithful finite-dimensional density matrix and $\rho_\varepsilon = \rho_0 + \varepsilon\,\delta\rho + O(\varepsilon^2)$ a normalized perturbation, with $K_0 = -\log\rho_0$. Then $\delta S = \delta\langle K_0\rangle$. An infinite-dimensional extension requires entropy differentiability and control of the modular-energy variation; faithfulness alone is insufficient. For a Type III local algebra the statement must instead use Araki relative entropy.*

Proof sketch. $S(\rho_\varepsilon\|\rho_0) = -S(\rho_\varepsilon) + \langle K_0\rangle_{\rho_\varepsilon}$ plus a constant is nonnegative and stationary at $\varepsilon = 0$. □

Conjecture 7.2 posits a family $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(\lambda)$. Modular theory treats two kinds of perturbation differently. For a state perturbation on a fixed algebra the Connes cocycle $u_t = (D\omega' : D\omega)_t$ relates the automorphism groups by $\sigma_t^{\omega'}(A) = u_t\sigma_t^{\omega}(A)u_t^*$ [50, 40]. This is not a left-multiplication formula for the two full modular operators. If the algebra family is generated by the positive translation of an HSMI, $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(a) = e^{iaP}\mathcal{A}_{\mathrm{geo},\mathbf{c}}(0)e^{-iaP}$,

and the vacuum is fixed by this translation, its full modular generator obeys $K(a) = e^{iaP}K(0)e^{-iaP}$. On a common smooth domain,

$$\left.\frac{dK(a)}{da}\right|_{a=0} = i[P, K(0)]. \tag{39}$$

Equation (39) concerns an HSMI-generated family and a derivative of matrix elements on that domain, not arbitrary subalgebra changes or an operator-norm Taylor expansion. The one-sided CHM charge used for a ball below is distinct from the full modular generator; their relevant local-unitary expectation differences agree, but the operators themselves do not.

Theorem 8.5 (Linearized gravity from entanglement, [39])**.** *In a holographic CFT dual to Einstein gravity in asymptotically AdS spacetime, the entanglement first law $\delta S = \delta\langle K_{\rm mod}\rangle$ for all boundary balls is equivalent to the linearized bulk Einstein equations $G^{(1)}_{\mu\nu} + \Lambda g^{(1)}_{\mu\nu} = 8\pi G_N T^{(1)}_{\mu\nu}$.*

The derivation uses the ball modular Hamiltonian [47]

$$K^B_{\rm mod} = 2\pi \int_B d^{d-1}x\, \frac{R^2 - |\mathbf{x} - \mathbf{x}_0|^2}{2R}\, T_{00}(x), \tag{40}$$

together with the Ryu–Takayanagi formula and the relation between bulk and boundary modular Hamiltonians [48]; complementary arguments are due to Jacobson [49].

Theorem 8.6 (Conditional holographic matching for an HSMI-generated family)**.** *Let $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ be a modular-stable candidate in a holographic CFT vacuum representation, and assume:*

(H1) the AdS/CFT correspondence holds;

(H2) the Ryu–Takayanagi formula and its quantum corrections hold;

(H3) the JLMS relation between bulk and boundary modular Hamiltonians holds;

(H4) the compared algebra family is an explicitly supplied HSMI/ball-region family whose generator P implements the boundary-region variation used in the holographic first-law argument;

(H5) the regulated entropy variation, or its relative-entropy replacement, satisfies the hypotheses of the Faulkner et al. theorem for all required balls.

Then Eq. (39) *describes the modular-generator variation within that family, and the holographic first law matches the linearized Einstein equations by the imported result:*

$$\begin{aligned} \textit{HSMI/ball family (H4)} &\Longrightarrow \delta K_{\rm mod} \xrightarrow{\textit{(H5)}} \delta S = \delta\langle K_{\rm mod}\rangle, \\ \textit{(H1)–(H3)} &\Longrightarrow \delta(G_{\mu\nu} + \Lambda g_{\mu\nu}) = 8\pi G_N T^{(1)}_{\mu\nu}. \end{aligned} \tag{41}$$

The theorem does not show that a generic stability-selected algebra variation is an HSMI translation, nor that HAFF algebra flow is gravitational dynamics.

8.4 The circularity issue

Jacobson's thermodynamic derivation [34] presupposes a background geometry, while HAFF wants geometry to emerge from algebra. Three strategies are discussed in the literature. (A) In the algebraic programme, wedges are defined through modular inclusions without a background metric, and Theorem A.5 recovers a positive translation from an HSMI; constructing a sufficient net and showing that the candidates of this paper form such inclusions remain open tasks. (B) A bootstrap starts from a seed geometry, derives linearized equations from the first law, updates the geometry and iterates. (C) In holography the boundary theory supplies the algebra and the bulk geometry is derived from boundary data. Theorem 8.6 uses strategy (C) and assumes the HSMI/ball interface. A geometry-free reconstruction from the present candidate class remains open.

9 Selection and response over growing windows

9.1 A conjecture on a sustainable selection–response loop

The unresolved question is whether one reused local body can combine selection, renewal, sharp exhaust and informative geometric feedback. Fix two physical preparations on a logical receiver A and its current purifier R:

$$|\Phi_Z\rangle = \sqrt{1-\epsilon}|00\rangle + \sqrt{\epsilon}|11\rangle, \qquad |\Phi_X\rangle = (H_A \otimes I_R)|\Phi_Z\rangle, \quad 0 < \epsilon < 1/2.$$

The logical labels denote Q-RAIF C's neutral pair codes, with their spectator sea retained. Here R is the paid, resettable current purifier in the third and fourth body species, available to joint neutral gates. Old references outside this source quartet are denoted R_{old} and are not acted upon. The quartet is a resettable working source alongside the series' protected unknown content H, not a replacement for it; access to the current R comes from those paid body species and joint neutral gates, not from an old-purifier permission. Each cycle's inaccessible environment applies the same pointer-aligned channel

$$\mathcal{D}_q(\rho) = (1-q)\rho + qZ_A\rho Z_A, \qquad 0 < q < 1/2, \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

No noise error label is available to the apparatus. Renewal returns the complete quartet, including the current purifier.

Proposition 9.1 (Pointer alignment and the complete renewal debit)**.** *For $\rho_p = \mathcal{D}_q(|\Phi_p\rangle\langle\Phi_p|)$, return to its specified pure quartet removes entropy $h_p = S(\rho_p)$, where*

$$h_Z = h\left(\frac{1-\sqrt{1-16q(1-q)\epsilon(1-\epsilon)}}{2}\right) > 0, \qquad h_X = h(q) > h_Z. \tag{42}$$

The receiver-only debits are 0 for Z and $h(\epsilon + q(1-2\epsilon)) - h(\epsilon)$ for X. Approximate quartet return subtracts its residual joint entropy from h_p, with finite-dimensional entropy continuity controlling the error. Program updates, sensors and usable controller registers enter the full maintenance debit in Proposition 4.1 as well.

Proof. The mixture of $|\Phi_p\rangle$ and $Z_A|\Phi_p\rangle$ has nonzero eigenvalues $(1 \pm d_p)/2$, with $d_p^2 = 1 - 4q(1-q)(1 - |\langle\Phi_p|Z_A|\Phi_p\rangle|^2)$. The overlaps are $1-2\epsilon$ and zero. Strict monotonicity of h on $(0, 1/2)$ proves the contrast. Tracing R gives the receiver debits. The reference marginal is unchanged by the noise, so the difference between each complete and marginal debit equals the receiver–purifier mutual information lost to the noise. □

Pointer alignment lowers the complete renewal price without making it zero. For $\epsilon = .1, q = .05$, $h_Z = .0877516288$ and $h_X = .1985152433$ nats per cycle. The marginal Z state is unchanged, but its purifier correlations are not. The loop must establish that its additional charged operations preserve an actual complete-debit contrast $\bar{s}_X > \bar{s}_Z$.

Conjecture 9.2 (Pointer-aligned selection with a sharp complete exhaust)**.** *Fix $\nu > 0$, $c_{\mathrm{ex}} > 0$ and thermal input at $T_H = 1/(2\pi)$ in the running rapidity clock, in natural units. For this dephasing family there are local models with one reused body during each N-cycle run, for arbitrarily large N, restoring both preparations to the declared error and realizing the following requirements jointly.*

Renewal acts on the current resettable A/R source, with joint control of both pairs explicitly permitted. Independently held unknown content H and its inaccessible old reference R_{old} remain preserved by the specified content-holding task throughout the loop. No reset of H or operation on R_{old} is granted by source renewal. Compatibility of that task with sustained renewal and the response controller is part of the conjecture.

The selected coupling supplies a complete chiral output carrying the charged heat and entropy. Every retained usable register, sensor and program update enters $s_{1:N}$, with $s_{1:N}/N \to \bar{s} > 0$ for the actual adaptive run. Initial usable purity and correlation capacity are $o(N)$; there is no positive rate of uncharged purity or nonequilibrium resource consumption. The excess above the complete-output floor

$$\limsup_{N\to\infty} \left[\frac{2\pi Q_{1:N} - s_{1:N}}{N} - \frac{6\nu\bar{s}^2}{c_{\mathrm{ex}}} \right] \tag{43}$$

can be made arbitrarily small along the family. This is sharpness for the whole adaptive output, including correlations and every atypical label.

The physical comparison has a declared block and decision clock. Let L_N be the sorting block length, $W_N \geq L_N$ the number of cycles used for one reading, and $d_N \geq 0$ the causal transport, sorting and sensing delay, all in cycle units. The program is held through each frame of $B_N = W_N + d_N$ cycles. Its outgoing observation interval $I_j = [\eta_j^{\mathrm{out}}, \eta_j^{\mathrm{out}} + W_N/\nu]$ is physically assigned to that program, and its record is available only by the frame's end, before the next decision. Require $K_N = \lfloor N/B_N \rfloor - 1 \to \infty$, so each tested decision and its following program are inside the run; when block sorting is used, $L_N \to \infty$. For the block converter, start at a sorting boundary and choose $W_N = m_N L_N$, $m_N \in \mathbb{N}$. Pad the output latency to $r_{\mathrm{out}} L_N$ and the subsequent readout latency to $r_{\mathrm{read}} L_N$, choosing $r_i = \max\{1, \lceil \delta_i/L_N \rceil\}$ for the physical latencies δ_i in cycle units, $i \in \{\mathrm{out}, \mathrm{read}\}$; these may depend on N. The window I_j contains m_N complete output blocks after the r_{out} blocks; the final r_{read} blocks allow its record to arrive before the next decision. Thus $d_N = (r_{\mathrm{out}} + r_{\mathrm{read}}) L_N$ aligns all frames and windows. Buffers, padded waiting, operation during delay and the final flush enter the complete entropy and energy account.

For fixed $Q_{\mathrm{ref}} > 0$ the normalized reading is

$$\overline{Y}_j = \frac{S_h(\eta_j^{\mathrm{out}} + W_N/\nu) - S_h(\eta_j^{\mathrm{out}})}{2\pi Q_{\mathrm{ref}} W_N}. \tag{44}$$

Whole-frame X and Z program interventions, made at the same pre-program state, history and geometric data and averaged over the declared actual-run pre-frame histories, give $\mathbb{E}(\overline{Y}_X - \overline{Y}_Z) \geq \delta_Y > 0$ uniformly as implementation and sharpness errors vanish. This is a per-cycle-normalized window contrast, not a one-cycle switch inside a pooled block.

The response changes the next program toward Z when the measured maintenance response is larger. To isolate this edge, let $P_j^{\mathrm{on/off}}(Z \mid \mathcal{F}_j)$ be the next-program probabilities with the physical response input present or blocked, respectively. The paired intervention keeps the upstream history and executed commands fixed up to that decision, including all non-response controller inputs. With $\mathcal{F}_j$ drawn from the actual run before this tested input, require

$$G_N = \frac{1}{K_N} \sum_{j=1}^{K_N} \mathbb{E}\big[P_j^{\mathrm{on}}(Z \mid \mathcal{F}_j) - P_j^{\mathrm{off}}(Z \mid \mathcal{F}_j)\big] \geq g_0 > 0. \tag{45}$$

The denominator includes every declared decision opportunity, not only selected informative or exploration windows. The response-channel replacement used in the test and its fixed reference are specified as part of the sensor. Its finite resolution, range, stress and entropy are charged. Neither a counterfactual fixed-program contrast nor a conditional gain on rare events substitutes for this operational test. For binary X/Z, the same opportunity/history average obeys $\langle P^{\mathrm{off}}(X)\rangle_N = \langle P^{\mathrm{on}}(X)\rangle_N + G_N \geq \langle P^{\mathrm{on}}(X)\rangle_N + g_0$. These are blocked-input probabilities on matched actual-run histories, not frequencies of a separately evolved off-history; if the on average tends to zero, the blocked average remains at least g_0. The actual adaptive output still satisfies Eq. (43).

Each model supplies a conserved complete stress tensor, the heat transport and a timelike worldline in a positive dilaton domain. Its actual clock has $0 < z_{\min} \leq d\tau/d\eta \leq z_{\max} < \infty$ uniformly along the family, with physical inlet temperature $T_\tau = 1/(2\pi z)$. Thermal preparation, reference stress and matching errors give a sublinear accumulated remainder in the horizon account. Drives are recovered, stored or routed away from the heat reading, or enter as explicit nonthermal sources in a matched background comparison. Their program-dependent residual remains in the contrast. Support work, clock preparation, thermal banks, buffer stages, ground offsets and final stored energy belong to the same ledger. Size, installed energy, control capacity and the dilaton domain may grow with N; each run keeps one body and a controller whose initial entropy and usable purity and correlation capacity are $o(N)$. Temperature in the rapidity frame, external exhaust capacity and cycle frequency stay fixed.

The conjecture keeps sustained renewal, sharp complete export and an informative causal return in one dynamics. At fixed $\nu, c_{\mathrm{ex}}, T_H, \bar{s}$, the quadratic excess over bare Landauer remains. Proposition 4.3 excludes direct purifier dumping from the sharp target. Arbitrary-unitary reservoir constructions approach Landauer's bound [13, Section 6], but do not keep this exhaust capacity fixed.

Proposition 9.3 (Finite-carrier sorting of the dephased source)**.** *Q-RAIF C's charge-sector sorter [14] applies to L independently realized copies of either ρ_p under its finite-carrier hypotheses: the source marginal is $\rho_p^{\otimes L}$, initially a product with the complete thermal carrier; their external purifications are retained; joint source operations act on both the receiver and its current R pair, while identity on old references refers to systems outside that quartet; the source is neutral in the original zero-mode quartet sector; the original neutral gates and paid loading, unloading and pipeline stages are available; and weighted energy tails control the implementation limit. It returns the source near its blank with entropy slack $o(L)$ and complete output Gibbs deficit $o(V)$, $V = L/\nu$, at*

$$T_{*,p} = T_H + \frac{6\nu h_p}{\pi c_{\mathrm{ex}}}, \qquad c_{\mathrm{ex}} = 4.$$

The same spectral argument applies to a supplied deterministic word with product marginal $\bigotimes_{j=1}^{L}\rho_{p_j}$ and actual entropy $s_L=\sum_j h_{p_j}$. The finite-bank target uses s_L plus the same subextensive margin. If $s_L/L\to s$, its temperature tends to $T_H+6\nu s/(\pi c_{\mathrm{ex}})$, with no convergence rate required; otherwise there need not be one limiting temperature. This is a block converter, not an adaptive moving apparatus or a per-cycle heat-reading law.

The spectral proof is in Appendix D.1. It uses a supplied product marginal and controlled energy tails, not a local reset certificate for an adaptive history.

This converter supports specified joint-source renewal, not correction of unknown content against an inaccessible purifier. For example, if R were instead forbidden to all operations, the noisy Z pair would have negativity $(1-2q)\sqrt{\epsilon(1-\epsilon)}$, below its target $\sqrt{\epsilon(1-\epsilon)}$. A channel on A and independent thermal ports cannot increase that negativity: partial transpose on R commutes with the channel, and trace norm contracts on Hermitian operators. It therefore cannot restore the entangled target. Preserving the distinct held content H/R_{old} alongside this converter requires the stated separate content-holding interface.

In a long fixed-program calibration window, suppose its full debit is $\bar{s}_p$ and converter and horizon remainders are sublinear. For $F(s)=s+6\nu s^2/c_{\mathrm{ex}}$ the matched mean reading at the floor is

$$\overline{Y}_p=\frac{F(\bar{s}_p)}{2\pi Q_{\mathrm{ref}}}+o(1),\qquad \Delta\overline{Y}=\frac{F(\bar{s}_X)-F(\bar{s}_Z)}{2\pi Q_{\mathrm{ref}}}+o(1). \tag{46}$$

For the quartet alone, $\nu=1, c_{\mathrm{ex}}=4$ give $F(h_Z)=.0993021514$, $F(h_X)=.2576276961$ and $\Delta\overline{Y}=.02519829/Q_{\mathrm{ref}}$. Here $\overline{Y}_p$ is the normalized window reading (44), under the stated fixed-program calibration assumptions, not a consequence of the adaptive sharpness clause. A single changed preparation within a pooled L-cycle block changes its mean debit by $O(L^{-1})$ and its normalized heat by $O(L^{-1})$; it does not produce this whole-window contrast.

Separate fixed-program sharpness does not prove the whole adaptive target. If time fractions a_p are occupied by long individually sharp phases with debits $\bar{s}_p$, their combined heat has the excess

$$\sum_p a_p F(\bar{s}_p)-F\left(\sum_p a_p\bar{s}_p\right)=\frac{6\nu}{c_{\mathrm{ex}}}\operatorname{Var}_a(\bar{s}_p). \tag{47}$$

This is positive for nonzero fractions and unequal debits. Joint block sorting can remove that energy gap at the channel level, but the present proof supplies no local program contrast or causal return after that sorting. Correlations with retained program history also enter the slack in Eq. (33). These are joint obstacles to Eq. (43). Sparse exploration with mean X-window share $a_N\to 0$ removes this phase-variance term but settles neither window gain. For independent noisy window tests with a fixed nonzero information per trial, vanishing estimation error requires the expected test count $K_N a_N\to\infty$. If window accuracy improves with N, the requirement concerns the accumulated physical detection information instead; counting windows alone is insufficient. Let b_N be the frequency of response-sensitive opportunities under the same opportunity/history measure as G_N. If on/off kernels coincide elsewhere, $|G_N|\le b_N$. When updates occur only in X frames, $b_N\le a_N$ with a_N evaluated over the same tested opportunities and actual-run histories. For equal-length windows, a_N is the expected X share of observed cycles, not the fraction of the full run covered by X observations. Hence $a_N\to 0$ violates Eq. (45) despite fixed

conditional contrast. Exploration and update duty are not identical in general: continuous monitoring during Z windows could keep b_N nonvanishing. Its compatibility with charged sensing, whole-output sharpness and a causal return is unproved, not excluded.

Under isotropic depolarization the two noisy quartet spectra and full debits instead coincide. Their separately entropy-optimal matched heat outputs have zero asymptotic program contrast; unsorted source contrast can become recovered work. This excludes that particular sharp-heat mechanism, not different full control debits or every causal realization using depolarization. A fixed initial record also obeys $I(X : Y_{1:N}|Z) \le H(X|Z)$ for charged baseline Z; sublinear initial record entropy cannot supply a positive new-information rate indefinitely.

The exact marginal-horizon balance gives $\overline{Y}_j = Q_{I_j}/(W_N Q_{\mathrm{ref}})$ when the entire matched source is charged heat. Drive, other and reference-source integrals add $[E_{\mathrm{drive},I_j} + E_{\mathrm{other},I_j} + E_{\mathrm{ref},I_j}]/(W_N Q_{\mathrm{ref}})$ in the same normalization. A common background cancels in program comparisons only with a justified common clock and transport; program-dependent residuals remain. Table 1 separates the available finite readouts from this joint inlet, renewal, sensor and outlet task. In particular, Q-RAIF C's massless sorter and massive-outside full-angle gate are different layouts [14].

The sorting-based route fails without source entropy concentration or controlled energy tails. Any joint realization must also avoid positive asymptotic output deficit or slack, loss of program contrast after drive and sensing charges, and failure of causal routing, temperature matching or the operating worldline.

Fresh, independently discarded quadrature drives have a separate accuracy cost [21]: at error ϵ_{drv} their input energy is at least $\omega_h[\pi/(4\sqrt{\epsilon_{\mathrm{drv}}})+O(1)]$, where $\omega_h^{-1} = \int |f|^2\omega^{-1}d\omega$. Reused drives retain correlations and correlated failures; recovery is not established by fresh-drive accuracy. At $\epsilon_{\mathrm{drv}} = .01$ the finite bound is $6.74238\omega_h$. Choose the source parameter $\epsilon = .01$ independently of ϵ_{drv}. With $q = .05$, $\nu = 1$ and $c_{\mathrm{ex}} = 4$, the complete quartet debits are $(h_Z, h_X) = (.01370657, .19851524)$. Their heat floors $F(h_p)/(2\pi)$ are $(.00222632, .04100272)$. With the additional benchmark choice $\omega_h = 1$, the drive/heat ratios are about $(3028, 164)$; unrecovered fractions $(.0003302, .0060813)$ already equal one corresponding heat unit. At the main example's $\epsilon = .1$, the Z floor is .01580443 and that ratio is about 427, while the X values are unchanged. These quartet-only benchmarks exclude additional charged controller debits and do not identify the drive with the quartet's full renewal device. Gate spectrum, duration and debit set the ratio; no universal $\omega_h \ge \nu$ or universal drive/heat multiplier follows. Outgoing coherent drive energy uses the nonthermal ledger in Eq. (34). Recoverable energy, stored energy and exported heat are separate entries; the same outgoing energy is never charged twice.

The exact moving-clock conversion is

$$z = \frac{Ly}{R}, \qquad y = \sqrt{R_b^2 + L^2\dot{\Phi}_b^2} - L\dot{\Phi}_b, \quad R_b^2 = \Phi_b^2 - R^2, \tag{48}$$

with proper-time dot. The tracking-clock conversion is derived in Appendix D.1, Eq. (94). Thus $z = L\rho$ is a stationary limit. The physical input needs $zT_\tau = T_H$, and support power is $(L^{-1} - K)\dot{\Phi}_b/\kappa$ in Eq. (88), including its actual sign and source. Prescribing the trajectory or transforming temperature constructs neither support nor inlet. When the monitored source carries heat plus a sublinear reference remainder, its source–clock sector has $R = R_0 + \kappa Q_{1:N} + o(N)$ at a positive bounded mean heat; an exterior operating domain must therefore reach $\Phi_b > R = \Omega(N)$. With bounded z, an isolated reserve paying all per-cycle losses also requires initial energy $\Omega(N)$; replenished charged work is an alternative.

The family must report its upper as well as lower size, installed-energy, clock-domain and compiler scalings. A stronger resource target would be $O(N)$ installed energy and dilaton range, with polynomial control capacity; this is an optional benchmark, not a hypothesis of Conjecture 9.2. Faster growth would miss that benchmark without refuting the stated conjecture. Fixed-gap and fixed-cutoff examples alone give no uniform growing-run certificate.

This loop would connect selection to its source and backreaction. Its finite pointer algebra still needs a map to the modular-stable geometric algebra, and Conjecture 7.2 still asks for the origin of the gravitational action.

9.2 The two routes and a common source principle

The finite return demonstrates that a law supplied to a model can feed back into the physical operations that select access. It does not explain that law. Modular reconstruction supplies a constraint on the law, but does not manufacture the apparatus. Their union requires a further mechanism: the same physical history must determine both the accessible structure and its complete source response.

Conjecture 9.4 (Common-action selection and geometric consistency)**.** *There is a local matter–controller theory in which readable histories form and maintain a nontrivial selected geometric structure, and whose causal effective action uses the same prepared fields, support and clock in its source and response. In a low-energy sector with a common cone, its selected ball data satisfy the geometric consistency condition of Consciousness II. Its first-order response is the corresponding Einstein response with one source normalization; its higher-order response includes the interaction, contact and clock work that produced the selection. Selection changes this response through physical couplings and sources, while a change of description at fixed matter history leaves the source unchanged. The sustainable family of Conjecture 9.2 is a finite operating sector of this theory, with a response-to-operation gain that survives the growing-window limit.*

The conjecture has tests beyond assigning gravity to a massive apparatus. Removing the history's access to a command must remove the record-mediated component of the source, while holding the commanded interaction fixed and merely discarding a log must preserve it. Source identification must survive switched physical references; logical noise records alone fail this test in Proposition B.1. The same calibrated source must then pass the all-ball consistency test with fixed initial or boundary data. A law that reproduces only the JT dilaton or one binary return has not passed the higher-dimensional test. At second order the source, geometry and changed preparation must be varied together; entropy Hessians do not become SLD or spacetime metrics by name.

Q-RAIF C supplies a local common action with dynamical support, memory and clock fields and a specified positive finite canonical ordering [14]. This permits a concrete attack on the missing source: freeze the texture only as a controlled limit, quantize the retained material and controls together, and compare the resulting local gates with the actual acquired-history Hamiltonian. Direct substitution fails already because dynamical support and clock introduce kinetic cross terms and forces absent from a fixed-profile gate. Moreover a spatial canonical cutoff preserves its positive form but does not by itself provide a covariant renormalized Ward identity. Classical on-shell conservation therefore cannot certify the complete quantum source. The coupled tolerance above closes the

trajectory/reader comparison in the retained mean-action branch; it cannot remove these two operator and continuum obstacles.

For the real-quartet carrier, Proposition F.9 supplies that Ward identity and the actual full-mode bulk state. Its projective and Weyl conversion is varied with the clock and director; adding a complete vacuum effective action to the retained canonical stress would double count the quantum response. This bulk construction does not yet transport the entire reset protocol through dynamical walls.

The finite receiver-renewal family of Q-RAIF C holds clock mean energy and norm fixed while its space, time and paid Gibbs bank grow. It is a real finite update result, with old references retained. It renews a receiver rather than the entire interacting material. In the 31-site full-program diagnostic, the receiver occupation .157712 exceeds the target .13 by .027712; this is not a full-program state error. The independently reported completion probability is .857524. An additional complete-ring construction renews the same body to fixed per-cycle tolerance through four thermal outlet species, retaining old correlations and restoring the original mixed receiver and its current purifier by local neutral pumps [14]. At the same incoming history, changing the maintained logical axis changes the stress emitted by that reset, with a strictly positive finite-error bound; no separate program-tagged emitter is added. Its prescribed drives are paid, and its energy-constrained storage stays bounded at fixed parameters. Its finite smooth schedule can also be driven by a work-clock field whose source includes the old incoming correlations [14, Section on a spatial work clock]. That result has static endpoints and a supplied frame. The bulk covariant construction and this fixed-frame schedule have not yet been joined with those endpoints and the sharp excess-heat limit required here. The finite-program diagnostic remains evidence against combining successful components without a common state and dynamics. The conjectured common theory would be refuted by an unavoidable extra extensive source or renewal price, by a persistent mismatch of its selected modular data, or by a response gain that vanishes under the stated fixed resources. These are different failure modes, not a general impossibility claim.

10 Relation to Existing Approaches

AdS/CFT [31] and its area–entropy relation [27] supply the dictionary used in Theorem 8.6; it is not derived here or granted to arbitrary selected algebras. MERA [32, 33] implements coarse-graining through disentanglers and isometries, with network depth resembling a radial AdS direction. Different networks can describe the same boundary data. Relating either construction to physical selection requires the explicit algebra-to-geometry map.

Jacobson's horizon Clausius argument [34] shares the entropy-based approach to gravitational dynamics. Section 2 addresses its source-side input: which selecting processes produce the stress whose response is computed. Loop quantum gravity [29, 35] instead quantizes the metric with background independence. Its relation to a path of selected algebras would require a representation map from spin-network states. These connections motivate the proposal without supplying its missing selection law.

11 Conclusion

Physical selection reaches gravity through the selecting interaction's source. Calibration changes activation rates; selected projections and learned protection change stress at fixed schedule. Logical noise records alone can miss that source. Prepared references or signed field readings remove specific ambiguities, with their resources charged.

Within matched JT data, a finite return retains the quantum reader, old reference, record spring, clock energy, mechanics, radiation and recoil in one mean action. Acquisition and feedback are separately necessary for its source contrast, and joint error bounds preserve the test under finite perturbations. The spatial construction includes support and synchronization, with a nonzero source–dilaton–sensor path and a subsequent complete material-force response. The latter also contains ordinary material and clock effects; it is not entirely gravitational. The full-mode quartet bulk supplies a covariant, actual-state source with clock and director work, rather than replacing the retained state by a vacuum.

Continued maintenance is a different result. The entropy of the complete retained system, consumed purity, correlations and conserved charges fixes the export debit. A thermal chiral channel of fixed capacity imposes its quadratic rate price. Matched horizon transport records that price; the actual thermal-source calculation also retains the moving clock's reference stress. Renewal accuracy alone does not establish sharp exhaust or indefinite operation.

The modular route constrains the law through a supplied geometric family and dictionary; the source route computes within an assigned law. The common-action and sustainable-loop conjectures ask for formed access, geometric consistency, complete causal sources and persistent return gain in one dynamics. Pointer alignment gives a complete-debit contrast; finite-carrier sorting can approach its channel floor. Individually sharp program phases still leave a positive variance price at nonzero duty fractions, so joint sharpness with an informative return remains open. Endpoint and gate matching, moving support, sustained thermal renewal and the origin of geometric selection remain construction tasks.

A Modular definitions and candidate checks

A.1 Preliminaries

Standard references are Takesaki [40], Bratteli–Robinson [41] and Haag [42].

Definition A.1. *A **von Neumann algebra** $\mathcal{M}$ on $\mathcal{H}$ is a $*$-subalgebra of $\mathcal{B}(\mathcal{H})$ containing the identity and closed in the weak operator topology; equivalently $\mathcal{M} = \mathcal{M}''$, where $\mathcal{M}'$ is the commutant. It is a **Type III$_1$ factor** if $\mathcal{M} \cap \mathcal{M}' = \mathbb{C}\mathbf{1}$, it admits no semifinite normal trace, and its Connes spectrum is $\mathbb{R}_{\geq 0}$.*

Under standard conditions, local algebras of quantum field theories are Type III_1 factors [42, 43].

A vector $|\Omega\rangle$ is **cyclic** for $\mathcal{M}$ if $\mathcal{M}|\Omega\rangle$ is dense, and **separating** if $M|\Omega\rangle = 0$ implies $M = 0$. For cyclic and separating $|\Omega\rangle$ the antilinear map $S_\Omega : M|\Omega\rangle \mapsto M^*|\Omega\rangle$ is closable, and its closure has the polar decomposition $S_\Omega = J_\Omega \Delta_\Omega^{1/2}$ with modular conjugation J_Ω and modular operator Δ_Ω.

Theorem A.2 (Tomita–Takesaki [40]). *With $\mathcal{M}$, $|\Omega\rangle$, J_Ω, Δ_Ω as above, $J_\Omega \mathcal{M} J_\Omega = \mathcal{M}'$, and the modular group $\sigma_t^\Omega(M) := \Delta_\Omega^{it} M \Delta_\Omega^{-it}$ satisfies $\sigma_t^\Omega(\mathcal{M}) = \mathcal{M}$ for all real t.*

A state ω is KMS at parameter β in the modular-time convention used here if for all $A, B \in \mathcal{M}$ there is a function $F_{A,B}$ analytic in $0 < \operatorname{Im} z < \beta$ and continuous on the closure, with $F_{A,B}(t) = \omega(\alpha_t(A)B)$ and $F_{A,B}(t + i\beta) = \omega(B\alpha_t(A))$. This ordering gives $\beta = 1$ for $\sigma_t^\omega(A) = \rho^{it} A \rho^{-it}$ in a density-matrix representation; the usual future physical Hamiltonian time has the opposite modular orientation.

Theorem A.3 (Takesaki [40]). *A faithful normal state ω on $\mathcal{M}$ is KMS at $\beta = 1$ for σ_t^ω, and σ_t^ω is the unique one-parameter automorphism group with this property.*

The modular flow is thus the unique time evolution for which the given state looks thermal. In the Rindler wedge it is the Lorentz boost, and the KMS condition at $\beta = 2\pi$ gives the Unruh temperature $T_U = a/(2\pi)$.

Definition A.4 ([44]). *An inclusion $\mathcal{N} \subset \mathcal{M}$ with common cyclic and separating vector $|\Omega\rangle$ is a **half-sided modular inclusion** (HSMI) if $\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N}$ for all $t \leq 0$.*

Theorem A.5 (Wiesbrock [44]). *For an HSMI in the stated $t \leq 0$ convention, the closure of $P = (\log \Delta_{\mathcal{N}} - \log \Delta_{\mathcal{M}})/(2\pi)$ is positive. The canonical unitary group is $U(a) = e^{iaP}$, fixes $|\Omega\rangle$, and obeys $\mathcal{N} = U(1)\mathcal{M}U(1)^*$ and $\Delta_{\mathcal{M}}^{it} U(a) \Delta_{\mathcal{M}}^{-it} = U(e^{-2\pi t} a)$ [15, Thm. 2.1].*

The translation from $\mathcal{M}$ to $\mathcal{N}$ is therefore encoded algebraically, without background geometry. This is the standard tool for extracting translation structure from modular data.

A.2 Example 1: Rindler wedge

Consider a free massless scalar field in $(1 + 3)$-dimensional Minkowski spacetime with Fock space $\mathcal{H}_U = \mathcal{F}(L^2(\mathbb{R}^3))$ and vacuum $|\Omega\rangle$. Let $\mathcal{W}_R = \{x > |t|\}$ be the right Rindler wedge and $\mathcal{R}(\mathcal{W}_R) = \{e^{i\phi(f)} : \operatorname{supp} f \subset \mathcal{W}_R\}''$. For this check we take $\mathcal{M}_U = \mathcal{R}(\mathcal{W}_R)$, which makes the ambient modular action well defined without performing a nontrivial selection.

The Reeh–Schlieder theorem [42] makes $|\Omega\rangle$ cyclic and separating for $\mathcal{R}(\mathcal{W}_R)$, so (U1) holds. By the Bisognano–Wichmann theorem [45, 46], $\Delta_\Omega = e^{-2\pi K}$ with K the global boost generator in the x-direction. The modular action is $\sigma_t^\Omega(X) = e^{-2\pi i t K} X e^{2\pi i t K}$; its spacetime action preserves the wedge, giving $\sigma_t^\Omega(\mathcal{R}(\mathcal{W}_R)) = \mathcal{R}(\mathcal{W}_R)$, which is (U2). No decoherence semigroup is specified, so (U3) is not verified. Haag duality, $\mathcal{R}(\mathcal{W}_R)' = \mathcal{R}(\mathcal{W}_L)$, does not imply that every proper extension of the wedge algebra contains the opposite wedge algebra, nor any maximality statement.

On the $t = 0$ Cauchy slice the full modular generator is

$$K_{\mathrm{mod}} = -\ln \Delta_\Omega = 2\pi K, \qquad K = \int_{\mathbb{R}^3} x\, T_{00}(0, x, y, z)\, d^3x. \tag{49}$$

Formally this is $2\pi(K_R - K_L)$, with $K_R = \int_{x>0} x T_{00}\, d^3x$ and $K_L = \int_{x<0} (-x) T_{00}\, d^3x$. The right-wedge expression $2\pi K_R$ alone is not $-\log \Delta_\Omega$: in a factorizing regulator it is the right reduced-state modular Hamiltonian, up to its normalization constant. In the Type III theory the separate one-sided charges require an appropriate quadratic-form or regulator interpretation. For a local unitary excitation in the right wedge, the

opposite-wedge expectation is unchanged, so the change of the full boost charge equals the right-sided change when those differences exist. This does not identify the two operators. The example reproduces the Bisognano–Wichmann geometric modular action; it does not show that (U1)–(U3) select the wedge algebra.

A.3 Example 2: Qubit chain with dephasing

Consider n qubits with Hamiltonian $\hat{H} = -J\sum_i \sigma_z^{(i)}\sigma_z^{(i+1)} - h\sum_i \sigma_x^{(i)}$, $J > 0$, coupled to an environment through

$$\mathcal{T}_t(\rho) = e^{t\mathcal{L}}(\rho), \qquad \mathcal{L}(\rho) = -i[\hat{H}, \rho] + \gamma \sum_i \left(\sigma_z^{(i)} \rho\, \sigma_z^{(i)} - \rho\right), \tag{50}$$

with Heisenberg dual $\mathcal{E}_t := \mathcal{T}_t^*$ and global state $\omega = Z^{-1}e^{-\beta\hat{H}}$. For an exact check we take $h = 0$; for $h \neq 0$ the diagonal algebra is stable only approximately in a strong-dephasing regime.

For $h = 0$ the diagonal observables are fixed by the Heisenberg semigroup, and the candidate is the maximal abelian algebra

$$\mathcal{A}_{\mathrm{geo},\mathbf{c}} = \{f(\sigma_z^{(1)}, \ldots, \sigma_z^{(n)})\}'' = \mathcal{A}_{\mathrm{diag}}. \tag{51}$$

The thermal state is positive definite at finite β, so $\omega(|s\rangle\langle s|) = \langle s|\rho_{\mathrm{th}}|s\rangle > 0$ for every computational basis state, which gives (U1); diagonal elements of a positive definite matrix are positive in any basis. The ambient modular flow is $\sigma_t^{\omega,U}(X) = \rho_{\mathrm{th}}^{it} X \rho_{\mathrm{th}}^{-it}$; for $h = 0$, ρ_{th} is diagonal and commutes with $\mathcal{A}_{\mathrm{diag}}$, giving (U2). For $h \neq 0$ the conjugation generally leaves $\mathcal{A}_{\mathrm{diag}}$. Under the dephasing dual, off-diagonal matrix units decay as $e^{-2\gamma\, d(s,s')\, t}\, |s\rangle\langle s'|$ with Hamming distance d; for $h = 0$ every diagonal observable is fixed, so (U3) holds and $\mathcal{A}_{\mathrm{diag}} \subset \mathrm{Fix}(\mathcal{E})$. This does not prove maximal setwise invariance, since the full matrix algebra is always mapped into itself.

The modular flow acts trivially on the abelian pointer algebra, and the entropy of a block A within $\mathcal{A}_{\mathrm{diag}}$ is the Shannon entropy of the marginal distribution $p(s_A)$. The example checks one modular- and semigroup-stable candidate, which under exact fixation is the pointer algebra; it neither verifies the Type III regime nor selects the candidate uniquely by (U1)–(U3). Selection of another type is available here: for $h = 0$ the fixed algebra is the commutant of the $\sigma_z^{(i)}$, and the window theorem of Paper B shows that every sufficiently long observation window selects it as the unique maximal stable algebra [20].

B Source identification, carriers and profiles

B.1 Continuing calibration and identification of the source

Source control need not interrupt stored quantum content. With an independently prepared vacuum mode and full output collection, Q-RAIF B's corrected parity instrument has branches

$$\mathcal{I}_\pm(\rho) = \tfrac{1}{2}(1 \pm e^{-2k(1-\alpha a)^2})\rho.$$

It yields the same calibration statistic without preparing a new body state. The coherent controlled-parity identity $Re^{-imZ_L Q_f}R = I_L \otimes e^{-imQ_f}$ goes further: it protects the body's marginal for an arbitrary mode state by transferring the interaction to the field alone [21].

For a centered independent input the coupling-added stress is still $m^2 F_f$. The meter can be returned coherently; retaining a calibration outcome is a separate read and memory operation. Partial collection loses coherence during the inquiry itself. Brain Anchor prices that loss, bounds continuing retuning under drift and gives an exact posterior criterion for when an extra inquiry pays for its source and apparatus price [57]. No geometric response or entropy export is needed for these preservation identities.

The inquiry and maintenance choices can also share one running body. Brain Anchor tracks both the interface posterior b_x and the coherence-weighted mass c_x in a finite hidden-state model. Equal b_x and even equal reduced body states can require different next commands, because past loss has different correlations with the interface. Q-RAIF B realizes collection by finite resonant contacts, including successive contacts with a finite packet. In the matched field realization, the source at the next encounter is still $m_j^2 F_f$, irrespective of the accessible correction, at the interaction output before collection. A collector can redistribute that energy, so this is not its full downstream stress. For a history-dependent policy the coefficient is computed from that policy's hidden-path weights, not from a new independent gain draw at every step. The four-encounter joint example gives $\sum_j \mathbb{E} m_j^2 = .588965$, against .640000 with permitted maintenance but no retained calibration reads. These are source accounts; the supplied objective's prices are not an entropy-export measurement. Using these sources in the JT response also requires their profiles and emission times and any collector contribution. The continuous collection route below specifies a free-field remainder, but does not supply the wall, thermal exhaust or entire sensing apparatus of the response model.

B.1.1 Continuous collection changes the outgoing profile

Q-RAIF B also gives an excitation-preserving, positive-frequency collection model [21]. A retained oscillator of frequency Ω couples to a continuum $\omega \in [\Omega - B, \Omega + B]$, with real even form factor $g(\omega - \Omega)$. A prescribed finite source drive prepares the matched packet. The one-particle propagator S then determines both the collected fraction and the outgoing wave. Its finite waveform certificate allows clock, arrival and switching errors; it is not a white-noise limit. The field interaction is band-smeared, so its spatial tails are part of the model. Parameters are fixed within each encounter.

Let $z_{\mathrm{out}}(\omega)$ denote the asymptotic free outgoing amplitude for a unit source drive, at the same null-time origin as the incoming packet, and let the body-conditioned amplitudes be $\pm m\beta z_{\mathrm{out}}$. With the current normalization of Eq. (3), put

$$\chi(u) = -i \int_{\Omega - B}^{\Omega + B} \sqrt{\frac{\omega}{4\pi}} z_{\mathrm{out}}(\omega) e^{-i\omega u} d\omega, \quad F_{\mathrm{out}}(u) = 4m^2\beta^2 [\mathrm{Re}\, \chi(u)]^2. \tag{52}$$

For independent vacuum preparation, this is the outgoing free-field stress for every logical input. Tracing the body gives a mixture of opposite coherent amplitudes, whose normal-ordered squared currents agree. Accessible parity correction does not change this marginal stress. For a capture pulse ending at T, the rotating-frame amplitude is $z_{\mathrm{out}}(\omega) = e^{i(\omega - \Omega)T} [P_{\mathrm{field}} S(0, w)](\omega)$, with the prescribed source carrier phase included. Removing the final free propagation phase fixes the common time origin; it must not be counted again when placing the source in the JT model. The input occupation is $k_{\mathrm{eff}} m^2 = \beta^2 m^2 \|w\|^2$. For real envelopes and an even band, the source, receiver and free remainder energies in units $\hbar = 1$ are, respectively,

$$\Omega k_{\mathrm{eff}} m^2, \qquad \Omega \xi k_{\mathrm{eff}} m^2, \qquad E_0^{\mathrm{out}} = \Omega(1 - \xi) k_{\mathrm{eff}} m^2. \tag{53}$$

Capture moves energy into the receiver. A later vacuum swap exports it to a separate exhaust. The net prescribed capture-switch work is zero for this symmetric packet family, while source preparation supplies the first energy in Eq. (53). The receiver, exhaust, controls and supports do not disappear from a gravitational account.

Frequency regularity matters for using this source in the JT moment formulas. A hard flat band generally gives $F_{\mathrm{out}}(u)$ tails of order u^{-2}, so finite energy alone does not ensure a finite second moment. Instead choose $g(\delta) = \sqrt{J_0}[1 - (\delta/B)^2]$ inside the band and zero outside. For a finite source envelope and finite collection time, $z_{\mathrm{out}}(\omega)$ is g times a smooth spectral function. Thus $\sqrt{\omega} z_{\mathrm{out}} \in H_0^1$ and Plancherel gives

$$E_2^{\mathrm{out}} = \int u^2 F_{\mathrm{out}}(u) du = m^2 \beta^2 \int \left| \partial_\omega (\sqrt{\omega} z_{\mathrm{out}}) \right|^2 d\omega < \infty. \tag{54}$$

E_1^{out} is finite by Cauchy–Schwarz. This specifies a legitimate free-field source for the response dictionary, rather than reusing the uncaptured packet unchanged. It does not determine the total apparatus stress or assert compact temporal support. The JT background, boundary access and matter normalization remain the declared inputs of the response model.

In the worked collection setting $\Omega = 20$, $B = 5$, $T = 16$ and $J_0 = 1/(2\pi)$, a fixed 8721-bit waveform table and stated clock tolerances certify $1 - \xi < 4.43 \times 10^{-7}$ and a relative preparation-strength uncertainty below .001. This is enough to keep the collection-induced loss below .001 in the preceding 4096-per-setting calibration example. The bound does not include parity-gate or detector errors. Repeated encounters require independent vacuum fields and exhausts, or a separately proved return/reset bound on one reused field. A fixed-capacity thermal cycle and the source–sensor return of Section 4 are not supplied by this finite-bank construction.

With several Gaussian field species, however, logical calibration alone may not identify the actual source. Let the species have the same switch profile and independent centered inputs, with quadrature covariance kW in their species coordinates. Put $G = (\mathbf{g}_1, \mathbf{g}_2)$ and $\mathbf{v}(a) = \mathbf{g}_1 - a\mathbf{g}_2$. The logical coherence and coupling-added stress are respectively

$$r_W(a) = e^{-2k\mathbf{v}(a)^T W \mathbf{v}(a)}, \qquad T_{\mathrm{add}}(u) = \|\mathbf{v}(a)\|^2 F_f(u). \tag{55}$$

The first measures $D_W = G^T W G$; the second needs $C = G^T G$. The incoming stress of the reference preparation is an additional source.

Proposition B.1 (Protection records do not always identify stress)**.** *Take* $W = \mathrm{diag}(1, 4)$ *and the two coupling families*

$$\mathbf{g}_1 = (1, 0), \quad \mathbf{g}_2 = (1, 1), \qquad \widetilde{\mathbf{g}}_1 = (0, 1/2), \quad \widetilde{\mathbf{g}}_2 = (2, 1/2).$$

Every command gives the same logical channel, with weighted residual $1 - 2a + 5a^2$*. Their stress coefficients are* $1 - 2a + 2a^2$ *and* $1/4 - a/2 + 17a^2/4$*. Thus no finite adaptive experiment using only these fresh-port logical channels distinguishes them. The noise-minimizing command is* $a = 1/5$ *for both, but their source-minimizing commands are* $1/2$ *and* $1/17$*.*

Proof. Insert each vector pair in Eq. (55). Equality of conditional channels at every command gives equality of every adaptive logical record law by induction, also with retained ancillas. Minimizing each positive quadratic gives the stated commands. □

At the noise optimum their added stress coefficients are 17/25 and 8/25. This is an ambiguity of the specified logical access, not an inability to observe stress through another instrument. It persists with the same incoming preparation and does not come from a changed background.

Proposition B.2 (Switched references identify the physical source)**.** *In the same two-species family, suppose the apparatus can independently prepare known centered Gaussian references with covariances kW_1, kW_2, where $W_1 = \mathrm{diag}(1,4)$ and $W_2 = \mathrm{diag}(4,1)$. At each reference measure the logical coherence means at $0, +h, -h$, $h > 0$. They determine the physical source Gram matrix exactly:*

$$D_{j,11} = -\frac{\log r_j(0)}{2k}, \qquad D_{j,12} = \frac{\log r_j(h) - \log r_j(-h)}{8kh},$$
$$D_{j,22} = -\frac{\log r_j(h) + \log r_j(-h) - 2\log r_j(0)}{4kh^2}, \qquad C = \frac{D_1 + D_2}{5}. \tag{56}$$

More generally, known reference weights with $I = \sum_j t_j W_j$ suffice, with $C = \sum_j t_j D_j$.

For independent ± 1 trials, N at each of the six settings, suppose all true means are at least $r_{\min} > 0$. Let $\eta = \sqrt{2\log(12/\zeta)/N} \le r_{\min}/2$, clip sample means to $[r_{\min}/2, 1]$ and use the same formulas. With probability at least $1-\zeta$, uniformly on $|a| \le A$,

$$|\widehat{V}(a) - \|\mathbf{v}(a)\|^2| \le \Delta_C := \frac{2\eta}{5kr_{\min}}\left(1 + \frac{A}{h} + \frac{2A^2}{h^2}\right). \tag{57}$$

Minimizing $\widehat{V}$ on any supplied finite command grid incurs actual excess source coefficient at most $2\Delta_C$ over its best point.

Proof. The log characteristic function is the quadratic form $-2kD_j$. Its three coefficients give Eq. (56); $W_1 + W_2 = 5I$ gives C and proves the general sufficient condition. The union Hoeffding bound covers all six means. Each log error is at most $2\eta/r_{\min}$, so the errors of C_{11}, C_{12}, C_{22} are at most $2\eta/(5kr_{\min})$, $\eta/(5khr_{\min})$ and $4\eta/(5kh^2 r_{\min})$. Insert them in $V = C_{11} - 2aC_{12} + a^2 C_{22}$ to prove the uniform bound. Two applications of that bound compare the chosen grid point with the best one. □

These are $6N$ new preparations, couplings and reads, not six free exact means. The reference covariances, their switches and their incoming energies have to be calibrated and charged. Separate logical test preparations are permitted here; applying these references to held content would also require the protection operations of Q-RAIF B and their collection account. Equation (56) identifies the coupling-added source under the declared profile and species architecture, not the full stress of an unknown apparatus. There is another route when the apparatus has an oriented field readout rather than only logical records.

Proposition B.3 (Signed field calibration with held content)**.** *For d independent input modes put $Q_s = b_s + b_s^\dagger$ and $P_s = i(b_s^\dagger - b_s)$. Assume centered independent Gaussian modes, initially independent of the body and its retained reference, a known common k, and $\mathrm{Var}(P_s)/(4k) \le \sigma_{\max}^2$. Controlled total parity conjugates $Z_L\sqrt{k}\sum_s v_s(a)Q_s$ to $I_L\sqrt{k}\sum_s v_s(a)Q_s$. After this coherent echo the signed readings $Y_s = -P_s/(2\sqrt{k})$ have mean $v_s(a)$ while the logical state remains unchanged. Two commands, $a = 0, h$ with $h \ne 0$, identify*

$$\mathbf{g}_1 = \mathbb{E}\mathbf{Y}(0), \qquad \mathbf{g}_2 = \frac{\mathbb{E}\mathbf{Y}(0) - \mathbb{E}\mathbf{Y}(h)}{h}.$$

This does not require the Gaussian variances as labels.

Suppose $|v_s(0)|, |v_s(h)| \le B$ and use a finite readout that clips Y_s to $[-Y, Y]$, $Y > B$, then rounds to the nearest point of a prescribed uniform grid of spacing $q > 0$ in that interval, adding the two endpoints if needed. This alphabet has at most $\lceil 2Y/q \rceil + 2$ symbols and absolute rounding error at most $q/2$. For N fresh queries at each command, define

$$\eta = Y\sqrt{\frac{2\log(4d/\zeta)}{N}} + \frac{q}{2} + 2\sigma_{\max}\varphi\left(\frac{Y-B}{\sigma_{\max}}\right), \qquad \varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}.$$

With probability at least $1-\zeta$, the estimated residual on $|a| \le A$ has vector error at most $e_v = \sqrt{d}\,\eta(1 + 2A/|h|)$. If its true norm is bounded by B_v, then

$$|\widehat{V}(a) - \|\mathbf{v}(a)\|^2| \le 2B_v e_v + e_v^2. \tag{58}$$

The readout uses a finite alphabet, with at most $\lceil \log_2(\lceil 2Y/q \rceil + 2) \rceil$ bits per species per query.

Proof. Total parity reverses every Q_s, so the coherent parity calculation of Q-RAIF B [21] gives a field-only displacement. The canonical relation $[Q_s, P_s] = 2i$ shifts P_s by $-2\sqrt{k}v_s(a)$ and leaves its variance unchanged. For a Gaussian with $|\mu| \le B$ and standard deviation at most $\sigma_{\max}$, the clipping bias is at most $2\sigma_{\max}\varphi((Y-B)/\sigma_{\max})$, by integrating the two tails. Hoeffding for the bounded clipped readings and a union over the $2d$ means give the first term of η; rounding adds at most $q/2$. Thus each coordinate error of $\mathbf{g}_1$ is at most η and of $\mathbf{g}_2$ at most $2\eta/|h|$. The Euclidean bound follows. Expanding $\|\mathbf{v} + \delta\mathbf{v}\|^2$ proves Eq. (58). □

The sign reference, species-resolved quadrature detector and complete mode control are its additional permissions. Two commands here mean $2N$ new field queries, each with d digitized readings; mode preparation, coupling, capture and memory remain charged. Unknown incoming means would confound the shift. Together with the matched JT equations either identified source predicts a response from actual records rather than from a supplied coupling label. Boundary sensing still checks the physical response itself. Neither route replaces the wall, clock and thermal-export construction in a closed loop.

B.2 The command carrier matters

How strongly the gravitational field records the command statistics depends on how the command is physically carried.

Proposition B.4 (Phase-carried commands)**.** *Keep the coupling switched on at every opportunity and carry the command in the drive phase: a correction uses the input $|\sqrt{N}\rangle$ and no correction uses $|i\sqrt{N}\rangle$. Then (i) the no-correction branch flips the bit with probability $1 - s_N$ and the correction branch with probability s_N; (ii) both inputs carry the same field energy $N\bar{\omega}$, and the branch sources are $4N[\mathrm{Re}\,\psi]^2 + F_g$ and $4N[\mathrm{Im}\,\psi]^2 + F_g$; (iii) for the worked packet both branch sources have the same E_0, E_1, E_2, so $\mathcal{C}_\infty$ does not depend on the command rate, while the dilaton coefficients $A(u)$, $B(u)$ at finite u do.*

Proof. With $Q = b + b^\dagger$, the bit channel multiplies coherences in the X basis by $\langle\beta|e^{-2i\lambda Q}|\beta\rangle = e^{-2\lambda^2}e^{-4i\lambda\,\mathrm{Re}\,\beta}$. For $\beta = i\sqrt{N}$ this factor is $e^{-2\lambda^2}$, real and positive, which gives (i). The stress of a coherent input β is $(\psi\beta + \psi^*\beta^*)^2$, which is $4N[\mathrm{Re}\,\psi]^2$ for

$\beta = \sqrt{N}$ and $4N[\operatorname{Im}\psi]^2$ for $\beta = i\sqrt{N}$, and the coupling adds F_g in both branches. The equal moments of the two quadrature profiles give (iii) through Eq. (14); the difference of A between branches is proportional to $\operatorname{Re}[a + i(u - u_0)]^{-2}$, which is nonzero for generic u. □

The hiding has a price. With phase-carried commands the uncorrected steps also err, so the world mean obeys $m \mapsto c(2s_N - 1)[(1 - h)m + h\nu]$ on the calibrated branch. At $N = 16$ and a known interface the stationary loss rises from 0.1896 with gated couplings to 0.2334, and the switching work per opportunity rises from $0.0039\bar{\omega}$ to $0.0386\bar{\omega}$. The phase must also be set on the incoming packet before the encounter, which for full-line packets has the same causal limitation as on-demand emission. At $N = 16$ and $\kappa = 10^{-4}$ the coefficient A at the packet centre differs between the two branch sources by 3.4×10^{-4}, and the difference decays as u^{-2}.

So the gravitational trace of a selection is not universal. With gated couplings it sits in the rate of an energy-carrying event; with phase-carried commands it sits in the shape of the source and disappears from the asymptotic invariant of this example. A general law connecting selection to gravitational response must therefore name the carrier of the selection.

B.3 A worked packet

Take the normalized mode $f(\omega) = 2a\sqrt{\omega}\, e^{-a\omega} e^{i\omega u_0}$, for which $\bar{\omega} = 1/a$ and

$$\psi(u) = -\frac{ia}{\sqrt{\pi}\,[a + i(u - u_0)]^2}. \tag{59}$$

The two quadrature profiles $4[\operatorname{Re}\psi]^2$ and $4[\operatorname{Im}\psi]^2$ have different shapes but the same normalized moments $(E_0, E_1, E_2) = (1, u_0, u_0^2 + a^2)/a$. They have power-law tails: $[\operatorname{Im}\psi]^2 \sim a^2/(\pi u^4)$ and $[\operatorname{Re}\psi]^2 = O(u^{-6})$. The absolute third moment of F_g diverges logarithmically; the first three moments are finite and are exactly those used in Eq. (14). The switch determined by $g' = 4\lambda \operatorname{Im}\psi$ and $g(\pm\infty) = 0$ is

$$g(u) = -\frac{4\lambda a(u - u_0)}{\sqrt{\pi}\,[a^2 + (u - u_0)^2]}. \tag{60}$$

Both g and F_g have full-line support. The formulas and table below are exact for this scattering idealization with asymptotic coefficient data. A command decided at a finite time cannot gate its entire switch; finite-time feedback uses the following compact approximation.

Proposition B.5 (Causal compact approximation)**.** *Choose a fixed even $\eta \in C_c^\infty(\mathbb{R})$ with $0 \le \eta \le 1$, $\eta = 1$ on $[-1, 1]$ and $\eta = 0$ outside $[-2, 2]$, and set $g_T(u) = \eta((u - u_0)/T)g(u)$. For $T \ge a$, write $F_T = (g_T')^2/4$. The unrescaled switching stresses obey*

$$\begin{aligned} \left| \int (F_T - F_g)du \right| &= O(\lambda^2 a^2 T^{-3}), \\ \int (u - u_0)(F_T - F_g)du &= 0, \\ \left| \int (u - u_0)^2 (F_T - F_g)du \right| &= O(\lambda^2 a^2 T^{-1}). \end{aligned} \tag{61}$$

The one-particle smearing vectors converge in norm, with difference $O(|\lambda|a/T)$. Hence the corresponding coherent-drive gate channels converge. A gate chosen before $u_0 - 2T$ can be applied causally. Its duration and any delay between the read and the gate are part of its task schedule.

Proof. With $w = u - u_0$ and $C = 4|\lambda|a/\sqrt{\pi}$, the tails satisfy $|g| \le C/|w|$ and $|g'| \le C/w^2$ for $|w| \ge T \ge a$. The derivative of the cutoff obeys the same w^{-2} bound on its annulus, up to a constant depending only on η. Integrating w^{-4} and w^{-2} gives the first and third bounds; parity gives the second. For $h = g_T - g$, $\|h\|_2 = O(|\lambda|aT^{-1/2})$ and $\|h'\|_2 = O(|\lambda|aT^{-3/2})$. Plancherel and Cauchy–Schwarz bound $\int_0^\infty \omega|\hat{h}|^2 d\omega/(4\pi)$ by $\|h\|_2\|h'\|_2/2$, giving the one-particle estimate. The coherent-state characteristic function is continuous in this norm. □

At finite T, the drive mode, its overlap with the incoming beam and the gate error must be recomputed from g_T before assigning values to a finite-cutoff table. If the drive is matched to the new mode, its quadrature profiles and first three stress moments converge as well. The real and imaginary profiles are Hilbert-transform pairs. For $r = (g_T - g)'$, the identity $u\mathcal{H}r = \mathcal{H}(ur) + \pi^{-1}\int r$ in the convention $\mathcal{H}r(u) = \pi^{-1}\,\mathrm{pv}\int r(v)/(u - v)dv$ has zero last term. The L^2 isometry of $\mathcal{H}$ therefore gives $\|u\mathcal{H}r\|_2 = \|ur\|_2 \to 0$. Together with convergence of the mode normalization, this proves the weighted profile convergence. Equation (6) holds exactly for each compact implementation with its actual gate parameters. The same discrete feedback law applies when the target is held between its read and gate; noise during a new delay must instead be included in the transition law.

With $A_{\mathrm{tot}} = N + p\lambda^2$ the asymptotic invariant is $\mathcal{C}_\infty = H_0^2 + 2\kappa H_0 u_0 A_{\mathrm{tot}}/a - \kappa^2 A_{\mathrm{tot}}^2$. Relative to the same beam without couplings,

$$\delta\mathcal{C}(p) = p\lambda^2\Big[\frac{2\kappa H_0 u_0}{a} - \kappa^2\big(2N + p\lambda^2\big)\Big], \tag{62}$$

which has an explicit quadratic term in p. Table 2 uses $a = 1$, $u_0 = 3$, $H_0 = 1$, $\kappa = 10^{-4}$, the task parameters $\nu = 0.8$, $c = 0.9$, $h = 0.4$ of Ref. [21], and the uncalibrated ($q = \frac{1}{2}$) and calibrated (32 passive blocks, $q = 0.0404888$) coupling rates.

N	gate error	p, $q = \frac{1}{2}$	p, q_{32}	$\delta\mathcal{C}$, $q = \frac{1}{2}$	$\delta\mathcal{C}$, q_{32}	beam share	quad./lin.
1	0.3544	0.15525	0.11413	5.746×10^{-5}	4.224×10^{-5}	7.0×10^{-2}	-1.2×10^{-6}
4	0.1327	0.15153	0.10698	1.402×10^{-5}	9.897×10^{-6}	4.1×10^{-3}	-2.7×10^{-7}
16	0.0371	0.15033	0.10468	3.476×10^{-6}	2.420×10^{-6}	2.5×10^{-4}	-6.7×10^{-8}
64	0.0095	0.15002	0.10408	8.657×10^{-7}	6.006×10^{-7}	1.6×10^{-5}	-1.7×10^{-8}

Table 2: JT invariant response to the coupling-added source of the worked packet. "Beam share" is $\delta\mathcal{C}(p)$ at q_{32} divided by the beam's own change $\mathcal{C}_\infty(N) - \mathcal{C}_\infty(0)$; "quad./lin." is the ratio of the p^2 term of Eq. (62) to the term linear in p.

At $N = 16$ the ratio of the two responses is 0.696342074, against the coupling-rate ratio 0.696342054. For $\kappa = 0.01$ and 0.05 ($\kappa N = 0.16$ and 0.8) the response ratios are 0.6963442 and 0.6963560; the quadratic term stays small because $p\lambda^2$ is small, not because κN is. The calibration-dependent part is a small fraction of the beam's own response, and that fraction varies roughly as N^{-2}: it is 7% at $N = 1$, where the gate fails 35% of the time, and 1.6×10^{-5} at $N = 64$. Its size is therefore a property of the chosen

drive and background, not a general feature of selection, and a field-energy ratio such as $p\lambda^2/N$ is not a general invariant response ratio. In this example the bracket of Eq. (62) is positive, so better calibration lowers $\delta\mathcal{C}$; for other backgrounds the bracket can vanish or change sign.

The mean-source solution is not the mean of branch solutions. For a single command $I \sim \text{Bernoulli}(p)$, $\mathbb{E}\,\mathcal{C}_\infty(N + \lambda^2 I) - \mathcal{C}_\infty(N + p\lambda^2) = -\kappa^2\lambda^4 p(1-p)$, which counts only the classical command randomness and not field fluctuations within a branch. With M independent copies of the whole experiment, each with its own hidden instrument, calibration records and drives, and with $G = G_1/M$, the classical source-fraction variance of smooth stress readings falls as $1/M$. A common hidden label shared by all copies does not give this independence, and a coherent drive still has nonzero energy variance. The exact mean-source solution does not show that a single quantum world has a definite classical geometry.

B.4 What remains input

The construction takes as given the JT action and its anti-de Sitter background, the matter field and its central-charge matching, the boundary data and the identification of the downstream null coordinate with the preparation coordinate, the semiclassical use of mean stress, and the actuator's field, switch, clock and fresh ports. The two-dimensional sector theorem permits either one or several cones; it neither derives this matter–gravity identification nor forbids a dynamical route to it [23]. The source is computed outside the apparatus, for the free outgoing field; the stress of the switch and its support is not included, and a covariant detector stress tensor shows that it cannot simply be dropped in the coupling region [59]. Within these inputs the response is exact.

C Finite boundary sensing and returned records

C.1 A conditional readback route

The matter action in Section B.4 does not directly couple to Φ. A bulk detector of the matter metric alone therefore does not acquire a local dilaton readout. Boundary motion offers a different route if a physical reflecting boundary with a known dilaton condition is available.

Proposition C.1 (A dilaton-blind record law)**.** *Compare two dilaton solutions with the same matter metric, the same accessible matter input states and the same initial controller state. Suppose every instrument and conditional evolution used by the controller depends on this metric and its earlier records, but not on the dilaton or a boundary that senses it. Then every finite adaptive record sequence has the same distribution in the two solutions.*

Proof. The first instrument has the same input and conditional outcome law. Given any earlier record sequence, the next instrument and input evolution are again the same. Induction gives equality of the joint record laws. Private random bits can be included in the controller state and the induction. □

This statement does not exclude inference from previously recorded sources. It excludes a new dilaton signal from an otherwise identical metric-only experiment. A reflecting boundary evades its hypothesis because its position depends on Φ.

Proposition C.2 (Reconstruction from three boundary echoes)**.** *In a source-free interval let a, b, γ be constant, $\mathcal{C} = a^2 + b\gamma > 0$, and suppose the apparatus can time reflections at the supplied boundary $\Phi = \Phi_b > \sqrt{\mathcal{C}}$. Use one calibrated null coordinate frame. Three reflection pairs (u_i, v_i), with distinct u_i and $d_i = (a + \Phi_b)u_i - \gamma \neq 0$, determine a, b, γ uniquely:*

$$\begin{pmatrix} 1 - u_1 v_1 & u_1 & v_1 \\ 1 - u_2 v_2 & u_2 & v_2 \\ 1 - u_3 v_3 & u_3 & v_3 \end{pmatrix} \begin{pmatrix} a \\ b \\ \gamma \end{pmatrix} = \Phi_b \begin{pmatrix} 1 + u_1 v_1 \\ 1 + u_2 v_2 \\ 1 + u_3 v_3 \end{pmatrix}. \tag{63}$$

Writing this matrix as M, its determinant is

$$\det M = \frac{2\Phi_b(\mathcal{C} - \Phi_b^2) \prod_{i<j}(u_j - u_i)}{\prod_i d_i} \neq 0.$$

If the u_i are exact and the measured returns are $v_i + \delta v_i$, an invertible measured matrix $\widetilde{M}$ yields the exact coefficient error

$$\delta(a, b, \gamma)^T = \widetilde{M}^{-1}(\delta v_i d_i)_{i=1}^3, \qquad \|\delta(a, b, \gamma)\|_2 \leq \|\widetilde{M}^{-1}\|_2 \|(\delta v_i d_i)_i\|_2. \tag{64}$$

Proof. The boundary is $v = V(u) = (a + bu - \Phi_b)/[(a + \Phi_b)u - \gamma]$, with $V' = (\Phi_b^2 - \mathcal{C})/[(a+\Phi_b)u-\gamma]^2 > 0$, so it is timelike. Its equation gives Eq. (63). Multiplying row i of M by d_i leaves evaluations of three quadratic polynomials; their coefficient determinant is $2\Phi_b(\mathcal{C} - \Phi_b^2)$, giving the displayed Vandermonde factor. In the noisy equation the difference between the measured right side and $\widetilde{M}(a, b, \gamma)^T$ is exactly $(\delta v_i d_i)_i$. □

Null signals can carry these timestamps back to a receiver before the next command. They reconstruct the coefficients, hence $\mathcal{C}$, in a declared source-free interval. The inverse becomes ill-conditioned near $\Phi_b = \sqrt{\mathcal{C}}$ or coincident sample times. Exact reconstruction is a test-probe limit, not three cost-free finite-energy measurements. A usable protocol must supply the reflecting boundary, its support, calibrated clocks, timing resolution, causal waiting time and probe stress; the last changes the background beyond that limit. Six timestamps stored with b bits each occupy $6b$ logical bits, not a determined physical energy or complete memory cost. This gives an explicit sensing interface to implement, rather than granting the controller direct access to Φ.

C.2 Probe backreaction and a returned program

A finite probe contributes stress, including the reflected component. The inverse can retain both chiral components instead of treating the probe as free. Work in the mean-source JT equations of Section B.4, with the same trace-anomaly subtraction. Suppose the complete additional bulk source in the sensing interval is calibrated as $T_{uu} = f(u)$, $T_{vv} = h(v)$, $T_{uv} = 0$. Define

$$\begin{aligned} A_f(u) &= -\kappa \int_{u_*}^{u} (u - z) f(z)\, dz, & B_f(u) &= -\kappa \int_{u_*}^{u} z(u - z) f(z)\, dz, \\ A_h(v) &= -\kappa \int_{v_*}^{v} (v - z) h(z)\, dz, & B_h(v) &= -\kappa \int_{v_*}^{v} z(v - z) h(z)\, dz, \\ K(u, v) &= A_f(u) + vB_f(u) + A_h(v) + uB_h(v). \end{aligned} \tag{65}$$

The matching convention fixes the lower limits and the homogeneous coefficient data. These are calibrated sources, not a derived reflection law for the wall. If reflection or

support stresses introduce other components or unknown source terms, the following inverse must be changed accordingly.

Proposition C.3 (Source-compensated echo inverse)**.** *For a known K as in Eq.* (65)*, the solution is*

$$\Phi(u,v) = \frac{a + bu + \gamma v - auv + K(u,v)}{1 + uv}. \tag{66}$$

Three points on the supplied boundary $\Phi = \Phi_b$ determine the coefficients $\boldsymbol{\theta} = (a, b, \gamma)^T$ whenever their matrix M in Eq. (63) *is invertible:*

$$M\boldsymbol{\theta} = \big(\Phi_b(1 + u_i v_i) - K(u_i, v_i)\big)_{i=1}^{3}. \tag{67}$$

The source-free determinant formula need not hold for these altered echoes; invertibility persists under sufficiently small perturbations of an invertible source-free matrix.

Suppose the measured u_i, v_i have errors at most δ_u, δ_v, the calibrated K has pointwise error at most η_K, and the measured matrix $\widetilde{M}$ is invertible. On a domain joining true and measured points, bound the derivatives of

$$R(u,v) = (\Phi_b - a) + (\Phi_b + a)uv - bu - \gamma v - K(u,v)$$

by $|R_u| \le L_u$ and $|R_v| \le L_v$. Then

$$e := \|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|_2 \le \sqrt{3}\|\widetilde{M}^{-1}\|_2(L_u\delta_u + L_v\delta_v + \eta_K), \tag{68}$$

$$|\widehat{\mathcal{C}} - \mathcal{C}| \le \sqrt{4a^2 + b^2 + \gamma^2}\, e + e^2, \qquad \widehat{\mathcal{C}} = \widehat{a}^2 + \widehat{b}\widehat{\gamma}. \tag{69}$$

If the next projection angle is $\vartheta_{\text{next}} = F(\widehat{\mathcal{C}})$ for a supplied Lipschitz program F, its error relative to $F(\mathcal{C})$ is at most $\mathrm{Lip}(F)$ *times Eq.* (69)*.*

Proof. The f contribution is the retarded solution already obtained in Section 2; interchange u, v for the h contribution. Their sum satisfies all three JT equations and gives Eq. (66). Evaluation on the boundary yields Eq. (67). The true residual $R(u_i, v_i)$ is zero. The residual at each measured point, including the calibration error, is bounded by $L_u\delta_u + L_v\delta_v + \eta_K$. Multiplication by $\widetilde{M}^{-1}$ gives Eq. (68). Expanding $\widehat{a}^2 + \widehat{b}\widehat{\gamma}$ leaves a linear term with gradient $(2a, \gamma, b)$ and the quadratic term $(\delta a)^2 + \delta b\delta\gamma$, whose magnitude is at most e^2. The program bound follows from its Lipschitz property. □

An apparatus needs a certified parameter range to bound L_u, L_v and the conditioning of the inverse before use. Neither the assumed calibration of K nor this error bound supplies a timing POVM or the wall's support energy.

Here is a causal one-step return in that declared geometry. Take the selected projection of Section 2.4, a fresh vacuum field and the independent bit state $(I + 0.8Z)/2$. Use a smooth compact switch

$$g(u) = A\exp\left[-\frac{1}{1 - ((u-3)/0.25)^2}\right] \quad (|u-3| < 0.25), \qquad g(u) = 0 \quad \text{otherwise,} \tag{70}$$

with A fixed by $\int (g')^2/4\, du = 0.25$. Unlike the full-line packet of Section 2, this source finishes before the readout begins. Set $H_0 = L = 1$, $\kappa = 0.05$ and $\Phi_b = 3$. The source moments are $E_0 = 0.25$, $E_1 = 0.75$, $E_2 = 2.257177876$; a coupled population r gives

$$(a, b, \gamma) = (1 + \kappa r E_1, -\kappa r E_0, \kappa r E_2).$$

The projections P_0 and $P_{\pi/2}$ give $r = 0.9$ and 0.5 respectively, and hence $\mathcal{C} = 1.067496366$ and 1.037498878.

Supply a timelike receiver at $v = -\alpha/u$, $\alpha = 1/4$. Its proper clock is $\tau = 2L\sqrt{\alpha}\ln u/(1-\alpha)$, up to an additive constant. Emit at receiver coordinates $t_i = 4, 6, 9$, with $v_i = -\alpha/t_i$. The outward ray has fixed v_i and meets the boundary at

$$u_i = \frac{a - \Phi_b + \gamma v_i}{(a + \Phi_b)v_i - b} > t_i.$$

The reflected ray has fixed u_i; reception at the receiver therefore occurs at the same coordinate u_i, with $v = -\alpha/u_i > v_i$. For $r = 0.9$ the receive coordinates are $8.18983, 12.56501, 19.53471$; for $r = 0.5$ they are $8.10371, 12.30538, 18.81553$. Throughout these intervals the receiver obeys $\sqrt{\mathcal{C}} < \Phi < \Phi_b$ and $1 + uv > 0$, inside the usual cutoff and outside the stationary horizon. A new compact coupling can start after $u = 20$. Writing $\mathcal{C}_{\mathrm{mid}}$ for the midpoint of the two invariants, choose

$$F(\mathcal{C}) = \frac{\pi}{4} + 10(\mathcal{C} - \mathcal{C}_{\mathrm{mid}}). \tag{71}$$

The two returned angles are 0.935385602 and 0.635410725. If the inverse error stays below the corresponding separation margin, the measurement changes the next coupled projection and its represented algebra $\{P_{F(\mathcal{C})}\}'$. This is a response-dependent return map, not a proof that this program is optimal or that its interface formed through learning. The controller uses the measured invariant; computing one from assumed initial data is a different protocol.

The listed times use the source-free geometry. With finite probes, the return must instead use Eq. (67) and the altered boundary. As a bulk check, two calibrated positive impulsive fluxes of integrals 10^{-5} in u and 2×10^{-5} in v give a coefficient bias of about 5.7×10^{-6} if ignored; source compensation removes it to numerical precision. This is not a physical reflection or timing experiment. For a translation-only readout with the listed proper-delay difference $\Delta\tau = 0.05001338$, the sharp signed-delay bound in Q-RAIF B requires at least $11.04\hbar$ in this clock's energy units for binary error 0.1 [21]. A real echo can also carry redshift and amplitude information, so that restriction is not a universal bound on geometric sensing. It shows why one cannot assign both negligible probe stress and a finite resolution without constructing the readout. The boundary, receiver, timing instrument, full probe-induced stress and support costs remain explicit apparatus inputs to this conditional return; mean-source evolution does not bound stress fluctuations.

C.3 A quantum boundary record without echo timestamps

A current detector gives a second readback route. It measures fluctuations over a prescribed interval of proper time, rather than the arrival time of a reflected signal. The boundary condition still does essential work: the detector is carried by the supplied timelike curve $\Phi = \Phi_b$. A detector on the same fixed bulk worldline in both solutions would remain subject to Proposition C.1.

Use a transparent chiral current interface, with the vacuum defined in u and the current normalization of Section 2. Its input on the interval to be sensed has zero mean and covariance $\langle j(u)j(u')\rangle = -[4\pi(u-u'-i0)^2]^{-1}$. The apparatus supplies a proper-time clock, a degenerate detector bit initially in $|0\rangle$, and a real compact switch $\zeta(\tau)$. The finite interaction is

$$U_s = e^{-iX_sQ_g}, \qquad Q_g = \int g(u)j(u)\,du, \qquad g(u(\tau)) = \zeta(\tau). \tag{72}$$

Equivalently its integrated interaction is $\int \zeta(\tau) X_s \dot{u}(\tau) j(u(\tau))\, d\tau$. The fixed detector operator and c-number current commutator make the time-ordering correction a scalar phase. This is the same gapless-detector method used for exact field channels [8]; the finite switch defines a smeared current, not an unsmeared instantaneous Hamiltonian. We read Z_s after contact and retain the bit $R = 0, 1$, with $R = 1$ the flipped outcome. Clock calibration, preparation and recording permissions are supplied resources.

Proposition C.4 (Invariant-dependent finite detector law)**.** *On a source-free background with $0 < \mathcal{C} < \Phi_b^2$, a future boundary segment has*

$$\dot{u} = \frac{bu^2 + 2au - \gamma}{2L\sqrt{\Phi_b^2 - \mathcal{C}}} > 0, \qquad \lambda_b = \frac{\sqrt{\mathcal{C}}}{L\sqrt{\Phi_b^2 - \mathcal{C}}}. \tag{73}$$

In the test-background calculation, Eq. (72) *has the record law*

$$k_b(\mathcal{C}) = \frac{1}{4\pi} \int_0^\infty \omega \coth \frac{\pi\omega}{\lambda_b} |\widehat{\zeta}(\omega)|^2 \, d\omega, \qquad \widehat{\zeta}(\omega) = \int e^{i\omega\tau} \zeta(\tau)\, d\tau, \tag{74}$$

$$\Pr(R = 1 \mid \mathcal{C}) = \frac{1 - e^{-2k_b(\mathcal{C})}}{2}. \tag{75}$$

For every nonzero switch with finite variance, the probability is strictly increasing in $\mathcal{C}$. It depends on the invariant, not on the three coefficients separately. For two equally likely backgrounds the optimal decision from this one bit has error $[1 - |p_1 - p_0|]/2$, where $p_i = \Pr(R = 1 \mid \mathcal{C}_i)$.

Proof. Pulling back the metric to the level curve gives $d\tau/du = 2L\sqrt{\Phi_b^2 - \mathcal{C}}/(bu^2 + 2au - \gamma)$. The Riccati equation in Eq. (73) makes $u(\tau)$ a fractional-linear transform of $e^{\lambda_b \tau}$ on each regular segment. Möbius invariance of the current two-point function therefore gives

$$\langle \dot{u}(\tau) j(u(\tau)) \dot{u}(\tau') j(u(\tau')) \rangle = -\frac{\lambda_b^2}{16\pi \sinh^2[\lambda_b(\tau - \tau' - i0)/2]}.$$

Its symmetrized spectrum yields Eq. (74); the pullback vacuum is KMS at temperature $\lambda_b/(2\pi)$ in this proper clock. For a centred Gaussian field, $\langle e^{2iQ_g} \rangle = e^{-2\langle Q_g^2 \rangle}$, hence the detector's Z_s expectation is e^{-2k_b}. The factors λ_b and $\coth(\pi\omega/\lambda_b)$ strictly increase with $\mathcal{C}$ for $\omega > 0$. A nonzero compact switch has Fourier weight on a set of positive measure, so the integral strictly increases. The binary decision formula is the total-variation formula for two Bernoulli laws. □

This response uses the familiar Hawking–Unruh covariance. It does not add a direct matter–dilaton coupling, and the KMS temperature here is the boundary proper-time temperature, not an automatically thermal exhaust in the running rapidity clock. Without calibration of L, Φ_b and the proper-time scale, these records identify λ_b rather than $\mathcal{C}$. A clock-scale error can mimic an invariant change; this instrument alone cannot distinguish them.

C.4 The sensing interaction changes the background

The same detector interaction adds the chiral mean stress $f(u) = [g'(u)]^2/4$. To include it, choose a boundary-source completion: this current is transmitted into the JT region, the coupling-added bulk source is precisely $T_{uu} = f(u)$, and the other state-dependent chiral

source is zero. Keep the trace-anomaly subtraction of Section B.4. This is a specified boundary injection model, not the reflected-probe model of Section C.1. No unknown reflected component is fitted as a known function. Forces supporting the boundary, an ultraviolet realization of the contact and the clock's stress require a further apparatus model.

Proposition C.5 (Self-consistent sensing source in mean JT geometry)**.** *In this completion, put* $R_b = \sqrt{\Phi_b^2 - \mathcal{C}}$ *and let dots denote proper-time derivatives. Starting with the matched coefficient data, solve*

$$\dot{u} = \frac{bu^2 + 2au - \gamma}{2LR_b}, \qquad f(u(\tau)) = \frac{\dot{\zeta}^2}{4\dot{u}^2},$$
$$(\dot{a}, \dot{b}, \dot{\gamma}) = \frac{\kappa\dot{\zeta}^2}{4\dot{u}}(u, -1, u^2). \tag{76}$$

On a regular finite segment this gives the boundary trajectory and the complete coupling-added bulk source together. Its exact balances are

$$\dot{\mathcal{C}} = \frac{\kappa L}{2} R_b \dot{\zeta}^2, \qquad R_b(\tau) = R_b(0) - \frac{\kappa L}{4}\int_0^\tau \dot{\zeta}(s)^2\, ds, \tag{77}$$

$$E_s^{(\tau)} := \int f\dot{u}^2\, d\tau = \frac{1}{4}\int \dot{\zeta}^2\, d\tau = \frac{R_b(0) - R_b(\tau_f)}{\kappa L}, \quad E_s^{(u)} := \int f\, du = \frac{1}{4}\int \frac{\dot{\zeta}^2}{\dot{u}}\, d\tau. \tag{78}$$

For the actual, generally nonstationary map $u(\tau)$*, the detector variance is*

$$k[g] = \frac{1}{4\pi}\int_0^\infty \omega|\widehat{g}(\omega)|^2 d\omega = \frac{1}{8\pi}\int_{\mathbb{R}^2} \frac{[g(u) - g(v)]^2}{(u-v)^2}\, du\, dv. \tag{79}$$

Its record probability remains $(1 - e^{-2k[g]})/2$*; replacing this variance by the stationary KMS integral during contact would neglect backreaction.*

Proof. The retarded JT solution has running coefficients $a' = \kappa u f$, $b' = -\kappa f$ and $\gamma' = \kappa u^2 f$, with primes in u. In the derivative of the boundary level equation the source contributions cancel: its numerator derivative is b and its denominator derivative is $a+\Phi_b$. Thus its slope is still $(\Phi_b^2 - \mathcal{C})/[(a+\Phi_b)u - \gamma]^2$ with running coefficients. The induced metric gives the first equation in Eq. (76), while $g' = \dot{\zeta}/\dot{u}$ gives the second. Differentiating $a^2 + b\gamma$ and substituting $\dot{u}$ gives Eq. (77). The energy balances follow by change of variable. The source f follows from $U_s^\dagger j U_s = j + X_s g'/2$ and the initially independent, centred detector with $\langle X_s\rangle = 0$. The incoming current on the deterministic mean trajectory is Gaussian; its vacuum variance has the Fourier expression in Eq. (79). Parseval's identity and $\int(1 - \cos\omega x)/x^2\, dx = \pi|\omega|$ give the second. □

In this model the mean source fixes a classical trajectory before the detector outcome is conditioned on. It is not a theory of outcome-conditioned quantum geometry. The inequality $\kappa L\int \dot{\zeta}^2/4 < R_b(0)$ prevents closure of the boundary gap; one must also check $\dot{u} > 0$ and $1 + uv > 0$ over the chosen window. The two energies in Eq. (78) refer to different clocks. Neither is the total preparation, support, readout or erasure cost. For any sequence obeying this positive-source completion at unchanged Φ_b, L, κ, the cumulative proper-clock source charge satisfies

$$\sum_n E_{s,n}^{(\tau)} < \frac{\sqrt{\Phi_b^2 - \mathcal{C}_{\text{initial}}}}{\kappa L}. \tag{80}$$

This follows directly from $R_b > 0$ and Eq. (77). A nonzero charge per contact therefore rules out infinitely many such contacts at a fixed cutoff. It is an upper budget for this positive-source boundary family, not a universal lower bound on sensing energy or a prohibition of protocols with a different net source.

Proposition C.6 (Finite sensing survives its own added source)**.** *Take two source-free initial backgrounds with different invariants, with the same Φ_b, L and a common proper-time contact interval on which both unperturbed curves are regular. Fix a nonzero C^2 compact switch ζ_0. At fixed κ, the completion in Proposition C.5, with $\zeta = \epsilon\zeta_0$, has distinct record laws for all sufficiently small $\epsilon > 0$. The added energies and invariant changes are finite and of order ϵ^2.*

Proof. On the compact regular segment the trajectory equations are locally Lipschitz, and their source-dependent terms are $O(\kappa\epsilon^2)$. Continuous dependence preserves the domain and gives $u_\epsilon - u_0 = O(\kappa\epsilon^2)$ in C^1. The corresponding compact smears obey $\|g_\epsilon - \epsilon g_0\|_0 = O(\kappa\epsilon^3)$: the inverse clocks and their first derivatives have the same perturbative order, and the H^1 bound controls the vacuum $H^{1/2}$ seminorm. Thus $k[g_\epsilon] = \epsilon^2 k[g_0] + O(\kappa\epsilon^4)$ and $p_\epsilon = \epsilon^2 k[g_0] + O((1+\kappa)\epsilon^4)$. The leading variances differ strictly by Proposition C.4. Their record difference cannot vanish for all sufficiently small nonzero ϵ. Equation (78) gives the energy order. □

C.5 A finite record returns to a physical selection

The two compact-source branches of Eq. (70) can be sensed by the same contact, without giving their coefficient values to the program. Start the boundary clock at $u = 4$, keep $\kappa = 0.05$, and use the C^2 compact profile

$$\zeta(\tau) = \frac{4}{3}\left[1 - \left(\frac{\tau - 4}{4}\right)^2\right]^3 \quad (0 \le \tau \le 8), \qquad \zeta = 0 \quad \text{otherwise.} \tag{81}$$

It has finite variance and energy; smooth approximants give the same limiting quantities. Solving Eq. (76) and integrating Eq. (79) gives

Original population r	Initial $\mathcal{C}$	Final $\mathcal{C}$	$\Pr(R = 1)$	Final u
0.5	1.037498878	1.1206720	0.31748037	57.481
0.9	1.067496366	1.1505122	0.31771688	52.458

The contact charge is exactly $E_s^{(\tau)} = 512(4/3)^2/(385\cdot 8) = 0.29552670$. The corresponding null energies are 0.06577751 and 0.06774638. These are deterministic numerical evaluations with converged quadratures, not certified intervals. The single-bit total variation is about 2.3650×10^{-4}, so one contact is a weak discriminator, with equal-prior error about 0.49988175. It is a nonzero physical return, not a high-confidence reconstruction or an optimal sensing strategy. Many repeats would need a new account of field correlations, preparation and accumulated source; they cannot be declared independent by multiplying this record law. Even ignoring all other operations, Eq. (80) allows at most 190 contacts with this unchanged charge on either initial branch. This is a necessary budget, not a 190-contact construction.

Retain R and let it select the next coupled projection: $P_0 = (I + Z)/2$ if $R = 0$ and $P_{\pi/2} = (I + X)/2$ if $R = 1$. Use the independent bit state $(I + 0.8Z)/2$ and a later

compact switch ζ_{next}, scheduled by the same proper clock after $\tau = 8$. Its realised null smear is $g_{\mathrm{next}}(u(\tau)) = \zeta_{\mathrm{next}}(\tau)$. The program supplies that proper-time profile and selects its projection using only R; it need not know $u(\tau)$ or $\mathcal{C}$. The mean population coupled to the next field is

$$r_{\mathrm{next}} = 0.9 - 0.4\Pr(R = 1), \qquad T^{\mathrm{added}}_{uu,\mathrm{next}} = r_{\mathrm{next}}\frac{(g'_{\mathrm{next}})^2}{4}. \tag{82}$$

It is 0.77300785 and 0.77291325 in the two branches. Thus the earlier selection changes a source, that source changes the boundary record law, and the recorded bit changes both the next pointer algebra and the next mean source. The controller uses no numerical dilaton input. For example, use the profile in Eq. (81) rescaled to amplitude $1/4$ and the proper-clock interval $9 \le \tau \le 11$. Its added proper-clock source charge is $r_{\mathrm{next}}\, 512(1/4)^2/(385 \cdot 2)$. The source kernel in u is the one on each branch's actual mean trajectory; the population difference alone is not a comparison of two complete fixed-shape stresses. The same trajectory equations apply with $\dot\zeta^2$ replaced by $r_{\mathrm{next}}\dot\zeta^2_{\mathrm{next}}$, continuing the causal mean solution beyond readout.

There is a same-current subtlety in this return. On the union of the sensor and later null intervals, take the input to have the vacuum restriction. A preceding compact Weyl source in a disjoint earlier interval leaves this restriction unchanged, since the relevant smeared currents commute. Conditioning on the sensor bit applies $\cos Q_g$ or $\sin Q_g$ to that restriction. Both resulting states are invariant under simultaneous current parity, so every later current has zero conditional mean. The next bit is independently prepared, and the linear stress cross term vanishes in each record branch, proving Eq. (82) without resetting the field. Its conditional covariance generally changes. The later field is not a fresh vacuum port, and a vacuum or thermal gate-error formula cannot be carried over to it.

For a realised switch $\widetilde{g}$, define the vacuum seminorm $\|g\|_0 = \sqrt{k[g]}$. If $\|\widetilde{g} - g\|_0 \le e$, then

$$|\widetilde{p} - p| \le |k[\widetilde{g}] - k[g]| \le 2\sqrt{k[g]}e + e^2.$$

This prices the needed control of the actual smeared interaction; clock and worldline errors must be included in $\widetilde{g}$, not only in its amplitude. Two record laws remain distinct if the sum of their probability-error bounds is below their stated separation. Section C.6 adds the independent source requirement, and Section C.8 gives a finite thermal replacement. The clock, contact and continuous exhaust with their complete stress remain the joint apparatus task of Conjecture 9.2. The finite run consumes a prepared detector bit and an independently prepared next bit. Resetting the record without retained side information must export its binary entropy; it is not supplied by coherent field return.

C.6 A joint certificate for records and gravitational sources

A detector can be accurate as an instrument and inaccurate as a source. The vacuum seminorm controls a half derivative of its switch, whereas the injected energy controls a full derivative. This distinction is already visible in a single contact, before questions of thermal reuse arise.

Proposition C.7 (Record accuracy does not control source energy)**.** *Let* $g_n(u) =$

$n^{-3/4}e^{-u^2/2}\cos(nu)$, $n \geq 1$, *and* $E[g] = \int (g')^2 du/4$. *Then*

$$k[g_n] = \frac{n^{-3/2}}{8}[n\sqrt{\pi}\ \mathrm{erf}(n) + 2e^{-n^2}],$$
$$E[g_n] = \frac{n^{-3/2}\sqrt{\pi}}{8}[n^2 + \tfrac{1}{2} + \tfrac{1}{2}e^{-n^2}]. \tag{83}$$

Its detector flip probability tends to zero while its source energy diverges. Every member has finite absolute stress moments through order two.

Proof. The Fourier transform is the sum of two shifted Gaussians. Integrating its squared modulus with weights ω and ω^2 gives the two formulas. Consequently $k[g_n] \sim \sqrt{\pi}/(8\sqrt{n})$ and $E[g_n] \sim \sqrt{\pi n}/8$. Gaussian decay gives the finite moments. □

This is a comparison on a fixed null-coordinate field. A member with too much energy can leave a chosen gravitational operating domain. A fixed bandwidth excludes this sequence, but that bandwidth is then part of the source certificate. The controlled weak-switch result above uses stronger regularity and is unaffected.

Proposition C.8 (Simultaneous record, source and response control)**.** *Write the realised switch as* $g + h$, *put* $w(u) = 1 + u^2$, *and assume* $g, h \in H^1$ *with weighted square-integrable derivatives. Define*

$$e_0 = \sqrt{k[h]}, \qquad A = \|\sqrt{w}\, g'\|_2, \qquad e_1 = \|\sqrt{w}\, h'\|_2, \qquad D_s = \tfrac{1}{2}Ae_1 + \tfrac{1}{4}e_1^2.$$

For the independent vacuum input of the current detector,

$$|\widetilde{p} - p| \leq 2\sqrt{k[g]}e_0 + e_0^2,$$
$$\int w\,|\widetilde{f} - f|du \leq D_s, \qquad f = (g')^2/4, \quad \widetilde{f} = (g' + h')^2/4. \tag{84}$$

The second line bounds each of the three stress-moment errors by D_s. *For matched retarded data, at each common* u *the running coefficients satisfy* $|\delta a|, |\delta b|, |\delta\gamma| \leq \kappa D_s$, *hence*

$$|\delta\mathcal{C}| \leq \kappa D_s(2|a| + |b| + |\gamma|) + 2\kappa^2 D_s^2,$$
$$|\delta\Phi(u,v)| \leq \frac{\kappa D_s(1 + |uv| + |u| + |v|)}{|1 + uv|}. \tag{85}$$

For a single contact, even when the detector has a retained reference, the half trace distance of the complete output from its ideal output is at most $\sqrt{1 - e^{-k[h]}} \leq e_0$.

Proof. Cauchy–Schwarz applied to $\widetilde{f} - f = g'h'/2 + (h')^2/4$ proves the weighted bound. The inequalities $1, |u|, u^2 \leq w$ cover both full and prefix moments; the retarded coefficient equations then give the stated errors. Expand $(a+\delta a)^2 + (b+\delta b)(\gamma+\delta\gamma)$ and the dilaton numerator for the response bounds. For the final claim, purify the initial detector and reference. The Weyl relation makes $U_g^\dagger U_{g+h}$ equal to $e^{-iX_sQ_h}$ up to one scalar phase, since $X_s^2 = I$. Its vacuum overlap is $e^{-k[h]/2}$ independently of the detector state. The pure-state trace-distance formula and partial-trace contraction give the bound. This claim concerns the declared vacuum input, not an arbitrary field left by earlier instruments. □

The coefficient bounds also control the clock on a regular window. If both compared fixed-cutoff solutions have $R_b \geq r_* > 0$, $P \geq p_* > 0$, and $R_b \leq r^*$, put $\Delta_C = \sup|\delta\mathcal{C}|$ and $\Delta_P = \sup \kappa D_s(1+|u|)^2$ on a null interval of length ℓ. Matching the initial proper time gives

$$\sup|\delta\tau(u)| \leq 2L\ell\left(\frac{\Delta_C}{2r_*p_*} + \frac{r^*\Delta_P}{p_*^2}\right).$$

An inverse-clock bound follows by dividing by a positive lower bound on $d\tau/du$. The interval and gap conditions must hold in both solutions. They cannot be inferred from fidelity alone.

A finite timing tolerance has its own source condition. For a translation error $h_\delta(u) = g(u-\delta) - g(u)$ and g with the indicated second derivative norms finite,

$$\begin{aligned}\sqrt{k[h_\delta]} &\leq |\delta|\sqrt{k[g']},\\ \|\sqrt{w}\,h_\delta'\|_2 &\leq |\delta|\|\sqrt{w}\,g''\|_2 + \tfrac{1}{2}\delta^2\|g''\|_2.\end{aligned} \tag{86}$$

The first inequality follows from $|e^{i\omega\delta}-1| \leq |\omega\delta|$. For the second, integrate $g''(u-s)$ between 0 and δ and use $\sqrt{1+u^2} \leq \sqrt{1+(u-s)^2} + |s|$. This is a null-clock translation certificate. A proper-clock error must be pulled through its actual worldline map; a finite quantum clock also needs control of its tails and correlations. Small arrival error alone does not supply a continuum stress bound.

C.7 Boundary energy, work and clock conversion

Choose the timelike Dirichlet boundary term, with the normal pointing toward increasing dilaton,

$$I_\partial = \frac{1}{\kappa}\int_{\partial M}\sqrt{-h}\,\Phi(K - L^{-1})\,dt. \tag{87}$$

Its lapse variation fixes the proper-time quasilocal energy

$$E_b = \frac{1}{\kappa}(\Phi_b/L - n^\mu\partial_\mu\Phi).$$

This is the finite-cutoff JT energy with a stated counterterm, rather than a total apparatus energy [60, Sec. 6]. For example, in Gaussian normal coordinates $ds^2 = dr^2 - N^2dt^2$, integrating the bulk curvature term by parts and adding Eq. (87) leaves the lapse boundary term $N(\partial_r\Phi - \Phi/L)/\kappa$. Its variation is $-E_b\delta N$. The dilaton boundary momentum is $(K - L^{-1})/\kappa$.

Proposition C.9 (Energy and work on a running level curve)**.** *In the traceless single-chiral source completion let $\Phi_b = \Phi_b(\tau)$ be a smooth prescribed boundary value. On a future timelike segment put*

$$R_b^2 = \Phi_b^2 - \mathcal{C} > 0, \quad S_b = \sqrt{R_b^2 + L^2\dot\Phi_b^2}, \quad y = S_b - L\dot\Phi_b > 0, \quad q_\tau = f\dot u^2.$$

Then

$$\begin{aligned}\dot u &= \frac{P}{2Ly}, & \dot{\mathcal{C}} &= 2\kappa Lyq_\tau, & \dot R_b &= \frac{\Phi_b\dot\Phi_b - \kappa Lyq_\tau}{R_b},\\ E_b &= \frac{\Phi_b - S_b}{\kappa L}, & K &= \frac{\Phi_b + L^2\ddot\Phi_b + \kappa L^2 q_\tau}{LS_b}, & \dot E_b &= q_\tau + \frac{L^{-1} - K}{\kappa}\dot\Phi_b.\end{aligned} \tag{88}$$

At a fixed cutoff these reduce to $E_b = (\Phi_b - R_b)/(\kappa L)$ and $\dot E_b = q_\tau$. On the source-free stationary family, $dE_b = T_b\,dS_h$ at fixed Φ_b, with $T_b = R/(2\pi LR_b)$.

Proof. Writing the boundary as $v = V(u)$, its dilaton equation and the induced metric give $1 = 4L^2R_b^2\dot{u}^2/P^2 - 4L^2\dot{\Phi}_b\dot{u}/P$. Its positive root is the first equation. Use $\mathcal{C}' = \kappa f P$ for the next two. The gradient identity gives $Ln^\mu\partial_\mu\Phi = S_b$. Along the unit tangent, the JT equation reads $\ddot{\Phi}_b = -\Phi_b/L^2 - \kappa q_\tau + KS_b/L$. This gives K; differentiating E_b proves the work balance. The stationary first law follows by differentiating $E_b = (\Phi_b - \sqrt{\Phi_b^2 - R^2})/(\kappa L)$ and $S_h = S_0 + 2\pi R/\kappa$. □

For example, retaining a constant gap $R_b = r > 0$ under positive flux requires

$$y = \frac{r}{\sqrt{1 + 2\kappa L^2 q_\tau/\Phi_b}}, \qquad \dot{\Phi}_b = \frac{\kappa L y q_\tau}{\Phi_b}.$$

These equations give a nonempty family on regular finite patches. Increasing the cutoff carries the work term in Eq. (88); it is not an uncharged extension of the fixed-cutoff budget. The boundary datum is still externally prescribed. A physical support must produce it and carry its own stress.

The same calculation fixes the conversion to the running horizon clock:

$$z := \frac{d\tau}{d\eta} = \frac{Ly}{R}, \qquad q_\eta = z^2 q_\tau. \tag{89}$$

For a fixed cutoff $y = R_b$, and the stationary temperature is $T_b = (2\pi z)^{-1}$. During a driven contact this is not a global KMS identification. If a local thermal-current regime has separately been justified, entropy flux transforms as $s_\eta = z s_\tau$ and its actual input temperature must satisfy $zT_\tau = 1/(2\pi)$ to use Proposition D.2. Changing coordinates does not prepare that input.

C.8 A thermal detector can be reused without refreshing its field

A finite Gibbs swap provides a physical replacement channel for the detector. Its limitation is equally useful: reset accuracy, retained correlations and heat cannot be specified independently.

Proposition C.10 (Reference-uniform thermal replacement)**.** *During sensing the detector Hamiltonian is degenerate. For reset raise its excited level to $\Delta > 0$ and supply an independent auxiliary bit in $\tau_\beta = \mathrm{diag}(1-\epsilon, \epsilon)$, with $\epsilon = (1 + e^{\beta\Delta})^{-1}$. A resonant SWAP, followed by lowering the detector gap, gives for every joint detector–reference state*

$$\rho_{SA} \longmapsto \tau_\beta \otimes \rho_A.$$

Its half-diamond distance from pure-blank replacement is exactly ϵ. If $p = \Pr(S = 1)$ just before reset, the auxiliary heat and net gap-modulation work are both

$$Q_{\mathrm{reset}} = W_{\mathrm{gap}} = \Delta(p - \epsilon). \tag{90}$$

A finite resonant swap can use $H_{\mathrm{sw}}(\tau) = \chi(\tau)(I - \mathrm{SWAP})$ with $\int \chi\, d\tau = \pi/2$; it commutes with the equal-gap free Hamiltonian. Its net switch work is zero when χ vanishes at both endpoints.

Proof. The swap transfers the whole previous detector state and its reference correlations to the auxiliary, not just its populations. The distance between the two replacement states is ϵ, for every reference, proving the channel statement and attainability. Raising the gap costs Δp, the resonant swap conserves total bare energy, and lowering it returns $\Delta\epsilon$. The outgoing auxiliary has excitation probability p. The expectation of $I-\mathrm{SWAP}$ is constant during the specified resonant contact, so its switching work integrates to zero. □

This replacement process is the standard swap benchmark for Landauer accounting [13, Example 1 and Sec. 3.3]. The auxiliary must be prepared and transported. The old state remains correlated with retained matter; it has not disappeared. If a copy of the outcome is kept, that copy is part of the maintained state in Eq. (26). At finite temperature this finite device never supplies an exact pure blank.

Proposition C.11 (A bounded-heat Gibbs ladder)**.** *Fix $\beta > 0$, a target $0 < \epsilon < 1/2$, and m finite thermal stages. First swap the degenerate detector with an independent zero-gap Gibbs bit $I/2$. Put $x_* = \ln[(1-\epsilon)/\epsilon]$, $x_j = jx_*/m$ and $p_j = (1+e^{x_j})^{-1}$. At stage j raise the detector gap from x_{j-1}/β to x_j/β and resonantly swap with an independent Gibbs bit of that gap. Finally lower its gap to zero. The complete replacement is exactly $\tau_\beta \otimes \rho_A$, with target excitation ϵ, and*

$$\beta Q_{\mathrm{ladder}} = \sum_{j=1}^{m} x_j(p_{j-1} - p_j) \le \ln 2 - h(\epsilon) + \frac{x_*}{m}(\tfrac{1}{2} - \epsilon). \tag{91}$$

The net gap-modulation work equals this heat. Choosing $m \ge x_*/(2\eta)$ gives $\beta Q_{\mathrm{ladder}} \le \ln 2 + \eta$ for every initial detector state and every retained reference.

Proof. The first swap transfers all old correlations to its output. Each later swap leaves an independent Gibbs detector. The heat formula follows from the thermal populations, and summation by parts gives the same net work, including final gap lowering. The sum is a right-endpoint approximation to $\int_0^{x_*} x[-p'(x)]dx = \ln 2 - h(\epsilon)$; its excess is at most $(x_*/m)(p_0 - p_m)$. □

This is an explicit finite swap construction of the type used in Ref. [13, Sec. 6], with a distribution-independent starting step and a bound on its stages. It consumes $m+1$ prepared Gibbs bits, including the zero-gap port, and raises the maximum gap to x_*/β. The finite reset does not hide a pure auxiliary. The first port can leave with less entropy than it had initially; it must be included along with all later outputs and their correlations. For a fixed supplied sequence of operations, N replacements differ from the pure-blank retained-system process by at most the sum of their ϵ_i, including arbitrary intervening instruments on the field. Thus high reset accuracy need not cause divergent heat, but the number of stages and the maximum gap grow. If the replacement changes the realised geometry or timing of those operations, their source and clock errors require the separate joint certificate of Section C.6. The timing, Gibbs preparation, transport and capacity required to execute these stages are additional resources.

Here is an exact two-read law on one field. Let g_1, g_2 have ordered, disjoint compact supports, put $k_i = \langle Q_i^2 \rangle$ and $c_{12} = \langle Q_1 Q_2 \rangle$ in the initial vacuum, and reuse the detector after the above swap. Both contacts start with its state τ_β. Read Z_s and use it for the subsequent command before each reset. A nondemolition dephasing of the pointer can be

implemented by a controlled Z_s using a separate degenerate Gibbs bit $I/2$. Those ports are supplied resources, not pure memories. Since $[Q_1, Q_2] = 0$, the joint outcomes obey

$$\begin{aligned}\Pr(R_1 = r, R_2 = t) = \frac{1}{4}\big[1 + \alpha\{(-1)^r e^{-2k_1} + (-1)^t e^{-2k_2}\} \\ + \alpha^2 (-1)^{r+t} e^{-2(k_1+k_2)} \cosh(4c_{12})\big].\end{aligned} \tag{92}$$

Here $\alpha = 1 - 2\epsilon$. Indeed each conditional pointer probability at a fixed Q_i is $[1 + \alpha(-1)^r \cos(2Q_i)]/2$. The commuting Gaussian characteristic function then gives the displayed law. The same field remains conditioned by the first read: thermal replacement of the detector does not restore a vacuum field. On a stationary test background the contrast of two invariant-dependent single-contact laws is multiplied by α, so it stays nonzero for $\epsilon < 1/2$. The weak-source proof of Proposition C.6 also survives this factor.

A retained body makes the two commands physically consequential. Prepare the body once in $(I + 0.8Z)/2$. After read r, couple it through $P_0 = (I + Z)/2$ or $P_1 = (I + X)/2$ to an independent vacuum control current. The first source population is $\rho_1 = 0.9 - 0.4 \Pr(R_1 = 1)$. If that contact's realised variance is k_c, the first X projection attenuates the body's Z component by $e^{-k_c/2}$. With p_{rt} denoting Eq. (92), its second source population is therefore

$$\rho_2 = \tfrac{1}{2} + 0.4\{p_{00} + e^{-k_c/2} p_{10}\}. \tag{93}$$

This uses the body left by the first command, not a newly prepared copy.

For a finite boundary run take $L = 1$, $\Phi_b = 3$, $\kappa = .01$, initial $(u, a, b, \gamma) = (1, 1, 0, 0)$, $\epsilon = .01$ and $\beta = 1$ in the proper clock. Use the profile $A[1 - (2(\tau - \tau_0)/d - 1)^2]^3$ on an interval of duration d. The two sensor contacts have amplitude .5 on $[0, 2]$ and $[4, 6]$; the two selected control contacts have amplitude .3 on $[2.5, 3.5]$ and $[6.5, 7.5]$. The gaps leave finite windows for readout and reset. Integrating the same mean boundary equations gives

$$k_1 = .068080013, \quad k_2 = .068082993, \quad c_{12} = -.000926643, \quad k_c = .024462001,$$

$$\rho_1 = .871049781, \quad \rho_2 = .870722307, \quad \mathcal{C}_{\text{final}} = 1.030570773.$$

The proper-clock coupling-source charge is .540937293. The two reset auxiliaries separately absorb .573257937, using $\Delta = \ln 99$; this heat has not been inserted as a chiral bulk source. The two outgoing reset bits have joint entropy .519488204, against .112003069 initially, and relative entropy .165772802 to the product Gibbs reference. Thus their heat obeys exactly $\beta Q = \Delta S(B_1 B_2) + D(\rho_{B_1 B_2} \| \tau_\beta^{\otimes 2})$. Other output ports and retained matter belong to the full entropy account. The source charge, reset heat and apparatus work are distinct entries.

Starting instead at $a = 1.4$ with the other initial coefficients unchanged changes both recorded laws and returned source populations on the same proper schedule. The detector and body are reused in both runs. The completion transmits the sensor current and two independent control currents into the bulk. The reset ports, clock and support still require a conserved transport completion; their cost is not removed by the displayed bulk balance. A long sequence of simple Gibbs swaps is also expensive: choosing $\epsilon_N = \varepsilon/N$ to certify total blank-replacement error at most ε requires $\beta\Delta_N = \ln(N/\varepsilon - 1)$. If detector excitation probabilities stay above a fixed positive constant, its heat per reset grows as $\log N$. This particular construction cannot attain the fixed-rate sharp-exhaust conjecture. The ladder of Proposition C.11 avoids this heat divergence, but fitting its stages and transported thermal units into a fixed capacity and operating rate remains necessary.

D Running horizons, source states and clocks

D.1 Source sorting and tracking-clock support

The finite-carrier sorting statement remains Proposition 9.3, with its source and old-reference permissions unchanged.

Proof of Proposition 9.3. The source enters Q-RAIF C's proof in three places: concentration of input eigenlabel strings, the source's fixed neutral charge, and the final source–carrier entropy identity. Its detailed depolarizing eigenvalues are not otherwise used. Each ρ_p has rank two in the same neutral quartet sector and finite information variance at fixed q, ϵ. Independent eigenlabel strings have information variance $O(L)$, including a nonidentical product over the two-symbol alphabet. The sorter's subextensive typical-label margin, thermal-sector state count and physical neutral-sector unitary compiler therefore apply with entropy $\sum_j h_{p_j}$. The source has zero total charge, leaving the carrier's sector weights and sublinear sector correction unchanged. Completing the eigenlabel injection to a sector unitary keeps all bad labels and acts as the identity on external references. The source–carrier entropy identity and controlled full-carrier energy tails give the deficit and slack limits through the Gibbs identity. Trace error alone would not suffice. Uniform local reset error alone does not establish the supplied product marginal for an adaptive history. □

Equation (48) gives the actual clock, not a static-wall substitution. For a tracking datum $R_b = \rho R$, $\Phi_b = \alpha R$, $\alpha = \sqrt{1+\rho^2}$ and $\delta = R^{-1} dR/d\eta$, it gives

$$z = L\sqrt{\rho^2 - 2\alpha\delta}, \qquad \rho^2 > 2\alpha\delta. \tag{94}$$

D.2 The running horizon and its entropy

Long-time entropy limits need an operating geometry, not just a source integral. Positive steady flux can exhaust a fixed u chart. At the boundary $v = -1/u$ the numerator of the retarded dilaton is

$$N_b(u) = 2H_0 - \frac{\kappa}{u}\int^u (u-z)^2 F(z)\, dz.$$

For a constant flux $F_0 > 0$ switched on at $u = 0$, this becomes $2H_0 - \kappa F_0 u^2/3$ and vanishes at $u_* = \sqrt{6H_0/(\kappa F_0)}$. A device at a fixed exterior position cannot therefore be assumed to run at fixed u frequency indefinitely. Finite observation windows and families with larger operating domains are different from an infinite-time protocol in one fixed background.

There is an exact horizon balance inside the retarded solution. In this subsection we use $\hbar = k_B = 1$ and the classical Wald entropy, not an event-horizon generalized entropy. Write the running coefficients as

$$a(u) = H_0 + \kappa\int^u zF(z)\, dz, \qquad b(u) = -\kappa\int^u F(z)\, dz, \qquad \gamma(u) = \kappa\int^u z^2 F(z)\, dz,$$
$$\Phi = \frac{a + bu + \gamma v - auv}{1+uv}, \qquad \mathcal{C}(u) = a^2 + b\gamma, \qquad R(u) = \sqrt{\mathcal{C}(u)}. \tag{95}$$

The lower limits and any source-free initial coefficients are fixed by the matching data. Put $P = bu^2 + 2au - \gamma$ and $\xi = P/(2R)$.

Proposition D.1 (Exact marginal-horizon balance)**.** *On a patch with $\mathcal{C} > 0$, $a + R \neq 0$, $P > 0$ and $1 + uv_h \neq 0$, the curve*

$$v_h(u) = \frac{b}{a + R}$$

is a future outer marginal horizon: $\partial_u \Phi = 0$, $\partial_v \Phi < 0$ and $\partial_v \partial_u \Phi < 0$ there. Its dilaton and Wald entropy obey

$$\Phi_h = R, \qquad S_h = S_0 + \frac{2\pi R}{\kappa}, \qquad \frac{dS_h}{du} = 2\pi F \xi, \qquad \frac{dS_h}{d\eta} = 2\pi F \xi^2, \quad d\eta = \frac{du}{\xi}. \tag{96}$$

For $F \geq 0$ this horizon is spacelike or null and its entropy is nondecreasing.

Proof. The coefficient derivatives $a' = \kappa u F$, $b' = -\kappa F$ and $\gamma' = \kappa u^2 F$ cancel from the first dilaton derivative. With $D = 1 + uv$,

$$\Phi_u = \frac{b - 2av - \gamma v^2}{D^2}, \qquad \Phi_v = -\frac{P}{D^2}.$$

Substitution of v_h gives $\Phi_h = R$ and $\Phi_{uv}|_h = -2R/(1 + uv_h)^2$. Also $\mathcal{C}' = \kappa F P$ and $v_h' = -\kappa F (1 + uv_h)^2/(2R)$. The metric then makes the tangent spacelike for $F > 0$ and null for $F = 0$. Finally $R' = \kappa F \xi$ and $\kappa = 8\pi G$ give Eq. (96). □

The identification of this classical marginal horizon and its area law is a standard JT result [53, Sec. 2.3.1]. Here it assigns an actual horizon observable to the running invariant used in the source account. Negative quantum excess stress need not be monotone, and replacing S_h by S_{gen} would require the matter entanglement contribution and a separate quantum argument.

Proposition D.2 (Conditional entropy-export cost at the horizon)**.** *Suppose the charged heat in the rapidity clock is exactly $\dot{Q}_\eta = F\xi^2$, and the exhaust satisfies Proposition 4.2 in that same clock at $T_H = 1/(2\pi)$. On a window of rapidity duration τ, let X_τ be the net exported maintenance entropy. In the extensive rate regime,*

$$\frac{\Delta S_h}{\tau} \geq \frac{X_\tau}{\tau} + \frac{6}{c_{\mathrm{ex}}} \left(\frac{X_\tau}{\tau} \right)^2 + o(1).$$

Writing bars for the corresponding mean rates gives

$$\overline{\dot{S}_h} \geq \dot{s} + \frac{6}{c_{\mathrm{ex}}} \dot{s}^2. \tag{97}$$

The same pointwise formula requires a justified local-current limit; it does not follow for arbitrary fluctuating packets. A mean horizon-entropy budget $\overline{\dot{S}_h} \leq B$ permits at most

$$\dot{s} \leq \frac{c_{\mathrm{ex}}}{12} \left(\sqrt{1 + \frac{24B}{c_{\mathrm{ex}}}} - 1 \right).$$

Proof. Multiply Eq. (30) at $T = T_H$ by 2π, use the integrated Eq. (96) and the long-window Gibbs account, then invert the increasing quadratic. □

This transfers the entropy removed from the full maintained state to a specified gravitational entropy, including the finite-rate excess. The heat-routing condition is substantive: incoming nonequilibrium energy, other stress sources, charge transfer and the clock's own entropy cannot be omitted. Nor does a change of clock alone prepare a thermal reservoir.

Proposition D.3 (Schwarzian of the running clock)**.** *On a smooth running patch put* $q = F\xi^2$ *and* $\delta = R^{-1}dR/d\eta = \kappa q/R$*. Then*

$$\{u;\eta\} = -\frac{1}{2} - \frac{d\delta}{d\eta} - \frac{\delta^2}{2}. \tag{98}$$

For constant $q > 0$*,* $R(\eta) = R_0 + \kappa q\eta$ *and*

$$\{u;\eta\} = -\frac{1}{2} + \frac{(\kappa q)^2}{2R(\eta)^2}. \tag{99}$$

Proof. Write $j = (bu + a)/R$, and let a dot denote $d/d\eta$. The coefficient equations and $h^2 - 2bR\xi = R^2$, with $h = bu + a$, give

$$\frac{\dot{\xi}}{\xi} = j - \delta, \qquad \dot{j} = \frac{j^2 - 1}{2} - j\delta.$$

Substitute these into $\{u;\eta\} = (\dot{\xi}/\xi)^{\cdot} - (\dot{\xi}/\xi)^2/2$. For constant q, $\dot{\delta} = -\delta^2$, which recovers Eq. (99). □

Constant positive heat current changes the vacuum clock correlation, not merely the horizon size. Positive growth itself, however, does not rule out an exact thermal reference. The following two histories solve the clock condition.

Proposition D.4 (Growing horizons with exact geometric KMS clocks)**.** *Use natural units and the prescribed-mean-source JT equations. An independent chiral conformal reference sector is in its vacuum; its state is not the state of the driven matter supplying* F*. On every finite smooth operating window, the following positive-flux profiles admit the initial matching* $(a, b, \gamma) = (R_0, 0, 0)$ *at* $\eta = 0$*,* $u = u_0 > 0$*:*

1. *Constant fractional growth* $\delta_0 > 0$*:* $R = R_0 e^{\delta_0 \eta}$*,* $q = \delta_0 R/\kappa$*. The reference pulled back to the* η *clock is geometric KMS at* $T = \sqrt{1 + \delta_0^2}/(2\pi)$*.*

2. *Fixed rapidity temperature: for* $\eta_0 > 0$*,*

$$R = R_0(1 + \eta/\eta_0)^2, \qquad q = \frac{2R_0(\eta + \eta_0)}{\kappa\eta_0^2}. \tag{100}$$

   The same reference is exactly geometric KMS at $T_H = 1/(2\pi)$ *despite the growing horizon.*

Here KMS refers to the pulled-back reference algebra and its complete η*-translation dynamics. A finite experiment samples its restriction.*

Proof. Both profiles make Eq. (98) constant, $-s^2/2$, with $s = \sqrt{1+\delta_0^2}$ in the first case and $s = 1$ in the second. Let $\delta(0) = \delta_0$ also in the second case, where $\delta_0 = 2/\eta_0$. An explicit solution with the required initial coordinate and derivatives is

$$u(\eta) = u_0 \left[1 + \frac{e^{s\eta} - 1}{s + m(e^{s\eta} - 1)}\right], \qquad m = \frac{s - 1 + \delta_0}{2}. \tag{101}$$

It has $\xi = \dot{u} > 0$, $\xi(0) = u_0$ and $\dot{\xi}(0)/\xi(0) = 1 - \delta_0$. Set $j = \dot{\xi}/\xi + \delta$ and

$$b = \frac{R(j^2 - 1)}{2\xi}, \qquad a = Rj - bu, \qquad \gamma = bu^2 + 2au - 2R\xi.$$

The constant-Schwarzian equation gives $\dot{j} = (j^2 - 1)/2 - j\delta$. Differentiation then gives $\dot{b} = -\kappa q/\xi$, $\dot{a} = \kappa u q/\xi$ and $\dot{\gamma} = \kappa u^2 q/\xi$. Thus these are the retarded coefficient equations with $F = q/\xi^2$, $a^2 + b\gamma = R^2$ and $P = 2R\xi$. At zero they give $(R_0, 0, 0)$; $a + R$ and the horizon denominator remain nonzero on a sufficiently short initial operating patch, which can be continued while those patch conditions hold.

Equation (101) is a Möbius transform of $e^{s\eta}$. Möbius invariance of the reference vacuum and the geometric KMS construction therefore give inverse temperature $2\pi/s$ [9, Sec. 2.8, Proposition 2.4 and Theorem 2.5]. For a free current the complete Gaussian correlation test is explicit: if $\langle j(u)j(u')\rangle = -C_j/(u - u' - i0)^2$, then

$$\langle \xi(\eta) j(u(\eta))\, \xi(\eta') j(u(\eta'))\rangle = -\frac{C_j s^2}{4\sinh^2[s(\eta - \eta' - i0)/2]}.$$

This is stationary and obeys the KMS boundary condition; the conformal-net argument establishes the state on bounded observables, not just on one stress expectation. □

These are actual nonstationary solutions of the mean-source equations, rather than frozen-background substitutions. The reference sector's thermality is a kinematic input that can now be made compatible with growth. It is not a thermal exhaust realization: the driving sector can be coherent, its state is not KMS merely because the independent reference is, and preparation, heat transport and the physical clock still need their own account. In the second profile q grows linearly, so it does not realize the fixed mean export in Conjecture 9.2. As $\eta \to \infty$, u approaches a finite endpoint and $F = q/\xi^2$ diverges. Each finite smooth window has finite source energy; these formulas assert no finite-resource infinite run.

D.3 The actual thermal source changes its clock reference

A thermal spectator field does not make the exhaust thermal. Here the state condition is imposed on the source field itself. Work with the vacuum-subtracted chiral flux of the preceding JT construction, with the state-independent trace background absorbed consistently into the renormalized JT parameters. A complementary chirality may remain in its reference state. This is a semiclassical conformal-matter completion, not a microscopic construction of a thermal emitter.

For the inner patch let $u = f(v_-)$ be its Poincaré null coordinate. The coordinate change $\rho = \Phi/f' - L^2 f''/(f')^2$ gives

$$\{u; v_-\} = -\frac{\mathcal{C}_-}{2L^4}.$$

Specify the source state as the pullback of a geometric KMS state at temperature T_o in η, with $T_H = 1/(2\pi)$ as the comparison temperature. Define the charged excess current and entropy current by

$$q = \frac{\pi c}{12}(T_o^2 - T_H^2) > 0, \qquad s = \frac{\pi c}{6}(T_o - T_H), \qquad 2\pi q = s + \frac{6s^2}{c}. \tag{102}$$

This state condition is imposed on the finite operating patch. Section D.4 gives a local preparation route from supplied thermal reservoirs and also treats nonthermal outputs; the reservoir and contact's complete material realization remain inputs.

The physical stress is a tensor. The Schwarzian enters when the state is expressed relative to the Poincaré vacuum, through the chiral stress transformation law [7, Eq. (2.14)]:

$$\begin{aligned} F_{vv} &= \left(\frac{d\eta}{dv_-}\right)^2 \left(\frac{c}{48\pi} + q\right) - \frac{c}{24\pi}\{\eta; v_-\} + \frac{c}{24\pi}\{u; v_-\} \\ &= \frac{R^2 q}{L^4} - \frac{c}{24\pi}\{\eta; v_-\}. \end{aligned} \tag{103}$$

The last term in the first line is essential. On a static patch, the T_H thermal state is the zero-flux vacuum reference; retaining only its positive thermal energy would falsely insert a source. Conversely, on a growing patch that same running-clock reference need not have zero physical stress.

Proposition D.5 (State–clock–source consistency)**.** *Put $a = \kappa c/(24\pi)$ and $\delta = R'/R$, where primes in this proposition denote η derivatives. The actual source state and the JT equation are jointly satisfied by*

$$R' = R\delta, \qquad \delta' = \frac{\kappa q - R\delta}{a} - \frac{\delta^2}{2}. \tag{104}$$

The physical flux in this clock and the horizon entropy account are

$$\begin{aligned} F_\eta &= q + q_{\rm ref} = R'/\kappa, & q_{\rm ref} &= -\frac{c}{24\pi}(\delta' + \delta^2/2), \\ \Delta S_{h,-} &= 2\pi q \Delta\eta - \frac{c}{12}\Delta\delta - \frac{c}{24}\int \delta^2 d\eta. \end{aligned} \tag{105}$$

For $R_0^2 > 2a\kappa q$ and $\kappa q/(2R_0) \le \delta_0 \le 2\kappa q/R_0$, the source–clock ODE has a global positive solution in η with

$$\frac{\kappa q}{2R} \le \delta \le \frac{2\kappa q}{R}, \quad R = R_0 + \kappa q \eta + O(1), \quad \int_0^\infty \delta^2 d\eta \le \frac{8\kappa q}{R_0}. \tag{106}$$

Thus $\Delta S_{h,-}/\Delta\eta \to 2\pi q = s + 6s^2/c$. This global statement concerns the source and clock, not the lifetime of a finite material energy reserve.

Proof. Since $dv_-/d\eta = L^2/R$, its Schwarzian is $\{v_-; \eta\} = -\delta' - \delta^2/2$. The chain rule in Eq. (103) gives $F_\eta = q - c(\delta' + \delta^2/2)/(24\pi)$. Together with $\partial_v \mathcal{C}_- = 2\kappa L^2 F_{vv}$ this is Eq. (104); integration gives Eq. (105). At the upper boundary $\delta = 2\kappa q/R$, the derivative of $\delta - 2\kappa q/R$ is $-\kappa q/a + 2(\kappa q)^2/R^2 < 0$. At the lower boundary the derivative of $\delta - \kappa q/(2R)$ is $\kappa q/(2a) + (\kappa q)^2/(8R^2) > 0$. The strip is invariant and $R' \ge \kappa q/2$. Its upper bound gives $\int \delta^2 \le \int_0^\infty 4(\kappa q)^2 (R_0 + \kappa q\eta/2)^{-2} d\eta$. Finally $R' + a\delta' + a\delta^2/2 = \kappa q$ gives the bounded correction to linear growth. All variables stay finite on finite η intervals. □

The charged heat q and physical source F_η differ by an explicitly accounted reference stress. Proposition D.2 remains conditional on their equality; it cannot be applied pointwise to q after discarding q_{ref}. Here the reference correction is nonextensive, so the sharp long-time rate is recovered by the stronger state–clock calculation. A finite patch does not establish a globally finite-energy state across a limiting Poincaré coordinate endpoint.

For any fixed finite proper-time window, nonempty operating solutions also follow without relying on the numerical table. At $q = 0$, start from the positive static support, $e > 0$, $R_0 > 0$ and $\delta = 0$. This is a regular equilibrium of the combined equations. Continuous dependence on q, with $\delta_0 = \kappa q/R_0$, preserves positive mass, energy reserve and exterior domain on that window for sufficiently small $q > 0$. This is not a fixed-rate infinite-lifetime statement.

Combining Eq. (104) with Eq. (125) uses $P_{\mathrm{out}} = R'/[\kappa z^2]$ and $P_{\mathrm{in}} = 0$. For the support above, initial $e = 2.34$ and $\delta_0 = \kappa q/R_0$, all three examples below run for proper duration 80 with positive reserves and $\Phi > R$:

q	η_{end}	Φ_{end}	e_{end}
.005	85.53862	3.003858	1.882696
.010	85.60190	3.007607	1.424039
.020	85.73040	3.015345	.502569

The mass-junction residual is below 1.2×10^{-12} on the checked grid. These solve the moving mean support and the state condition together. The internal reserve's decline is part of the solution. It has not been derived from the seven-qubit core's local release Hamiltonian.

D.4 A local thermal contact supplies the actual source

The source condition can be replaced by a scattering calculation. Use two complex chiral fermion ports, each with $c = 1$, at a local conformal contact. This is the doubled free-fermion defect of Bernard, Doyon and Viti [5, Sections 3 and 5]. The zero-charge input streams are jointly independent and thermal at T_H and T_o in the same supplied η frame. Their fields mix by $S_\theta = \cos\theta I - i \sin\theta \sigma_x$ at the contact. Put $\chi = \sin^2\theta$. For positive frequency, the source occupation is

$$n_\chi(\omega) = (1-\chi) n_{T_H}(\omega) + \chi n_{T_o}(\omega), \qquad n_T(\omega) = \frac{1}{1 + e^{\omega/T}}. \tag{107}$$

The actual output is hot KMS for $\chi = 1$. For $0 < \chi < 1$ it is stationary but nonthermal: its high-frequency tail has coefficient χ rather than one. Its return-port cross covariance is nonzero. These are retained field correlations, not fresh independent thermal maps. Q-RAIF C derives the complete contact law and its energy account [14, Section on a local thermal contact].

Proposition D.6 (Stationary transfer without a KMS output)**.** *For a constant contact angle set*

$$q_\chi = \frac{\pi\chi}{12}(T_o^2 - T_H^2), \qquad s_\chi = \frac{1}{\pi}\int_0^\infty [h(n_\chi) - h(n_{T_H})]\, d\omega. \tag{108}$$

Here $h(x) = -x \ln x - (1-x)\ln(1-x)$. *The state–clock equations, positive growing family and finite reference correction in Proposition D.5 hold with* $c = 1$ *and* $q = q_\chi$.

Their asymptotic horizon rate is

$$\frac{\Delta S_{h,-}}{\Delta\eta} \longrightarrow 2\pi q_\chi \geq s_\chi + 6s_\chi^2. \tag{109}$$

The inequality is strict for $0 < \chi < 1$ *and* $T_o > T_H$.

Proof. The stationary output's stress in η is $1/(48\pi) + q_\chi$. Its coordinate/state transformation has the same central charge and projective-vacuum subtraction as Eq. (103). That derivation uses only this stress expectation, not the KMS property. The source–clock ODE and its invariant strip therefore follow unchanged. At fixed energy the Gibbs state maximizes the fermion entropy current. Its temperature is $T_{\rm eff} = \sqrt{T_H^2 + 12q_\chi/\pi}$ and its entropy excess is $\pi(T_{\rm eff} - T_H)/6$. Since the partially transmitted state is not that Gibbs state, the entropy inequality is strict. Eliminate $T_{\rm eff}$ and use $T_H = 1/(2\pi)$. □

For a hot-input excess .02 and $\chi = .5$, the output has $q_\chi = .01$, $s_\chi = .04818703$ and thermal-floor excess $q_\chi - T_H s_\chi - 3s_\chi^2/\pi = .000113460$. A KMS output with the same heat has $s = .04863795$. Their mean stress, geometry and clock can be identical despite their different states and entropy currents. A geometric response alone does not establish thermalization or identify the source's complete state.

A program that changes the contact also changes its stress. For a smooth prescribed angle, point splitting the full unequal-time covariance gives

$$q_{\rm eff}(\eta) = \sin^2\theta(\eta) q_{\rm hot} + \frac{\theta'(\eta)^2}{4\pi}, \qquad F_\eta = q_{\rm eff} + q_{\rm ref}. \tag{110}$$

Each output port receives the same pump term. The total field work is $\int \theta'^2 d\eta/(2\pi)$ and is at least $(\Delta\theta)^2/(2\pi\Delta\eta)$ over a switching interval. Thus switching the selected interface has a source cost in the same interaction, rather than triggering an unrelated emitter. The running clock equations use $q_{\rm eff}(\eta)$, retaining their reference stress. They also permit a stronger statement for changing programs.

Proposition D.7 (Positive growth under bounded changing sources)**.** *Suppose* $q_{\rm eff}$ *is smooth,* $0 < q_{\min} \leq q_{\rm eff}(\eta) \leq q_{\max}$ *and* $R_0^2 > 2a\kappa q_{\max}$*, with* $\kappa q_{\min}/(2R_0) \leq \delta_0 \leq 2\kappa q_{\max}/R_0$*. Then the source–clock ODE with* $q_{\rm eff}$ *has a global positive solution satisfying*

$$\begin{aligned} \frac{\kappa q_{\min}}{2R} \leq \delta \leq \frac{2\kappa q_{\max}}{R}, \qquad & R' \geq \frac{\kappa q_{\min}}{2}, \\ \int_0^\infty \delta^2 d\eta \leq \frac{8\kappa q_{\max}^2}{q_{\min} R_0}, \qquad & R - R_0 = \kappa \int q_{\rm eff} d\eta - a\Delta\delta - \frac{a}{2}\int \delta^2 d\eta. \end{aligned} \tag{111}$$

The reference correction stays bounded. If the actual source has a long-time mean, the mean horizon entropy rate is 2π *times that mean. This concerns the source–clock sector, not an unlimited body energy reserve.*

Proof. At the upper barrier, the relative derivative is at most $-\kappa q_{\max}/a + 2(\kappa q_{\max})^2/R^2 < 0$. At the lower barrier it is at least $\kappa q_{\min}/(2a) + (\kappa q_{\min})^2/(8R^2) > 0$. Thus the strip is invariant. Integrate its upper bound using $R \geq R_0 + \kappa q_{\min}\eta/2$ to obtain the integral estimate. Integration of $R' + a\delta' + a\delta^2/2 = \kappa q_{\rm eff}$ gives the last identity. The strip also prevents finite-time blowup. □

A bounded clock-reference correction does not remove the program's work. If N disjoint contact changes each have angle increment at least $\Delta_{\min}$ and duration at most ν^{-1}, their total pump work is at least $N\nu\Delta_{\min}^2/(2\pi)$. At $N/\Delta\eta = \alpha\nu$ the mean work current is at least $\alpha\nu^2\Delta_{\min}^2/(2\pi)$, half in each output. This is an additional switching-charge requirement of this contact architecture, not a universal heat bound to be added again when that same output energy has already been counted in the entropy account.

The source is prepared by the contact, but the input reservoir is still prepared in a specified clock frame. The contact's local scattering law does not supply a microscopic moving material body or its controller. Its other port, reservoir preparation, switching work, clock conversion and recoil must be included before it can replace the complete source in the mass-10 table. Nor may a stationary entropy formula be applied instantaneously during the switch. For a stationary frequency cutoff Λ with the common vacuum retained, a separate current bound is

$$|\Delta q| \le \frac{1}{\pi} \sum_{T=T_H,T_o} T(\Lambda + T)e^{-\Lambda/T}.$$

This controls omitted thermal source energy, not the field's global trace distance or the stress of turning the contact on.

E Finite preparation and quantum material source certificates

E.1 Finite preparation on a predicted running clock

The finite interval need not be supplied by an infinite-energy KMS state. Q-RAIF C constructs a compact diffeomorphism of the vacuum that agrees with $\exp[2\pi T\eta(v_-)]$ on the regular operating interval [14, Section on thermal correlations from a finite-energy preparation]. Its full local covariance equals the thermal η covariance, while the global state is pure and has finite energy and its second moment. Two independent copies give the local contact's required inputs, retaining each copy's outside purification. Their stress is exactly Eq. (103) after the same Poincaré subtraction. A local bounded regularization and positive history clock approximate this preparation and its smooth stress to any fixed finite tolerance. The field-energy gain is paid by the clock-interaction energy; this is not a free reservoir renewal.

The prepared states and inertial contact now have a common fermion carrier [14, Proposition on a positive contact on the full CAR carrier]. A compact-gauge compensation proves the single-contact Hamiltonian positive and self-adjoint even when its static dressing is not implemented. The flavor rotation retains the original two fields' correlations instead of replacing them by independent current baths. On this carrier the actual reaction and the exterior smooth-source certificate apply together. Matching that quantum contact to the moving material action remains a separate requirement.

A record selected by earlier local operations can also choose the next thermal preparation on this same field [14, Proposition on record-selected local thermal preparation]. If its past field protocol lies in an interval of length a, the next preparation support has length b, and their gap is d, then the two complex chiral fields give $I_0 = \frac{1}{3}\log[1+ab/(d(d+a+b))]$. The probability-weighted conditional thermal error is at most $\sqrt{I_0/2}$. A smooth source test with actual and comparison second-moment norms M_a, M_i has error at most

$\sqrt{I_0}(M_a + M_i)$, in addition to the preparation-gate and clock errors. The outside purification and prior records remain in the device. The same relative-entropy bound includes a controller and its initially retained references obtained by the local past protocol; it passes through later joint channels on that controller and window. Additional incoming fields must be included rather than treated as independent. Separation, physical record transport and a new preparation are charged; the comparison state is not a physical reset of a correlated field. This permits a conditional finite update, while causal transport in the moving material and renewal of its clock, blanks and reference remain to be realized together.

The field contact can also acquire a finite-mass quantum position [14, Section on the contact position]. The same full-CAR Hamiltonian then conserves body-plus-field momentum; a finite anchor contributes its own recoil. Its moving-frame term is $(p - H_f)^2/(2M)$, so the fixed contact is a fixed-momentum large-mass limit, not a free prescribed trajectory. Position tails require a generator-weighted locality error before using the record-controlled thermal certificate. This support is nonrelativistic and does not yet provide the covariant material stress or the physical clock required below. Prescribing a constant velocity is a different limit: the harmonic support has a stiffness-dependent stability boundary, while the full finite-mass recoil Hamiltonian remains positive.

Energy conservation in the unfolded carrier does not identify its translation generator with bulk momentum. Folding its two flavors into right- and left-moving fields makes the finite-width mixing a mirror coupling. A local bidirectional current instead needs two flavors per direction, with $P = H_R - H_L$ and recoil $(p - P)^2/(2M)$ [14, Proposition on bidirectional current and physical recoil]. Its harmonic prescribed-velocity threshold is $\sqrt{a_0/(a_0 + d_R + d_L)}$, while the full finite-mass energy remains positive. The added preparations and purifications enter the source account, and each exhaust capacity counts its actually routed fields. For the same interval lengths a, b and gap $g > 0$, their product free vacuum gives $I_0 = \frac{2}{3}\log[1+ab/(g(g+a+b))]$, twice the two-field renewal value above; the conditional errors use that value. This supplies a physical momentum and local field density; deriving the propagating material's renormalized quantum stress is still needed before inserting that density as the complete matter in a JT apparatus.

For a flat-coordinate interval of length ℓ, the compact vacuum-orbit preparations at temperature T have infimum energy

$$E_{\min} = \frac{\pi c T^2 \ell}{12} - \frac{cT}{6}\tanh\frac{\pi T \ell}{2}.$$

On a changing clock the actual map, including its compact completion, sets the physical preparation cost. Its waveform cannot be redefined after a read changes the future trajectory.

Proposition E.1 (Source error from a predicted thermal clock)**.** *Prepare with $\widehat{\eta}$ while the actual clock is η. On a fixed regular source interval suppose both first derivatives are at least $m > 0$, both jth derivatives have absolute value at most M_j, and $\|\eta^{(j)} - \widehat{\eta}^{(j)}\|_\infty \le \epsilon_j$ for $j = 1, 2, 3$. Put $J_T = \pi c T^2/12$. The prepared one-point stress differs from the exact η-thermal stress by at most*

$$B_{\text{clock}} = 2J_T M_1 \epsilon_1 + \frac{c}{24\pi}\left(\frac{\epsilon_3}{m} + \frac{M_3\epsilon_1}{m^2} + \frac{3M_2\epsilon_2}{m^2} + \frac{3M_2^2\epsilon_1}{m^3}\right). \tag{112}$$

A constant two-port contact combines its two input bounds with weights $1-\chi, \chi$. A smooth source test f has additional clock error at most $\|f\|_1 B_{\text{clock}}$, separate from the autonomous execution's state and stress error.

Proof. Apply $|a^2 - b^2| \le 2M_1|a-b|$ to the squared Jacobian. The ratio of third to first derivative changes by at most $\epsilon_3/m + M_3\epsilon_1/m^2$. Factoring the difference of the squared second-to-first ratios gives $2M_2\epsilon_2/m^2 + 2M_2^2\epsilon_1/m^3$. The Schwarzian's factor $3/2$ yields Eq. (112). The same coordinate-vacuum subtraction cancels between the two preparations, not between different geometric backgrounds. □

This tolerance makes a record-selected prediction testable without an exactly known future geometry. It controls mean source, not complete state thermality or entropy. Learning a predictor that maintains the stated derivative tube, and embedding the local gate, clock and two-port stress in the moving material action, remain physical tasks. The thermal restriction has an exterior purification; it cannot be inserted unchanged into a finite-window entropy-reset argument.

E.2 A matched quantum material source

Positivity of an excitation Hamiltonian leaves its vacuum contribution to the source undetermined. A local Gaussian material makes the required comparison explicit. On a finite interval with Dirichlet boundaries, let $a \in C_c^\infty$ vanish near the endpoints and $m > 0$. The spatial form for an active field ϕ and a normalized angle Θ is

$$q_a = \int \left[|\phi' + a\Theta|^2 + |\Theta'|^2 + m^2|\Theta|^2\right] dx.$$

It is closed and coercive. Its operator, with the same boundary conditions for $a = 0$, is

$$K_a = \begin{pmatrix} -\partial_x^2 & -\partial_x a \\ a\partial_x & -\partial_x^2 + a^2 + m^2 \end{pmatrix}.$$

Here $-\partial_x a$ denotes $-\partial_x \circ a$, including the derivative of the field. A constant positive angle–storage mass matrix and free spectator fields leave the following coefficient unchanged.

Proposition E.2 (Matched material vacuum coefficient)**.** *For the proper-time cutoff comparison*

$$\Delta E_\Lambda = -\frac{1}{4\sqrt{\pi}} \int_{\Lambda^{-2}}^\infty t^{-3/2} \operatorname{Tr}(e^{-tK_a} - e^{-tK_0})\, dt,$$

and a fixed reference scale $\mu > 0$,

$$\Delta E_\Lambda = \frac{\log(\Lambda/\mu)}{8\pi} \int a^2 dx + C_\mu[a] + o(1). \tag{113}$$

The finite part requires a common material normalization condition.

Proof. Write $K_a = -\partial_x^2 + B\partial_x + C$, where $B = \left(\begin{smallmatrix} 0 & -a \\ a & 0 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 0 & -a' \\ 0 & a^2+m^2 \end{smallmatrix}\right)$. The antisymmetric connection $A = -B/2$ gives $K_a = -(\partial_x + A)^2 + P$, with

$$P = C - B'/2 + B^2/4 = \begin{pmatrix} -a^2/4 & -a'/2 \\ -a'/2 & m^2 + 3a^2/4 \end{pmatrix}, \qquad \operatorname{tr}(P - P_0) = a^2/2.$$

The heat coefficient for $-(\nabla^2 + E)$ uses $E = -P$ [3, Eq. (4.27)]. Matched boundary jets cancel, so the trace difference is $-\sqrt{t}\int a^2/(4\sqrt{\pi}) + O(t^{3/2})$. Its proper-time integral gives Eq. (113). Coercivity controls the large-t end of that integral. □

The coefficient can be assigned to a covariant material action. For the positive target metric

$$ds^2_{\rm mat} = (d\phi + \lambda\theta f(Z)dZ)^2 + m_Z^2 dZ^2 + I d\theta^2,$$

linearization about a static texture Z_0, with $\theta = 0$, gives $a = \lambda f(Z_0)Z_0'/\sqrt{I}$. Direct curvature calculation yields $\mathrm{Ric}_{ZZ} = -\lambda^2 f(Z)^2/(2I)$ there. The usual one-loop kinetic counterterm in the Euclidean action [4, Section 7.2] is, in the present normalization,

$$S_{\mathrm{ct},E} = \frac{\log(\Lambda/\mu)}{4\pi} \int \sqrt{g}\, \mathrm{Ric}_{AB}\, \partial_\mu X^A \partial^\mu X^B.$$

On the texture it cancels the logarithm in Eq. (113). Other potential, boundary and metric counterterms must be matched as well. The finite Gaussian model is not the full charged-fermion contact or an all-order continuum construction.

The charged carrier can nevertheless be retained at finite material resolution. Q-RAIF C's positive material construction [14] uses the full fixed-spin charge/current basis, including the unused oscillator modes, and keeps the material momentum squares. Its boson–fermion correspondence is an operator equivalence, not a neutral-sector replacement. This supplies a positive nonlinear regulated Hamiltonian. The local continuum, its thermal states and its renormalized source remain to be matched; the correspondence does not determine a new vacuum subtraction for each contact. Smoothing a permitted bounded gate under this interacting Hamiltonian also supplies the energy regularity needed for autonomous record-controlled operation. Its positive regulated energy tests have a source certificate, while their covariant continuum stress still requires a separate limit. A local positive binding potential also produces finite normalizable angle and storage modes with the finite contact's mechanical coefficients [14]. Their omitted material modes give an explicit self-energy and a filtered causal-response error. This avoids an infinite uniform angle, but an exact local light front and quantum thermal state are not supplied by the collective-mode projection. Local record-selected drives can instead be compared on the same complete Gaussian Hamiltonian and initial state [14], Proposition on local actuation with the initial state retained. The actual field's characteristic function controls the joint state error with references retained, and a centred Gaussian source changes by the classical full retarded response. Its force work is charged; an old mean contributes a cross term. This supplies a quantum actuation/source bridge in the fixed-texture sector, not a new vacuum for the projected body or a proof of the moving clock's stress and thermal renewal.

A source can be derived from this driven material rather than specified as a separate emitter. In the corresponding flat Gaussian sector, ξ_Θ is the complete record-driven angle response, including its coupling to the carrier. Outside the compact contact, the actual right and left profiles satisfy

$$\begin{aligned} \xi_R(u) &= -\frac{1}{2}\int a(y)\xi_\Theta(y, u+y)dy, & \xi_L(v) &= \frac{1}{2}\int a(y)\xi_\Theta(y, v-y)dy, \\ \Delta T_{uu} &= (\partial_u \xi_R)^2, & \Delta T_{vv} &= (\partial_v \xi_L)^2. \end{aligned} \tag{114}$$

Here $u = t - x$, $v = t + x$, the past displacement is zero and the initial Gaussian mean is zero. The carrier equation is $(\partial_t^2 - \partial_x^2)\xi_\phi = \partial_x(a\xi_\Theta)$. Its retarded Green function, integrated once in y, proves the two profiles. Unchanged covariance and common vacuum subtraction prove the stress identities. The integrated two-side carrier radiation is at most the actual force work; the rest is retained material energy. Different finite drive records

can therefore change the source through actual material motion. This is not injective: opposite drives have equal quadratic stress on centred input, and $a = 0$ emits no carrier source while material energy can remain. Putting these flat-background data into JT requires a common quantum-state and coordinate map plus the driver, moving support and clock's stress; the formulas alone do not supply that map.

Finite autonomous execution also needs a source certificate for this material, rather than the free-carrier estimate. On a fixed Dirichlet interval, Q-RAIF C derives a relative form bound for each compact-time quadratic energy or exterior null-stress test O:

$$\|(H+\Lambda)^{-1/2}O(H+\Lambda)^{-1/2}\| \le C_O, \qquad H = d\Gamma(\sqrt{K}), \quad \Lambda > 0.$$

The complete Gaussian mode tower and its common vacuum subtraction are retained. If actual and target output purifications have $e = \|\psi - \varphi\|$, $g_H = \|H(\psi - \varphi)\|$, then

$$|\langle O\rangle_\psi - \langle O\rangle_\varphi| \le C_O\sqrt{\Lambda e^2 + e g_H} \times \left[\sqrt{\Lambda + \|H\psi\|} + \sqrt{\Lambda + \|H\varphi\|}\right]. \tag{115}$$

Energy-regular return under this same H therefore certifies a common output source reading, with old references retained. Uniform physical rms bounds M_a, M_i give $g_H \le M_a + M_i$, so the certificate converges with its state error despite the growing history clock. The bound does not use ordinary trace distance on an unbounded stress. It is an output comparison under the agreed subsequent evolution, not an account of the driver and clock stress during their interaction. Its constants depend on the finite interval. Matched source tests can then enter the response estimate of this paper only after the state, coordinate and total-source identification is supplied; neither a uniform real-line limit nor the full moving JT apparatus follows from this Gaussian certificate. The unbounded-time weighted moments in the JT source hypothesis also need their own tail control; finite smooth tests alone do not certify them.

A physical change of the coupling or texture changes this fluctuation operator. It must use the same measured material parameters and vacuum normalization as the original configuration. Independently setting each contact's vacuum energy to zero would also discard its response to those changes. Conversely, the bare logarithm supplies no finite universal gravitational price. A covariantly renormalized material functional Γ, with $T_{\mu\nu} = -2\delta\Gamma/(\sqrt{-g}\,\delta g^{\mu\nu})$ and $E_i = \delta\Gamma/(\sqrt{-g}\,\delta X^i)$ for its scalar arguments, obeys

$$\nabla_\mu T^\mu{}_\nu = E_A\partial_\nu X^A + E_\Phi\partial_\nu\Phi,$$

where E_A, E_Φ are its variational equations. On the material equations this retains the compensating dilaton force. The same functional must be varied for the material and gravity; a static Euclidean vacuum comparison cannot replace the causal, initial-state response of a controller.

F Supported apparatus and an acquired-record source return

F.1 A material support with its own conserved core

A boundary action gives its conjugate energy but not the matter that holds the boundary. A metric-only local apparatus is also blind to the dilaton on a fixed AdS_2 patch. We now

change the matter model explicitly, without applying the preceding boundary readout laws to this new apparatus. In natural units add a timelike worldline Γ to the JT action:

$$I = \frac{1}{2\kappa}\int \sqrt{-g}\,\Phi(R+2/L^2)\,d^2x + I_{\rm outer} + I_{\rm fields} - \int_\Gamma M(\Phi)\,d\tau,$$
$$M = m_{\rm rest} + U(\Phi) + \langle H_{\rm core}\rangle. \tag{116}$$

The usual outer boundary terms are included. Take $m_{\rm rest} > 0$ and $U \geq 0$. The core has worldline quantum action $\int d\tau\,[i\psi^\dagger D_\tau\psi - \psi^\dagger(m_{\rm rest}+U+H_{\rm core})\psi]$; its expectation provides a semiclassical worldline source; the apparatus's internal spatial geometry is not resolved in this approximation. Q-RAIF C constructs a finite time-independent, nonnegative $H_{\rm core}$ containing the battery, clock-dependent interactions, detector, retained body and every thermal output cell [14]. Its mean energy is conserved. Output cells remain inside the apparatus until a release operation is supplied.

Let δ_Γ be the covariant line delta and $M_\Phi = \partial M/\partial\Phi$. Variation gives

$$R + 2/L^2 = 2\kappa M_\Phi \delta_\Gamma, \tag{117}$$
$$E_{\mu\nu} := g_{\mu\nu}\Box\Phi - \nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Phi/L^2 = \kappa T^{\rm total}_{\mu\nu}. \tag{118}$$

The material stress is not separately divergence-free:

$$\nabla_\mu T^{{\rm total}\,\mu\nu} = -M_\Phi\delta_\Gamma\nabla^\nu\Phi, \qquad \nabla_\mu E^{\mu\nu} = -\tfrac{1}{2}(R+2/L^2)\nabla^\nu\Phi. \tag{119}$$

The identities agree by Eq. (117), including energy exchange with the dilaton. Before radiation forces are added, $Ma^\mu = -(g^{\mu\nu}+u^\mu u^\nu)\nabla_\nu M$. Direct dilaton coupling therefore changes the curvature equation at the wall.

Proposition F.1 (Positive static support)**.** *In a symmetric thin-wall limit choose a common normal toward increasing dilaton. At a static wall let $\Phi = \phi > 0$, with derivatives $d_\pm = \partial_n\Phi|_\pm$ and extrinsic curvatures $K_\pm$. The vacuum junction and constraint equations are*

$$[d] = -\kappa M, \quad [K] = -\kappa M_\Phi, \quad K_\pm d_\pm = \phi/L^2, \quad M\overline{K} + M_\Phi\overline{d} = 0. \tag{120}$$

For $M > 0$ and $D > \kappa M/2$, they have the solution

$$d_\pm = D \mp \kappa M/2, \qquad K_\pm = \frac{\phi}{L^2 d_\pm}, \qquad M_\Phi = -\frac{\phi M}{L^2 d_+ d_-}.$$

A positive quadratic material potential realizes these data whenever its stiffness a obeys $M - M_\Phi^2/(2a) > \langle H_{\rm core}\rangle$. If a positive inner invariant is required, also choose $\phi > Ld_-$. For every finite conserved core mean energy, a choice of M, D and sufficiently large a gives positive rest mass and linear radial stability.

Proof. In Gaussian normal coordinates $ds^2 = dr^2 - N(r)^2dt^2$, $R = -2N''/N$ gives the curvature jump. The singular tt metric equation gives $[d] = -\kappa M$, and the regular rr equation gives $Kd = \phi/L^2$. Its jump gives the averaged force equation. The displayed data satisfy all four conditions. Choose

$$\phi_* = \phi - M_\Phi/a, \quad U(\Phi) = \tfrac{a}{2}(\Phi-\phi_*)^2, \quad m_{\rm rest} = M - \frac{M_\Phi^2}{2a} - \langle H_{\rm core}\rangle > 0.$$

The regular patches near the wall are $N_\pm = \cosh(r/L) + LK_\pm\sinh(r/L)$ and $\Phi_\pm = \phi\cosh(r/L) + Ld_\pm\sinh(r/L)$. They have positive lapse there and solve the vacuum equations. □

For $L = 1$, $\kappa = .01$, $M = 10$, $\phi = 3$, $D = 2$ and $a = 100$,

$$d_+ = 1.95, \quad d_- = 2.05, \quad \mathcal{C}_+ = 5.1975, \quad \mathcal{C}_- = 4.7975, \qquad \mathcal{C}_\pm = \phi^2 - L^2 d_\pm^2.$$

Here $M_\Phi = -7.50469043$, $\phi_* = 3.07504690$, and $m_{\rm rest} + \langle H_{\rm core}\rangle = 9.71839811$. The two-cycle core with 39 battery levels has mean 2.34, including its clock-dependent interaction, so $m_{\rm rest} = 7.37839811 > 0$. The 135-level core has mean 7.14 and still fits with positive rest mass. Support and a running finite internal circuit thus have jointly satisfiable semiclassical source data, with clock and battery included in the mass.

The reference support is locally stable in radial vacuum dynamics. On the positive-normal branch let $\beta_\pm = \sqrt{\dot\Phi^2 + (\Phi^2 - \mathcal{C}_\pm)/L^2}$. The moving-wall junction $\beta_- - \beta_+ = \kappa M(\Phi)$ gives

$$\dot\Phi^2 + V(\Phi) = 0, \qquad V = \frac{\Phi^2 - \mathcal{C}_+}{L^2} - \left[\frac{\mathcal{C}_+ - \mathcal{C}_-}{2\kappa L^2 M(\Phi)} - \frac{\kappa M(\Phi)}{2}\right]^2. \tag{121}$$

At the reference, $V = V' = 0$, $V'' = 388805871/5681780 > 0$, and the proper-time linear frequency is 5.84937147. Increasing $\mathcal{C}_+$ by 10^{-5} gives a small oscillation between approximately 2.99755 and 3.00244. This proves linear radial stability at one vacuum support, not stability through an arbitrarily long export history.

The stability construction extends beyond the numerical example. At the supported point, differentiation of Eq. (121) with fixed $\mathcal{C}_\pm$ gives

$$V''(\phi) = \frac{2}{L^2} - \frac{2\phi^2}{L^4 d_+^2} - \frac{4D\phi^2}{L^4 d_+ d_-^2} + \frac{2d_+ d_-}{M}\, a. \tag{122}$$

Choose $M > \langle H_{\rm core}\rangle$ and $D > \kappa M/2$ first. The coefficient of a is positive, so finite stiffness can satisfy both $m_{\rm rest} > 0$ and $V'' > 0$. This supports any fixed finite compiled core; increasing accuracy or operating length may require a larger material device.

F.2 An output pulse must recoil against its source

A thermal cell becomes radiation only through a release mechanism. Even an idealized event must conserve momentum as well as energy. Suppose the initially static wall emits an inward null pulse of energy q in its initial local rest frame. Take an isolated instantaneous event with $0 < q < M/2$, with no external holding impulse. Local four-momentum conservation fixes

$$M_f = \sqrt{M^2 - 2Mq}, \quad v_f = \frac{q}{M-q}, \quad \dot\Phi_f = \frac{d_+ q}{M_f}, \quad \beta_{+,f} = \frac{d_+(M-q)}{M_f}. \tag{123}$$

Proposition F.2 (The source–support charge balance)**.** *Keeping $\mathcal{C}_+$ fixed and imposing the moving-wall junction after the event gives*

$$\mathcal{C}_{-,f} - \mathcal{C}_- = 2\kappa L^2 d_- q. \tag{124}$$

The wall's Killing charge $E_\Gamma = (\mathcal{C}_+ - \mathcal{C}_-)/(2\kappa L^2)$ loses exactly $d_- q$, the pulse's initial interior Killing energy. For an internal mass drop $\Delta = M - M_f$, $q = \Delta - \Delta^2/(2M)$ and recoil energy is $\Delta^2/(2M)$.

Proof. Initially $E_\Gamma = Md_+ + \kappa M^2/2$; after emission it is $M_f\beta_{+,f} + \kappa M_f^2/2$. Equation (123) gives their difference $(d_+ + \kappa M)q = d_- q$. Solving the mass shell for q gives the last claim. □

At $q = .06$ in this support, $M_f = 9.93981891$ and $\Delta\mathcal{C}_- = .00246$. Reducing mass while pinning the wall would introduce the spurious term $-\kappa^2 L^2 q^2$. The inner horizon entropy has infinitesimal response

$$\left.\frac{dS_{h,-}}{dq}\right|_{q=0} = \frac{2\pi L^2 d_-}{\sqrt{\mathcal{C}_-}}, \qquad S_{h,-} = \frac{2\pi}{\kappa}\sqrt{\mathcal{C}_-}.$$

The number $q = .06$ is an emission example, not the finite core's net heat automatically sent into the bulk. Its cells contain gross energy and an initial thermal contribution. The reference and all correlated ports must be transported before an excess source can be identified with charged heat. The finite energy-sector gates do not establish a local field release Hamiltonian or prepare a KMS exhaust on a running horizon clock. This branch supplies a positive support and its charge balance while leaving those transport operations explicit.

F.3 Continuous radiation moves the supported apparatus

The mass, recoil and radiation can be evolved together. On each side of the material wall use an ingoing Eddington–Finkelstein coordinate:

$$ds_\pm^2 = -f_\pm dv_\pm^2 + 2dv_\pm d\Phi, \qquad f_\pm = \frac{\Phi^2 - \mathcal{C}_\pm(v_\pm)}{L^2}.$$

Keep the normal toward increasing Φ and write $w = d\Phi/d\tau$, $\beta_\pm = \sqrt{w^2 + f_\pm}$. Let $P_{\rm in}$ be the proper power of inward radiation absorbed from the outer side, and $P_{\rm out}$ the inward power sent into the inner side. These are the actual renormalized source powers; an excess over a thermal reference is specified separately below. Put $M = m_{\rm rest} + U(\Phi) + e$ and $A = U'(\Phi)$, where e is usable internal mean energy. The mean worldline approximation is that of Eq. (116); junctions with dilaton-coupled surface matter are standard gravitational constructions [6, Sec. V].

Proposition F.3 (Radiating material junction)**.** *For smooth supplied powers and initial data with $M > 0$, $\beta_- - \beta_+ = \kappa M$ and $\beta_\pm - w > 0$, the coupled equations are*

$$\begin{aligned}
\dot e &= P_{\rm in} - P_{\rm out}, & \dot M &= Aw + P_{\rm in} - P_{\rm out},\\
\dot w &= \frac{\beta_+ P_{\rm out} - \beta_- P_{\rm in} - A\beta_+\beta_-}{M} - \frac{\Phi}{L^2}, & &\\
\dot{\mathcal{C}}_+ &= 2\kappa L^2 P_{\rm in}(\beta_+ - w), & \dot{\mathcal{C}}_- &= 2\kappa L^2 P_{\rm out}(\beta_- - w),\\
\dot v_\pm &= (\beta_\pm - w)^{-1}. &&
\end{aligned} \tag{125}$$

Dots denote proper time. These equations preserve the mass junction and obey $[K] = -\kappa A$. They have a local solution while the stated domain and energy reserves hold.

Proof. The regular JT equation gives $T_{v_\pm v_\pm} = \partial_{v_\pm}\mathcal{C}_\pm/(2\kappa L^2)$. Its contraction with the wall velocity gives the two invariant equations. The tangent and normal projections of

the material force balance give $\dot{M} - Aw = P_{\rm in} - P_{\rm out}$ and $M\overline{K} + A\overline{\beta} = P_{\rm out} - P_{\rm in}$. Direct evaluation of the extrinsic curvature gives

$$K_+ = \frac{\dot{w} + \Phi/L^2 + \kappa P_{\rm in}}{\beta_+}, \qquad K_- = \frac{\dot{w} + \Phi/L^2 + \kappa P_{\rm out}}{\beta_-}.$$

Their jump and the mass junction give Eq. (125). Differentiating $\beta_- - \beta_+ - \kappa M$ using these equations gives zero. The right-hand sides are smooth inside the domain, giving local existence and uniqueness. Positivity of quantum excess stress is not used in this algebraic consistency statement. □

For $P_{\rm in} = 0$, $R = \sqrt{\mathcal{C}_-} > 0$ and $d\eta/dv_- = R/L^2$, the actual clock conversion is

$$z := \frac{d\tau}{d\eta} = \frac{L^2(\beta_- - w)}{R}, \qquad \frac{dS_{h,-}}{d\eta} = 2\pi z^2 P_{\rm out}. \tag{126}$$

Recoil enters both the source and the clock; keeping the wall fixed while its mass decreases does not solve these equations.

F.4 Stored mechanical energy, radiation and the moving support

The angle need not be specified externally. Q-RAIF C gives a positive inertial current Hamiltonian with a storage coordinate, its retarded reaction and force spectrum [14, Section on an inertial contact]. A displacement selected by a record changes the actual excess current stress, not merely a reconstructed parameter. That local rest-frame model still needs its infrared representation, moving ports and complete quantum stress to be matched to the material action. In particular, $\gamma\langle\dot{\theta}^2\rangle$ alone is not its net heat: the noise–velocity correlation contributes to the work.

A distinct classical branch makes a common moving action explicit. Put a real massless scalar on each side of the wall, with actions $-\frac{1}{2}\int_{\mathcal{M}_\pm}\sqrt{-g}(\nabla\varphi_\pm)^2$. Append the mechanical worldline action

$$S_\Gamma = \int d\tau \left[-m_{\rm rest} - U(\Phi) + \frac{I}{2}\dot{\theta}^2 + \frac{J}{2}\dot{y}^2 - \frac{k}{2}(\theta - y)^2 - \frac{K}{2}y^2\right], \qquad I, J, k, K > 0, \tag{127}$$

and impose $\varphi_\pm|_\Gamma = g_\pm\theta$ with worldline Lagrange multipliers. The wall, storage and angle are all varied. These are full nonchiral scalar fields, not two auxiliary right movers reinterpreted as an anomaly-free bulk theory. With no classical incoming waves, their outgoing solutions are $\varphi_\pm(\xi_\pm) = g_\pm\theta(\tau_{\rm ret})$. Their proper powers and the mechanical equations are

$$\begin{aligned} P_\pm &= g_\pm^2\dot{\theta}^2, & \gamma &= g_+^2 + g_-^2, \\ I\ddot{\theta} + k(\theta - y) &= -\gamma\dot{\theta}, & J\ddot{y} + k(y - \theta) + Ky &= 0, \\ e &= \tfrac{I}{2}\dot{\theta}^2 + \tfrac{J}{2}\dot{y}^2 + \tfrac{k}{2}(\theta - y)^2 + \tfrac{K}{2}y^2, & \dot{e} &= -P_+ - P_-. \end{aligned} \tag{128}$$

Boundary variation gives the mechanical force; characteristics give the powers. There is no independent supplied radiation function. Incoming waves would add their force and work and are excluded in this branch. Quantizing a moving boundary also requires its vacuum stress and fluctuations; the classical formulas do not supply them.

The outer radiation now moves outward, unlike the outer input of Eq. (125). Use outgoing EF coordinate u_+ outside and ingoing v_- inside. Set $M = m_{\rm rest} + U + e$, adding any fixed probe offset, and keep $\beta_- - \beta_+ = \kappa M$. The same action gives

$$\begin{aligned} \dot w &= \frac{\beta_+ P_- - \beta_- P_+ - U'\beta_+\beta_-}{M} - \frac{\Phi}{L^2}, \\ \dot{\mathcal{C}}_+ &= -2\kappa L^2 P_+(\beta_+ + w), & \dot{\mathcal{C}}_- &= 2\kappa L^2 P_-(\beta_- - w), \\ \dot u_+ &= (\beta_+ + w)^{-1}, & \dot v_- &= (\beta_- - w)^{-1}. \end{aligned} \tag{129}$$

The opposite invariant signs follow from $T_{u_+u_+} = -\partial_{u_+}\mathcal{C}_+/(2\kappa L^2)$ and $T_{v_-v_-} = \partial_{v_-}\mathcal{C}_-/(2\kappa L^2)$. The tangent Ward equation is $\dot M = U'w - P_+ - P_-$ and the normal equation is $M\overline{K} + U'\overline{\beta} = P_- - P_+$. Here $K_\pm = (\dot w + \Phi/L^2 + \kappa P_\pm)/\beta_\pm$. These identities give both $[K] = -\kappa U'$ and conservation of the mass junction. They do not follow by putting a negative outer input into Eq. (125); the null direction also changes.

Proposition F.4 (A finite mechanical-source window)**.** *Equations* (128) *and* (129) *have a unique local solution in the domain* $M > 0$, $\beta_\pm > 0$, $\beta_+ + w > 0$, $\beta_- - w > 0$, $\mathcal{C}_- > 0$. *They preserve the mass junction,* $e \ge 0$, *and*

$$\int_0^\tau (P_+ + P_-)\, ds = e(0) - e(\tau) \le e(0). \tag{130}$$

For every fixed finite proper-time window, positive static-support data admit a sufficiently small positive coupling with nonzero radiation throughout a nontrivial subinterval and a solution inside that domain.

Proof. The right-hand side is smooth in the stated domain. Direct differentiation of the junction gives zero, and the positive quadratic mechanical energy obeys the displayed budget. At $g_\pm = 0$ the supported geometry is static and the internal oscillator energy is conserved. Continuous dependence on the couplings preserves its strict domain margins on any specified compact time interval for sufficiently small positive $g_\pm^2$. Nonstationary mechanical data give nonzero angle velocity. □

Balanced outgoing power also gives a necessary operating-lifetime bound for this same apparatus. If its fixed operating domain satisfies $0 < \Phi \le \Phi_{\max}$, $\Phi^2 > \mathcal{C}_-$ and $\beta_- - \dot\Phi \ge b_{\min} > 0$, then

$$\begin{aligned} \mathcal{C}_-(\tau) - \mathcal{C}_-(0) &\ge \kappa L^2 b_{\min} Q_{\rm out}(\tau), \\ Q_{\rm out}(\tau) &< \frac{\Phi_{\max}^2 - \mathcal{C}_-(0)}{\kappa L^2 b_{\min}}. \end{aligned}$$

Equation (129) integrates this inequality for $P_- = P_+ = P/2$. If each complete episode exports at least $q_{\min} > 0$, its count is below the latter bound divided by $q_{\min}$, as well as limited by the installed program and prepared bank. The two acquisitions in Foundation III are one episode, not two renewals. Enlarging the domain or changing its null-clock margin changes this necessary bound; it is not a general impossibility theorem.

The common energy and geometry can be held fixed across records. Use $I = 10$, $J = 20$, $k = 1$, $K = .1$, $g_+^2 = g_-^2 = 1/(4\pi)$ and $e(0) = .95$. Install $\theta(0) = \dot\theta(0) = 0$, $y(0) = A_R$ and $\dot y(0) = \sqrt{2[.95 - .55A_R^2]/J}$ for $A_R = .5, 1.3$. Both records then use the same mass-10

support, potential and initial invariants of Section F.1. With the positive probe offset .02 below, their common m_{rest} is 8.74839811. At proper time 80 the two solutions give

A_R	.5	1.3
$\int (P_+ + P_-)d\tau$	.428637558	.522051307
$e(80)$	.521362442	.427948693
$\mathcal{C}_-(80) - \mathcal{C}_-(0)$	.008793748	.010712134
$\Phi(80)$	3.003737133	3.004545378

(131)

Both remain outside the inner horizon with positive reserves. Integrating both invariants independently preserves the junction to 8.3×10^{-15}; another integrator agrees to 3.7×10^{-12}. The phase probe below, at $\lambda = .02$, yields Y-outcome probabilities .503428742 and .504187486. The difference is created during evolution, not by a different initial mass or an externally selected source. The numerical 80-window check is not an interval-arithmetic domain proof.

The inner rapidity clock is determined by the same solution: $\dot{\eta} = \sqrt{\mathcal{C}_-}/[L^2(\beta_- - w)]$ and $dS_{h,-}/d\eta = 2\pi(d\tau/d\eta)^2 P_-$. This source is coherent classical radiation, not an η-KMS heat output, so its horizon response cannot be substituted into the thermal entropy bound. Installing the mechanical record has a preparation cost. The finite return in Section F.6 instead lets an acquired quantum record exert force on the same storage coordinate. Preparing these initial mechanical data and renewing the whole apparatus are distinct from that return.

Proposition F.5 (A rest-frame mean-stress bridge)**.** *Use Q-RAIF C's quadratic current–mechanical contact on a fixed inertial line, with the same I, J, k, K and a common centered Gaussian state for mechanics and incoming currents. Displace only the mean mechanical data; take the mean initial angle to be zero. Keep all centered correlations and old references. For independent physical outgoing directions $s = \pm$, let the contact coefficients obey $\kappa_s = \sum_i q_{si}^2$ and $\gamma_s = \kappa_s/(4\pi)$, where q_{si} are the effective angle-coupling coefficients, not a change of the carrier's physical charge sectors. Assume the renormalized current products exist on the common domain. In the zero-width limit of the retarded mean solution, away from initial/contact singularities, the outgoing stress difference and mean mechanical equations are*

$$\Delta\langle T_{\mathrm{out},s}\rangle = \gamma_s v(\tau_{\mathrm{ret}})^2, \qquad I\ddot{\theta} + k(\theta - y) = -(\gamma_+ + \gamma_-)\dot{\theta}, \quad v = \dot{\theta}. \tag{132}$$

Thus $g_s^2 = \gamma_s$ identifies this excess mean stress and its mechanical energy loss with Eq. (128) *in that fixed local frame. The balanced scalar example requires $\kappa_+ = \kappa_- = 1$. Q-RAIF C's unscaled two-flavor bidirectional contact has $\kappa_+ = \kappa_- = 2$ and twice that total damping; scaling its angle coupling by $1/\sqrt{2}$ gives the balanced normalization.*

Proof. Take $f_a(x) = a^{-1}f(x/a)$ with nonnegative $f \in C_c^\infty(\mathbb{R})$ and $\int f = 1$. At finite width the displacement adds the classical linear solution without changing the common covariance. In direction s its current mean shift is $q_{si}h_a/(2\pi)$, where h_a is the retarded convolution of v with f_a along that characteristic. The Sugawara stress $\pi \sum_i : j_{si}^2 :$ has zero cross term with the centered baseline. Its difference is therefore $\gamma_s h_a^2$. Eliminating the mean currents gives the retarded kernel $\Gamma_{a,s}(t) = \kappa_s \int f_a(x) f_a(x - t)dx/(2\pi)$, whose one-sided integral is γ_s. For smooth mean histories its convolution error for a smoothly extended velocity is bounded by $\|v'\|_\infty \int_0^\infty t\Gamma_{a,s}(t)dt = O(a)$. For the initial-value solution, the initial kernel tail has integrated size $O(a)$ as well. The common kernel mass bounds

the linear Volterra solution uniformly on fixed windows; Gronwall then gives convergence to the displayed damped equation. Beyond the contact, h_a tends to the retarded velocity, proving the stress statement. The resulting positive quadratic displacement energy obeys $\dot{e} = -(\gamma_+ + \gamma_-)v^2$, the scalar-source balance. Incoming noise–velocity work and centered quantum stress remain in the common baseline, rather than being set to zero. □

This identifies an excess mean radiation channel. Absolute quantum stress, a quantum zero-width channel limit, dilaton-dependent mass, moving-frame work, physical support and recoil, acquired-record spring forces and stress fluctuations require further matching. In particular a statistical mixture of mechanical records has branch covariance and cannot be replaced by one Gaussian displacement in this proposition.

F.5 A quantum phase reads the moving dilaton

The material model also permits a direct finite sensor. For $\lambda > 0$, add a probe with

$$H_P(\Phi) = \epsilon_P I + \lambda(\Phi - \phi_0)Z_P, \qquad \epsilon_P > \lambda \sup_\Gamma |\Phi - \phi_0|, \tag{133}$$

initially in $|+\rangle$. This is the same kind of explicit dilaton coupling as the support, now in an internal quantum degree of freedom. It does not follow from metric-only matter. During the reading, $\langle Z_P \rangle = 0$ and Z_P is conserved. In the mean worldline approximation its energy is ϵ_P and its mean additional dilaton force is zero. Count ϵ_P in the mass, reducing m_{rest} by that amount in the displayed examples. The probe's phase gives

$$\Pr(Y_P = +1) = \frac{1 + \sin[2\lambda \int (\Phi - \phi_0) d\tau]}{2}. \tag{134}$$

With $\lambda = \epsilon_P = .02$, the $q = .01$ and $.02$ histories give .50617417 and .51230214 at the same proper duration. Their actual bit laws have total variation distance .00612797. No numerical dilaton value is supplied to the controller.

The terminal read itself can be implemented. Q-RAIF C gives a positive autonomous clock whose last two gates write the actual Y result into a record, then apply a true SWAP with a paid coherent blank [14]. The probe is renewed; all old correlations move into the blank and record. For a supplied finite $\Phi(\tau)$ trajectory, its joint error relative to the actual phase at $\tau_0 < T$ is bounded by

$$d_{\text{joint}} \le 2A\tau_0\sqrt{P_0} + 2\sqrt{P_0} + \lambda D(T - \tau_0), \quad D = \sup |\Phi - \phi_0|, \tag{135}$$

where P_0 is the clock's terminal binomial tail and A is the sum of its last two edge strengths. Finite resources suffice for each specified error; the clock mean energy is Jm, and must enter the supported mass. The sufficient $m = 10^5$, $T = 80$, $\tau_0 = 78$ example uses clock mean energy 1963.4954, not the mass-10 apparatus above. During writing and renewal the mean additional force need not vanish: the material equation uses $A_{\text{mat}} = U' + \lambda\langle Z_P \rangle$, with $|A_{\text{mat}} - U'| \le \lambda$. The probe, clock, blank and record belong in the same matter action. If e now includes their Φ-dependent internal energy, its equation is $\dot{e} = \lambda\dot{\Phi}\langle Z_P \rangle + P_{\text{in}} - P_{\text{out}}$; then $\dot{M} = A_{\text{mat}}\dot{\Phi} + P_{\text{in}} - P_{\text{out}}$ as required by the material Ward balance.

A smaller clock can be included directly in this mean material action, instead of using the large worst-case certificate. Take $m = 128$, $J = \pi/160$, $\epsilon_P = \lambda = .02$, and retain the complete 129-state clock, probe, record and blank throughout the proper-duration

80 evolution. Its initial internal mean energy is 2.5532741. Choose $m_{\mathrm{rest}} = 4.8251240$ so that the initial mass 10 already includes it and the declared source reserve 2.34. Evolve the joint Schrödinger equation for

$$H_{\mathrm{int}}(\Phi) = H_{\mathrm{hist}} + 2\epsilon_P I + \lambda(\Phi - \phi_0)Z_P$$

with the radiating junction and its actual η clock. The two contact outputs $\chi = .5$ and 1, at hot-input excess .02, give

q_χ	actual $\Pr(R = 1)$	final source reserve	probe reset distance
.010	.50529766	1.42403868	.00075428
.020	.51055894	.50256908	.00150041

The last column is the distance to the closest pure product of a renewed probe and a normalized retained internal reference, not an error certificate against the ideal meter's entire reference state. The actual record laws differ by .00526128. Norm, mass-junction and internal dilaton-work residuals are below 4×10^{-11} in this integration; a second integrator agrees on the displayed trajectory and record values within 1.3×10^{-9}. These are numerical checks, not interval-certified error bounds. For these gates and initial data, the vanishing mean force has an exact reason.

Proposition F.6 (A force-neutral read and replacement)**.** *Let K denote conjugation in the clock and computational bases and set*

$$\mathsf{C} = (-1)^{N_{\mathrm{clock}}} X_P X_B K, \qquad H_0 = H_{\mathrm{int}} - (Jm + 2\epsilon_P)I.$$

For the identity-edge clock with terminal C_Y and true SWAP, $\mathsf{C}H_0(\Phi)\mathsf{C}^{-1} = -H_0(\Phi)$ and $\mathsf{C}Z_P\mathsf{C}^{-1} = -Z_P$. With initial clock state 0 and data $|+\rangle_P|0\rangle_R|+\rangle_B$, every real dilaton trajectory has $\langle Z_P\rangle = 0$ and $\langle H_{\mathrm{int}}\rangle = Jm + 2\epsilon_P$ throughout the read.

Proof. Conjugation exchanges the two Y projections, and X_P exchanges them again. Thus $X_P X_B C_Y^* X_P X_B = C_Y$; the real SWAP commutes with $X_P X_B$. Clock parity reverses every off-diagonal edge, while X_P reverses Z_P. This proves both operator identities. Remove the constant Hamiltonian phase. Since C is antiunitary, it reverses both i and H_0 in the Schrödinger equation, and so maps its solution to another solution with the same initial state. Uniqueness makes that solution C-invariant. Expectations of the two odd Hermitian operators must vanish. □

The clock energy is therefore included from the start without changing the previously supported mean trajectory. This does not set the quantum force operator or its fluctuations to zero, and different blanks or gates need not have the symmetry. The general coupled force equation, not a zero-force constraint, was used in the numerical calculation. The thermal reservoir's quantum degrees, both-port material stress and its local preparation in η are not thereby included in that body.

The same finite meter also reads the mechanical-source branch of Section F.4. Its complete mean energy 2.5532741, rather than just the probe offset, is included from the start by using the common positive rest mass 6.21512398 for both equal-energy records. Proposition F.6 leaves their mean mechanical and geometric trajectories unchanged. Their actual acquired record probabilities are .502991099 and .503655046, a total variation distance .000663947. Keeping clock, blank and record gives distances

.000376618 and .000458340 to the closest renewed-probe pure product with its normalized internal reference. These are not errors against an ideal reference state. The full vector evolution preserves norm to 1.6×10^{-11} and internal mean energy to 2.6×10^{-12}; a second integration agrees on the record probability to 3.1×10^{-14}. This finite episode obtains the record through physical gates, rather than reading the numerical solution into a controller. In this force-neutral example the record does not act on the mechanics. The next section adds that interaction and includes its reaction during acquisition. The scalar source has no thermal reservoir whose regeneration could be inferred from either construction.

For input streams jointly independent of the record and prepared in a common clock frame, the record can select either of two contact angles in the next stationary window. If their transmissions are $\chi_0 \neq \chi_1$, its source-current mean is

$$q_{\mathrm{next}} = q_{\mathrm{hot}}[\chi_0 + (\chi_1 - \chi_0)\Pr(R = 1)].$$

It depends on the actual phase bit, not a supplied numerical dilaton. The angle change's pump work in Eq. (110) is also charged. This gives the conditional response-to-source map; it does not yet autonomously prepare the reservoirs or renew the entire controller. Quantum fluctuations of the probe force also lie outside the mean-source approximation. The phase sensor and the earlier fixed-boundary current sensor are different realizations, not two freely composable laws for one apparatus.

F.6 An acquired record drives a new material source

A record can act without first being converted into a supplied mechanical initial condition. Keep the two classical scalar fields and the material action of Section F.4, and append $i\langle\Psi, \dot{\Psi}\rangle - \langle\Psi, H_c\Psi\rangle$ to its proper-time Lagrangian. The normalized state Ψ contains a finite history clock, probe P, record R, replacement blank B and an old purification Q. Set

$$H_c(\Phi, y) = H_{\mathrm{hist}} + E_{\mathrm{off}}\mathbf{1} + \lambda(\Phi - \phi_0)Z_P + \frac{\chi_R}{2}P_R(y-d)^2, \qquad P_R = |1\rangle\langle 1|_R, \quad \chi_R \geq 0. \tag{136}$$

H_{hist} is the positive gate-conjugated clock of Q-RAIF C. Its first nonidentity edge performs C_Y, the next performs a true SWAP of P and B, and the remaining edges are identities. There is no proper-time switch. Initially $R = 0$, so the spring has zero expectation. The actual acquisition opens its force channel. $H_c \geq 0$ while $E_{\mathrm{off}} > |\lambda(\Phi - \phi_0)|$. The target d, couplings and initial clock are prepared hardware. The scalar fields, mechanical coordinates and geometry use an Ehrenfest mean-action approximation [61, Section 2]; the finite quantum core and its old reference are retained in full, without sampling or collapsing R.

Let $p = \langle P_R\rangle$, $z = \langle Z_P\rangle$ and $E_c = \langle H_c\rangle$. Variation of this one action gives

$$\begin{aligned} i\dot{\Psi} = H_c\Psi, \qquad & F_y = \chi_R(y-d)p, \qquad & F_\Phi = \lambda z, \\ J\ddot{y} + k(y-\theta) + Ky = -F_y, \qquad & \dot{e} = -P_+ - P_- - F_y\dot{y}, \qquad & \dot{E}_c = F_y\dot{y} + F_\Phi w. \end{aligned} \tag{137}$$

The angle equation and $P_\pm = g_\pm^2\dot{\theta}^2$ are unchanged. Here p is an expectation obtained by variation, not a decoded number fed to a separate control policy. The mass and its tangent balance are

$$M = m_{\mathrm{rest}} + U + e + E_c, \qquad \dot{M} = (U' + F_\Phi)w - P_+ - P_-. \tag{138}$$

Consequently Eq. (129) holds with U' replaced by $U' + F_\Phi$ in its acceleration equation; both invariant and null-clock equations retain their directions. Differentiating the mass junction with these equations gives zero. The separate work balances are

$$e(\tau) + Q_{\text{out}}(\tau) + \int_0^\tau F_y \dot{y}\, ds = e(0), \tag{139}$$
$$E_c(\tau) - E_c(0) = \int_0^\tau (F_y \dot{y} + F_\Phi w)\, ds.$$

The clock interaction and conditional spring are included in E_c. The spring breaks the force-neutral symmetry used above; F_Φ cannot be set to zero in this coupled calculation.

Proposition F.7 (A nonzero record-mediated source return)**.** *Take two initial geometries with the same $\Phi = \phi_0$, $w = 0$, mass, mechanical data and quantum input, but different $\mathcal{C}_-$; set $\mathcal{C}_+$ by the mass junction. In the positive exterior domain, either $\lambda = 0$ or $\chi_R = 0$ makes their proper emitted powers identical. With both interactions present, there are positive finite-clock data for which the emitted powers differ on a nonempty finite interval. Their record, mechanical response, source and geometry obey the same action and work account.*

Proof. For $\chi_R = 0$, the mechanical subsystem is independent of the reader and geometry. For $\lambda = 0$, the joint mechanical and quantum equations are independent of geometry. Either condition fixes identical mechanical sources for the two initial geometries.

For a nonzero example choose $\theta(0) = \dot{\theta}(0) = 0$, $y_0(y_0 - d) \neq 0$ and probe state $(I + \upsilon X_P)/2$, $\upsilon \neq 0$, with a blank R and its old purification held. Let $r \geq 1$ be the C_Y edge, $a_j > 0$ the clock edge strengths and $G_r = \prod_{j=1}^r a_j$. The initial acceleration difference δa follows from the junction equations, so that $\delta\Phi(t) = \delta a\, t^2/2 + O(t^3)$. The shortest-path Dyson expansion gives

$$\delta p(t) = K_r t^{2r+3} + O(t^{2r+4}), \qquad K_r = \frac{\upsilon G_r^2 \lambda \delta a}{6r!(r-1)!(r+3)}. \tag{140}$$

An insertion of $\lambda\delta a\, t^2 Z_P/2$ after $j < r$ forward edges contributes the time integral $t^{r+3}(j+1)(j+2)/(r+3)!$. Their sum is $t^{r+3}r/[3(r+3)r!]$. Using $C_Y^* P_R C_Y = (I + Y_P)/2$ in the interference with the unperturbed shortest path gives the coefficient above. The insertion after all r edges has Hermitian $P_R Z_P$ in that interference and zero real part. Further graph paths and spring insertions raise the order; the spring annihilates the blank record before acquisition.

Integrating the mechanical equations then yields

$$\delta(P_+ + P_-)(t) = -\frac{2\gamma k^2 y_0 \chi_R (y_0 - d) K_r}{I^2 J (2r+4)(2r+5)(2r+6)} t^{2r+7} + O(t^{2r+8}). \tag{141}$$

The baseline angle velocity is $k y_0 t/I + O(t^2)$. Different initial $\mathcal{C}_-$ give a nonzero δa for generic supported data, since initially $\dot{w} = -U'\beta_+\beta_-/M - \phi_0/L^2$. The displayed coefficient is nonzero for the stated nonzero parameters. The coupled finite-dimensional equations are analytic on their strict positive exterior domain. Local existence and continuity therefore give a nonempty admissible family, not just a numerically resolved difference. □

A complete integration uses the earlier I, J, k, K, $\gamma = 1/(2\pi)$, $e(0) = .95$, $y_0 = .5$ and $d = -.5$, with $\lambda = .2$, $\chi_R = .05$ and $E_{\mathrm{off}} = .04$. The 33-state clock has edge scale $\pi/160$, $r = 3$, mean energy .628318531 and spectral maximum 1.256637061. The initial probe has $\upsilon = .8$, its two-dimensional old purification is retained, and B starts in $|+\rangle$. The quadratic support has stiffness 100 and $U'(3) = -30/(1.95 \cdot 2.05)$, as in the static mass-10 support. Both geometries have initial $\Phi = 3$, $w = 0$, $M = 10$ and $\mathcal{C}_- = 4.7975 \pm .1$, with the outer invariant fixed by the junction. The initial core energy .668318531 is already in the mass; the common rest mass is 8.100079577. At proper time 100:

$\mathcal{C}_-(0) - 4.7975$	$\Pr(R = 1)$	Q_{out}	$\mathcal{C}_-(100) - \mathcal{C}_-(0)$
$-.1$	.506528009	.518808111	.010786169
$+.1$	.496292018	.517763845	.010480933

The largest proper-power difference on this window is 1.0350613×10^{-4}; with either interaction removed it is below 3.2×10^{-16}. With feedback removed the two record laws still differ, while their scalar sources agree. Thus acquisition and actuation are separately tested. The nonzero-source result concerns the emitted power, not the already different initial geometries.

The full core–reference vector is evolved along with both invariants. Norm, junction and the three work balances, including U's dilaton work, have residuals below 3.7×10^{-14} in the primary integration. A separate tensor-neighbour implementation with another integrator agrees on the record and source to 6.0×10^{-13} and 2.0×10^{-15}, respectively, and on the mass to 1.2×10^{-8}. The largest probe force is about .0082 and is retained. Clock energy changes during acquisition; it is not an uncharged timer. Only the C_Y edge changes P_R, giving

$$|\dot{p}| \le 2a_r \|\Psi_{r-1}\| \|\Psi_r\|.$$

This bound is checked on the interacting clock, not a free-clock tail. It supplies finite record retention when that edge is sparsely occupied, not permanent halt. The construction closes an actual finite geometry–record–force–source return in the mean material branch. Full quantum recoil, covariant thermal transport and renewal of the clock, blank and reference remain required for the sustainable loop below.

F.7 A spatial return with a physical clock

The geometry and support below are semiclassical mean fields. The internal oscillators and both retained spins are quantum. The regulator is part of the physical action; the statement does not remove its bandwidth cutoff.

Take JT gravity with radius R and coupling $\kappa > 0$, a full material label a, and a full synchronization scalar θ, with

$$S_b = -\int \sqrt{-g}\, M \left[mn + \frac{K}{2}(n - n_0)^2 \right], \quad n^2 = (\nabla a)^2, \qquad S_\theta = -\frac{J}{2} \int \sqrt{-g} (\nabla \theta)^2. \tag{142}$$

Require $m > K n_0 > 0$, n near n_0, and a time function $\theta_t > 0$ with timelike gradient. Use the unrestricted inverse embedding $\theta(X(\tau, \xi)) = \tau$, $a(X(\tau, \xi)) = \xi$ and $h = X^* g$. Only the internal expansions are finite: $\sqrt{I}\chi = \sum q_{c\alpha} e_\alpha(\xi)$ and $u = \sum q_{u\alpha} e_\alpha(\xi)$, for smooth supported real e_α. Substitute them into the two scalar kinetic actions and the potential

$$V = \frac{I \mu^2 \chi^2}{2} + \frac{1}{2} \left[\omega u + \frac{A\chi}{\omega} \right]^2 + E_o \rho_o - D_o, \quad A = \lambda + g w Y_P + \gamma w_2 Y_S, \tag{143}$$

before Legendre transform and metric/embedding variation. Here

$$D_o = \frac{C_\chi}{\omega^2}(\lambda g w Y_P + \lambda\gamma w_2 Y_S + g\gamma w w_2 Y_P Y_S), \qquad C_\chi(\xi) = e^T Q_c e / I.$$

Q_c is the fixed prepared clock covariance. Choose the paid scalar offset to dominate $|D_o|$. Quantize the resulting Hamiltonian with Weyl ordering and a joint-state Berry term on the actual θ leaves. Its kinetic matrices are

$$C = -\int \sqrt{-h}\, h^{\tau\tau} e e^T, \quad D = -\int \sqrt{-h}\, h^{\tau\xi} e (e')^T, \quad K_{\rm grad} = \int \sqrt{-h}\, h^{\xi\xi} e' (e')^T,$$

giving $(p - Dq)^T C^{-1}(p - Dq)/2 + q^T K q/2$ plus the paid offset/calibration terms. A proper-number sensor contributes

$$H_S = \int n_S(\xi) N [E_S + h_s \Phi(X) Z_S]\, \mathrm{d}\xi, \quad N = \sqrt{-h_{\tau\tau}}, \quad \int n_S \mathrm{d}\xi = 1. \tag{144}$$

n_S is smooth, symmetric, and has nonzero width. Include both spins, their Berry terms, all profile reactions, and any old reference in the same retained state. Restrict $|h_s| \sup|\Phi| < E_S/2$. No profile is an external time-dependent signal.

Proposition F.8 (Finite-bandwidth spatial return)**.** *There are finite positive I, M, J, R, smooth finite-width preparations, and nonzero finite h_s, γ for which this one mean-action system has a positive local solution and two actual histories with the same initial geometric constraints, differing only in their retained P record, whose next-program probabilities for $(1 + Y_S)/2$ differ at the same central proper time. The retained Y_S enters the subsequent actual material Hamiltonian, with a nonzero mixed sensing/return response of the material mean acceleration. All clock, support, interaction and probe sources are metric variations of this action, and satisfy the full Ward identity. This is an existence result, not a uniform gain, state-independent CPTP map, continuum-controller or thermal-renewal statement.*

Proof. Prepare $P = |\pm Y_P\rangle$, $S = |+X_S\rangle$, a common free product Gaussian field state with $\langle q\rangle = 0$, zero symmetrized qp covariance, and clock momentum $\sqrt{I}\Omega F$. References are retained, not reset. At $\gamma = h_s = 0$, calibration cancels the initial expected record-dependent potential as a functional of X. Thus the complete initial metric stresses coincide, not just their integrals. Take common static AdS data, $\theta = t$ on a finite prepared plateau, and hydrostatic bar density $n = n_0 + (m/K)[\cos(x/R) - 1]$ on a small causal patch. The common initial JT constraint is a linear spatial equation for Φ; adding a sufficiently large finite positive homogeneous solution $B\sec(x/R)$ makes $\Phi > 0$. This changes initial geometric data, not the charged positive source.

We first check the physical time equation. At rest, varying $k = X^t_\tau$ gives $C = \int x' e e^T/k$, $K_{\rm grad} = \int k e'(e')^T/x'$ and $K_{\rm mass} = \int e^{2\rho} k x' e e^T \mathcal{M}_\pm$. Hence the local time-momentum current $\delta H/\delta X^t_\tau$ contains only q^2, p^2 and commuting spin constants. Its commutator with the initial Hamiltonian is Weyl qp; its expectation, and the mixed time/spatial current expectation, vanish. The corresponding initial mixed principal block vanishes too. The full time EL equation therefore gives $\theta_{tt}(0) = 0$, or $\theta = t + O(t^3)$. It changes no potential trace coefficient of order t^2. Nevertheless, if $\theta = t + \theta_3 t^3/6 + \cdots$, the actual $T_{\theta,tt} = T_{\theta,xx} = J/2 + J\theta_3 t^2/2 + \cdots$ is retained; only its trace is identically zero.

Here is a nonempty analytic trace certificate. Choose one normalized even smooth mode e, constant and positive in an open central neighborhood, $w = 1$ on its support, and

$F = 1$. Put $k = \int (e')^2$, $\beta = \int \sec^2(x/R) e^2$, $Q_c = [2\sqrt{k + \mu^2 \beta}]^{-1}$, $Q_u = [2\sqrt{k + \omega^2 \beta}]^{-1}$. The exact static central trace-contrast coefficient is

$$d_2 = -D_0 + B_1/I + B_2/I^2, \quad D_0 = 4\lambda g \Omega^2 e(0)^2/\omega^2,$$
$$B_1/D_0 = \beta[\mu^2 Q_c + \omega^2(Q_c + Q_u)]/\Omega^2, \tag{145}$$
$$B_2/D_0 = 2\beta(\lambda^2 + g^2) Q_c/(\omega^2 \Omega^2). \tag{146}$$

These follow by $[t^2]Q = -\{K_\pm - K_0, Q_0\}/2$; the off-diagonal species covariance is included. The initial spatial momentum current is locally proportional to $ee' \operatorname{sym}(qp)$, so its force contrast and spatial derivatives vanish on the central plateau. No bar recoil trace term of order $1/I$ survives there. For $(\mu, \omega, \lambda, g, \Omega) = (.3, 1.4, .6, .2, .8)$, $B_1/D_0 \le 6.432292\beta$ and $B_2/D_0 \le 1.062926\beta$. Take finite R with $\beta < 1.01$, and $I = 100$. Then $d_2 < -0.9349 D_0$. Scale $M = \alpha L$, $J = \beta_b L$ together, with fixed positive ratios. Actual common-acceleration corrections to (146) are $O_I(L^{-1})$; hydrostatic preparation cancels the order-M force, not the quantum reaction. A sufficiently large finite L preserves $d_{2,\text{total}} < -D_0/2$.

For completeness the actual coupled equations have a local solution. Let $\mathcal{A} = 1 + \sum(q_j^2 + p_j^2)$. A degree-two Hamiltonian gives $\|\mathcal{A}^s U_z \psi\| \le e^{C_s T} \|\mathcal{A}^s \psi\|$ and

$$\|\mathcal{A}^2 (H_z - H_{\hat{z}})\psi\| \le C \|z - \hat{z}\|_{H^{r-1}} \|\mathcal{A}^3 \psi\|, \qquad r \ge 4. \tag{147}$$

Here z consists of first-order classical variables. Quadratic principal/source moments and quartic differentiated-current forces are Lipschitz in $D(\mathcal{A}^2)$ on this graph ball. The latter degree, which arises from noncommuting spin coefficients, is not suppressed. The full bar and synchronization principal forms have size L and strict positive kinetic/spatial margins; their finite quantum corrections, including smooth finite-rank kernels, have bounded norm. Large finite L preserves those margins.

To express geometry on actual leaves, use ambient-frame derivatives $P = (\rho_t, \rho_x)$ and $Q = (\Phi_t, \Phi_x)$ pulled back to X. Both first-order wave systems have the symmetric matrices

$$A_\tau = \begin{pmatrix} \theta_t & -\theta_x \\ -\theta_x & \theta_t \end{pmatrix} > 0, \qquad A_\xi = \begin{pmatrix} a_t & -a_x \\ -a_x & a_t \end{pmatrix}.$$

Their sources are respectively $(\kappa e^{2\rho} J_\Phi - e^{2\rho}/R^2, 0)$ and $(-2e^{2\rho}\Phi/R^2 - \kappa e^{2\rho} T, 0)$. No second embedding derivative enters this geometric principal block. The first-order classical energy in an H^r ball contracts in H^{r-1}; (147) contracts the state in $D(\mathcal{A}^2)$ using only a uniform $D(\mathcal{A}^3)$ bound. Both difference estimates have a factor T. Time derivatives of principal quadratic moments are at most quartic, so there is no higher-graph difference cycle. Picard iteration, followed by closedness and finite smooth-data bootstrap, gives the full unprojected EL solution. Smooth preparations with the additional finite set of graph moments give six time derivatives and the two coupling derivatives used below. Unrestricted X, including its time equation, gives $\nabla_\mu T^{\mu\nu} = -J_\Phi \nabla^\nu \Phi$; consequently the JT constraints propagate. The sources include the variation of the regulator and proper-number probe. At nonzero γ, retain $J_\Phi = h_s n_S \langle Z_S \rangle / \ell_\perp$ and the resulting $R(g) + 2/R^2 = 2\kappa J_\Phi$.

At $\gamma = 0$, Z_S is conserved with mean zero; the source evolution is independent of h_s, and $\langle Y_S \rangle = \sin(2h_s \Theta)$, where $\Theta = \int n_S N \Phi \, \mathrm{d}\tau \mathrm{d}\xi$. At the parity-fixed center, $X = (T(\tau), 0)$ and $N = T'$. Composing with each history's endpoint $\tau_\pm(t) = \theta_\pm(t, 0)$ therefore gives exactly $\Theta_\pm(t) = \int_0^t \Phi_\pm(v, 0) \mathrm{d}v$, with t central proper time. The wave equation with common initial data yields

$$\Delta\Phi(t, 0) = -\kappa d_{2,\text{total}} t^4/12 + O(t^5), \quad [t^5] \partial_{h_s} \Delta \langle Y_S \rangle \big|_0 = -\kappa d_{2,\text{total}}/30 \neq 0. \tag{148}$$

Finite-width proper-time phases converge in this fifth jet to (148). Scaling M, J together keeps background-dilaton times recoil jets bounded; the center endpoint contribution cancels exactly. Choose a small but nonzero symmetric width preserving the coefficient, then finite L preserving all margins. Lower-order recoil jets are allowed. Thus some finite short $t > 0$ and small nonzero h_s give nonzero next-program contrast. Smooth parameter dependence preserves it for sufficiently small finite $\gamma \neq 0$, when the retained program enters the actual material square (143).

One can also check a complete mean-force response, without attributing every term to geometry. Put $w_2 = 1$ on $\operatorname{supp} e$ and write $\bar{\Phi}_0 = \int n_S N \Phi(0) > 0$, $b = \beta$, $c = \langle q_c \rangle / \sqrt{I}$, $y = \langle q_u \rangle$, $a_\pm = \lambda \pm g$. At the static-support large-L limit the unperturbed means satisfy $\ddot{c} = -\nu_c^2 c - a_\pm^2 bc/(I\omega^2) - a_\pm by/I$, $\ddot{y} = -\nu_u^2 y - a_\pm bc$. For $\delta = \partial_\gamma \partial_{h_s}|_0$, independence at $\gamma = 0$ and $\partial_{h_s}\langle Y_S \rangle = 2\bar{\Phi}_0 t + O(t^3)$ give

$$\begin{aligned} \delta\ddot{c} = & -(\nu_c^2 + a_\pm^2 b/(I\omega^2))\delta c - a_\pm b \delta y/I - 2\bar{\Phi}_0 t(by/I + 2a_\pm bc/(I\omega^2)), \\ \delta\ddot{y} = & -\nu_u^2 \delta y - a_\pm b \delta c - 2b\bar{\Phi}_0 tc. \end{aligned}$$

The common $O(t^3)$ sensor term cancels in the branch fourth-jet. With $\dot{c}(0) = \Omega$, all other initial means zero, these equations include the clock/state variation and give

$$[t^4]\delta(\ddot{y}_+ - \ddot{y}_-) = \bar{\Phi}_0 \left[\frac{8\lambda g b^2 \Omega}{3I\omega^2} + O_I(L^{-1}) \right] \neq 0. \tag{149}$$

The finite-L correction includes proper-time pullback: mixed synchronization displacement can start at t^4, spatial displacement at t^5, while mixed curvature starts at t^5. Their fourth-jet mean-force contributions, divided by $\bar{\Phi}_0$, are $O_I(L^{-1})$ by the scaled positive EL inverse. Initial mixed source differences vanish. Ordinary phase terms are present: (149) is not a purely geometry-induced coefficient. Geometry's nonzero path was separately certified by (148). Choose finite L also preserving the strict sign in (149), then small finite h_s, γ and a short nonzero event. All parameters and the positive operating window are finite; no external read/reinstall is performed. □

F.8 Covariant bulk evolution of a retained quartet

The full-mode carrier requires a covariant source, not just a conserved finite-mode Hamiltonian. Use a physical metric $g = e^{2\rho}(-dt^2 + dx^2)$ and an independent unit material director $U = e^{-\rho}(\cosh\eta, \sinh\eta)$, with orthogonal unit V. The work field χ and director have actions

$$S_\chi = \frac{I_\chi}{2} \int \sqrt{-g}[(U\nabla\chi)^2 - C_\chi(V\nabla\chi)^2], \qquad S_\eta = \frac{I_\eta}{2} \int \sqrt{-g}[(U\Omega)^2 - C_\eta(V\Omega)^2], \tag{150}$$

where $\Omega = d\eta$ plus the physical spin connection and $0 < C_i < 1$. A positive elastic bar with coordinate a anchors this director through $-\Lambda \int \sqrt{-g}(U\nabla a)^2/2$. Prepare a strict bar domain $B - \Lambda > K_a > 0$ and finite alignment overcoming the work-clock orientation curvature $I_\chi(C_\chi - 1)$ and the quantum pressure debit. These are paid material parameters. Deriving U from ∇a instead would put second bar derivatives into the optical coefficients and lose the same-order estimate below.

The quartet comprises two nonchiral Majorana sectors of central charge $c_s = 1/2$, and all three spectator bosons of $c_s = 1$. Each sector has both chiralities and $\gamma_s = c_s/(24\pi)$; the total central charge is four. Its actual velocity $0 < v_s(\chi) < 1$ defines $\bar{g}_s = -v_s^2 U^\flat U^\flat +$

$V^\flat V^\flat$. Supply the original global representation, density operator and every old reference. Choose optical-metric covariant point splitting with no additional finite two-derivative medium counterterm. The anomaly normalization is the conventional one [2, Section 2]; the actual causal state, rather than a vacuum inverse-curvature functional, determines its homogeneous stress.

Proposition F.9 (Actual-state covariant quartet bulk)**.** *Supply a consistent global full-mode representation and a smooth actual initial state with the required energy moments and old references. Prepare smooth local clock/bar/director data with strict paid margins, the causal Hessian conditions* (160) *and the JT constraints* (163)*. The paid fields, retained quartet state and renormalized constrained JT equations then have a smooth common local-time bulk solution. Its mean solution and source are independent of admissible exterior data continuations inside the physical domain of dependence. Equality of its quantum restrictions additionally requires equality of the initial local joint-state restrictions, not merely stress means. This result does not prescribe arbitrary local quantum means or a physical moving boundary.*

Proof. For one sector, set

$$d = \cosh^2\eta - v^2\sinh^2\eta, \quad N = v/d, \quad \beta = (v^2-1)\sinh\eta\cosh\eta/d, \quad \omega = \rho + \tfrac{1}{2}\log d. \tag{151}$$

Thus $\bar{g} = e^{2\omega}\hat{g}$ with $\hat{g} = -N^2dt^2 + (dx+\beta dt)^2$. Its canonical chiral coefficients are $h_\sigma = N - \sigma\beta > 0$. Write $D_t = \partial_t - \beta\partial_x$, $k = \beta_x/N$, $E = T_+ + T_-$ and $P = T_+ - T_-$. The actual Virasoro evolution gives

$$D_tE + NP_x + 2N_xP - 2\beta_xE = -2\gamma\beta_{xxx}, \tag{152}$$

$$D_tP + NE_x + 2N_xE - 2\beta_xP = 2\gamma N_{xxx}. \tag{153}$$

The exact conversion to the orthonormal $\hat{g}$ frame is

$$\hat{E} = E - \gamma k^2, \quad \hat{P} = P + 2\gamma k_x, \quad \hat{S} = \hat{E} + \gamma\hat{R}, \qquad \hat{R} = -2(N_{xx} + D_tk - Nk^2)/N. \tag{154}$$

Substitution gives both covariant conservation equations with zero residual and trace $\gamma\hat{R}$. The lapse and shift variations of $-\gamma\int\beta_x^2/N$ give precisely this projective conversion. The Weyl cocycle $\gamma\int\sqrt{-\hat{g}}[(\hat{\nabla}\omega)^2 + \omega\hat{R}]$ then yields

$$\bar{T}_{ab} = \hat{T}_{ab} + 2\gamma\left[\hat{\nabla}_a\hat{\nabla}_b\omega - \omega_a\omega_b - \hat{g}_{ab}\hat{\Box}\omega + \tfrac{1}{2}\hat{g}_{ab}(\hat{\nabla}\omega)^2\right]. \tag{155}$$

Its trace is $\gamma\bar{R}$ and its $\bar{g}$ divergence vanishes. In the bulk, the combined conversion is the first-jet density

$$\mathcal{L}_{\mathrm{loc},s} = \gamma_s\left[-\frac{(D_t\omega_s - \beta_{s,x})^2}{N_s} + N_s\omega_{s,x}^2 + 2\omega_{s,x}N_{s,x}\right]. \tag{156}$$

The integrated-by-parts difference is $-2\gamma_s[\partial_x(\omega_sN_{s,x}) + \partial_t(\omega_sk_s) - \partial_x(\omega_s\beta_sk_s)]$. It cannot be dropped at a physical wall.

Define the canonical closed-time-path functional by $Z_{\mathrm{can}}[z_+, z_-] = \mathrm{tr}\big(U[z_+]\varrho_0U[z_-]^\dagger\big)$. Add $\Gamma_{\mathrm{loc}}[z_+] - \Gamma_{\mathrm{loc}}[z_-]$, with $\Gamma_{\mathrm{loc}} = \sum_s\int\mathcal{L}_{\mathrm{loc},s}$. Variation at coincident branches gives the actual canonical expectation and the local forces in (156). No new homogeneous Polyakov field, independent purifier or in-out state is introduced. Positive h_σ makes each canonical

Hamiltonian lower bounded, not nonnegative: diffeomorphism conjugation includes the debit $-\gamma_s \int h_{\sigma,x}^2/h_\sigma\,/2$ and the chosen Casimir term. On the prepared smooth coefficient ball these have a finite uniform bound. One common paid scalar offset covers it; its stress is included in both histories.

An exact local reduction supplies the same-order estimate. For each $u_j = \sigma_j h_j$, solve

$$\begin{aligned} w_{j,t} + u_j w_{j,x} + u_{j,x} w_j &= -u_{j,xx}, & w_j(0,x) &= 0, \\ A_{j,t} + u_j A_{j,x} + 2u_{j,x} A_j &= 0, & A_j(0,x) &= T_j(0,x). \end{aligned} \tag{157}$$

Differentiation gives $(\partial_t + u_j\partial_x + 2u_{j,x})(w_{j,x} - w_j^2/2) = -u_{j,xxx}$. Hence, with $Z_j = \gamma_j w_j$,

$$T_j = A_j - Z_{j,x} + Z_j^2/(2\gamma_j). \tag{158}$$

This is the exact Ward solution. If $f_{j,t} + u_j f_{j,x} = 0$, $f_j(0,x) = x$, then $w_j = \partial_x \log f_{j,x}$, $A_j = f_{j,x}^2 T_j(0, f_j)$ and $T_j = f_{j,x}^2 T_j(0,f_j) - \gamma_j \mathcal{S}(f_j)$. These are the actual characteristic connection and transported initial stress, not additional quantum fields or a new source action.

Put $q = (\chi, \eta)$, $P_q = q_t$, $Q_q = q_x$. Let K_{tt}, K_{tx}, K_{xx} be the q Hessians of the complete first-jet classical action plus (156), with $K_{tx} = \mathcal{L}_{P_q Q_q} + \mathcal{L}_{Q_q P_q}$, and let G have columns $g_j = \nabla_q h_j$. The matter principal matrix and its symmetrizer are

$$A_m = \begin{pmatrix} K_{tt}^{-1}K_{tx} & K_{tt}^{-1}K_{xx} & -K_{tt}^{-1}G \\ -1 & 0 & 0 \\ 0 & \operatorname{diag}(\gamma_j\sigma_j)G^T & \operatorname{diag}(\sigma_j h_j) \end{pmatrix}, \quad H_m = \begin{pmatrix} K_{tt} & 0 & 0 \\ 0 & -K_{xx} & G \\ 0 & G^T & \operatorname{diag}(h_j/\gamma_j) \end{pmatrix}. \tag{159}$$

Multiplication gives $H_m A_m = (H_m A_m)^T$. The Schur conditions for H_m and $H_m \pm H_m A_m$ are exactly

$$\begin{aligned} K_{tt} > 0, \qquad -K_{xx} - \sum_j \gamma_j g_j g_j^T/h_j > 0, \\ K_{tt} + K_{xx} \pm K_{tx} \pm \sum_j \frac{\gamma_j \sigma_j g_j g_j^T}{1 \pm \sigma_j h_j} > 0. \end{aligned} \tag{160}$$

At rest the director stiffness is corrected by $\Delta D = \sum_s 2\gamma_s(1 - v_s^2)^2/v_s$. Finite large inertias with strict $0 < C_i < 1$ give a nonempty domain; (160) uses its actual nonlinear coefficients, including nonzero director boost. These inequalities imply a uniform matter cone strictly inside the physical light cone.

The metric and dilaton each have light speeds $\lambda = \pm 1$. There is no resonant trace coupling between them. For the full (χ, η, ρ) Hessians of a quantum sector, direct substitution of (151) gives the exact identity

$$(K_{tt,s}^3 - \lambda K_{tx,s}^3 + K_{xx,s}^3) \begin{pmatrix} 0 \\ \lambda \\ 1 \end{pmatrix} = \sum_{\sigma=\pm 1} \frac{\gamma_s \sigma \lambda h_{\sigma,s,\eta}}{\lambda - \sigma h_{\sigma,s}} \nabla_{(\chi,\eta,\rho)} h_{\sigma,s}. \tag{161}$$

The right side has zero ρ component. The director connection also vanishes on this light mode; its clock component is zero. The corresponding stress is $Z_{\sigma,s} = \gamma_s \sigma h_{\sigma,s,\eta} \lambda p/(\lambda - \sigma h_{\sigma,s})$, with $P_\eta = -p$, $Q_\eta = \lambda p$ and $P_\chi = Q_\chi = 0$. Hence the complete trace residue is zero. The strict matter cone solves the matter-to-light Sylvester equations; the zero

resonant projection solves the remaining metric-to-dilaton equations. Pulling back the block positive form produces a smooth full symmetrizer. The ten A_j variables append the diagonal principal block $\mathrm{diag}(u_j)$ and positive form 1_{10}; they introduce no new light resonance.

For $m > 5/2$, use $q \in H^{m+1}$ and $P_q, Q_q, Z, A \in H^m$; (158) lies in H^{m-1}. The exact local split is

$$\begin{aligned} Z_{j,t} + u_j Z_{j,x} + \gamma_j \sigma_j g_j^T Q_{q,x} &= -\sigma_j (g_j^T Q_q) Z_j - \gamma_j \sigma_j h_{j,qq}[Q_q, Q_q], \\ A_{j,t} + u_j A_{j,x} &= -2\sigma_j (g_j^T Q_q) A_j. \end{aligned} \tag{162}$$

The force is $\sum_j g_j Z_{j,x} - \sum_j g_j [A_j + Z_j^2/(2\gamma_j)]$. All remainders are smooth local products in the same H^m algebra. The conversion contributes first-jet Hessian principal terms and quadratic first-jet remainders. No spatial inverse, omitted mean or higher derivative of the quantum state enters the iteration. Symmetric-hyperbolic iteration gives $d\mathcal{E}_m/dt \le C_R(1 + \mathcal{E}_m)$ on the declared operating ball. Tame higher-order estimates keep smooth data smooth on that same local time. The actual full-mode propagator therefore exists for the solved coefficients; uniqueness of the distributional Ward transport identifies its stress with the solved means on the interior patch, with the same initial state and all references retained. A norm-controlled exterior extension of the reduced initial variables can be chosen for the local iteration. It does not assign artificial means to that quantum state outside the initial interval, and no common-action solution is inferred there. Its bound fixes the local existence time; arbitrary exterior continuations are compared only on their common interval of existence.

This local reduction also proves physical bulk causality. Let two admissible continuations agree initially on (a, b), including scalar values and actual stress means. Their local difference energy on $(a + t, b - t)$ has endpoint fluxes $-V^T(H - HA)V$ on the left and $-V^T(H + HA)V$ on the right. They are nonpositive; the interior is bounded by its energy. Gronwall therefore gives identical mean solutions until the interval closes or the operating domain is left. The interval boundaries are the null curves of g. For quantum restrictions require the same initial joint local state on the causal interval and the compared old references. The actual sector diffeomorphisms pull local observables into that interval, giving the corresponding restriction equality. Stress means alone do not give quantum-state equality, nor is a Type-III local density matrix assumed.

Finally write the minimally coupled JT action as $(2\kappa)^{-1} \int \sqrt{-g}\Phi(R - 2K_{\mathrm{geo}})$. Its metric curvature equation is $\rho_{tt} - \rho_{xx} = K_{\mathrm{geo}} e^{2\rho}$. Set $w = \Phi_x$, $\pi = \Phi_t$. The initial constraints are

$$\Phi_x = w, \qquad w_x = \rho_x w + \rho_t \pi - K_{\mathrm{geo}} e^{2\rho}\Phi - \kappa T_{00}, \qquad \pi_x = \rho_x \pi + \rho_t w - \kappa T_{0x}. \tag{163}$$

Here T is the complete source just constructed, including initial conversion jets; needed accelerations come from its invertible time matrix. This spatial ODE has a local solution for three integration constants, with positive Φ on a sufficiently small interval. Conservation propagates its two traceless constraints locally along physical light directions. Arbitrary periodic positive-energy JT data are not assumed. A consistent global representation and state realize the quantum solution; the causal interior restriction no longer depends on an arbitrary exterior coefficient continuation. □

This closes the quartet's nonlinear bulk source–clock–director–JT interface. It supplies neither moving-wall boundary terms nor the other gate windows' errors. The original

mixed target and its purifier remain in the actual state, but clock and geometry changes alter gate areas. Their same-history finite-horizon comparison and the physical exhaust still have to be incorporated before the complete thermal apparatus is obtained.

F.9 A physical wall retains its thermal friction

A finite rest mass cannot repair a perfect mirror by ignoring its high-frequency reaction. Partial transmission has a causal passive linear-response formulation [1]. For the four actual Dirac flavours, the corresponding force can be derived from a smooth material mass layer rather than imported from a scalar mirror. In a static material frame let $\mathsf{h}_z = -iv\sigma_z\partial_x + M(x-z)\sigma_x$, with $M \in C_c^\infty$, $M \geq 0$, and positive bare mechanical mass m_b. The same mass action gives the translation force $F = \int M'(x-z)\psi^\dagger\sigma_x\psi\,dx + F_{\mathrm{ct}}$, distributed bar reaction and metric mass stress. Use one fixed local counterterm action, an actual static interacting Hadamard ground or KMS preparation, and smooth flat-start displacement tests.

Proposition F.10 (Passive Dirac-layer response)**.** *For the static ground-state linear response, the full-continuum induced vacuum inertia is finite. With positive bare mass and total positive elastic stiffness, the unreduced causal mechanical impedance is positive-real. In the static zero-charge KMS continuum the layer instead has positive ohmic friction. Its vacuum-style low-frequency inertia integral diverges, while the unreduced thermal impedance remains convergent and positive-real. These are linear statements about the specified passive preparation, not arbitrary driven quartet states.*

Proof. Let $u_{E,a}$, $a = L, R$, be energy-normalized scattering modes with incoming amplitude $(2\pi v)^{-1/2}$. Define $A = \int M/v$, $Q = \|M'\|_1$ and $B = e^A/\sqrt{\pi v}$. The exact transfer equation $u_x = (iE\sigma_z + M\sigma_y)u/v$ bounds $\|u\| \leq B$. Integration by parts gives

$$f_{ab}(E,E') := \int M' u^\dagger_{E,a}\sigma_x u_{E',b} = -(E+E')\int (M/v)u^\dagger_{E,a}\sigma_y u_{E',b},$$
$$|f_{ab}(E,E')| \leq B^2 \min\{Q, A|E+E'|\}. \tag{164}$$

The mass anticommutator vanishes and M removes the endpoint term. For $N = 4$ flavours the filled-negative-state force spectrum obeys

$$\chi_I(\nu) = \pi N \sum_{a,b}\int_0^\nu |f_{ab}(E, E-\nu)|^2\,dE \leq 4\pi N B^4 \min\{Q^2\nu, A^2\nu^3/3\},$$
$$\mu = \frac{2}{\pi}\int_0^\infty \frac{\chi_I(\nu)}{\nu^3}\,d\nu \leq \frac{16NB^4AQ}{\sqrt{3}}. \tag{165}$$

At zero energy the exact transfer gives reflection $\tanh^2 A$, and hence $\chi_I(\nu) = N\tanh^2 A\,\nu^3/(6\pi v^2) + O(\nu^4)$. Retain the complete causal memory. For $\mathrm{Re}\,s > 0$ its impedance is

$$Z(s) = m_b s + k/s + \frac{2}{\pi}s\int_0^\infty \frac{\chi_I(\nu)}{\nu(\nu^2+s^2)}\,d\nu. \tag{166}$$

The measure is positive and $s/(s^2+\nu^2) = \frac{1}{2}[(s-i\nu)^{-1} + (s+i\nu)^{-1}]$ has positive real part. Thus $\mathrm{Re}\,Z(s) > 0$; the response has no right-half-plane pole. The observed vacuum inertial mass is $m_{\mathrm{phys}} = m_b + \mu$, not the action's bare m_b. For two separated layers the same supportwise identity gives a positive matrix spectral measure. Adding their

actual finite static polarization/Casimir Hessian to the paid elastic stiffness extends this argument when that total stiffness and the bare mass matrix are positive.

In the KMS continuum, with Fermi function $n_T(E)$,

$$\chi_{I,T}(\nu) = \pi N \sum_{a,b} \int_{\mathbb{R}} [n_T(E) - n_T(E+\nu)] |f_{ab}(E+\nu, E)|^2 \, dE. \tag{167}$$

Its weights are positive. On shell, $\sum_{a,b} |f_{ab}(E,E)|^2 = 2E^2 R(E)/(\pi^2 v^2)$, where $R(E)$ is the actual reflection probability. Therefore

$$\chi_{I,T}(\nu) = \gamma_T \nu + o(\nu), \qquad \gamma_T = \frac{2N}{\pi v^2} \int E^2 R(E) [-n_T'(E)] \, dE > 0. \tag{168}$$

The vacuum μ integral now diverges in the infrared. Equation (166) itself stays finite: its infrared integrand uses this ohmic law, while $\int [n_T(E) - n_T(E+\nu)] dE = \nu$ and (164) give an ultraviolet linear bound. Its positive-real proof is unchanged. A finite prepared exterior has the corresponding discrete positive Kubo measure; this continuum friction limit is not an uncharged infinite thermal reservoir. □

All transmitted exterior channels, old purifiers, static polarization and motion work remain in this construction. Smooth spatial profiles alone do not justify applying an instantaneous force to the old free sea as a finite-energy vector; the valid response tests are also smooth in time. More importantly, the actual driven body need not be passive. Its opposite-chirality mass mixing and non-Gaussian quartet require a joint local-source and task-error estimate. The common-velocity Ward closure inside a massless gate does not close that boundary hierarchy. The passive wall result establishes a physical linear margin, without substituting that margin for the retained mixed-target reset proof.

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Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout

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Abstract

A measurement context supplies probes, environmental dynamics and physical record ports. These determine different objects: implementable effects, persistent content and readable records. We compute their relations in a fresh-memory collision model. For two pure conditional fragment states of overlap r, the exact pointer-effect permission set is $\sqrt{ab} + \sqrt{(1-a)(1-b)} \geq r$; the closest permitted sharp pointer test has error $(1 - \sqrt{1-r^2})/2$. Its operator span does not determine this resolution. Holevo and single-measurement information count different numbers of readable records: one environment has 12 Holevo records but only 8 certified readable ones. Conditional product fragments give explicit public-agreement laws, and pooled one-sided local tests attain the joint pure-state discrimination exponent. Recording does not require a pure memory. We derive a purity-deficit budget and exhibit partially mixed qubits whose exact permissions form a parallelogram, whose local measurement saturates Holevo information, and whose reduced source channel is the same complete dephasing for every memory polarization. At zero polarization this channel produces no record. Reusing a spent port can erase its record and restore coherence, so freshness cannot be inferred from a channel label. For a common unknown reader orientation, agreement alone cannot certify correctness; prepared-pointer references give a finite calibration bound and an explicit residual common-error term. The same actual fragment readout supports Paper A's finite correlation-geometry certificate. The construction accounts for preparation, readout and shared standards without selecting a unique global outcome or deriving the Born rule.

1 Introduction

1.1 Three questions about measurement

Standard quantum mechanics gives rules for outcome probabilities but does not say how definite records arise from states that assign amplitudes to several possibilities. Collapse models change the dynamics, many-worlds accounts deny that one outcome is selected, and decoherence explains the suppression of interference without selecting a result. This

paper does not add to that list. It asks a set of narrower questions that every account has to answer in some form, and it answers them inside a declared physical context.

The first question is one of *permission*. A probe is a physical channel. Which effects on the system can the probe actually implement, given its coupling and initial state? The second concerns *stability*: which effects, or which record values, survive the environmental dynamics over the relevant time window? The third concerns *public readout*: when can several readers, each with access to a different part of the environment, obtain the same value, and how often is a shared value wrong?

The three answers are different kinds of object. Each probe's implementable effects form a convex set inside an operator system; a family of probes gives their union, which is convex only if the permitted family also includes the required randomized choices. Exact fixed points of a channel form an operator system, and under further hypotheses a von Neumann algebra. Records concern a classical label and the information that environmental fragments carry about it. Treating the three as one "accessible algebra" obtained by intersection loses the information that matters, and the paper keeps them apart.

The central claim of the paper is that measurement, in the operational sense used here, is a property of an interaction context: what a declared probe family can implement, what the declared dynamics leaves fixed, and what declared fragments record. We do not explain why one global outcome occurs, and we assume the Born rule.

1.2 Results

Section 2 fixes the typed data of a context. Section 3 states the three tests and separates two notions of redundancy, one based on Holevo information and one based on what a measurement can read. Section 4 introduces a record model in which fresh memories collide once with a pointer, and computes:

- the exact set of pointer effects that a fragment can implement (Theorem 4.2) and the distance from a sharp pointer projection to that set (Corollary 4.3);
- a nesting of the three tests: fragment effects lie in the pointer algebra, which equals the fixed algebra of the induced dephasing (Proposition 4.4);
- the gap between Holevo and accessible information, with closed-form fragment lengths for both notions of redundancy (Propositions 4.5 and 4.6);
- a purity-deficit budget and exact nonzero readout from partially mixed memories (Propositions 4.7 and 4.8);

Section 5 turns to several readers: conditional independence, the error exponents of joint, pooled local and majority readout, and a finite calibrated public-readout bound when the readers share an unknown orientation. Section 6 collects the model's conditions and interfaces. Section 6.2 compares the approach with decoherence theory, quantum Darwinism and other frameworks, and Section 6.3 identifies the actual record interface consumed elsewhere.

Several ingredients are standard: the exact Holevo quantity of a branching state, the Helstrom bound, and the role of redundancy in quantum Darwinism [10, 11, 12]. What the paper adds is the typed separation, the exact permission set of a fragment, the nesting of the tests in one dynamics, the counting difference between Holevo and readable records, the mixed-memory realization and a calibration-sensitive public-readout certificate.

2 Typed accessibility data

2.1 Algebraic background

In the algebraic formulation of quantum theory a system is described by an algebra of observables, and states are positive normalized functionals on it [8, 9]. In algebraic quantum field theory local algebras are attached to regions without a global tensor product, and tensor product structures can be induced by the accessible observables [20, 22]. HAFF builds on this view and asks which operator structures a physical context selects. Paper A treats the reduction of a global state by physical channels and shows that two channels can generate the same algebra while resolving different things [1]. The present paper applies the same caution to measurement.

2.2 The objects of a context

Let $\mathcal{M}_U$ be the ambient von Neumann algebra. A context $\mathbf{c}$ specifies allowed probe channels, an environmental dynamics, a record algebra, time scales and resolution thresholds.

Definition 2.1 (Probe channel and implementable effects)**.** *For each allowed probe α, let*

$$\mathcal{R}^*_{\alpha,\mathbf{c}} : \mathcal{B}(\mathcal{K}_\alpha) \longrightarrow \mathcal{M}_U$$

be its normal unital completely positive Heisenberg map. The set of ***implementable effects*** *of the context is*

$$\mathrm{Eff}(\mathbf{c}) := \{\mathcal{R}^*_{\alpha,\mathbf{c}}(F) : 0 \le F \le \mathbf{1},\ \alpha \text{ allowed}\},$$

and the ***probe-reachable operator system*** *is its ultraweakly closed linear span,*

$$\mathrm{Eff}_{\mathrm{span}}(\mathbf{c}) := \overline{\mathrm{span}}^{\,uw}\,\mathrm{Eff}(\mathbf{c}).$$

The family $\mathfrak{R}_{\mathrm{probe}}(\mathbf{c}) := \{\mathcal{R}_{\alpha,\mathbf{c}}\}_\alpha$ is part of the context, not an observable algebra.

For a single probe Eff is convex, contains 0 and $\mathbf{1}$, and is closed under $E \mapsto \mathbf{1} - E$. The span is the natural object for linear reconstruction of expectation values from many runs. It is not a list of effects that one run of an allowed probe implements: Section 4 gives probes whose span is the whole pointer algebra while the sharp pointer projections are not implementable. The same distinction appears for reductions in Paper A, where the effects a channel reads out, $\Phi^*([0, \mathbf{1}])$, are in general fewer than the positive contractions in the range of Φ^* [1].

Definition 2.2 (Environmental fixed set)**.** *For a normal UCP semigroup $\{\mathcal{E}_t\}_{t\ge 0}$ on $\mathcal{M}_U$, set*

$$\mathrm{Fix}(\mathcal{E}) := \{O \in \mathcal{M}_U : \mathcal{E}_t(O) = O \text{ for every } t \ge 0\}.$$

This is always an ultraweakly closed operator system. It is a von Neumann subalgebra, for example, when the semigroup admits a faithful normal invariant state: equality in the Schwarz inequality then places every fixed element in the multiplicative domain. A discrete-time channel is treated in the same way with $t \in \mathbb{N}$.

Definition 2.3 (Pointer and record algebra)**.** *A* ***pointer/record algebra*** $\mathcal{P}_{\mathrm{env}} \subset \mathcal{M}_U$ *is a designated commutative von Neumann algebra generated by a classical record PVM* $Z(B)$*, or by a finite-resolution coarse graining. For a finite label set, a fragment record channel acts on label states by*

$$\Gamma_{k,*}\big((p_x)_x\big) = \sum_x p_x \rho^x_{E_k}.$$

Its normal UCP Heisenberg dual is

$$\Gamma_k^* : \mathcal{B}(\mathcal{H}_{E_k}) \longrightarrow \mathcal{P}_{\mathrm{env}}, \qquad \Gamma_k^*(F) = \sum_x \mathrm{Tr}(\rho^x_{E_k} F) Z_x.$$

More generally the state map acts on the predual of the commutative record algebra. Approximate pointer observables are treated by their distance to $\mathcal{P}_{\mathrm{env}}$*.*

There is no default equality among Eff, $\mathrm{Fix}(\mathcal{E})$ and $\mathcal{P}_{\mathrm{env}}$. The declared maps are the probe maps, the dynamics, the inclusions of chosen subalgebras into $\mathcal{M}_U$, and the record channels. A model may establish relations among the objects, for example that every implementable effect lies in $\mathcal{P}_{\mathrm{env}} \cap \mathrm{Fix}(\mathcal{E})$; such relations are results or assumptions of that model.

Definition 2.4 (Measurement-accessibility profile)**.** *The* ***measurement-accessibility profile*** *of a context is the typed datum*

$$\mathbf{Acc}(\mathbf{c}) := \big(\mathrm{Eff}(\mathbf{c}),\ \mathrm{Eff}_{\mathrm{span}}(\mathbf{c}),\ \mathfrak{R}_{\mathrm{probe}}(\mathbf{c}),\ \mathcal{P}_{\mathrm{env}}(\mathbf{c}),\ \mathrm{Fix}(\mathcal{E}_{\mathbf{c}});\ \Gamma_{k,*}, T_{\mathbf{c}}, \epsilon, \delta\big). \tag{1}$$

The operator-system and fixed-set statements above hold for general von Neumann algebras under the hypotheses shown. Entropic statements are made for finite alphabets or finite-resolution labels and specified states.

3 Three tests

3.1 Probe reachability

Constraint 3.1 (Probe reachability)**.** *An effect* $M(B) \in \mathcal{M}_U$ *is implementable in context* $\mathbf{c}$ *if* $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F)$ *for some allowed probe* α *and some probe effect* $0 \le F \le \mathbf{1}$*. A POVM* M *is implementable if one probe and one probe POVM* F *give* $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F(B))$ *for every outcome set* B*.*

Implementability is a property of a channel and an effect. A probe Hamiltonian may realize the channel, but the condition $[O, H_{\mathrm{probe}}] \neq 0$ is neither necessary nor closed under sums and products; ideal nondemolition couplings commute with the measured pointer observable. Effects outside $\mathrm{Eff}(\mathbf{c})$ are not ruled out as physical quantities. The context simply supplies no implementation of them, and enlarging the context, for example by allowing more fragments or repeated runs, is a change of context.

3.2 Dynamical stability

Constraint 3.2 (Stability)**.** *An effect* O *passes the* ***finite-window stability test*** *if*

$$\sup_{0 \le t \le T_{\mathbf{c}}} \|\mathcal{E}_t(O) - O\| < \epsilon. \tag{2}$$

Exact stability is the separate statement $O \in \mathrm{Fix}(\mathcal{E})$*.*

In decoherence theory the pointer observables are those left stable by the system–environment interaction [10]. The threshold ϵ and the window $T_{\mathbf{c}}$ are set by the recording mechanisms and time scales of the context. An implementable effect need not be stable, and an exact fixed point need not be implementable by any allowed probe.

3.3 Environmental redundancy: Holevo and readable records

Let X be the classical label of a pointer PVM, or of a finite coarse graining of a POVM, with distribution (p_x). For each environmental fragment E_k form the classical–quantum state $\rho_{XE_k} = \sum_x p_x |x\rangle\langle x| \otimes \rho^x_{E_k}$. Two quantities measure what E_k carries about X. The Holevo information

$$\chi(X : E_k) = S\Big(\sum_x p_x \rho^x_{E_k}\Big) - \sum_x p_x S(\rho^x_{E_k})$$

bounds what any measurement on E_k can extract. The accessible information

$$I_{\mathrm{acc}}(X : E_k) = \sup_{\text{POVM } N \text{ on } E_k} I(X : Y_N)$$

is what a single measurement on the fragment does extract at best, where Y_N is the outcome. Always $I_{\mathrm{acc}} \le \chi$ [14].

Constraint 3.3 (Redundancy)**.** *Fix $0 < \delta < 1$ and a family of disjoint fragments. The label X has R **Holevo records** if R disjoint fragments satisfy $\chi(X : E_k) \ge (1-\delta)H(X)$, and R **readable records** if R disjoint fragments satisfy $I_{\mathrm{acc}}(X : E_k) \ge (1-\delta)H(X)$. For a declared measurement N_k on each fragment one may use $I(X : Y_{N_k})$ in place of I_{acc}.*

Readable records are Holevo records, but not conversely, and the two counts can differ (Proposition 4.6). For continuous spectra X denotes a declared finite-resolution partition. No von Neumann entropy is assigned to an operator, and no expression such as $I(\hat{O} : E_k)$ is used.

Redundancy is what allows several observers to read the same label from different fragments without disturbing one another. It does not by itself say how often their common value is wrong, nor which global outcome is realized. Section 5 computes the former in the model.

4 A record model in which the three tests can be computed

4.1 Fresh memories colliding once with a pointer

The system S is a qubit with pointer projections $P_x = |x\rangle\langle x|$, $x \in \{0, 1\}$. The environment consists of n memory qubits $E_1, \ldots, E_n$, each prepared in $|0\rangle$. The k-th memory collides once with S through

$$U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k, \qquad R_k = \begin{pmatrix} c_k & -\sqrt{1-c_k^2} \\ \sqrt{1-c_k^2} & c_k \end{pmatrix}, \qquad 0 \le c_k \le 1, \tag{3}$$

and does not interact again. Starting from $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$ with $0 < p < 1$, the state after n collisions is

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_{k=1}^{n} |e_{x,k}\rangle, \tag{4}$$

with $|e_{0,k}\rangle = |0\rangle$ and $|e_{1,k}\rangle = R_k|0\rangle$, so that $\langle e_{0,k}|e_{1,k}\rangle = c_k$. A fragment F is a declared set of memories; write

$$r_F = \prod_{k\in F} c_k, \qquad H_p(r) = h\left(\frac{1+\sqrt{1-4p(1-p)(1-r^2)}}{2}\right),$$

where $h(x) = -x\ln x - (1-x)\ln(1-x)$. All entropies are in nats. The label X is the value a pointer measurement on S would return.

The inputs are explicit: the pointer basis, the split into system and memories, the order of collisions, the fragment partition, and the purity of the memories. None of them is derived here. The model is a version of the standard branching model of quantum Darwinism [11, 12]; its role in this paper is to make all components of the profile of Definition 2.4 computable at once.

Proposition 4.1 (Record quantities)**.** *For every fragment F,*

$$\chi(X : E_F) = H_p(r_F), \qquad (\rho_S)_{01} = \sqrt{p(1-p)}\, r_{[n]},$$
$$S(\rho_S) = H_p(r_{[n]}), \qquad I(S : E_{[n]}) = 2H_p(r_{[n]}).$$

Proof. Given $X = x$ the fragment is in the pure state $|e_x^F\rangle = \bigotimes_{k\in F} |e_{x,k}\rangle$, and $\langle e_0^F|e_1^F\rangle = r_F$. The average state $p|e_0^F\rangle\langle e_0^F| + (1-p)|e_1^F\rangle\langle e_1^F|$ has two nonzero eigenvalues with sum 1 and product $p(1-p)(1-r_F^2)$, so its entropy is $H_p(r_F)$, and the conditional entropies vanish. Tracing the memories out of (4) multiplies the off-diagonal element by $\langle e_1|e_0\rangle = r_{[n]}$, and the global state is pure, so $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ and $I(S : E_{[n]}) = 2S(\rho_S)$. □

The reduced dynamics of S is the dephasing channel $\mathcal{D}_k(\rho) = P_0\rho P_0 + P_1\rho P_1 + c_k(P_0\rho P_1 + P_1\rho P_0)$ at each step. Its fixed algebra, for any $c_k < 1$, is the diagonal algebra $\mathcal{P} = \mathrm{span}\{P_0, P_1\}$, which is also the pointer algebra.

4.2 Measurement permissions of a fragment

A reader with access to fragment F implements effects on S through the Heisenberg map

$$\Phi_F^*(A) = \langle e_0^F|A|e_0^F\rangle\, P_0 + \langle e_1^F|A|e_1^F\rangle\, P_1, \qquad A \in \mathcal{B}(\mathcal{H}_F), \tag{5}$$

obtained by coupling S to F through the collisions and tracing out the rest. The span of the image is all of $\mathcal{P}$ as soon as $0 < r_F < 1$, since it contains $\mathbf{1}$ and the operator $\Phi_F^*(|e_0^F\rangle\langle e_0^F|) = P_0 + r_F^2 P_1$. The implementable effects are fewer.

Theorem 4.2 (Permission set of a fragment)**.** *Let $|e_0\rangle, |e_1\rangle$ be unit vectors with $|\langle e_0|e_1\rangle| = r$. Then*

$$\big\{(\langle e_0|A|e_0\rangle, \langle e_1|A|e_1\rangle) : 0 \le A \le \mathbf{1}\big\}$$
$$= L_r := \big\{(a,b) \in [0,1]^2 :\ \sqrt{ab} + \sqrt{(1-a)(1-b)} \ge r\big\}.$$

Hence $\mathrm{Eff}_F := \Phi_F^*([0,\mathbf{1}]) = \{aP_0 + bP_1 : (a,b) \in L_{r_F}\}$.

Proof. Inclusion. For an effect A put $a = \langle e_0|A|e_0\rangle$ and $b = \langle e_1|A|e_1\rangle$. By the Cauchy–Schwarz inequality applied to $A^{1/2}e_0$ and $A^{1/2}e_1$, $|\langle e_0|A|e_1\rangle| \le \sqrt{ab}$, and likewise $|\langle e_0|(\mathbf{1}-A)|e_1\rangle| \le \sqrt{(1-a)(1-b)}$. Adding and using the triangle inequality gives $r \le \sqrt{ab} + \sqrt{(1-a)(1-b)}$.

Equality. The function $f(a,b) = \sqrt{ab} + \sqrt{(1-a)(1-b)}$ is a sum of geometric means, hence concave, so L_r is a compact convex set. The set of attained pairs is convex as the image of the convex set of effects under an affine map. At $r = 0$ the states are orthogonal and all pairs in the square are attained by diagonal effects; at $r = 1$ the states are identical up to phase and the attained set is the diagonal. For $0 < r < 1$, a compact convex subset of the plane is the convex hull of its boundary, so it suffices to attain every boundary point of L_r. Choose phases and an orthonormal basis of $\mathrm{span}\{e_0, e_1\}$ with $e_0 = (1, 0)$ and $e_1 = (\cos\theta, \sin\theta)$, $\cos\theta = r$, $0 \le \theta \le \pi/2$; effects supported on this span suffice. For $u = (\cos\varphi, \sin\varphi)$ with $\theta - \pi/2 \le \varphi \le 0$, the projection $|u\rangle\langle u|$ gives $a = \cos^2\varphi$ and $b = \cos^2(\theta - \varphi)$, and since $\cos\varphi \ge 0$, $\sin\varphi \le 0$, $\cos(\theta-\varphi) \ge 0$ and $\sin(\theta-\varphi) \ge 0$,

$$f(a,b) = \cos\varphi\cos(\theta-\varphi) - \sin\varphi\sin(\theta-\varphi) = \cos\theta = r.$$

This traces the arc of $\{f = r\}$ from $(1-r^2, 0)$ to $(1, r^2)$. The map $A \mapsto \mathbf{1} - A$ sends it to the arc from $(r^2, 1)$ to $(0, 1-r^2)$. The remaining boundary consists of segments of the square: $b = 0$ with $0 \le a \le 1 - r^2$, attained by $A = t|e_1^\perp\rangle\langle e_1^\perp|$; $a = 1$ with $r^2 \le b \le 1$, attained by $A = |e_0\rangle\langle e_0| + t|e_0^\perp\rangle\langle e_0^\perp|$, $0 \le t \le 1$; and their images under $A \mapsto \mathbf{1} - A$. On each edge of the square f reduces to $\sqrt{1-a}$, $\sqrt{b}$, $\sqrt{1-b}$ or $\sqrt{a}$, so these segments are exactly the parts of the edges in L_r. For $r < 1$, $f(a,a) = 1 > r$, so the level set $\{f = r\}$ inside the square consists of the two arcs. Every boundary point is therefore attained. □

For $0 < r < 1$ the permission set is bounded by two curved arcs and four square-edge segments. The corners $(0,0)$ and $(1,1)$ lie on those segments, not on the arcs $f = r$. It fills the square when $r = 0$ and collapses to the diagonal $\{a = b\}$, the trivial effects $a\mathbf{1}$, when $r = 1$. For $0 < r < 1$ it contains no sharp pointer projection: $(1,0) \notin L_r$ because $f(1,0) = 0$.

Corollary 4.3 (Closest implementable pointer test)**.** *For $0 \le r \le 1$,*

$$\min_{E \in \mathrm{Eff}_F} \|E - P_0\| = \varepsilon(r) := \frac{1 - \sqrt{1-r^2}}{2},$$

with $r = r_F$, attained at $E = (1-\varepsilon)P_0 + \varepsilon P_1$. The value $\varepsilon(r)$ is the minimal error probability for distinguishing $|e_0^F\rangle$ from $|e_1^F\rangle$ with equal priors.

Proof. For $E = aP_0 + bP_1$ one has $\|E - P_0\| = \max(1-a, b)$. If $\max(1-a, b) \le t < 1/2$, then $a \ge 1 - t > 1/2 > t \ge b$. In that region $\partial f/\partial a = \frac{1}{2}\big(\sqrt{b/a} - \sqrt{(1-b)/(1-a)}\big) < 0$ and $\partial f/\partial b = \frac{1}{2}\big(\sqrt{a/b} - \sqrt{(1-a)/(1-b)}\big) > 0$, so $f(a,b) \le f(1-t,t) = 2\sqrt{t(1-t)}$. Since $2\sqrt{\varepsilon(1-\varepsilon)} = r$ and $t \mapsto 2\sqrt{t(1-t)}$ is increasing on $[0, 1/2]$, the region contains points of L_r only if $t \ge \varepsilon(r)$, and $(1-\varepsilon, \varepsilon) \in L_r$. The identification with the Helstrom error is the formula $(1 - \sqrt{1 - 4p(1-p)r^2})/2$ at $p = 1/2$ [15]. □

For a fragment of m memories with equal overlaps c, $r_F = c^m$ and $\varepsilon(c^m) = c^{2m}/4 + O(c^{4m})$. Measurement permission for the pointer grows with the number of memories a reader controls, and an η-approximate pointer test becomes available once $c^{2m} \le 4\eta(1-\eta)$. Enlarging F is a change of context, in the sense of Constraint 3.1.

4.3 The three tests in one dynamics

Proposition 4.4 (Nesting)**.** *Let a probe act through a pointer-controlled unitary* $U = \sum_x P_x \otimes V_x$ *on* $\mathcal{H}_S \otimes \mathcal{K}$ *with probe state* $|\phi\rangle$*. Then for every* $A \in \mathcal{B}(\mathcal{K})$

$$\mathcal{R}^*(A) = \sum_x \langle\phi|V_x^* A V_x|\phi\rangle\, P_x \in \mathcal{P}_{\text{env}}.$$

In the record model, for every fragment F *with* $0 < r_F < 1$ *and every* $c_k < 1$*,*

$$\text{Eff}_F \subsetneq \text{Eff}_{\text{span},F} \cap [0,\mathbf{1}] = \mathcal{P} \cap [0,\mathbf{1}] = \text{Fix}(\mathcal{D}_k) \cap [0,\mathbf{1}].$$

Proof. $U^*(\mathbf{1}\otimes A)U = \sum_{x,x'} P_x P_{x'} \otimes V_x^* A V_{x'} = \sum_x P_x \otimes V_x^* A V_x$, and the partial expectation in $|\phi\rangle$ gives the formula. In the model $\text{Eff}_{\text{span},F} = \mathcal{P}$ as noted before Theorem 4.2, the fixed algebra of $\mathcal{D}_k$ is $\mathcal{P}$, and Eff_F is the lens of Theorem 4.2, which omits P_0. □

In this model every effect that the environment can report is a stable pointer effect, and the stable pointer effects are exactly the positive contractions of the span. The tests still differ. The lens measures resolution, the fixed algebra measures persistence, and redundancy, below, counts how many disjoint readers have access. A probe that is not pointer-controlled, for example one that rotates the system before recording, can reach effects outside $\text{Fix}(\mathcal{D}_k)$; those effects are then not stable under the environment that produced the records.

4.4 Holevo records and readable records

Proposition 4.5 (Accessible information of a fragment)**.** *Let*

$$P_e(r) = \frac{1 - \sqrt{1 - 4p(1-p)r^2}}{2}$$

be the Helstrom error for two pure states with overlap r *and prior* $(p, 1-p)$*. For every fragment,*

$$I_{\text{acc}}(X : E_F) = h(p) - h(P_e(r_F)) \le \chi(X : E_F) = H_p(r_F).$$

For $p = 1/2$*,*

$$I_{\text{acc}}(X : E_F) = \ln 2 - h\big(\varepsilon(r_F)\big), \qquad \chi(X : E_F) = h\Big(\frac{1 + r_F}{2}\Big),$$

and the two differ whenever $0 < r_F < 1$*.*

Proof. This binary pure-state optimum is standard [16, Sec. 3.5.4]; the following argument proves it for all POVMs. Write their conditional outcome laws as a_y, b_y, put $q_y = pa_y + (1-p)b_y$, and omit zero-weight outcomes. Their posterior probabilities are $\pi_y = pa_y/q_y$; set $C_y = 2\sqrt{\pi_y(1-\pi_y)}$ and

$$E(C) = h\Big(\frac{1 - \sqrt{1 - C^2}}{2}\Big).$$

Then $H(X \mid Y) = \sum_y q_y E(C_y)$. For $t = \sqrt{1-C^2}$, $E'(C) = C\,\text{artanh}(t)/t \ge 0$ and $E''(C) = (\text{artanh}(t) - t)/t^3 \ge 0$ for $0 < C < 1$; continuity extends convexity to the endpoints. The outcome-wise Cauchy–Schwarz argument of Theorem 4.2 gives

$$\sum_y q_y C_y = 2\sqrt{p(1-p)} \sum_y \sqrt{a_y b_y} \ge 2\sqrt{p(1-p)}\, r_F.$$

Jensen and monotonicity imply $H(X \mid Y) \geq E(2\sqrt{p(1-p)}r_F) = h(P_e(r_F))$. Diagonalizing $p|e_0^F\rangle\langle e_0^F| - (1-p)|e_1^F\rangle\langle e_1^F|$ shows that its Helstrom measurement has posterior error $P_e(r_F)$ on every nonzero outcome, so it attains the bound. The cases $r_F = 0, 1$ follow directly. The Holevo bound and value are those of Proposition 4.1. At equal priors the displayed forms agree only at $r_F = 0, 1$. □

At $p = 1/2$ and $r_F = 0.8$, $\chi = 0.325083$ while $I_{\text{acc}} = 0.192745$. With $\delta = 0.6$ the threshold $(1-\delta)\ln 2 = 0.277259$ lies between them: the fragment is a Holevo record and not a readable one. The threshold for a Holevo record and the threshold for a measurement-based record are therefore not equivalent, even when the measurement is optimal; the difference concerns one measurement on one fragment, and joint coding over many independent preparations is a different task.

Proposition 4.6 (Fragment lengths)**.** *Let all $c_k = c \in (0,1)$ and $0 < \delta < 1$, and let h^{-1} denote the inverse of h on $[0, 1/2]$. Put*

$$q_\delta = h^{-1}\big((1-\delta)h(p)\big), \qquad a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)},$$

$$e_\delta = h^{-1}\big(\delta h(p)\big), \qquad a_{\text{op}} = \frac{e_\delta(1-e_\delta)}{p(1-p)}.$$

A fragment of m memories is a Holevo record if and only if $c^{2m} \leq a_\chi$, and it is a readable record if and only if $c^{2m} \leq a_{\text{op}}$. The smallest such lengths are

$$m_\chi = \left\lceil \frac{\ln a_\chi}{2\ln c} \right\rceil, \qquad m_{\text{op}} = \left\lceil \frac{\ln a_{\text{op}}}{2\ln c} \right\rceil,$$

so m_{op} is the smallest readable length for every $0 < p < 1$.

Proof. $H_p(r)$ is a decreasing function of r^2, and $H_p(r) = (1-\delta)h(p)$ exactly when the larger eigenvalue $\lambda = (1+\sqrt{1-4p(1-p)(1-r^2)})/2$ equals $1 - q_\delta$, that is when $r^2 = a_\chi$. For the readable case, Proposition 4.5 gives $I_{\text{acc}} \geq (1-\delta)h(p)$ exactly when $h(P_e(r)) \leq \delta h(p)$, that is when $P_e(r) \leq e_\delta$, which rearranges to $4p(1-p)r^2 \leq 4e_\delta(1-e_\delta)$. Since $e_\delta \leq \min(p, 1-p)$, $a_{\text{op}} \leq 1$. □

p	c	n	δ	m_χ	m_{op}	Holevo records	certified readable
0.5	0.80	60	0.10	5	7	12	8
0.2	0.80	60	0.10	5	7	12	8
0.5	0.95	100	0.05	27	38	3	2

Table 1: Fragment lengths and the number of disjoint equal-length records in n memories. The binary pure-state optimum in Proposition 4.5 makes every listed m_{op} minimal over all POVMs. At $p = 0.2$, I_{acc} is .439321 for $m = 6$ and .458263 for $m = 7$, across the threshold .450362.

When fragments are unequal, the counts depend on the chosen partition. With equal overlaps, m_χ gives the largest Holevo count. m_{op} gives the largest readable count for the same pure binary ensemble at every prior.

4.5 The preparation resource for a record

Proposition 4.7 (Purity-deficit budget)**.** *Replace the memories by systems of dimension d_k in product states $\bigotimes_k \sigma_k$, and let the collisions be pointer-controlled unitaries $P_0 \otimes \mathbf{1} + P_1 \otimes W_k$. Then for every fragment*

$$\chi(X : E_F) = S\Big(p \bigotimes_{k\in F} \sigma_k + (1-p) \bigotimes_{k\in F} W_k \sigma_k W_k^*\Big) - \sum_{k\in F} S(\sigma_k) \le \sum_{k\in F} \big[\ln d_k - S(\sigma_k)\big],$$

while each collision multiplies the off-diagonal element $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. If $\sigma_k = \mathbf{1}/2$ and $W_k = R_k$, no fragment carries any information about X, and the coherence of S is still multiplied by c_k.

Proof. The two conditional states of the fragment are unitarily related products, so both have entropy $\sum_{k\in F} S(\sigma_k)$; the average state has entropy at most $\sum_{k\in F} \ln d_k$. The block $P_0\rho_S P_1 \otimes \sigma_k$ is mapped to $P_0\rho_S P_1 \otimes \sigma_k W_k^*$, so tracing out the memory multiplies $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. For $\sigma_k = \mathbf{1}/2$ both conditional states equal $\mathbf{1}/2$, and $\mathrm{tr}(R_k^*)/2 = c_k$. □

The initial purity deficits bound the information that these unitary ports can record. They do not require every memory to be pure. Dephasing is not evidence of a readable record: a maximally mixed port can generate the same source channel as a pure port and carry no label information. Paper F uses this resource budget as a quantitative low-record boundary condition [5]. Preparation, correlation and renewal must then be accounted for separately; this entropy bound alone is not a universal work price.

4.6 Partially mixed memories: exact permission and readout

Let each fresh qubit memory start in $\eta_\alpha = (I + \alpha Z)/2$, $0 \le \alpha \le 1$, and use

$$U = P_0 \otimes I + P_1 \otimes X.$$

The conditional states are $\eta_0 = \eta_\alpha$ and $\eta_1 = X\eta_\alpha X = (I - \alpha Z)/2$. Put $\varepsilon_\alpha = (1-\alpha)/2$.

Proposition 4.8 (A mixed-memory record)**.** *For one fresh memory, its exact effect permission set is*

$$\mathrm{conv}\{(0,0), (1,1), (1-\varepsilon_\alpha, \varepsilon_\alpha), (\varepsilon_\alpha, 1-\varepsilon_\alpha)\}. \tag{6}$$

For $\alpha > 0$ the effect span is the full pointer algebra, but the closest permitted P_0 test has distance ε_α. At equal label priors,

$$\chi(X : E) = I_{\mathrm{acc}}(X : E) = \ln 2 - h((1+\alpha)/2) = \ln 2 - S(\eta_\alpha). \tag{7}$$

Measurement of Z_E attains this value and has symmetric error ε_α. For every α, tracing the memory completely dephases the source.

Proof. Only the diagonal entries $u, v \in [0,1]$ of a memory effect matter. The reported pair is $((1-\varepsilon_\alpha)u + \varepsilon_\alpha v, \varepsilon_\alpha u + (1-\varepsilon_\alpha)v)$. Diagonal effects attain every such pair, proving the parallelogram. For any reported pair (a,b), $a - b = \alpha(u - v) \le \alpha$; hence $\max(1-a, b) \ge (1-\alpha)/2$, with equality at $u = 1, v = 0$. The conditional states commute, have entropy $h((1+\alpha)/2)$ and average to $I/2$. Their common eigenbasis measurement attains the Holevo quantity. Finally the source coherence multiplier is $\mathrm{Tr}(\eta_\alpha X) = 0$. □

Records in partially mixed spin environments are already part of quantum Darwinism [12]. For $0 < \alpha < 1$ this construction gives their exact effect permissions and an explicit readout saturating the one-port purity-deficit budget. At $\alpha = 0$ its effects collapse to the trivial diagonal (a, a) and it carries no record, although its source channel is unchanged. At $\alpha = 1$ it records perfectly. Thus neither an effect span nor the complete source channel fixes record quality.

After m fresh collisions, the conditional memory states are $\eta_x^{\otimes m}$. Measuring every Z_E reads the entire commuting ensemble and still saturates Holevo information. The reports are binary independent errors conditional on x. Majority vote has exact error

$$e_m = \Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) > m/2\} + \tfrac{1}{2}\Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) = m/2\} \le e^{-m\alpha^2/2}. \tag{8}$$

The bound is Hoeffding's for the sample mean of signs in $[-1, 1]$. Fano's inequality then gives a sufficient readable-record length: choose m so that the bound is at most $1/2$ and $h(e^{-m\alpha^2/2}) \le \delta \ln 2$. This constructive mixed-state extension needs no classification of all mixed conditional-state permission sets.

4.7 A spent port is not a fresh channel input

The same controlled-X unitary also makes the port distinction exact. Since $U^2 = I$, applying it a second time to the same memory, without intervening measurement, export or dynamics, returns the complete state to its pre-collision value. The first collision dephases the source and writes the α-dependent record; the second removes that record and restores the original source coherence. Two independent fresh collisions instead preserve complete source dephasing and leave two conditional memory copies. This reversal does not erase an external measurement record or a copy already transferred to another port.

The one-step reduced channel cannot be applied again to the source while silently replacing the spent memory by η_α. After the first collision the source and memory are correlated. A second actual interaction acts on that joint state, including its old purification. Paper F studies repeated contact and joint memory updating [5]; the distinction also applies to the retained-port acquisition used by Q-RAIF C [4]. Storage, overwrite, reset and acquisition are different operations even when their isolated channel labels are similar.

5 Public readout

5.1 Agreement, correctness and common error

Proposition 5.1 (Conditional independence of readers)**.** *Let readers $j = 1, \dots, J$ apply POVMs $N^{(j)}$ to disjoint fragments F_j. Then the joint distribution of the pointer value and the outcomes is*

$$\Pr(x, y_1, \dots, y_J) = p_x \prod_{j=1}^{J} \langle e_x^{F_j} | N_{y_j}^{(j)} | e_x^{F_j} \rangle.$$

For two readers who each report a value $\hat{x}_j(y_j) \in \{0, 1\}$ with error probabilities $\varepsilon_j(x) = \Pr(\hat{x}_j \ne x \mid x)$, the agreement rate A, the correctness rates C_j and the common error rate $E = \Pr(\hat{x}_1 = \hat{x}_2 \ne x)$ are

$$E = \sum_x p_x \varepsilon_1(x)\varepsilon_2(x), \qquad A = \sum_x p_x \big[(1 - \varepsilon_1(x))(1 - \varepsilon_2(x)) + \varepsilon_1(x)\varepsilon_2(x)\big],$$

and $C_1 + C_2 = 1 + A - 2E$*, so that* $\Pr(\textit{correct} \mid \textit{agree}) = 1 - E/A$.

Proof. Tracing S out of (4) gives $\rho_E = \sum_x p_x |e_x\rangle\langle e_x|$ with $|e_x\rangle$ a product over memories. A pointer measurement on S commutes with the fragment measurements and returns x with probability p_x, leaving the memories in $|e_x\rangle$. The product form of $|e_x\rangle$ over disjoint fragments gives the factorization. The formulas for E and A follow by summing over the four combinations of right and wrong reports, and the identity $C_1 + C_2 = 1 + A - 2E$ holds event by event: an agreeing correct pair contributes two correct reports, an agreeing wrong pair none, and a disagreeing pair one. □

With equal priors, $c = 0.8$ and two five-memory fragments read with their Helstrom measurements, each reader errs with probability $\varepsilon(0.8^5) = 0.02761$, the readers agree with probability $A = 0.94631$, and an agreeing pair is wrong with probability $E/A = 8.05 \times 10^{-4}$. In this model agreement certifies correctness up to the product of private errors. The identity $C_1+C_2 = 1+A-2E$ and the separation of agreement from correctness are the subject of Foundation II, where a contamination shared by two instruments raises the common error far above the product of private errors [7]. Independent private errors already produce a nonzero common error. An excess beyond the conditional-product formula requires an additional shared influence or a failure of that record law, such as shared instrument contamination or calibration error. T-DOME III makes this precise: under a symmetry acting on all readers at once, their relative sign is detectable from agreement and a common inversion is not, so the error that agreement can never expose is the direction shared by the readers' standards [21].

5.2 Calibrated public readout with a common orientation

Conditional independence is a property of the prepared ports and their actual instruments. It is not implied by agreement or commuting effects. The mixed-memory family gives an explicit calibration problem. Encode the pointer value as $z = (-1)^x$. Suppose all readers share an unknown fixed orientation $C \in \{-1, 1\}$, so each reported elementary sign has conditional mean $C\alpha z$. The physical response $\alpha > 0$ is supplied and stable across reference and target runs. A common inversion leaves every agreement rate unchanged. With equal target priors and no independently anchored reference, the transformations $(C, z) \mapsto (-C, -z)$ give the same observable target-record law, so the absolute orientation is unidentifiable. A known asymmetric source prior would be additional orientation information and a different calibration problem.

The context must supply a preparation of the physical projection P_0 whose label is anchored independently of the unknown reader orientation. Its verification and preparation are resources. If the preparation label is itself inverted by the same unknown C, the reference cannot remove the ambiguity. Given that permission, prepare $z = +1$ in n_{ref} independent reference trials and measure fresh mixed ports with the same physical C response used for target readout. Estimate C by the sign of their mean, breaking a tie at random. Hoeffding gives

$$\Pr\Big(\widehat{C} \neq C\Big) \leq \delta_{\mathrm{cal}} := \min\{1, e^{-n_{\mathrm{ref}}\alpha^2/2}\}. \tag{9}$$

A change of orientation or response between reference and target runs violates the premise; the bound does not detect arbitrary shared contamination.

Proposition 5.2 (Finite public correctness after calibration)**.** *Let J readers each control m fresh mixed-memory ports of the same target pointer. They measure the memory Z values, take a majority with independent fair tie coins, and multiply their decision sign by $\widehat{C}$. The reference trials are independent of the target ports. With e_m from* (8)*,*

$$A_J := \Pr(\textit{all readers agree}) = (1-e_m)^J + e_m^J, \qquad \Pr(\textit{all agree wrongly}) \le \delta_{\mathrm{cal}} + e_m^J. \tag{10}$$

Consequently

$$\Pr(\textit{wrong} \mid \textit{all agree}) \le \min\{1, (\delta_{\mathrm{cal}} + e_m^J)/A_J\}.$$

Proof. Correcting by a common sign does not change agreement. Conditional on the pointer and the true orientation, the fresh ports give independent symmetric majority errors of probability e_m. On the event $\widehat{C} = C$, all reports are wrong with probability e_m^J; on its complement this probability is at most one. Reference–target independence gives the stated bounds. □

For $\alpha = 0.6$, $m = 5$, $J = 2$, the exact private error is $e_5 = 0.05792$ and $A_2 = 0.8908694528$. With $n_{\mathrm{ref}} = 100$ the calibration failure bound is e^{-18}. Even after calibration, the conditional common-error bound is about 0.003766. Increasing the number of readers suppresses their independent mistakes, but a fixed shared calibration error does not disappear through agreement.

The protocol consumes n_{ref} prepared-pointer references and fresh mixed ports, Jm target ports, all corresponding readout bits, access to a common orientation standard, and the communication needed for calibration and agreement. These are prepared resources, not an inference of correctness from consensus. The preparation cost is not identified with the entropy deficit alone. T-DOME III analyzes the remaining common-sign ambiguity [21]; Foundation II distinguishes a shared state or instrument from an independently checked source [7]. Here the reference permission and its finite error are part of one explicit record protocol.

5.3 Many readers: error exponents

Proposition 5.3 (Joint, pooled and majority readout)**.** *Let R disjoint fragments each have overlap $r \in (0,1)$, and let the prior be $(p, 1-p)$.*

1. *Any measurement on the union of the fragments errs with probability at least $P_e(r^R) = p(1-p)r^{2R} + O(r^{4R})$, and a joint measurement attains this value.*

2. *Let each reader measure $\{|e_0^{F_j}\rangle\langle e_0^{F_j}|,\ \mathbf{1} - |e_0^{F_j}\rangle\langle e_0^{F_j}|\}$ on its own fragment, and decide $\hat{x} = 1$ if at least one reader obtains the second outcome. This pooled local rule errs with probability $(1-p)\, r^{2R}$.*

3. *Let $p = 1/2$, let each reader use its single-fragment Helstrom measurement, and let the decision be the majority of the R reports, with ties broken by a fair coin. The error probability is $r^{R+o(R)}$, and no rule that uses only the R reports has a better exponent.*

Proof. (1) The conditional states of the union are pure with overlap r^R, and the Helstrom bound is attained by the projective measurement onto the positive part of $p|e_0\rangle\langle e_0| - (1-p)|e_1\rangle\langle e_1|$ [15]. (2) Given $x = 0$, every reader obtains the first outcome with certainty. Given $x = 1$, reader j obtains it with probability $|\langle e_0^{F_j}|e_1^{F_j}\rangle|^2 = r^2$, independently by

Proposition 5.1, so all R readers obtain it with probability r^{2R}. (3) Each report is wrong with probability $\varepsilon = \varepsilon(r)$ independently. With equal priors and symmetric errors the likelihood ratio of the reports depends only on the number of reports for each value, so the majority rule is optimal among rules that use the reports. Its error is $\Pr(\mathrm{Bin}(R,\varepsilon) > R/2)$ plus half the probability of a tie, which equals $\exp\big[-R\,D(\frac{1}{2}\|\varepsilon) + O(\ln R)\big]$ with $D(\frac{1}{2}\|\varepsilon) = -\ln\big(2\sqrt{\varepsilon(1-\varepsilon)}\big)$ by the standard large-deviation estimate for binomial tails. Finally $4\varepsilon(1-\varepsilon) = 1-(1-r^2) = r^2$. □

Local readers who only pool their results afterwards thus reach the joint error exponent $2|\ln r|$ per fragment, with a prefactor larger by $1/p$. At $p = 1/2$ and $r = 0.8$, twenty fragments give a joint error of 3.32×10^{-5}, a pooled one-sided error of 6.65×10^{-5}, and a majority error of 1.58×10^{-3}. Readers who are allowed to choose each measurement after learning the earlier outcomes can attain the joint Helstrom error exactly for pure conditional states [17].

The pooled rule requires a matched implementation of the projector onto $|e_0^{F_j}\rangle$. Knowledge of the coupling is one way to supply it; a trusted prepared reference or a previously calibrated fixed reader is another. An incorrect coupling model generally sets up the wrong test. When that relation is unknown, calibration records can change the actual readout permission. T-DOME I computes the value of a readout model in a specified thermodynamic task [6]; the present exponent does not require every successful fixed reader to carry an explicit self-model.

6 Conditions and interfaces

6.1 Preparation and reading permissions

The carrier, pointer basis, collision order, preparation maps and fragment partition enter explicitly. The pure-memory model gives a noncommuting two-state permission lens; the mixed-memory model gives a commuting parallelogram and readable records without complete purity. Both use independent fresh ports. Retained ports must instead be evolved with their actual old correlations, as Section 4.7 demonstrates.

Reader permissions also matter. A one-sided pure-state test requires access to its conditional state vector. The orientation protocol requires an independently anchored pointer preparation and the same stable response in reference and target trials. Neither capability follows from membership of an effect in a generated algebra. Arbitrary shared errors and preparation–reader co-inversions remain outside that calibration certificate.

The global branching state remains a superposition. Born probabilities determine the label distribution used in the readout laws; no unique global outcome is selected by these calculations. A reader can be any physical device controlling a fragment. The model does not privilege a conscious observer.

Einselection and Paper B's stability analysis address how a given interaction chooses a pointer or protected sector [10, 2]. Paper B's formation constructions add real shared recording ports and Hamiltonian links. Locality and the history of those links must still explain why a particular fragment partition is available. Paper F addresses updating and repeated contact [5]; Q-RAIF C supplies finite acquisition on a carrier retaining quantum history [4]. These are complementary realizations, not permission to replace one model's initial states by another's without a physical bridge.

6.2 Relation to existing approaches

Decoherence and quantum Darwinism. Decoherence theory explains the suppression of interference through entanglement with an environment and identifies pointer observables by their stability [10]. Quantum Darwinism adds that objectivity requires redundant records in many fragments [11], and the exact information content of branching states in spin environments has been analysed in detail [12]. The present paper uses these results and adds the typed distinctions: the permission set of a fragment, the nesting of permission, stability and redundancy in one dynamics, and the difference between Holevo and readable records.

Spectrum broadcast structures. The conditional product form used in Proposition 5.1 is one half of a spectrum broadcast structure, the class of states in which many observers can find the same value independently [13]. The other half, perfect distinguishability of the conditional fragment states, holds here only in the limit $r_F \to 0$. Proposition 5.3 measures how fast readers approach that limit as fragments are added.

QBism and relational quantum mechanics. QBism reads quantum states as an agent's degrees of belief [18], and relational quantum mechanics takes states to be relative to a reference system [19]. Here accessibility is fixed by the interaction structure of a context, and a single objective global state is assumed. The emphasis on relations is shared with relational quantum mechanics; the stable records computed here are candidates for the relata that its relations require.

Algebraic quantum field theory. AQFT treats observable algebras as primary and states as functionals on them [8]. The profile of Definition 2.4 adds typed selection data to that setting. Its entropic statements are finite-dimensional, and no Type III version of the redundancy counts is claimed.

6.3 The actual record interface

Paper A uses these pure fragments as actual readers of its noisy GHZ source [1]. Their visibility $v_i = \sqrt{1 - r_i^2}$ gives the collision-derived law

$$\mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}, \qquad c_{ij} = (1 - 2p_i)(1 - 2p_j).$$

The source sites are correlated; no independence of their answers is assumed. Calibrated finite samples then certify non-isometric shortest-path metrics for two equal-flip-budget noise allocations. This supplies the physical interface a dictionary statement alone lacks.

Paper B's block-constant recorder depends only on sector weights, under its specified commuting noise and recording permissions [2]. Its protected content, fixed algebra and finite channel range differ from this fragment-effect set. The mixed-memory family sharpens the distinction: one unchanged dephasing channel has permissions ranging from trivial to perfect. Paper D and Q-RAIF C add record-controlled physical sources and retained-port acquisition [3, 4]; they grant neither free sharp effects nor fresh inputs in place of spent ports.

The measured record law is also the output consumed by Paper F and T-DOME I's joint memory accounts and Foundation II's distinction between agreement and correctness [5, 6, 7]. Its entropy, preparation cost and action value retain their different resource

conditions. Shared geometry further requires source preparation, site labels, calibrated readers and a reconstruction rule; the finite orientation certificate here supplies one of those permissions.

7 Conclusion

Measurement permissions, stable content and public records are different physical data. In the fresh pure-memory model, the permission lens gives the sharp-test error, while the fixed pointer algebra gives persistence and fragment discrimination gives readable redundancy. Holevo and single-measurement thresholds differ. Pooled local tests attain the joint pure-state error exponent when their readers know the conditional fragment state.

The mixed-memory construction shows why preparation must be accounted for without demanding purity. Its actual readout saturates a purity-deficit budget, its exact permissions form a parallelogram, and its source channel remains complete dephasing even when no record is produced. Reusing the same port reverses its record rather than supplying another independent sample.

Public correctness additionally needs a physical standard. Conditional independence suppresses private common errors, but agreement cannot discover a shared orientation inversion. Prepared-pointer references calibrate that inversion with a finite failure probability; this probability remains in the public error bound and its resources remain in the protocol. Together with Paper A's actual record-to-geometry construction, these results identify a finite operational route from interaction to readable shared structure. They leave outcome selection, elementary factorization and general spacetime reconstruction as independent explanatory questions.

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Records, Dissipation and Intervention: Conditions for a Temporal Direction

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Abstract

Records, entropy production and intervention need not point in the same direction. We bound disjoint fragments' Holevo information by the purity supplied at a product boundary, bath work and a possible system contribution. Fair pointer copying into independent partly mixed ports saturates the pointer-controlled bound: each port's readable information equals its initial purity deficit. The same controlled operations yield an exact joint ledger, a posterior for action and, for one port, a transfer of its purity into the source without a bath. Retained correlations provide another resource: a conditional bound prices new ports relative to an old reference even when their marginal purity is zero. Fresh ports suppress coherence and accumulate records along a CP-divisible source dynamics. Reusing a port can instead return correlations and restore coherence. Reset, reversible relabelling and reversal of the complete recording history have different balances. For a bank degraded by independent flips, restoration of the complete retained state removes a specified joint entropy increment. Label-controlled thermal swaps realize the correction for nonzero target error and approach its entropy price; a declared thermal chiral exhaust adds a conditional finite-rate heat bound. Time, coupling order, gates and preparation are inputs. The results explain how records support later intervention and what must be restored for that capacity to continue.

1 Introduction

1.1 The problem

The fundamental dynamical laws are invariant under time reversal, or nearly so, while records, memories and the reach of actions are not: we have records of the past and can influence only the future. Since Boltzmann the standard explanation has been a special initial condition, a low-entropy past, together with the statistical tendency of entropy to grow [6, 7, 8, 9]. The explanation leaves several questions open. What exactly must be special about the initial condition for records to form? Is the growth of records the same arrow as the growth of entropy, or as the dissipation of a system's free energy? Why can an intervention affect later records and not earlier ones, and what would it take to undo a record?

The paper answers these questions for record-forming processes in which records are well defined. We do not derive a time parameter. The dynamics is given with an external parameter and an order of interactions. What the paper derives is a set of quantitative relations among four directions that are often treated as one: the growth of records, the growth of entropy, the dissipation of free energy, and the reach of interventions. It shows when they coincide, when they separate, and what each costs.

1.2 Results

Section 3 recalls the record model of Paper E [3], in which fresh memories collide once with a pointer, and the two counts of records it supports. The results are:

- a purity law valid for every record-forming process that starts from a product of system, memories and thermal bath: the Holevo information of disjoint fragments about a pointer label is at most the initial purity of the memories, plus the work not stored in the system in units of k_BT, plus the difference between the label entropy and the initial system entropy (Theorem 4.1). The records version of a low-entropy past is a stock of purity or free energy, and the bound is attained;

- in the collision model, an exponential bound on the system's coherence in terms of the number of records and the corresponding loss of free energy, which disappears when the system starts diagonal (Proposition 5.1); record growth along a CP-divisible reduced dynamics (Proposition 5.2); and a gap between local and global entropy that grows linearly with the number of records (Proposition 5.3);

- thermal reset prices: at least k_BT times local entropy for separate fragment erasers in the pure-reset limit, a joint-reset infimum at most k_BT times label entropy, and zero net endpoint work for complete dynamical reversal (Proposition 6.1);

- partly mixed ports that attain the pointer-controlled purity bound, an exact acquisition–readout–action ledger and a conditional record budget with a retained reference (Propositions 4.3 and 4.4);

- an exact intervention order: operations on the system cannot change memories that no longer interact, and change records formed later (Proposition 7.2); repeated interaction breaks the alignment (Proposition 7.3).

- a separate price for renewing a noisy record bank: a complete cyclic correction removes $R[h(q) - h(\varepsilon)]$ nats, while a protected old record can require no renewal. A thermal chiral exhaust gives a conditional finite-rate capacity for such a bank (Propositions 8.1 and 8.2).

Section 9 collects the comparisons, including the two models of T-DOME I in which correlations return to a system without records or no correlations return at all [5]. Section 10 relates the results to the thermodynamic arrow, the past hypothesis and timeless formulations, and Section 10 places the paper in the HAFF programme.

2 Time in physical theory

In Newtonian mechanics and in the Schrödinger equation time is an external parameter, and the equations have no intrinsic arrow. In canonical quantum gravity the Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ contains no time parameter [11]. The Page–Wootters construction recovers an effective time from correlations between a clock subsystem and the rest of a stationary universal state [12], but it explains an ordering rather than its asymmetry. The thermodynamic arrow supplies the asymmetry through a coarse-graining and a low-entropy boundary condition [9, 10].

The model below uses an external parameter and an externally given order of interactions. It asks how records become asymmetric relative to a boundary condition, and it states that boundary condition as a resource. It is not an input-free derivation of time.

3 Records and how to count them

3.1 The record model

The system S is a qubit with pointer projections P_x, $x \in \{0, 1\}$, and initial state $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$. Memory qubits $E_1, \ldots, E_n$ start in $|0\rangle$, and the k-th collides once with S through $U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k$, where R_k is a real rotation with $\langle 0|R_k|0\rangle = c_k \in [0, 1]$. After n collisions

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_k |e_{x,k}\rangle, \qquad \langle e_{0,k}|e_{1,k}\rangle = c_k. \tag{1}$$

For a fragment F put $r_F = \prod_{k\in F} c_k$ and

$$H_p(r) = h\left(\frac{1 + \sqrt{1 - 4p(1-p)(1-r^2)}}{2}\right), \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

Paper E shows that $\chi(X : E_F) = H_p(r_F)$, that the coherence of S is $\sqrt{p(1-p)}\,r_{[n]}$, and that $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ [3]. Here X is the value a pointer measurement on S returns. Entropies are in nats.

This pure-port model will be compared with the partly mixed controlled-X model below. The inputs are the pointer basis, the split into system and memories, the fragment partition, the order of collisions with its external parameter, and the preparation of the memories. The collisions can be implemented by couplings $g_k(t)P_1 \otimes \sigma_y$ that vanish at the start and end of each pulse and commute with a pointer Hamiltonian $H_S = \varepsilon P_1$; with degenerate memory Hamiltonians the recording unitary has zero net endpoint work. Preparation of the ports, calibrated pulse control and later bank renewal are separate resources. T-DOME I gives the full thermodynamic account of this protocol [5].

3.2 Redundancy and protocol-relative comparisons

Definition 3.1 (Redundancy index)**.** *For a classical label X, a classical–quantum state ρ_{XE} and a declared family of disjoint fragments $\{E_k\}$, define*

$$\mathcal{R}_\delta^{\chi} = \#\big\{k : \chi(X : E_k) \geq (1-\delta)H(X)\big\}, \qquad \mathcal{R}_\delta^{\mathrm{acc}} = \#\big\{k : I_{\mathrm{acc}}(X : E_k) \geq (1-\delta)H(X)\big\}.$$

The first counts Holevo records and the second records that a permitted single measurement can read; $\mathcal{R}_\delta^{\mathrm{acc}} \le \mathcal{R}_\delta^{\chi}$, and Paper E gives closed-form fragment lengths for both in the collision model. With $p = 1/2$, $c_k = 0.8$ and $\delta = 0.1$, blocks of five memories are Holevo records and blocks of seven are readable, so after n collisions $\mathcal{R}_\delta^{\chi} = \lfloor n/5 \rfloor$ and $\mathcal{R}_\delta^{\mathrm{acc}} = \lfloor n/7 \rfloor$. High redundancy supports public readability of the label; it does not select one global outcome. Here the quoted pure-port counts permit arbitrary fragment POVMs, as in Paper E. A restricted readout menu needs its own accessible-information optimization; ideal algebra closure does not grant those measurements.

Definition 3.2 (Protocol-relative comparisons)**.** *Fix a dynamics, a fragment family and a label X, and let α, β be configurations along the resulting history. Write $\alpha \preceq_{\mathcal{R}} \beta$ if $\mathcal{R}_\delta(\rho_\alpha) \le \mathcal{R}_\delta(\rho_\beta)$, and $\alpha \preceq_{\mathrm{inc}} \beta$ if the algebra of memories that have interacted at α is contained in that at β. Each relation is a preorder on its declared domain.*

In the collision model both comparisons follow the collision order. They need not agree in general: decoherence may increase the number of records while the coherence-sensitive effects of the system, whose expectation values scale with $r_{[n]}$, become unreadable on S itself.

3.3 Temporal inputs

The construction uses temporal orientation at several points: the parameter of the dynamics, the order of collisions, the comparison of records at different parameter values, and a boundary condition of fresh memories at one end of the history. The comparisons above therefore compress an already timed process. The aim of the following sections is to identify what that boundary condition consists of, what record growth costs and implies, and which other directions follow it.

4 The purity of the past

Bath balances below use finite entropies, relative entropies and mean energy changes. An infinite reservoir requires a controlled limit of finite reservoirs preserving those balances, not subtraction of divergent global entropies.

Theorem 4.1 (Purity law for records)**.** *Let a system S, memories $E_1, \dots, E_n$ of dimensions d_k with degenerate Hamiltonians, and a bath B with Hamiltonian H_B start in the product state $\rho_S \otimes \bigotimes_k \sigma_k \otimes \tau_B$, where τ_B is the Gibbs state of H_B at inverse temperature β. Let them evolve under a time-dependent Hamiltonian whose interaction terms vanish at the initial and final times and whose bath and memory terms are then H_B and the given degenerate Hamiltonians, let W be the work done on the whole and ΔE_S the change of the energy of S. Let X be the outcome of a final measurement of a PVM on S with rank-one projections, and let $F_1, \dots, F_R$ be disjoint sets of memories. Then*

$$\sum_{j=1}^{R} \chi(X : E_{F_j}) \le \sum_{k=1}^{n} \big[\ln d_k - S(\sigma_k)\big] + \beta\,(W - \Delta E_S) + \big[H(X) - S(\rho_S)\big]. \tag{2}$$

Without a bath the middle term is absent, and the bound holds for an arbitrary joint unitary on S and the memories, with no condition on their Hamiltonians.

Proof. Let ρ be the final state, and $\rho' = \sum_x (P_x \otimes \mathbf{1})\rho(P_x \otimes \mathbf{1})$ its pinching in the pointer basis. Pinching does not decrease entropy and unitary evolution preserves it, so $S(\rho') \geq S(\rho_S) + \sum_k S(\sigma_k) + S(\tau_B)$. Since the P_x have rank one, $S(\rho') = H(X) + S(E_{[n]}B \mid X)$, where the conditional entropy refers to the classical–quantum state of X, the memories and the bath. Write G for the memories outside $\bigcup_j F_j$. Subadditivity of conditional entropy gives

$$S(E_{[n]}B \mid X) \leq \sum_j S(E_{F_j} \mid X) + S(G \mid X) + S(B \mid X) \leq \sum_j S(E_{F_j} \mid X) + \ln d_G + S(\rho_B),$$

with ρ_B the final state of the bath. Nonnegativity of $D(\rho_B \| \tau_B)$ gives $S(\rho_B) \leq S(\tau_B) + \beta \Delta E_B$. Hence

$$\sum_j S(E_{F_j} \mid X) \geq S(\rho_S) + \sum_k S(\sigma_k) - H(X) - \ln d_G - \beta \Delta E_B.$$

Since $\chi(X : E_{F_j}) = S(E_{F_j}) - S(E_{F_j} \mid X) \leq \ln d_{F_j} - S(E_{F_j} \mid X)$, summing over j gives (2) with $\beta \Delta E_B$ in the middle. The memory Hamiltonians are degenerate and the interactions vanish at the endpoints, so the first law gives $\Delta E_B = W - \Delta E_S$. Without a bath the same argument applies with B omitted. □

The bound is attained without a bath. If S starts in $|0\rangle$, a single memory starts maximally mixed, and a controlled-NOT with the memory as control and S as target is applied, then the memories have no purity, $H(X) - S(\rho_S) = \ln 2$, and $\chi(X : E_1) = \ln 2$. In that example the label is created by copying the memory's randomness into a pure system, which is why the system can contribute one record's worth. For a bath-free pointer-controlled unitary, $U = \sum_x P_x \otimes V_x$, the system term is absent: the conditional states of the memories are unitary images of their initial state, and the argument of the proof gives $\sum_j \chi(X : E_{F_j}) \leq \sum_k [\ln d_k - S(\sigma_k)]$. If in addition each V_x is a product over memories, as in the collision model, each fragment is bounded separately by its own purity, $\chi(X : E_F) \leq \sum_{k \in F} [\ln d_k - S(\sigma_k)]$ (Paper E's purity-deficit bound). Without the product form the per-fragment bound can fail: with E_1 maximally mixed, E_2 pure, $p = \frac{1}{2}$ and V_1 the swap of E_1 and E_2, the fragment $\{E_1\}$ holds 0.216 nats about X.

Corollary 4.2 (Purity budget of a record history)**.** *If R disjoint fragments are Holevo records at level δ, then*

$$R(1-\delta)H(X) \leq \beta \sum_k \big[F(\sigma_k) - F(\mathbf{1}/d_k)\big] + \beta\,(W - \Delta E_S) + H(X) - S(\rho_S),$$

where F is the nonequilibrium free energy at temperature T. In particular, memories that start in thermal equilibrium can hold R records only if the work not stored in the system satisfies $\beta(W - \Delta E_S) \geq R(1-\delta)H(X) - [H(X) - S(\rho_S)]$. In a pointer-controlled process without a bath, $R(1-\delta)H(X) \leq \sum_k [\ln d_k - S(\sigma_k)]$.

Proof. Each record contributes at least $(1-\delta)H(X)$ to the left side of (2). For a degenerate Hamiltonian the thermal state of a d-dimensional memory is $\mathbf{1}/d$, and $\beta[F(\sigma) - F(\mathbf{1}/d)] = D(\sigma \| \mathbf{1}/d) = \ln d - S(\sigma)$. The pointer-controlled statement is the summed bound stated after Theorem 4.1. □

The acquisition boundary is thus a quantitative resource, not a requirement that every memory be pure. Stable old records can also serve as standards across time, provided their couplings and orientation connect the old and present instruments [15]. The bound counts simultaneously present information in declared disjoint fragments, not all records that could ever be made, read, erased and remade. With product maximally mixed ports, no bath and pointer-controlled recording, it gives zero; an old correlation can change that boundary, as the conditional bound below shows. The theorem specifies what the acquisition history must supply. It does not explain the primitive origin of that stock.

4.1 Partly mixed ports: acquisition, reading and action

Let a fair classical pointer X be held in S, initially $I/2$. Each of R fresh ports starts independently in $\eta_\alpha = (I+\alpha Z)/2$, with $0 \le \alpha \le 1$. Put $\varepsilon = (1-\alpha)/2$ and $i_\alpha = \ln 2 - h(\varepsilon)$. Apply controlled X from S to each port, using the same physical mixed-port coupling as Paper E. The resulting state is diagonal:

$$P(x, \boldsymbol{y}) = \tfrac{1}{2} \prod_{j=1}^{R} (1-\varepsilon)^{\mathbf{1}_{y_j = x}} \varepsilon^{\mathbf{1}_{y_j \neq x}}. \tag{3}$$

An actual Z reading obtains Y_j. The stated calibration fixes which port value labels which pointer; a common unknown reversal would not be removed by collecting more ports.

Proposition 4.3 (A saturated acquisition and action ledger)**.** *Every port has Holevo and accessible information i_α, equal to its initial purity deficit. Write $I_R = I(X : \boldsymbol{Y})$. Then*

$$\begin{aligned} S(SE) &= \ln 2 + Rh(\varepsilon), & S(E \mid X) &= Rh(\varepsilon), \\ S(E_j) &= \ln 2, & S(E) &= Rh(\varepsilon) + I_R, \\ \sum_j S(E_j) - S(E) &= Ri_\alpha - I_R, & H(X \mid \boldsymbol{Y}) &= \ln 2 - I_R. \end{aligned} \tag{4}$$

For $R = 1$, feedback controlled by the actual port, $D = X \oplus Y$, leaves the source in η_α and the port in $I/2$, independently. For degenerate source and port Hamiltonians, the source free energy increases by $k_B T i_\alpha$, at zero net endpoint work for the ideal recording and feedback unitaries.

Proof. Conditional ports have entropy $h(\varepsilon)$, while their fair-label averages are $I/2$. Their states commute, so Z reading attains Holevo information. Thus $\sum_j \chi = Ri_\alpha$ saturates the pointer-controlled bound for any number of independent ports, including partly mixed ones. Conditional product structure gives $S(SE)$ and $S(E \mid X)$; the definition of I_R gives $S(E)$. For one port, $Y = X \oplus N$, where N is independent with $\Pr(N = 1) = \varepsilon$. Fair X makes Y independent of N; feedback gives $D = N$. Both gates are basis permutations and preserve the complete joint entropy. The source entropy falls from $\ln 2$ to $h(\varepsilon)$, giving the free-energy increment. □

The posterior is available from the actual record, not just from the algebra it generates. For $k = \sum_j y_j$,

$$\Pr(X = 1 \mid \boldsymbol{y}) = \frac{(1-\varepsilon)^k \varepsilon^{R-k}}{(1-\varepsilon)^k \varepsilon^{R-k} + (1-\varepsilon)^{R-k} \varepsilon^k}, \tag{5}$$

with the deterministic limits interpreted on their support. A reversible gate controlled by $\boldsymbol{y}$ can send X to $X \oplus g(\boldsymbol{y})$. Majority is the minimum-error choice here; ties may be resolved by a fixed label. All original ports remain. Computing and retaining an extra classical copy or a decision register would require its own physical stock, so the ideal implementation uses the ports themselves as controls.

The one-port result locates the resource after action: initial memory purity has moved into the source, not disappeared into an unpriced bath. It supplies free energy for a subsequent task, not work already delivered to a work store. T-DOME I implements such a task and accounts for its reset. Neither the source nor the bank has completed a thermodynamic cycle. Paper C's blank-body transfer is another use of finite stock, with different dynamics; its resource account must be retained when the two mechanisms are composed.

4.2 Old references change the acquisition boundary

Proposition 4.4 (Conditional budget with a retained reference)**.** *Let X be classical, K a retained reference, and $E = E_1 \otimes \cdots \otimes E_R$, with $\dim E_j = d_j$. The initial state may correlate all of them. Recording applies $\sum_x |x\rangle\langle x| \otimes I_K \otimes V_x$, with V_x acting on E only. Then*

$$\sum_j I(X : E_j \mid K)_{\text{out}} \leq \sum_j \ln d_j - S(E \mid XK)_{\text{in}}. \tag{6}$$

For independent fresh product ports the right side is their summed purity deficit. This bounds final conditional information, not only its increment during recording.

Proof. The controlled unitary preserves $S(E \mid XK)$. Strong subadditivity gives $\sum_j S(E_j \mid XK)_{\text{out}} \geq S(E \mid XK)_{\text{out}}$. Also $S(E_j \mid K) \leq S(E_j) \leq \ln d_j$. Subtracting these inequalities proves the claim. □

Take independent fair bits X, L, an old reference $K = L$ and a new port $E = L$. Each port is marginally maximally mixed. Controlled X writes $Y = L \oplus X$ into E; E alone carries no information about X, but reading K, E recovers it exactly. Here $S(E \mid XK)_{\text{in}} = 0$ and $I(X : E \mid K)_{\text{out}} = \ln 2$, attaining Eq. (6). The usable stock is the old correlation and permission to read the reference jointly, not a hidden pure marginal. Creating and retaining the two correlated copies of L belongs to the earlier history's account.

For classical readable K, the conditional information is the Holevo quantity averaged over its records, and in this commuting example it is actually attained. A quantum reference need not admit that interpretation: the entropy inequality remains valid, but its conditional mutual information is not by itself a measurement permission or accessible information. This distinction connects acquisition to T-DOME I's retained side information without identifying unlike tasks.

5 Records and dissipation

Proposition 5.1 (Records bound the coherence that survives)**.** *In the collision model with $0 < p < 1$, let $0 < \delta < 1$, let q_δ satisfy $h(q_\delta) = (1-\delta)h(p)$ with $q_\delta \leq 1/2$, and put*

$$a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)} \in (0,1).$$

If R disjoint fragments, of any sizes and any collision overlaps c_k, are Holevo records, then

$$r_{[n]}^2 \le a_\chi^R, \qquad |(\rho_S)_{01}| \le \sqrt{p(1-p)}\, a_\chi^{R/2}.$$

If the pointer Hamiltonian is diagonal in the pointer basis, the system's nonequilibrium free energy falls by

$$F_S(0) - F_S(n) = k_B T\, H_p(r_{[n]}) \ge k_B T\, H_p\big(a_\chi^{R/2}\big).$$

If instead S starts in $pP_0 + (1-p)P_1$, every record quantity is unchanged and $F_S(n) = F_S(0)$.

Proof. A fragment is a Holevo record exactly when $r_F^2 \le a_\chi$ (Paper E's Holevo fragment criterion). The overlaps multiply over disjoint fragments, and the memories outside them contribute factors at most 1, which gives the first two bounds. The collisions preserve the pointer populations, so the energy of S is unchanged and $F_S(0) - F_S(n) = k_B T[S(\rho_S(n)) - S(\rho_S(0))] = k_B T\, H_p(r_{[n]})$; H_p decreases in r. For the diagonal initial state the conditional fragment states are the same, so all $\chi(X : E_F)$ are unchanged, and ρ_S does not change at all. □

Records and dissipation in the system therefore coincide only when the system has coherence to lose. As R grows, $H_p(a_\chi^{R/2}) \to h(p)$ and the system loses all the free energy carried by its coherence. A system that is already diagonal acquires the same records at no cost to itself. In the pure-port model, the marginal purity of written memories is transferred into correlations. The complete unitary preserves joint entropy; the mixed-port ledger makes that transfer explicit.

Proposition 5.2 (Records along a CP-divisible dynamics)**.** *The reduced dynamics of S after n collisions is $\mathcal{D}_n \circ \cdots \circ \mathcal{D}_1$, where $\mathcal{D}_k$ keeps the populations and multiplies the coherence by c_k. Every intermediate map $\mathcal{D}_m \circ \cdots \circ \mathcal{D}_{l+1}$ is completely positive and trace preserving, and for equal overlaps $c \in (0,1)$ applied at intervals Δt the family embeds in the dephasing semigroup generated by $\mathcal{L}(\rho) = \frac{\gamma}{2}(Z\rho Z - \rho)$ with $\gamma = -\ln c/\Delta t$. Along this dynamics $\mathcal{R}_\delta^\chi$ and $\mathcal{R}_\delta^{\mathrm{acc}}$ grow without bound in n.*

Proof. Each $\mathcal{D}_k$ is a dephasing channel, and compositions of channels are channels. The semigroup multiplies the coherence by $e^{-\gamma t}$, which equals c^m at $t = m\Delta t$. The record counts are $\lfloor n/m_\chi \rfloor$ and $\lfloor n/m_{\mathrm{op}} \rfloor$ for equal overlaps. Unbounded record counts refer to a family with growing fresh-port supply, not an unbounded yield from a fixed finite bank. □

Record growth thus does not require memory effects in the reduced dynamics of the system. The memories, once written, have a mixture-of-product marginal correlated through X and are independent conditional on a pointer reading; a Markovian system dynamics says nothing about whether the environment's outputs are correlated. The converse separation also holds. T-DOME I gives an autonomous two-qubit model in which correlation built between a system and an environment qubit returns to the system at a recurrence time while the environment holds no information at all about the pointer, and it shows that the pure-dephasing spin–boson model with a Lorentz–Drude spectral density has no backflow at any temperature [5]. Backflow and records are logically independent.

Proposition 5.3 (Local and global entropy)**.** *For disjoint fragments $F_1, \dots, F_R$ of the collision model,*

$$\sum_{j=1}^{R} S(\rho_{F_j}) = \sum_j \chi(X : E_{F_j}), \qquad S\big(\rho_{\cup_j F_j}\big) \le h(p).$$

If the F_j are Holevo records, the total correlation among them, $\sum_j S(\rho_{F_j}) - S(\rho_{\cup_j F_j})$, is at least $[R(1-\delta)-1]h(p)$.

Proof. The conditional states of every fragment are pure, so $S(\rho_{F_j}) = \chi(X : E_{F_j})$. The union of the fragments has conditional states that are pure as well, so $S(\rho_{\cup F_j}) = H_p(r_{\cup F_j}) \le H_p(0) = h(p)$. Subtracting gives the last claim. □

With $p = 1/2$, $c = 0.8$ and $n = 60$ memories in twelve blocks of five, the local entropy is 7.661 nats while the entropy of all sixty memories is 0.693147 nats, equal to $\ln 2$ to nine digits. A reader with access to single fragments sees entropy grow with every record. The global entropy of the environment saturates at the entropy of the label. Which of the two grows without bound is fixed by the access structure, not by the dynamics.

6 Undoing records

Proposition 6.1 (Local and joint reset)**.** *Let the pure-port model's memories have degenerate Hamiltonians. Each eraser couples the memories it controls to its own independent Gibbs bath at inverse temperature β, with interactions off at endpoints. A reset with final memory entropy tending to zero has:*

1. *for separate fragment erasers,*

$$\liminf W^{\text{local}} \ge k_B T \sum_j S(\rho_{F_j}) = k_B T \sum_j \chi(X : E_{F_j}) \ge R(1-\delta) k_B T h(p)$$

   when those fragments are Holevo records;

2. *for one eraser controlling all memories but not S, reset work infimum $k_B T S(\rho_{E_{[n]}}) = k_B T H_p(r_{[n]})$, approached with vanishing reset error and increasingly large, engineered thermal reservoirs;*

3. *for an agent controlling S and every memory, the inverse $U_n^{-1} \cdots U_1^{-1}$ restores the initial product state and source coherence at zero net endpoint work in the ideal recording model.*

Proof. For one controlled memory M and its initially independent bath, unitarity and subadditivity give $\Delta S_B \ge S(\rho_M) - S(\rho'_M)$. The Gibbs identity gives $\beta W = \beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$, since the degenerate memory energy is unchanged. Hence $\beta W \ge S(\rho_M) - S(\rho'_M)$. Apply this separately, or to the entire bank. Separate baths cannot exploit correlations between fragments; their marginal inputs fix these entropy decreases. The pure-port bank marginal has rank at most two. A unitary acting only on the degenerate bank maps its two eigenvectors to a logical qubit tensored with blank remaining bits. The logical qubit has eigenvalues $(1-z, z)$, $0 \le z \le 1/2$, and entropy $h(z) = H_p(r_{[n]})$. For $z > 0$, thermal swaps through decreasing positive error probabilities to $\epsilon > 0$ approach

$k_B T[h(z)-h(\epsilon)]$, as in Proposition 8.2. Taking $\epsilon \to 0$ gives the joint reset infimum; for $z = 0$ the bank unitary suffices. The same thermal-swap method applies more generally [14]. The inverse recording unitaries have the same unchanged endpoint energy as the forward gates. □

Exact purification of a mixed memory is not attainable with a finite-dimensional, full-rank Gibbs bath at finite energies and temperature. Its input joint rank is $\mathrm{rank}(\rho_M)d_B > d_B$, whereas a pure final marginal would force the joint state to have rank at most d_B. Unitary evolution preserves rank. Thus the work infimum is not an exact finite-bath reset guarantee; finite reset error uses the entropy decrease in the proof, with nonnegative bath relative-entropy and correlation terms.

For $p = 1/2, c = 0.8, n = 60$, the corresponding asymptotic reset bounds are $19.505\, k_B T$ for sixty separate ports, $7.661\, k_B T$ for twelve five-port fragments, $5.368\, k_B T$ for eight seven-port readable fragments, and $0.693\, k_B T$ for the joint bank. The local price reflects the access structure, not a larger amount of distinct label content.

Lemma 6.2 (Reset with an old reference)**.** *Let M, K have zero endpoint Hamiltonians and the bath be thermal and independent of MK. Allow a joint unitary on MKB that leaves the marginal of K unchanged. Then*

$$\beta W = \beta Q \geq S(M \mid K)_{\mathrm{in}} - S(M').$$

In the pure-reset limit the lower bound is $S(M \mid K)_{\mathrm{in}}$.

Proof. The decrease of the retained joint entropy is $S(MK)_{\mathrm{in}} - S(M'K')_{\mathrm{out}}$. Since $S(M'K') \leq S(M') + S(K') = S(M') + S(K)_{\mathrm{in}}$, it is at least $S(M \mid K)_{\mathrm{in}} - S(M')$. The independent-bath argument above applies to the joint retained state. □

The target here preserves the declared reference marginal. If other reference correlations must also be restored, they belong to the retained target and its ledger. Quantum conditional entropy is not a universal attained work price without an allowed protocol. For two classical copies $M = K = L$ of a fair bit, controlled XOR from K resets M exactly at zero endpoint work and leaves K unchanged. Its conditional entropy is zero, although $S(M) = \ln 2$. Permission to use the reference changes the reset task.

Changing a record is not resetting it. A unitary bit flip reverses its labels at zero endpoint work for a degenerate memory and retains all information. Swapping a record into an independent maximally mixed port makes the old memory uninformative at zero endpoint work, but transfers the record into the other port instead of destroying the joint history. Local thermal reset, such transfer, error correction and inversion of the complete source–bank unitary cannot be assigned one erasure bill. Exact history reversal is allowed when all required gates and systems are accessible. Neither its probability nor a suppression of reversed histories follows without an ensemble and measure [16].

7 Intervention and the agent's time order

Definition 7.1 (Protocol-relative causal-influence diagnostic)**.** *Fix a record-forming protocol with external parameter t, labels X_A, X_B, and a family of interventions on X_A. Write $X_A \rightsquigarrow_{\mathsf{P}} X_B$ when (i) both labels have records in the declared fragments, (ii) the X_A records form no later than the X_B records according to t, and (iii) interventions on X_A, with the rest of the protocol held fixed, change the distribution of X_B or its records.*

The diagnostic is not asserted to be transitive, antisymmetric or independent of the intervention family. Redundancy makes an influence readable across fragments; the intervention structure carries the causal content. The next two propositions compute the diagnostic in the collision model.

Proposition 7.2 (Interventions reach only later records)**.** *In the collision model let an arbitrary channel $\mathcal{T}$ act on S between the j-th and $(j+1)$-th collisions.*

1. *The final reduced state of $E_1, \dots, E_j$ is the same as without $\mathcal{T}$.*
2. *If $\mathcal{T}$ is the pointer flip $\rho \mapsto \sigma_x \rho \sigma_x$, the final state is $\sum_x \sqrt{p_x}\,|1-x\rangle|e_x^{\le j}\rangle|e_{1-x}^{>j}\rangle$: the earlier memories record the value x and the later ones record $1-x$.*
3. *If the Kraus operators of $\mathcal{T}$ are diagonal in the pointer basis, the joint distribution of the pointer value and all records is unchanged.*

Proof. (1) After the j-th collision, $\mathcal{T}$ and the later collisions act on $S \otimes E_{j+1} \otimes \cdots \otimes E_n$, and the partial trace over these systems is invariant under any operation on them. (2) σ_x maps the branch $\sqrt{p_x}|x\rangle|e_x^{\le j}\rangle$ to $\sqrt{p_x}|1-x\rangle|e_x^{\le j}\rangle$, and the later collisions then write the value $1-x$. (3) Diagonal Kraus operators commute with the pointer-controlled collisions and with the pointer measurement, and they leave the populations unchanged. □

For a flip after the second of four collisions with overlaps $(0.3, 0.5, 0.2, 0.4)$ and $p = 1/2$, the reduced states of the first two memories and of the last two are each unchanged, while the joint state of the first and third memories changes by 0.437 in its largest matrix element: the flip is visible only in the relation between earlier and later records. In this fresh-port protocol, interventions on the source can alter later records, while detached earlier marginals remain fixed. The flip example makes the change visible in joint relations even when single-fragment marginals do not change.

Proposition 7.3 (Repeated interaction lets a later intervention rewrite an earlier record)**.** *Let a memory collide with S twice through the same rotation with $c = \cos(\pi/4)$, and let $p = 1/2$. After the first collision the memory holds $H_{1/2}(2^{-1/2}) = 0.4165$ nats about the initial pointer value. If nothing happens between the collisions, the second collision makes the record perfect, $\ln 2$ nats. If S is flipped between the collisions, the second collision leaves the memory with no information about the initial pointer value.*

Proof. Write R for the rotation by $\pi/4$, so $R^2|0\rangle = |1\rangle$. Without the flip, the memory ends in $|0\rangle$ if $x = 0$ and in $R^2|0\rangle = |1\rangle$ if $x = 1$. With the flip, it is in $|0\rangle$ after the first collision if $x = 0$ and is then rotated once, ending in $R|0\rangle$; if $x = 1$ it is in $R|0\rangle$ after the first collision and is not rotated again. Both branches end in $R|0\rangle$. □

For the coherent source $|+\rangle$, the same two-qubit protocol also gives the entropy and population ledger:

$$|\Psi_1\rangle = (|0,0\rangle + |1\rangle R|0\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{noflip}}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{flip}}\rangle = |+\rangle \otimes R|0\rangle.$$

Source populations stay $1/2, 1/2$, and joint entropy stays zero. Put $a = h[(1+2^{-1/2})/2] = 0.4164955$. At first contact, $S(S) = S(E) = a$ and $I(S:E) = 2a$. The no-flip continuation gives $\ln 2, \ln 2, 2\ln 2$. The flip continuation gives $0, 0, 0$: source coherence and both marginal purities return while the earlier pointer information disappears. For a pointer-diagonal source Hamiltonian, energy stays fixed; source free energy first falls by $k_B T a$

and then recovers it on the flip branch. In fact $U(X_S \otimes I)U = X_S \otimes R$, so the echo decouples every source input. Reading or discarding the port after the first contact would change this coherent protocol. The source trace distance of the inputs $|+\rangle, |-\rangle$ falls to $2^{-1/2}$ at first contact and returns to 1 after the echo: this is distinguishability backflow, not merely an entropy change.

The alignment of the intervention order with the record order thus rests on a specific feature of the dynamics: memories that have recorded do not interact with the source again. Under repeated interaction, whether an early memory ends as a perfect record or as no record can be decided by an intervention made after it first recorded.

An agent can therefore date an intervention by its effect on records within the stated one-contact architecture. It cannot change a detached port by acting only on the source. Acting directly on that port can relabel, transfer or reset it, with the different balances above; reconnecting it to the source can also restore coherence. The alignment comes from the actual interaction menu, fresh stock and external order, not from a universal prohibition on changing the past.

8 Renewing records instead of making new ones

A written distinction and a renewable distinction are different resources. The purity law prices acquisition from its declared product boundary. It does not make an old record disappear when fresh blank memories run out. For example, a copied Z label remains unchanged under Z dephasing, although each fair-label record has zero marginal purity. Maintaining it costs no entropy removal in that noise model. Independent bit flips give a different, quantitative renewal problem.

Proposition 8.1 (The entropy price of a cyclic record bank)**.** *Let a classical label X be retained unchanged, and let R degenerate binary registers C_i be conditionally independent copies of X with error $\varepsilon \in [0, 1/2)$. Between corrections each undergoes an independent flip of probability $p \in (0, 1/2]$, whose noise port is inaccessible. The controller starts independent of XC and is unchanged by this noise. A correction returns the complete joint state of X, the bank and its controller to the pre-noise state. Its bath starts thermal and independent, and its interactions vanish at the endpoints. Put*

$$q = p + (1 - 2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0. \tag{7}$$

The entropy removed from the retained state is exactly $Rd_{p,\varepsilon}$ per correction, and its bath heat obeys $\beta Q \geq Rd_{p,\varepsilon}$.

If corrections occur at frequency ν and the sole exhaust has the thermal, zero-charge, extensive-current properties of HAFF-D's chiral exhaust [2], the mean heat current obeys

$$\overline{\dot{Q}} \geq k_B T \nu R d_{p,\varepsilon} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} (\nu R d_{p,\varepsilon})^2. \tag{8}$$

Thus an exhaust-power budget $\overline{\dot{Q}} \leq J$ permits at most

$$R \leq \frac{\sqrt{(k_B T)^2 + 4\alpha J} - k_B T}{2\alpha \nu d_{p,\varepsilon}}, \qquad \alpha = \frac{3\hbar}{\pi c_{\mathrm{ex}}}. \tag{9}$$

Here c_{ex} is dimensionless chiral capacity, and $\nu R d_{p,\varepsilon}$ is the dimensionless entropy removed per unit time, in nats per time. The current bound is a long-window extensive average,

not an individual pulse identity. No inference from vanishing budget to vanishing records follows when $d_{p,\varepsilon} = 0$.

Proof. Before noise $S(XC) = H(X) + Rh(\varepsilon)$, and after it $S(XC) = H(X) + Rh(q)$. Cyclicity of the entire retained state prevents a controller or a correlation with the label from absorbing that entropy without charge. For its independent Gibbs bath B, a joint unitary and subadditivity give $\Delta S_B \geq Rd_{p,\varepsilon}$. The Gibbs identity $\beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$ proves the heat bound. The actual net extensive entropy current is at least $\nu R d_{p,\varepsilon}$. Substitution into HAFF-D's increasing finite-rate bound gives Eq. (8); inversion gives Eq. (9). □

Proposition 8.2 (Thermal correction with an accessible label)**.** *Under the renewal proposition's conditional-iid error law, assume* $0 < \varepsilon < q \leq 1/2$*, and allow retained* X *to control reversible XOR gates on each bank register. Supply an independent Gibbs bath qubit with gap* $\Delta_\varepsilon = k_B T \ln[(1-\varepsilon)/\varepsilon]$ *for each register. XOR by* X*, swap with that bath qubit, and XOR by* X *again restore the complete retained* X, C*, controller target. Per register the bath heat and mean driven work are*

$$Q = W = k_B T (q - \varepsilon) \ln[(1-\varepsilon)/\varepsilon]. \tag{10}$$

With $q = t_0 > t_1 > \cdots > t_m = \varepsilon$*, successive swaps with independent Gibbs qubits of gaps* $\Delta_i = k_B T \ln[(1-t_i)/t_i]$ *give*

$$\beta Q = \beta W = \sum_i (t_{i-1} - t_i) \ln[(1-t_i)/t_i] \longrightarrow h(q) - h(\varepsilon) \tag{11}$$

as the partition mesh tends to zero. All stages have finite dimensions and finite gaps for a fixed positive target error. The finite-step excess is exactly

$$\beta Q - [h(q) - h(\varepsilon)] = \sum_i D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$$

per register; the spent baths end independent of the retained target.

Proof. The first XOR converts $C_i = X \oplus N_i$ into the independent error bit N_i of probability q, without changing X. The thermal swap replaces it by an independent error bit of probability ε; the second XOR restores the intended conditional law. The bath output has probability q, so its energy gain is $\Delta_\varepsilon(q - \varepsilon)$. Controllers remain untouched, and conditional independence is restored. For several swaps, the output of bath qubit i has probability t_{i-1}. Summing its energy gain gives Eq. (11). The right-endpoint Riemann sums tend to $\int_\varepsilon^q \ln[(1-t)/t]\, dt = h(q) - h(\varepsilon)$. The degenerate retained registers have no endpoint energy change, so the first law gives $W = Q$. Each swap's heat equals its error-bit entropy decrease plus $D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$. Telescoping gives the stated finite-step excess. All initial error and bath bits are independent after the first XOR, and the swaps merely permute them; hence the spent baths are independent of the final target. □

This is a label-preserving use of the thermal swap construction of Ref. [14]. These swaps are driven gates between degenerate bank bits and nondegenerate bath qubits, not spontaneous energy-conserving collisions. The label's control permission, thermal preparations, clock and switching belong to the implementation. Spent bath qubits retain the exported entropy and are not reset by the construction. It realizes cyclicity of

the declared retained bank, not an autonomous battery-complete apparatus or an indefinitely recycled bath. The vanishing-error limit $\varepsilon \to 0$ also requires growing gaps or an appropriate limiting reservoir. This thermal-qubit implementation does not establish the separate sole-chiral-exhaust conditions or sharpness of its finite-rate bound.

For $p = .12$ and $\varepsilon = .05$, one correction removes .2378232484 nat per register. Eight correlated copies of one fair label require 1.9025859874 nats of renewal, despite carrying only one bit of distinct label content. Their independent degradation, not the number of distinct labels, sets this maintenance bill. The cyclic target and the independent noise are substantive inputs. Jointly correlated errors, an accessible noise record, a noncyclic pure controller or a protected bank can change the cost. The heat-current claim additionally needs the actual exhaust assumptions; it is not a Landauer bound on every control pulse.

This gives the record side of a maintained observer. T-DOME III and Brain Anchor let retained history choose which access to repair [17]; Eq. (9) prices one implementation of the retained bank on which such a choice relies. Foundation's exact fixed algebras describe passive protection, not a requirement to export positive entropy indefinitely. Under HAFF-D's further matched horizon-flux conditions, the bank's removal rate can enter the conditional horizon-entropy bound. Those conditions connect a real renewal process to gravity; record existence alone does not.

9 Which arrows coincide

Table 1 collects the directions studied above in four specified settings: the collision model with a coherent or a diagonal initial system state, the collision model with one repeated collision, and the autonomous two-qubit model of T-DOME I, in which a system qubit and an environment qubit build and return correlation without any pointer record [5].

Direction	collision, coherent S	collision, diagonal S	one repeated collision	two-qubit recurrence
Record count grows	yes	yes	can fall	no records
Local entropy of fragments grows	yes	yes	can fall	constant E
Marginal memory purity	falls	falls	falls, then returns	no fresh bank
System free energy falls	yes	no	falls, then returns	falls, then returns
Interventions reach only later records	yes	yes	no	no records
Local thermal reset bound $\propto R$	yes	yes	no final record	—
Correlation flows back to S	no	no	yes, with flip	yes

Table 1: Specified protocols, not universal arrows. The repeated-contact column uses the coherent, unmeasured flip continuation of Proposition 7.3. Reset prices refer to blank targets, not arbitrary changes of record content.

With pure fresh ports interacting once, record accumulation, local entropy growth and decreasing marginal port purity align; detached marginals also fix which source

interventions can still reach them. Local thermal reset prices that bank relative to a blank target. These relations do not make every record update an entropy-producing operation. The mixed-port ledger shows that acquisition and action redistribute finite stock, while the echo reverses the local entropy and correlation changes. Source free-energy loss depends on initial coherence. Reduced-system backflow, record acquisition and bank renewal remain separate questions.

10 Relations and interfaces

The purity law gives the record boundary of the past hypothesis a quantitative form, while separating record accumulation from a source's loss of free energy [9, 10]. Mixed-environment redundancy [13] is one instance; the theorem also prices stock drawn from work and a bath. Page–Wootters correlations can supply ordering within a stationary state [12]. A direction of recording additionally needs the acquisition boundary and actual contact architecture specified here.

Complete-access reversal is compatible with retrocausal programmes [16]; restricted thermal reset is a different task. The causal-influence diagnostic computes supplied interventions in a physical recording process. It does not turn them into a derived causal partial order.

Paper E supplies the recording instruments [3]. The present posterior, intervention order and reference-dependent budgets are the record-side interfaces to T-DOME I's thermodynamic agent [5]; Paper C implements a separate maintained sensing body [1]. Paper D's heat and horizon bounds apply only under their matched source conditions [2]. The purity law supplies a physical input for Paper G's question of how selection permissions form [4]; it does not derive those permissions, clocks or a geometric algebra.

11 Conclusion

The product-boundary purity law bounds simultaneously present fragment information. Independent partly mixed ports attain the pointer-controlled bound, and their actual readout supplies a posterior for action. The complete ledger tracks where purity and correlations go; a one-port feedback transfers source free energy without already extracting work. A retained reference can supply conditional stock even when a new port is marginally maximally mixed.

Fresh one-contact ports align record acquisition, loss of coherence and restricted source intervention in a particular architecture. Recontact can instead erase the earlier record and recover source coherence. Reversible relabelling, transfer, thermal reset and exact history reversal are distinct operations. Their prices depend on the target, retained references and accessible gates, not on the word *undoing*.

A noisy record bank poses a further task. Independent errors increase its complete retained entropy; label-controlled thermal swaps restore that target for positive error and approach the entropy-removal bound. A specified chiral exhaust additionally limits the correction rate. This gives a nonempty maintenance mechanism without claiming a renewable autonomous universe or the origin of its clocks and couplings. A broader theory must explain how fragment and interaction structures form, how correlated noise changes renewal, and how actual entropy export participates in source and geometric response.

A Joint-state checks

The mixed-port check constructs the complete diagonal density operator and applies the recording CNOTs, not separately chosen conditional laws. For $\alpha = 0, .2, .8, 1$ and $R = 1, 2, 3, 5$, it verifies Eq. (4), the actual Z-record posterior and one-port feedback. The maximally mixed keyed example is an explicit three-bit permutation. Forty full-density examples with a quantum reference and two initially correlated ports test the conditional bound; the entropy proof establishes it generally.

The coherent echo check applies both collisions and the intermediate flip to the same pure state, verifying $UX_SU = X_S \otimes R$, all marginal and joint entropies, and the recovery of coherence. For renewal, an eight-input XOR–SWAP–XOR permutation verifies the full label–bank target. Thermal chains with $1, 4, 16, 64, 256, 1024$ stages verify the heat and its relative-entropy excess. At $p = .12, \varepsilon = .05$, one stage costs $0.31799941\, k_B T$ per register; 1024 stages cost $0.23789030\, k_B T$, approaching $0.23782325\, k_B T$. These checks certify the finite-dimensional specified circuits, not the physical chiral exhaust.

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Internal Selection and Revisable Organization: How History Changes Permissions and Physical Sources

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Abstract

A physical system can use its records to revise the operations through which it reads and acts. We construct this process with an internal program, a fixed executor and a record-driven update. Controlled couplings select protected algebras; overlap of their spectra permits coherent internal choices without a central program label. The same choice changes an outgoing matter source. A qubit loop has two configurations whose records select their own protection, and different initial histories produce late source weights 1/3 and 2/3. With rotation noise these choices persist for a calculable window and eventually lose their contrast. An arbitrary quantum record can also be converted into the two sustained source configurations: the largest achievable source-weight contrast equals the trace distance between the record states. Finite execution errors shorten the history window by their accumulated operational error. A separate price concerns sharp readouts. Selecting every sharp qubit measurement on a great circle to worst-case basis-state error δ requires program dimension of order $\delta^{-1/2}$; pure and mixed programs have the same feasible dimensions for this criterion. These results identify what history can form inside a supplied architecture. They support a broader mechanism in which learned relations become revisable permissions, while leaving primitive couplings and their origin as a distinct generation problem.

1 Introduction

A record can change what a system is able to do next. A damaged body may install a different protective encoding; a reader may move its probe; an internal estimate may lower the authority given to an unreliable response. These changes concern actual operations. To explain them, a theory of selected structure needs a physical account of how a history selects and revises the relations that carry its own activity.

Papers A–F separate parts of this problem. Channels determine which distinctions are read in Paper A; couplings and observation windows determine which survive in Paper B; instruments create records in Paper E; their retention and renewal have resource prices in Paper F [1, 2, 4, 5]. Paper C connects sensing and intervention to a body's viability, while Paper D asks how selected structure and physical sources constrain geometry [25, 3].

Here the question is whether the choice of operation can itself arise from records inside the system.

An operation, a stored description and the device that executes it are different physical objects. We specify an internal program register, a fixed executor and an update instrument. Under these inputs the system selects its couplings, maintains a choice through the records it protects, and changes the matter it emits. The choice need not be a central environmental record. Its origin in a distinguishable history and its effect on a source can be tested separately.

The main construction is a qubit with two record-selected protective configurations. Their history-dependent source weights remain different over a finite noisy window. We then identify the exact information that can pass from a quantum record to this source choice, including a construction that reaches the trace-distance bound. An error bound makes the same finite window robust to imperfect execution. These statements supply a causal interface from history to permission and source; they do not use a reconstructed geometry as a substitute for an actual change of matter.

The cost of selecting operations is separate from the cost of retaining their records. Sharp alternatives require orthogonal program supports. For the one-parameter great-circle qubit family the required register dimension grows as $\delta^{-1/2}$, and mixing cannot improve it under the two basis-state error criterion. Section 6 states this result; Appendix A contains the full proof and numerical bounds. Section 7 places the finite target-register boundary in its proper scope. Section 8 develops the broader formation mechanism and its failure conditions.

Breuer's state self-measurement limit [12] and Wolpert's physical inference limits [13] concern what an embedded system can know. Here we ask what it can choose and execute. The existence of self-reference is not in dispute: self-reproducing automata, recursive programs and fixed-point constructions provide positive examples [14, 21, 22]. What requires explanation is the physical relation between a description, its execution and the history that revises it.

2 A typed model of internal selection

An element of an observable algebra is an operator, not a description of an algebra, and a formula naming a selection rule is syntax until an encoding maps it into physical degrees of freedom. The following model makes that encoding explicit.

Definition 2.1 (Internal selection model)**.** *A finite internal selection model consists of*

1. *a finite-dimensional system S, on which selected operations act;*
2. *a program register P of dimension d_P and program states $|p\rangle \in P$, $p \in \mathcal{P}$, unit vectors that need not be orthogonal;*
3. *an executor, a fixed physical operation on $P \otimes S$, which for each program state determines an operation on S: a channel $\Phi_p(\rho) = \mathrm{Tr}_P\, \mathcal{G}(|p\rangle\langle p| \otimes \rho)$, a measurement E^p obtained by following $\mathcal{G}$ with a fixed measurement, or a program-controlled coupling to an environment;*
4. *records, the classical outcomes of instruments, and an update rule, a fixed instrument that prepares the next program state from records.*

A selection rule is internalized if the choice of operation is made by program states under a fixed executor and a fixed update rule. A configuration (p, ρ) is self-grounding if ρ is left stable by the operation selected by p and the update rule, applied to the records that ρ produces, returns p.

The space of descriptions is the set of program states, the encoding is the register, the evaluation map is the executor, and the admissible fixed points are the self-grounding configurations. The executor's dispatch map, $p \mapsto \Phi_p$, is physical dynamics; it is not itself a program. Whether it can be made one is the question of Section 7.

3 Selecting one's own accessible algebra

Paper B shows that a system coupled to its environment through commuting operators $A_1, \ldots, A_K$ at positive rates γ_k has, as its stable algebra, the commutant $\{A_k\}'$, and that environmental records can carry only the label of the block of that algebra. We now let the system choose its own couplings. Take an orthonormal program basis and program-controlled couplings

$$A_k = \sum_{p \in \mathcal{P}} |p\rangle\langle p| \otimes A_k^{(p)}, \tag{1}$$

where for each program p the operators $A_1^{(p)}, \ldots, A_K^{(p)}$ on S commute. Let $|p, x\rangle$ be the product of $|p\rangle$ with a joint eigenvector of the $A_k^{(p)}$, with eigenvalues $a_k^{(p)}(x)$.

Proposition 3.1 (Program selection and central records)**.** *For the couplings* (1)*:*

1. *they commute, and the algebra fixed by the generated dephasing semigroup is* $\{A_1, \ldots, A_K\}'$*.*

2. *Compressed to the subspace* $|p\rangle \otimes S$ *of a program* p*, this algebra is* $|p\rangle\langle p| \otimes \{A_1^{(p)}, \ldots, A_K^{(p)}\}'$*: each program selects the algebra that Paper B assigns to its own couplings.*

3. *The matrix element between* $|p, x\rangle$ *and* $|q, y\rangle$ *is multiplied by* $\exp[-\frac{t}{2}\sum_k \gamma_k (a_k^{(p)}(x) - a_k^{(q)}(y))^2]$*. Coherence between two programs survives exactly between joint eigenvectors with equal eigenvalue tuples. Disjoint joint spectra for* p *and* q *make every operator connecting those two programs decay. The label projector* $|p\rangle\langle p| \otimes \mathbf{1}_S$ *is central in the stable algebra if and only if the joint spectrum of* p *is disjoint from that of every other program. In that case environmental fragments can record its label.*

Proof. The operators are block diagonal in p and commute within each block, so they commute; the basis $|p, x\rangle$ diagonalizes all of them, with eigenvalues $a_k^{(p)}(x)$. Paper B's proposition on commuting couplings then gives the Schur-multiplier form in (3) and the fixed algebra in (1). Its compression to $|p\rangle \otimes S$ consists of the operators there that commute with the $A_k^{(p)}$, which is (2). A program label is central precisely when it is constant on each joint eigenspace, which gives the disjointness condition in (3). The record statement is Paper B's result that fragments carry only block labels, applied to the joint system. □

When the programs have disjoint joint spectra, their labels belong to the centre of the stable algebra and can become redundant environmental records. Shared eigenvalue tuples protect coherence between the corresponding configurations; the spectra need not coincide completely for this exception to occur. With $A = |0\rangle\langle 0| \otimes Z + |1\rangle\langle 1| \otimes X$ on a program qubit and a system qubit, both programs have eigenvalues ± 1, and the superposition $(|0,0\rangle + |1,+\rangle)/\sqrt{2}$ is an eigenvector of A. Replacing X by $X + 3$ makes the spectra disjoint, and the same superposition decays to an equal mixture of the two selections. Both cases are computed in the accompanying check script.

For more than two programs, disjointness from just one alternative is not enough. With $A = |p\rangle\langle p| \otimes Z + |q\rangle\langle q| \otimes (Z + 3\mathbf{1}) + |r\rangle\langle r| \otimes Z$, the spectra of p and q are disjoint, but $|p,0\rangle\langle r,0|$ commutes with A and not with $|p\rangle\langle p| \otimes \mathbf{1}_S$. Thus p is not a central label. The condition in Proposition 3.1 concerns the whole program family.

Internal selection can also change a material source. Use the field normalization of Paper D, in which a coupling through a projector R obeys $U^\dagger j(u) U = j(u) + R g'(u)/2$. The program controls that projector while the executor, drive and switching schedule stay fixed.

Proposition 3.2 (A program-selected material source)**.** *Let R_p be projectors on S and put*

$$A = \sum_p |p\rangle\langle p| \otimes R_p, \qquad U = \exp(-i\lambda A \otimes Q), \qquad Q = b + b^\dagger.$$

Take an initial state ρ_{PS} independent of the field state σ. Set $r = \mathrm{Tr}(\rho_{PS} A)$. The output field marginal is exactly

$$\sigma_{\mathrm{out}} = (1 - r)\sigma + r\, e^{-i\lambda Q} \sigma e^{i\lambda Q}.$$

When the displayed current moments exist, its vacuum-subtracted stress is

$$T_{\mathrm{out}}(u) = T_{\mathrm{in}}(u) + r \left[g'(u) \langle j(u) \rangle_\sigma + \frac{g'(u)^2}{4} \right].$$

In a fixed classical program branch p, a non-atomic quadrature spectral law and $\lambda \neq 0$ give the exact multiplicative domain $\{R_p\}'$ of the induced system channel. Thus changing the program can change both the represented algebra and the field source under the same executor.

Proof. Since $A^2 = A$, write $U = (1 - A) \otimes 1 + A \otimes e^{-i\lambda Q}$. The off-diagonal terms have zero partial trace over PS, because $\mathrm{Tr}(A \rho_{PS} (1 - A)) = 0$. This gives the field mixture even when the program is coherent or correlated with S. Squaring $j + A g'/2$ gives the stress formula: initial independence supplies the mixed expectation $\langle A j \rangle = r \langle j \rangle$, and $\langle A^2 \rangle = r$. Within branch p, the channel is the two-block dephasing channel of Paper D; a non-atomic spectral law makes its nonzero-gap characteristic value strictly smaller than one, so equality in the Schwarz inequality holds exactly on $\{R_p\}'$. □

For $R_0 = (1 + Z)/2$, $R_1 = (1 + X)/2$ and $\rho_S = (1 + 0.8Z)/2$, a program bit prepared in 0 or 1 gives $r = 0.9$ or 0.5, respectively. Their conditional pointer algebras differ, although the two coupling spectra are identical. The field stress therefore need not reveal the program label as a central sector to respond to its choice. With a classical program mixture π, $r = 0.9\pi_0 + 0.5\pi_1$. A record-driven program update can change this weight without an externally assigned emission probability. The global algebra $\{A\}'$ stays fixed; the selected branch algebra on S and the outgoing source vary. This separates internal

selection from a change of the executor. Paper D supplies the corresponding JT response for this same wavepacket with matched initial coefficient data; the current statement identifies its physical source, not the origin of the JT action or of a Type III geometric sector. The program, field preparation and switching remain resources of the executor.

4 A self-grounding selection

A selection rule can take as input a record that the selection itself protects. A small model gives two self-grounding fixed points and their history-dependent selection.

A memory qubit M is subject to rotation noise $H = \frac{\omega}{2}Y$. A classical program bit p selects a dephasing coupling of M with rate γ: through Z for $p = 0$ and through X for $p = 1$. Strong dephasing in a basis protects a record in that basis against rotation, by the quantum Zeno mechanism [19, 20]. Every time τ the memory is read in the protected basis, and the update rule sets p equal to the outcome. The rule is internalized in the sense of Definition 2.1, and it is self-applied: the record that sets the selection is a record in the algebra that the selection protects.

After a read the configuration is one of four: $F_0 = (p = 0, |0\rangle)$, $T_1 = (p = 1, |1\rangle)$, $F_1 = (p = 1, |-\rangle)$ and $T_0 = (p = 0, |+\rangle)$. In F_0 and F_1 the protected record agrees with the program it sets; these are the self-grounding configurations. In T_1 and T_0 the record has just switched the program to a basis in which that record is not sharp.

Proposition 4.1 (Self-grounding configurations)**.** *Without rotation noise, F_0 and F_1 are absorbing, and from T_1 and T_0*

$$\Pr(\textit{absorbed in } F_1 \mid T_1) = \tfrac{2}{3}, \qquad \Pr(\textit{absorbed in } F_1 \mid T_0) = \tfrac{1}{3}.$$

From any initial program and memory state the loop reaches F_0 or F_1 with probability one.

Proof. A read in the basis that is already sharp returns the same outcome, so F_0 and F_1 repeat. In T_1 the memory $|1\rangle = (|+\rangle - |-\rangle)/\sqrt{2}$ is read in the X basis, giving F_1 or T_0 with probability $\frac{1}{2}$ each; in T_0 the memory $|+\rangle$ is read in the Z basis, giving F_0 or T_1 with probability $\frac{1}{2}$ each. Writing a and b for the absorption probabilities into F_1 from T_1 and T_0, $a = \frac{1}{2} + \frac{1}{2}b$ and $b = \frac{1}{2}a$, so $a = \frac{2}{3}$, $b = \frac{1}{3}$. Every read from a transient configuration ends in an absorbing one with probability $\frac{1}{2}$, and the first read from an arbitrary initial state produces one of the four configurations. □

Corollary 4.2 (History selects a sustained source)**.** *In the zero-rotation loop, insert before each read a fresh field interaction of Proposition 3.2, using $R_0 = (1 + Z)/2$ and $R_1 = (1 + X)/2$. Each cycle uses a distinct tensor-factor port, in the same state and initially independent of the program, memory and all previous ports. The four-state transition law and its absorption probabilities are unchanged. In F_0 the output field is $e^{-i\lambda Q}\sigma e^{i\lambda Q}$; in F_1 it is σ. Thus the two self-grounding configurations carry different steady field sources under one executor. Starting from T_1 or T_0, the late-cycle mean source weight tends to $1/3$ or $2/3$, respectively.*

Proof. Within either program branch the field coupling is diagonal in the basis of the ensuing read. It changes no outcome probabilities, so the transition matrix of Proposition 4.1 is unchanged. The configurations $F_0 = (0, |0\rangle)$ and $F_1 = (1, |-\rangle)$ are eigenvectors of A with eigenvalues 1 and 0. They remain product states of program and memory, and

each pass gives the claimed field output. The source weight therefore converges to the indicator of absorption in F_0. Its expectation is one minus the corresponding probability of absorption in F_1, which gives the stated fractions. □

The read/update history generates the source difference under one executor. Fresh ports and memory operations sustain it. Each cycle supplies a fresh product-state field port, a protected-basis read, the program update and its protecting environment. For a centered field input the coupling's mean switching work is $r_n E_g$, where $E_g = \lambda^2 \bar{\omega} = \int g'^2 du/4$ and r_n is the actual source weight. Noncentered inputs add the displayed current cross term. Incoming beam preparation, memory retention and eventual reset have their own charges. The 1/3 and 2/3 weights arise from absorption of the two transient histories, not a continuing supply of new source information. Distinct modes of a common bosonic field must be orthogonal and prepared in a product state to realize these ports; mere time translation does not suffice. For Paper D's exact packet, a translation by Δ has overlap modulus $4a^2/(4a^2 + \Delta^2)$. Source accounting for such overlapping pulses requires their joint field state. The full-line packet is a scattering idealization; a record decided at a finite time needs a compact switch scheduled after that decision, as constructed in Paper D. With rotation, the following lifetime uses the stated continuous dephasing generator.

The dephasing plays no role without noise, because the reads are made in the protected basis. With noise it decides whether the self-grounding configurations last.

Proposition 4.3 (Lifetime of a self-grounding selection)**.** *With rotation noise, the probability of leaving F_0 (or F_1) in one cycle is $\varepsilon = \frac{1}{2}(1-z(\tau))$, where z solves $\ddot{z}+2\gamma\dot{z}+\omega^2 z = 0$, $z(0) = 1$, $\dot{z}(0) = 0$. For $\gamma > |\omega|$,*

$$z(\tau) = \frac{\lambda_+ e^{\lambda_- \tau} - \lambda_- e^{\lambda_+ \tau}}{\lambda_+ - \lambda_-}, \qquad \lambda_\pm = -\gamma \pm \sqrt{\gamma^2 - \omega^2},$$

so $\varepsilon \approx \omega^2 \tau/(4\gamma)$ when $\gamma \gg |\omega|$, $\gamma\tau \gg 1$ and $\omega^2\tau/\gamma \ll 1$. Without protection $\varepsilon = \sin^2(\omega\tau/2)$; with protection in the wrong basis ε approaches $\frac{1}{2}$ in the strong-dephasing regime.

Proof. In F_0 the Bloch vector starts at $z = 1$. The rotation gives $\dot{x} = \omega z$, $\dot{z} = -\omega x$, and dephasing through Z adds $-2\gamma x$ to $\dot{x}$. Eliminating x gives the stated equation; a Z read returns 1 with probability $\frac{1}{2}(1 - z)$. The case F_1 is the same with x and z exchanged, since rotation about y treats both records alike. For $\gamma \gg \omega$, $\lambda_+ \approx -\omega^2/(2\gamma)$ dominates. Without dephasing $z = \cos\omega\tau$. Dephasing in the wrong basis destroys the record at rate 2γ. □

For $\omega = 0.1$ and $\tau = 1$, the escape probability per cycle is 1.42×10^{-3}, 2.37×10^{-4} and 2.49×10^{-5} for $\gamma = 1$, 10 and 100, against 2.50×10^{-3} without protection. Protection in the wrong basis gives 0.433 at $\gamma = 1$ and approaches 0.5 at the larger rates. At $\gamma = 100$ a self-grounding selection lasts on average 4.0×10^4 cycles before the loop passes through a transient configuration and settles again, in F_0 or F_1.

The residence time alone does not determine how long two histories produce different sources. The same four-state process gives that distinction exactly.

Proposition 4.4 (History-dependent source window)**.** *Let $\gamma > 0$, $\omega \neq 0$ and $\tau > 0$. In each cycle first evolve for time τ with the selected rotation–dephasing generator, then*

apply the fresh field interaction of Corollary 4.2, and then read and update. Put

$$k = \sqrt{\gamma^2 - \omega^2}, \quad f = e^{-\gamma\tau}\left(\cosh k\tau + \gamma\frac{\sinh k\tau}{k}\right), \quad b = \omega e^{-\gamma\tau}\frac{\sinh k\tau}{k},$$

with analytic continuation when $\gamma < |\omega|$ and $\sinh(k\tau)/k = \tau$ at $k = 0$. Set $\varepsilon = (1-f)/2$, $q_+ = (1+b)/2$, $q_- = (1-b)/2$. In the post-read order (F_0, T_1, F_1, T_0) the transition matrix and the source-weight vector are

$$T = \begin{pmatrix} 1-\varepsilon & \varepsilon & 0 & 0 \\ 0 & 0 & q_+ & q_- \\ 0 & 0 & 1-\varepsilon & \varepsilon \\ q_- & q_+ & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1-\varepsilon \\ q_- \\ \varepsilon \\ q_- \end{pmatrix}. \tag{2}$$

For a post-read distribution μ_n, the next field output has mean stress

$$T_{\mathrm{in}}(u) + (\mu_n r)\left[g'(u)\langle j(u)\rangle_\sigma + g'(u)^2/4\right].$$

The chain has the unique stationary law

$$\pi = \frac{(q_-, \varepsilon, q_+, \varepsilon)}{1+2\varepsilon}, \qquad r_\infty = \pi r = \frac{1}{2} - \frac{b}{2(1+2\varepsilon)}.$$

For initial states T_1 and T_0, define

$$R_1^\circ = \varepsilon + f\frac{q_-^2}{1-q_+q_-}, \qquad R_0^\circ = \varepsilon + f\frac{q_-}{1-q_+q_-}.$$

Their actual source weights satisfy, for every integer $n \geq 0$,

$$|\mu_n r - R_i^\circ| \leq n\varepsilon + (q_+q_-)^{\lfloor n/2\rfloor}, \qquad \mu_0 = T_i. \tag{3}$$

Thus a small escape rate preserves different history-selected sources over $1 \ll n \ll \varepsilon^{-1}$, although both histories ultimately have the same stationary mean.

Proof. The two Bloch generators on (x, z) are

$$\begin{pmatrix} -2\gamma & \omega \\ -\omega & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & \omega \\ -\omega & -2\gamma \end{pmatrix}.$$

Exponentiating them gives (2). The source weight is the population of the selected projector immediately before the read, not its value in the preceding post-read state. The intervening field coupling preserves that population. It equals the probability that the next program is 0. For nonzero rotation and positive dephasing, the Bloch norm strictly decreases from each of the four pure initial states over any positive interval. Hence $0 < \varepsilon < 1$ and $0 < q_\pm < 1$. The graph of T is connected and has self-loops, giving uniqueness and convergence. Direct multiplication gives $\pi T = \pi$ and the stated source weight.

For the window bound, replace the two F rows by absorbing rows, retaining $q_\pm$ in the transient rows and retaining the same observable r. In this comparison chain the absorption probabilities into F_0 are $q_-^2/(1-q_+q_-)$ and $q_-/(1-q_+q_-)$; the corresponding limiting expectations of r are R_1° and R_0°. Transient survival after n steps is at most $(q_+q_-)^{\lfloor n/2\rfloor}$. Couple the true and comparison chains until their first escape from an F state. That event has probability at most $n\varepsilon$, and $0 \leq r \leq 1$, proving (3). □

The history window and the lifetime of one configuration are different. The characteristic polynomial is

$$(z-1)\left[z^3+(2\varepsilon-1)z^2+(\varepsilon^2-q_+q_-)z+\varepsilon^2-\varepsilon+q_+q_-\right].$$

For small ε the slow root is $1-\varepsilon/(1-q_+q_-)+O(\varepsilon^2)$, so the late history contrast decays on a scale of about $(1-q_+q_-)/\varepsilon$ cycles. In the strong-protection regime of Proposition 4.3, this is about $3\gamma/(\omega^2\tau)$, whereas a single configuration has mean residence $1/\varepsilon \approx 4\gamma/(\omega^2\tau)$. At $\gamma = 100$, $\omega = 0.1$, $\tau = 1$, the source weights from T_1 and T_0 are respectively 0.33844 and 0.66090 after 1000 cycles, and 0.49370 and 0.50579 after 10^5 cycles. Both approach 0.4997500.

The loop selects a protection because the record it protects says so, and its history selects a material source through the same program-controlled executor. Finite noise makes that selection metastable rather than permanent. The rotation–dephasing stage, field interaction and read are sequential operations here; simultaneous rotation during a finite field pulse would require a different transition law. The protecting environment, fresh packets, read instrument and update map are supplied. The loop generates the history-dependent choice, not these resources.

5 When a history changes the next source

The four-state loop starts with explicit configurations. An earlier history need not already be a classical choice among them. Let R be a retained physical record, and let $\Lambda : R \to P \otimes M$ be a fixed preparation channel acting after that history has formed. It can include a measurement, an outcome record and a controlled preparation. The next fresh field port remains independent of R, P, M and old emitted ports. This time ordering gives a direct criterion for what the history can cause.

Proposition 5.1 (Record-to-source distinguishability)**.** *For two record states ρ_0, ρ_1 and the projector A in Proposition 3.2, the two source weights satisfy*

$$r_0 - r_1 = \mathrm{Tr}[(\rho_0-\rho_1)\Lambda^*(A)], \qquad |r_0-r_1| \le D(\rho_0,\rho_1) := \tfrac{1}{2}\|\rho_0-\rho_1\|_1.$$

For the zero-rotation loop, the largest contrast over fixed preparation channels Λ equals $D(\rho_0,\rho_1)$. A channel attaining it prepares F_0 or F_1 from an actual binary record measurement. Both subsequent histories remain in their chosen configuration, so the same contrast occurs in every later fresh source port.

Proof. The first equality is the adjoint-channel identity. Complete positivity and unitality of Λ^* give $0 \le \Lambda^*(A) \le 1$. Writing the trace-zero Hermitian operator $\rho_0-\rho_1$ as its positive part minus its negative part bounds the expectation of any effect by D. To attain the bound, let H be the spectral projector onto its positive part. Measure $\{H, 1-H\}$ and prepare F_0 on outcome H, F_1 otherwise. The resulting source weight is $\mathrm{Tr}(\rho_i H)$ because F_0, F_1 have A-eigenvalues $1, 0$. Their difference is D. Proposition 4.1 preserves these configurations and Corollary 4.2 preserves their source weights. □

The discrimination identity is a standard trace-distance fact; its role here is to determine exactly how much record information can become a difference in protected configuration and actual source. A nonzero source contrast is possible precisely when the two record states differ, provided this binary measurement and the two preparations are

available. A specified executor can fail to use that difference: if $\Lambda^*(A)$ is orthogonal to $\rho_0 - \rho_1$ in trace pairing, its outgoing source is the same. Distinguishable histories need not yield distinguishable sources. Conversely, changing a conditional pointer algebra need not change r; a maximally mixed M gives $r = 1/2$ for both the Z and X selections.

This construction does not preserve an arbitrary coherent input record under measurement. For a classical history stored in orthogonal states the preparation can be controlled without deleting it. An independent old content–reference state Ω_{CE} is preserved jointly, even conditioned on the new outcome, when all preparation and loop operations act on R, P, M and leave C, E untouched. If R is already correlated with CE, the old marginal is preserved by the nonselective channel but conditioning can change its state. This is the distinction required when selection is claimed to retain an earlier body or history.

Ideal execution is not required for a finite history effect. Suppose each complete read/update instrument still outputs the four listed configurations, its actual sampled transition in cycle j is $\widetilde{T}_j$, and its actual source weight is $\widetilde{r}_n$, with

$$\max_x \mathrm{TV}(\widetilde{T}_j(x,\cdot), T(x,\cdot)) \le a_j, \qquad \|\widetilde{r}_n - r\|_\infty \le c_n.$$

These are errors of the stated four-state instrument and observable, not an identification of every physical error with a flip.

Corollary 5.2 (A robust finite source window)**.** *From either transient initial configuration T_i, the actual source weight obeys*

$$|\widetilde{\mu}_n \widetilde{r}_n - R_i^\circ| \le n\varepsilon + (q_+q_-)^{\lfloor n/2 \rfloor} + \sum_{j=1}^{n} a_j + c_n.$$

Consequently the contrast between the two histories is at least

$$\frac{|f| q_+ q_-}{1 - q_+q_-} - 2n\varepsilon - 2(q_+q_-)^{\lfloor n/2 \rfloor} - 2\sum_{j=1}^{n} a_j - 2c_n,$$

whenever the right-hand side is positive.

Proof. At each step, couple the true row and its ideal counterpart with mismatch probability at most a_j. After n steps their distributions differ in total variation by at most $\sum_j a_j$. Expectations of $r \in [0,1]$ differ by at most that amount, and changing the source observable costs at most c_n. Apply Eq. (3). Finally, $|R_0^\circ - R_1^\circ| = |f| q_+q_-/(1 - q_+q_-)$, and use the triangle inequality for the two source weights. □

For sufficiently small escape and execution errors, this bound certifies a nonempty interval of different physical consequences. Lower escape rates extend that interval only if execution errors and fresh-port correlations remain controlled. Holding a noisy selection indefinitely would need renewed resources or a stronger protection mechanism. The present bound supplies no free infinite stream.

6 The price of selecting sharp readouts

The fixed executor may be followed by a fixed measurement, with a fixed ancilla. A program selects effects E_z^p on a labelled target basis $\{|b_z^p\rangle\}$. We use the worst-case basis-state error

$$\delta_p = \max_z \left(1 - \langle b_z^p | E_z^p | b_z^p \rangle\right).$$

The program dimension counts the selectable register; a fixed ancillary state does not enlarge it. This criterion concerns the target outcome probabilities, and does not compare complete instruments in diamond norm.

Theorem 6.1 (Cost of sharp selection)**.** *1. If two pure program states select two distinct labelled sharp measurements exactly, they are orthogonal. Exact mixed programs for distinct sharp measurements have orthogonal supports. A register of dimension d_P selects at most d_P distinct labelled sharp measurements exactly.*

2. Let S be a qubit, and suppose that for every Bloch angle θ on a great circle some pure program state selects the sharp measurement along θ with error at most δ, where $0 < \delta < 1$. Then

$$d_P \;\geq\; L(\delta) := \max_{M\geq 1} \frac{M}{1 + \frac{1}{3}\big(2\sqrt{\delta} + \delta\big)^2 (M^2 - 1)}, \tag{4}$$

and $L(\delta)\sqrt{\delta} \to \sqrt{3}/4 \approx 0.433$ as $\delta \to 0$. Conversely, $N = \lceil \pi/(2 \arcsin\sqrt{\delta}) \rceil$ orthogonal program states with a controlled rotation achieve error δ, and $N\sqrt{\delta} \to \pi/2$.

3. Part (2) holds unchanged for mixed program states, with $0 < \delta < 1$. For each target a pure program in the same register has exactly the same two basis-state errors. Consequently $d_P \geq L(\delta)$ and $\liminf_{\delta\to 0} d_P\sqrt{\delta} \geq \sqrt{3}/4$; the same grid gives the upper bound. Mixing cannot reduce the required program dimension under this criterion.

The proof and representative numerical bounds are in Appendix A. The constants $\sqrt{3}/4$ and $\pi/2$ are lower and constructive upper bounds; the result establishes the exponent, not an optimal leading constant. Exact orthogonality is the measurement counterpart of the no-programming theorem for unitaries [15]. The equality of the pure and mixed costs here follows from the convex joint numerical range of two Hermitian operators [18]. Three independent errors need not admit the same pure replacement. The asymptotic exponent applies to this great-circle family and this criterion; universal processors have other resource bounds [17].

A two-axis reader needs only two orthogonal choices [11]. Tracking a continuum of axes to increasing accuracy needs a growing register. Storing and refreshing the chosen programs has a further purity and free-energy account [4, 5]. Coherently returning a program does not, by itself, assign it a reset heat. Conversely, a large register does not supply the entropy outlet needed to renew noisy records.

7 What the executor leaves specified

Rules, programs, their revision and self-grounding selections can be internal. Within Definition 2.1, a full copy of an executor acting on $P \otimes S$ acts on dimension $d_P d_S$, whereas each selected operation acts on S or part of it, of dimension at most $d_S < d_P d_S$ when $d_P \geq 2$. It cannot be that full executor. A meta-register P' can select among executors on $P \otimes S$ under a fixed meta-executor. At the top of each stipulated finite hierarchy the model still supplies an executor as physical dynamics. This is a feature of the architecture, not a general prohibition on a system containing its own control law.

The top dispatch map is specified dynamics in this architecture. A meta-register can make the current executor selectable, at a resource cost set by the family to be selected. For the one-parameter great-circle qubit family, Theorem 6.1 gives order $\delta^{-1/2}$ register dimension; it assigns no common exponent to every continuous family. Stored classical choices also use the record resources described above. The dimension argument excludes a full executor acting on the larger space from being an operation on the smaller target; it permits storing a description of that executor. In infinite dimension the counting argument fails, since a separable Hilbert space is isomorphic to its tensor product with $\mathbb{C}^2$. Kleene and Lawvere fixed points [21, 22], and the order-theoretic results of Scott and Tarski [23, 24], also permit self-reference. Existence of a self-representing physical executor in that setting remains open.

Every stipulated finite hierarchy specifies its top dispatch map. Self-application is different: the loop of Section 4 applies a rule to its own protected records and reaches fixed points without adding levels. A claim that every operation is selected faces a separate resource test. By Theorem 6.1, an executor selecting every sharp readout exactly would require infinitely many orthogonal program states.

The distinction between structure and knowledge has a concrete meaning here. The dispatch map is physical dynamics; program contents are records. When their coupling spectra are disjoint they can be public environmental records, while overlapping eigenvalue tuples can hide some coherence. Proposition 3.2 adds a further distinction: a program need not become a central label to change the field source through the operation it selects.

8 From selectable rules to formed permissions

The loop forms a choice among supplied couplings. A self-frame needs more: history must establish a relation between a task and the body's own reading or action channels. T-DOME II separates an aligned axis from a signed relation that can actually predict [26]. If opposite couplings produce the same retained axis, the axis can be perfectly aligned while its posterior prediction is zero. Outcome-bearing records supply the missing sign. A useful program must encode whichever distinctions its future task consumes.

Self-readout then becomes a problem of attribution. A changed residual can come from the world, from a damaged body, or from a miscalibrated interface. T-DOME III uses identifiable references and paid maintenance to decide among these causes [9]. Its action is based on retained value and body condition; confidence alone need not determine whether repair is worthwhile. Imprint W supplies a complementary mechanism in which the system uses its own reading history to choose a later inquiry [11]. These are mechanisms for revising programs, with their own record laws and budgets.

Brain Anchor locates formed relations in a body: diagnostic outcomes can alter routes and install a different protected encoding [29]. Such relations can be distributed among components. Calling their aggregate a program register does not imply that a single place stores a full description, or that every relation becomes a central environmental label. Proposition 3.1 already gives a finite counterexample to the latter inference. A mathematical encoding is an interface for analysis; a physical implementation must say where its records, actuators and update couplings reside.

Q-RAIF C gives a further, independent source implementation [27]. Its actual quantum history enters a complete positive Hamiltonian and changes material motion and

interacting field source. The fresh-port projector mixture used here is simpler and has different inputs. The two constructions support the same causal question, but their clocks, controllers and source objects cannot be exchanged merely because both have an internal program. Returning a field response to the same body requires the actual sensor and finite-resource operations, as studied in Foundation III [28]. A reconstructed geometry, a field stress and the total apparatus stress remain distinct.

The more ambitious generation problem concerns how these operational relations become available in the first place. A fixed meta-register only relocates a supplied dispatch map. A more informative mechanism is adaptation of the body: records change local couplings, routes or encodings, and the new relations alter which later records and acts are reliable. This can generate a permission from primitive interactions without generating the primitive interactions themselves.

Conjecture 8.1 (Formation of revisable permissions)**.** *Consider a body with locally adjustable couplings and a finite family of reversible encodings of the same logical content. Let actual diagnostic instruments act on relational degrees of freedom outside that logical algebra, and distinguish configurations with different task values. Suppose exploratory operations reach each necessary diagnostic relation, the best configuration has a nonzero value gap on each stationary interval, and paid update and maintenance resources suffice over that interval. There exist local record-driven update dynamics for which task-relevant relations become usable permissions from initially uncalibrated couplings, while previously encoded content remains within a stated error budget. When the task or body changes slowly enough, the same retained relations support detection and revision, rather than only repeating a fixed choice.*

The proposed mechanism is concrete: exploration creates diagnostic records; local retained relations select a route or encoding; an actuator installs it; subsequent errors and maintenance outcomes modify those relations. History-selected protection, signed self-frames and local body learning support different parts of this mechanism. Its quantitative test compares formed relations with supplied ones under the same preparation, diagnostic, installation and content-loss budgets. Formation must lose its advantage when diagnostics alias task-distinct configurations, exploration cannot reach the relevant coupling, or noise destroys old content before learning and migration finish. A distributed learner that changes only labels while all implemented instruments remain fixed also fails this test.

The conjecture asks for a common finite construction with these inputs and outcomes. It is stronger than the supplied two-choice loop, and is not used in its proof. An infinite stationary guarantee would additionally need a sustainable resource supply and control of rare failures. A finite family chosen by the theorist can demonstrate a formation mechanism; it does not explain the origin of every physical interaction or privilege one universal observable algebra.

9 Conclusion

History can select the relations through which a system continues to act. In the finite construction, records choose their own protection and alter the next field source under one executor. Shared coupling spectra show that an internal choice need not become a central label. The exact noisy transition law distinguishes residence in one configuration, a finite history-dependent source window and eventual loss of source contrast. Record-

to-source transfer reaches the trace-distance limit, while accumulated execution errors bound how much of that window remains usable.

Selecting accurate sharp readouts has its own dimensional cost. The one-parameter great-circle family requires order $\delta^{-1/2}$ dimension, with the same feasibility for pure and mixed programs under two target errors. This price and the finite target/executor distinction constrain a specified architecture. They allow stored descriptions, self-application and stronger models of executor formation.

The resulting account joins internal description to a physical consequence without making them identical. Its broader question is how a body's learned relations become revisable permissions while content, references and resources survive. The formation conjecture sets out a mechanism and interventions for that question; primitive dynamics, general geometric generation and first-person experience require their own arguments.

The selection registers and target systems are finite-dimensional; the source realization uses the specified bosonic field port of Paper D. The counting observation concerns this target-register architecture, not every formalism for physical self-reference, and fails in infinite dimension. Internal program changes select actual couplings and can alter their output stress. They do not by themselves explain why this executor exists, reconstruct a geometric sector or establish reflective experience.

Paper D's sustained selection–response conjecture additionally prices thermal entropy export and its gravitational reading [3]. The fresh-port program loop here supplies neither that complete thermal account nor its joint sharp-exhaust and causal-return implementation.

A Proofs and constructions for sharp selection

Here we give the full argument for the readout cost. A readout here is a measurement on S implemented by a fixed executor: a unitary on $P \otimes S \otimes A$, with an ancilla A in a fixed state, followed by a fixed measurement. Purifying the fixed ancilla and dilating the measurement gives projections F_z pulled back to the input space. The embedding $|p\rangle \mapsto |p\rangle|a\rangle$ keeps program overlaps and has span dimension d_P; the ancillary dimension is not a selectable program resource. Write G_z for the induced POVM effects on the original $P \otimes S$, after compression to the fixed ancilla state. A pure program $|p\rangle$ selects effects $E_z^p = (\langle p| \otimes \mathbf{1})G_z(|p\rangle \otimes \mathbf{1})$. Its target is a labelled orthonormal basis $\{|b_z^p\rangle\}$ of S, with selection error $\delta_p = \max_z\big(1 - \langle b_z^p|E_z^p|b_z^p\rangle\big)$. For a mixed program σ, use $E_z^\sigma = \mathrm{Tr}_P[(\sigma \otimes \mathbf{1})G_z]$ and the same error on the target basis states. This is a basis-state error criterion, not a diamond-norm distance between instruments.

Lemma A.1 (Near-orthogonality)**.** *For programs p, q and outcomes $z \neq z'$,*

$$|\langle q|p\rangle|\,|\langle b_{z'}^q|b_z^p\rangle| \leq \sqrt{\delta_p} + \sqrt{\delta_q} + \sqrt{\delta_p\delta_q}.$$

Proof. Put $v = |p\rangle|b_z^p\rangle|a\rangle$ and $w = |q\rangle|b_{z'}^q\rangle|a\rangle$. Then $\langle v|F_z|v\rangle = \langle b_z^p|E_z^p|b_z^p\rangle \geq 1 - \delta_p$, so $\|(\mathbf{1} - F_z)v\| \leq \sqrt{\delta_p}$, and likewise $\|(\mathbf{1} - F_{z'})w\| \leq \sqrt{\delta_q}$. Write $v = F_z v + (\mathbf{1} - F_z)v$ and $w = F_{z'}w + (\mathbf{1} - F_{z'})w$. Since $F_{z'}F_z = 0$, the inner product $\langle w|v\rangle = \langle q|p\rangle\langle b_{z'}^q|b_z^p\rangle$ consists of three cross terms bounded by $\sqrt{\delta_p}$, $\sqrt{\delta_q}$ and $\sqrt{\delta_p\delta_q}$. □

The two-expectation fact used below is the two-Hermitian-operator case of joint numerical-range convexity, equivalent to the Toeplitz–Hausdorff theorem [18]. A rank-reduction proof makes clear why the replacement stays in the original register.

Lemma A.2 (Matching two error rates)**.** *For any two Hermitian operators B_0, B_1 on a finite program register P and any density operator σ on P, there is a unit vector $|p\rangle \in P$ such that*

$$\langle p|B_z|p\rangle = \mathrm{Tr}(\sigma B_z), \qquad z = 0, 1.$$

Proof. If σ has rank $r \geq 2$, the Hermitian operators on its support form a real vector space of dimension $r^2 \geq 4$. The three real linear constraints

$$\mathrm{Tr}\, H = \mathrm{Tr}(HB_0) = \mathrm{Tr}(HB_1) = 0$$

therefore have a nonzero Hermitian solution H supported there. Its zero trace implies that it has both positive and negative eigenvalues. Since σ is positive definite on its support, the interval of t for which $\sigma + tH \geq 0$ contains a neighbourhood of zero and has finite endpoints. At either endpoint the rank strictly decreases, while the trace and both expectations remain unchanged. Repeat on the smaller support until the density operator has rank one. □

Theorem 6.1 is proved below with the same fixed ancilla and basis-state error criterion.

Proof. (1) With $\delta_p = \delta_q = 0$ the lemma gives $\langle q|p\rangle\langle b^q_{z'}|b^p_z\rangle = 0$ for all $z \neq z'$. If $\langle q|p\rangle \neq 0$, each $|b^p_z\rangle$ is orthogonal to every $|b^q_{z'}\rangle$ with $z' \neq z$, hence proportional to $|b^q_z\rangle$, and the two labelled measurements coincide. Each mixed program's errors are averages of nonnegative pure-component errors. Zero average forces every vector in its spectral decomposition to select the same measurement exactly. The pure result makes the supports for distinct measurements orthogonal. Their nonzero dimensions sum to at most d_P.

(2) Choose M targets at angles $\theta_j = 2\pi j/M$ with program states $|p_j\rangle$. For $z = 0$, $z' = 1$ and Bloch angle φ between the targets, $|\langle b^q_1|b^p_0\rangle| = |\sin(\varphi/2)|$, so the lemma gives $|\langle p_j|p_l\rangle| \leq (2\sqrt{\delta} + \delta)/\sin(\pi|j-l|/M)$. The Gram matrix Γ of the $|p_j\rangle$ has $\mathrm{Tr}\,\Gamma = M$ and, by the identity $\sum_{k=1}^{M-1} \csc^2(\pi k/M) = (M^2-1)/3$,

$$\mathrm{Tr}\,\Gamma^2 \leq M + M\,\tfrac{1}{3}(2\sqrt{\delta} + \delta)^2(M^2 - 1).$$

Then $d_P \geq \mathrm{rank}\,\Gamma \geq (\mathrm{Tr}\,\Gamma)^2/\,\mathrm{Tr}\,\Gamma^2$, which is (4). With $c = \frac{1}{3}(2\sqrt{\delta} + \delta)^2$ the maximum over M lies near $M = c^{-1/2}$ and equals $\frac{1}{2}c^{-1/2}(1 + O(\sqrt{c}))$, which gives the limit. For the converse, the executor $\sum_j |j\rangle\langle j| \otimes R(\theta_j)^\dagger$ followed by a measurement in the computational basis selects the sharp measurement along $\theta_j = 2\pi j/N$ exactly; using the nearest θ_j for a target θ misaligns the axes by at most π/N, which costs error $\sin^2(\pi/(2N)) \leq \delta$.

(3) Fix a target basis $\{|b_0\rangle, |b_1\rangle\}$ and define positive operators on the original register P by

$$B_z = \mathbf{1}_P - (\mathbf{1}_P \otimes \langle b_z|)G_z(\mathbf{1}_P \otimes |b_z\rangle), \qquad z = 0, 1.$$

A mixed program σ has target errors $\mathrm{Tr}(\sigma B_z)$. Lemma A.2 supplies a pure program in the original P matching both errors exactly. Choose it for each target angle. The fixed executor then has a pure-program family with the same error bound δ and dimension d_P, so part (2) applies. Conversely, pure programs are included among mixed programs. This equivalence concerns the two target errors, not preparation costs or equality of the complete instruments; arbitrary additional error constraints need not admit a rank-one replacement. □

For k positive error operators $B_1, \ldots, B_k$, a mixed program whose errors are all at most δ always has a pure spectral component with every error at most $k\delta$: some component has the sum of its errors no larger than the mixed sum. Exact matching of all errors need not hold when $k \geq 3$: the three positive operators $(I+\sigma_x)/2$, $(I+\sigma_y)/2$, $(I+\sigma_z)/2$ all have expectation $1/2$ in the maximally mixed qubit, while no pure state matches all three. The two-error equality is therefore stronger than this general factor-k bound.

δ	$L(\delta)$	$N(\delta)$	$L\sqrt{\delta}$	$N\sqrt{\delta}$
10^{-2}	4.15	16	0.415	1.600
10^{-4}	43.1	158	0.431	1.580
10^{-6}	432.8	1571	0.433	1.571

Table 1: Common pure- and mixed-program lower bound (4) and the grid construction for selecting every sharp qubit measurement on a great circle to worst-case basis-state error δ.

D'Ariano and Perinotti give finite exact programming for covariant measurements and polynomial approximate programming for general measurements [16]. This does not remove the sharp-selection constraint.

For N grid programs and a uniformly distributed angle the mean error is $\frac{1}{2} - \frac{N}{2\pi} \sin \frac{\pi}{N} \approx \pi^2/(12N^2)$. Doubling the register dimension therefore divides the leading mean error by four.

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Postscript: Persistence, Participation and an Unfinished Self

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A world can leave a record without leaving anyone able to use it. A reader can continue without knowing everything about itself. These two distinctions frame the question of closure in the Holographic Alaya–Field Framework: what can end, what can remain, and what would be required for participation to begin again?

The framework studies physical access rather than treating intelligibility as an attribute of matter in isolation. A channel resolves some differences and loses others. A stable relation can preserve classical labels or quantum content. A body can use retained history to change its next action, its own channels and the physical sources it produces [1, 2, 3, 4, 7, 5]. These claims connect world formation with agency, but they do not make the universe depend for its existence on a particular observer. Nor do they identify a stable observable algebra, a finite reconstruction of geometry and a spacetime metric. Each reaches a different part of the question.

1 A trace is not a continuing reader

Consider a fair classical label X copied perfectly into R two-level memories. Its joint state is

$$\rho_{XC} = \frac{1}{2}\sum_{x=0}^{1} |x\rangle\langle x| \otimes (|x\rangle\langle x|)^{\otimes R}.$$

Every memory alone is maximally mixed; the bank has no marginal purity. Nevertheless each copy determines X. Dephasing in the Z basis leaves this joint record unchanged. The product state with the same maximally mixed marginals carries no such record. The difference is in the correlations, not in the local inventories.

An archive of this kind may outlast the machinery that formed it. A later body with appropriate access can still read it. Without that machinery, however, the archive does not acquire a new observation, test an interpretation or correct its own readout. Passive protection and active participation are different physical achievements. Foundation's distinction between held content and return into another operation makes this difference constructive [9]. The continuing operation must actually be available; a comparison state used to bound an error is not a free preparation of another program.

This is also a philosophical distinction. What persists as a trace need not persist as an agent. The past can remain physically relevant through future interventions even after its

original reader is gone. Conversely, continuity of an agent need not mean preserving every past datum or every microscopic component. In the framework's agency account it means maintaining an organization that can carry usable relations into later choices, repair its channels and revise what those relations warrant [3, 8]. This is a proposed criterion of causal participation, not a claim that causal organization alone explains experience.

2 Acquisition and renewal have different limits

HAFF F bounds record acquisition at a declared initial resource boundary [6]. Its assumptions include initially independent, unused memories and a specified thermal work account. The initial purity

$$B = \sum_{k} [\ln d_k - S(\sigma_k)]$$

is then part of what can pay for new correlations, together with supplied resources and the system's own contribution. The example above explains why one cannot impose this product boundary anew at each time. Old correlations are resources even when $B = 0$. Running out of fresh purity does not erase them.

Renewal asks another question: what must be done when an existing useful relation degrades? In F's conditional-independent bank, $C_i = X \oplus N_i$, with error probability $\varepsilon < 1/2$, independent flips of probability $0 < p \le 1/2$ give

$$q = p + (1 - 2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0,$$

where $h(u) = -u \ln u - (1 - u) \ln(1 - u)$. Returning the complete retained label, bank and controller to their joint target removes $R d_{p,\varepsilon}$ nats. An independent Gibbs bath at temperature T must receive

$$Q \ge k_B T R d_{p,\varepsilon}.$$

Here the flip record is unavailable, the bath is initially independent of the noisy retained state, and the controller cannot absorb the entropy without being restored. For $p = .12$ and $\varepsilon = .05$, eight copies of one fair label require 1.9025859874 nats of correction. The number of distinct labels is one; the independent degradation of eight carriers still has a price.

A sole thermal chiral exhaust with the additional current assumptions in F and D has a quadratic finite-rate price. That result does not apply to every bath or every control pulse. Exact protection gives $d = 0$ in the bank model and defeats a blanket assertion that records require positive upkeep. Accessible noise records, correlated errors and a controller allowed to consume its own purity alter the boundary as well. The distinction matters before any cosmological extrapolation.

3 A finite-body extension

The bank is not the only setting in which a retained relation has a renewal cost. The following result treats an entire finite memory as one object, allows correlated and quantum content, and makes the noise assumption explicit. Entropies and mutual informations are in nats.

Proposition 1 (Renewing a useful relation under partial reset)**.** *Let θ_{XMC} be a target state of a retained reference X, a memory M of dimension d, and a controller C. Suppose $I_\theta(M:X) \geq \iota > 0$. During each degradation interval the unavailable noise acts as*

$$\mathcal{D}_\lambda(\theta) = (1-\lambda)\theta + \lambda\, \theta_{XC} \otimes \tau_M, \qquad \tau_M = I_M/d, \quad 0 < \lambda \leq 1.$$

A correction restores the complete target θ_{XMC}, using a bath initially independent of the noisy retained state and Gibbs at temperature T. If the correction is a possibly driven unitary on the retained system and bath, with no other entropy sink and vanishing endpoint interaction energy, then

$$Q \geq k_B T \lambda \iota. \tag{1}$$

If the total retained-system energy, including internal couplings, is equal immediately before and after the correction, and driven work and the counted bath heat are the only energy flows, then $W = Q$. A degenerate memory with additive, unchanged apparatus energies is one sufficient realization. For N renewals with $T_j \geq T_* > 0$, $\lambda_j \geq \lambda_* > 0$, and $I_{\theta_j}(M_j : X_j) \geq \iota_* > 0$, a finite total supplied work $\sum_j W_j \leq E_{\mathrm{av}}$ implies

$$N \leq \frac{E_{\mathrm{av}}}{k_B T_* \lambda_* \iota_*}. \tag{2}$$

For the last statement each interval must satisfy the preceding cyclic correction and energy conditions, even when its target differs.

Proof. Write $Z = XMC$ and $\eta = \mathcal{D}_\lambda(\theta)$. Entropy concavity gives

$$\begin{aligned} S(\eta) - S(\theta) &\geq \lambda[S(\theta_{XC}) + \ln d - S(\theta)] \\ &= \lambda[\ln d - S_\theta(M|XC)] \\ &= \lambda[\ln d - S_\theta(M) + I_\theta(M:XC)] \\ &\geq \lambda I_\theta(M:X) \geq \lambda\iota. \end{aligned}$$

The last line uses $S(M) \leq \ln d$ and monotonicity of mutual information under discarding C. The proof is valid for quantum X and for a controller correlated with the target.

Initially Z and the Gibbs bath B are independent. Entropy conservation under the correction and subadditivity at its end give $\Delta S_B \geq S(\eta) - S(\theta)$. For $\tau_B = e^{-\beta H_B} / \operatorname{Tr} e^{-\beta H_B}$,

$$\beta Q = \Delta S_B + D(\rho'_B \| \tau_B), \qquad \beta = (k_B T)^{-1}.$$

This proves (1). With vanishing endpoint bath coupling, the first law reads $W = \Delta E_{XMC} + Q$. The total-energy and energy-flow assumptions give $\Delta E_{XMC} = 0$ and hence $W = Q$; summing the lower bounds proves (2). □

The reset mixture models a lost alternative, not an accessible branch flag the controller can undo for free. It acts on the whole designated memory, so no protected subsystem within that memory escapes the stated channel. The cost is paid for restoring the joint relation, not for restoring a local marginal. A fair classical copy has a maximally mixed memory marginal and still satisfies the bound. An initially independent maximally mixed memory has $\iota = 0$ and does not.

This extends the renewal argument beyond independent copies. It does not establish that every physical body experiences this noise, that every update needs a fixed amount

of information, or that every target can be restored under a body's allowed access. The bound is necessary where such a correction exists. A consumed controller, an accumulating waste register or an accessible auxiliary noise record must be included among the resources. If the noise environment is retained, its correlations cannot be discarded when defining the target. Calling only the memory cyclic would hide the expenditure. The entropy proof permits an externally scheduled unitary; it is not a proof of autonomous apparatus closure. A physical battery that takes part in the entropy account must be included or separately shown not to supply an additional entropy sink.

Equation (2) bounds successful renewals, not necessarily calendar time. A system can postpone a last update arbitrarily far. A finite upper bound on an uninterrupted active interval additionally requires a positive minimum update rate and a bound on gaps. Hibernation, cooling toward zero temperature, decreasing task quality and improving protection are substantive alternatives, not semantic exceptions.

4 An unfinished self can continue

A demand for complete self-description makes participation seem impossible for the wrong reason. A self-model need only support the inquiries and actions on which continuation depends. T-DOME II gives a concrete example: history must retain the signed relation between the task and the body's channels. An accurately learned but unsigned axis can leave the prediction useless [8]. The useful self-frame therefore consists in how a system can reach its task, not in an inventory of everything it is.

Some distinctions are inaccessible to a body's legitimate references. Foundation II separates that failure of identifiability from disagreement, and separates agreement from correctness [9]. A hidden symmetry can make one parameter unknowable while leaving a valuable intervention completely identifiable. In other cases inquiry resolves the ambiguity, but its preparation and maintenance consume resources. A continuing body must decide which error matters and which probe can repair it before the inquiry destroys the content it was meant to help. Brain Anchor's learning-versus-survival tradeoff supplies one physical version of this problem [12].

Finite self-knowledge is consequently not equivalent to paralysis. A body may track residual errors, revise its references and retain a margin of viable action without ever obtaining a complete account of itself. The stronger philosophical claim is that an imperfect, historically formed organization can be a continuing subject precisely through this revisability. Its identity lies partly in the maintained relation between history and possible action, not in an immutable perfect image.

A rival account could identify continuation with persistence of the stored pattern. It then misses a relevant difference between two copies of the same archive: one has working probes and can revise its routing after an error, while the other's channels have permanently failed. Their stored descriptions agree; their possible next histories do not. The active criterion explains that difference without demanding exact microscopic identity. Two functioning copies may both inherit a causal history, so this criterion does not by itself select a unique first-person successor. That further identity question, like experience, cannot be decided by the survival of a data pattern alone.

That claim still needs an account of first-person experience. Consciousness III's explanatory gap cannot be crossed by renaming calibration as awareness [10]. A candidate bridge must explain why these particular internal relations matter to experience, dis-

tinguish alternatives that make the same reports, and expose a relevant intervention. Neither Proposition 1 nor a finite self-frame proves that a body feels anything. The thermodynamic argument addresses the support for continuation; the experience argument addresses what, if anything, that continuation is like.

5 A cosmological participation-window conjecture

The universe's duration and a body's active history need not have the same limit. Dyson's cooling and scaling scenario explicitly allows indefinitely continued life and communication from finite energy under its assumed laws [13]. Finite resources alone therefore do not establish a finite active window. In asymptotically de Sitter models a detector can encounter cosmological-horizon thermal radiation [14]. Krauss and Starkman examine how horizon restrictions, recoverable energy and memory requirements change the prospect of indefinite activity [15]. These are competing conditional mechanisms, not observational proof of the far future.

The framework suggests a more specific conjecture.

Conjecture 1 (Resource-limited participation)**.** *In a late-time causal domain with finite accessible usable work, a positive environmental temperature floor and unavoidable loss of task-relevant relations, any fixed-capacity body that continues at a fixed nonzero task quality can perform only finitely many successful inquiry–revision–action updates, even while passive traces and spacetime structure persist.*

The proposed mechanism is loss of correlations on which usable permissions depend. A retained history supports diagnosis; diagnosis changes routing or protection; the revised body supports another action and its physical source. Mixing damages the relations required to repeat that loop. Restoration exports entropy or consumes a fresh resource. A horizon may restrict replenishment and a thermal floor may prevent the cost per restoration from tending to zero. Exhaustion then ends this participation without erasing every trace.

The bank and partial-reset propositions establish this mechanism in two specified memory models. G supplies finite history-dependent source differences, while Imprint and Brain connect retained relations to subsequent inquiry and bodily change [7, 11, 12]. The conjecture reaches further: an interacting adaptive body can compress memories, migrate codes and replace its task. To prove it for a proposed body class one must obtain a uniform positive resource cost per successful update after optimizing those operations. Neither a count of bit erasures nor a total marginal purity inventory supplies that result automatically. The temperature floor, accessible work bound and irreducible relation loss must also be established for the actual cosmological domain. They are not consequences of selecting an algebra.

The conjecture is weakened or falsified by a body at the stipulated quality with an indefinitely usable protected relation, by vanishing renewal cost after admissible compression or migration, by a cooling law permitting a summable sequence of positive costs, or by unbounded replenishment. An accessible noise record can break the proposed loss mechanism. Discovery of one such regime would identify which part of the mechanism fails, rather than merely extending a numerical lifetime. A static surviving archive is not a counterexample to this active claim, but a continuing reader with the stated resources would be.

The sustained selection–response conjecture in D has different resource quantifiers [4]: it asks for families of longer runs whose installed energy, size and operating domain may grow, while one body is reused within each run. The participation conjecture here fixes capacity and limits the total accessible work. Neither quantifier refutes the other. Foundation III's finite semiclassical mean-action source–response construction [5] concerns a specified operating window; it does not establish indefinitely renewed thermal operation.

6 Origin, return and what closure would mean

An unused record port and an exhausted one can be locally indistinguishable and have different futures. The former may be independent of the source and backed by a usable gradient. The latter can carry correlations with an old history or lack the resources needed for another operation. Allowed future interventions can distinguish them. Equal local states therefore do not identify origin with terminus.

The stronger origin proposal in G is physical formation of permissions: a body with locally adjustable relations encounters diagnostic tasks, retains their results and uses them to install a useful coupling [7]. Exploration, a distinguishable value difference and enough survival time must be supplied by the primitive dynamics. This moves the problem beyond an external program name, without proving that every supplied executor arises in this way. Enlarging an executor introduces another joint interaction to explain; it does not prove a universal final boundary or forbid a finite system from carrying a useful description of itself.

A cosmological return would require more still. One concrete mechanism candidate is that contraction or another geometric transition creates newly accessible nonequilibrium gradients; interactions then allocate part of that work to fresh record ports and bodies, while old correlations and exported entropy remain accounted for in other degrees of freedom. A repeated active phase would be a renewal of resources and formation, not necessarily recurrence of the global state. The physical transition must have its own dynamics, show that the gradients can be used, and explain why old entropy does not obstruct the new phase. Local thermal swaps or recurrence of a finite register do not establish any of these cosmological steps. A bounce that restores density but no usable work would fail the mechanism; so would a transition requiring an uncharged deletion of the old history. No such return is forced by the participation-window conjecture.

Closure thus has several meanings that must not be confused. A record can become unreadable to one body; a body can lose the capacity to revise itself; a cosmic domain can cease to support new active readers. None entails the nonexistence of its remaining physical structures. Nor must an imperfect self finish knowing itself before it can live. The framework's most consequential prospect is that history changes what a body can do next, and that this relation can be maintained and revised for a time without being complete.

A history can remain after the means of continuing it have gone. A continuing self need not wait to become a finished account of itself.

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